Science Dept
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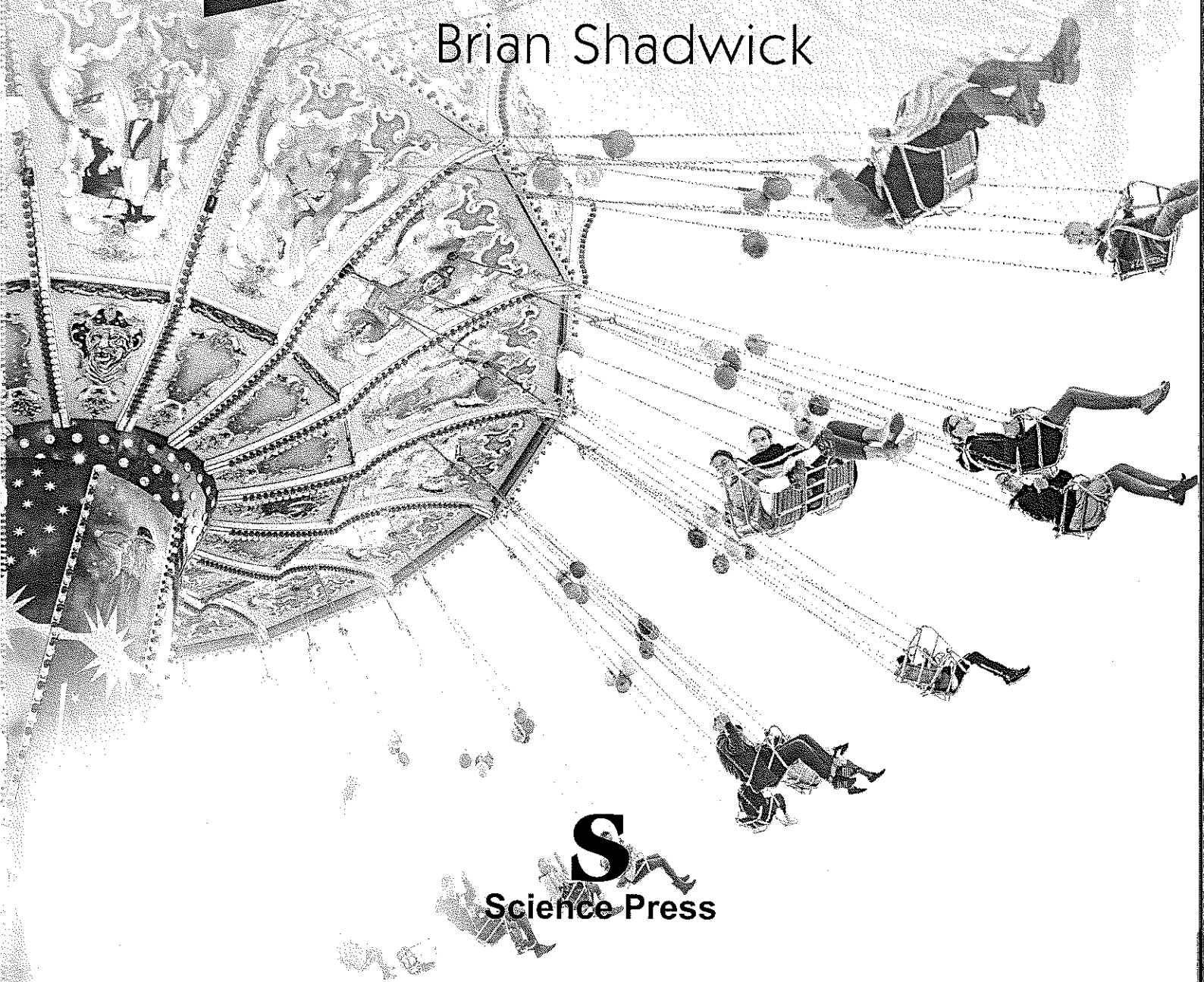
Q+A Questions and Answers

NATIONAL PHYSICS

Unit 3 Gravity and Electromagnetism

Brian Shadwick



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Words to Watch

account, account for State reasons for, report on, give an account of, narrate a series of events or transactions.

analyse Interpret data to reach conclusions.

annotate Add brief notes to a diagram or graph.

apply Put to use in a particular situation.

assess Make a judgement about the value of something.

calculate Find a numerical answer.

clarify Make clear or plain.

classify Arrange into classes, groups or categories.

comment Give a judgement based on a given statement or result of a calculation.

compare Estimate, measure or note how things are similar or different.

construct Represent or develop in graphical form.

contrast Show how things are different or opposite.

create Originate or bring into existence.

deduce Reach a conclusion from given information.

define Give the precise meaning of a word, phrase or physical quantity.

demonstrate Show by example.

derive Manipulate a mathematical relationship(s) to give a new equation or relationship.

describe Give a detailed account.

design Produce a plan, simulation or model.

determine Find the only possible answer.

discuss Talk or write about a topic, taking into account different issues or ideas.

distinguish Give differences between two or more different items.

draw Represent by means of pencil lines.

estimate Find an approximate value for an unknown quantity.

evaluate Assess the implications and limitations.

examine Inquire into.

explain Make something clear or easy to understand.

extract Choose relevant and/or appropriate details.

extrapolate Infer from what is known.

hypothesise Suggest an explanation for a group of facts or phenomena.

identify Recognise and name.

interpret Draw meaning from.

investigate Plan, inquire into and draw conclusions about.

justify Support an argument or conclusion.

label Add labels to a diagram.

list Give a sequence of names or other brief answers.

measure Find a value for a quantity.

outline Give a brief account or summary.

plan Use strategies to develop a series of steps or processes.

predict Give an expected result.

propose Put forward a plan or suggestion for consideration or action.

recall Present remembered ideas, facts or experiences.

relate Tell or report about happenings, events or circumstances.

represent Use words, images or symbols to convey meaning.

select Choose in preference to another or others.

sequence Arrange in order.

show Give the steps in a calculation or derivation.

sketch Make a quick, rough drawing of something.

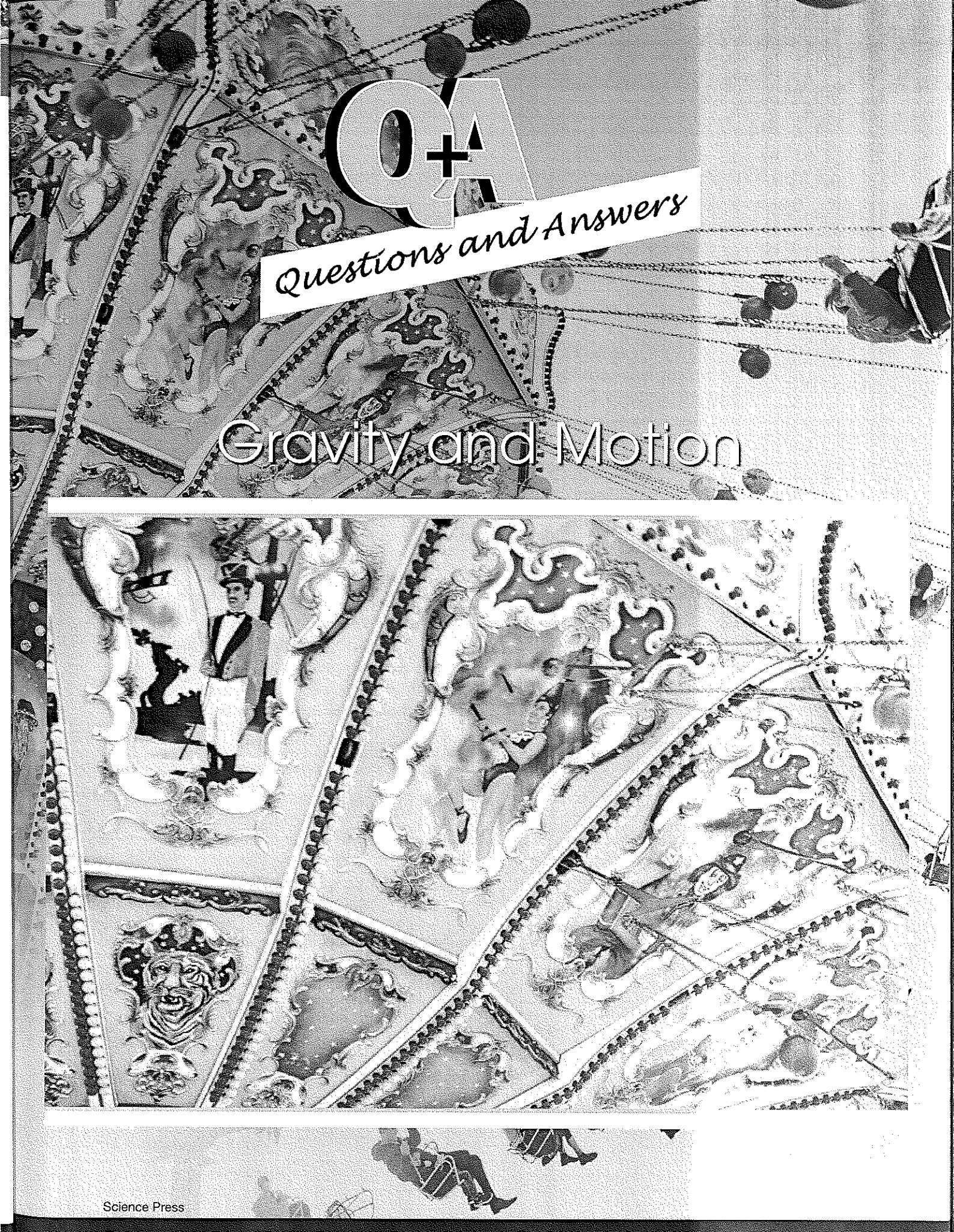
solve Work out the answer to a problem.

state Give a specific name, value or other brief answer.

suggest Put forward an idea for consideration.

summarise Give a brief statement of the main points.

synthesise Combine various elements to make a whole.



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SET 1 Mass and Weight

1. (a) Define mass.

(b) How does the mass of an object change when it is moved from one position in the Universe to another position?

(c) Give three different units used to measure mass, and their symbols.

2. (a) What is weight?

(b) How does the weight of an object change when it is moved from one position in the Universe to another position?

(c) Give the unit used to measure weight, and its symbols.

(d) Write down the equation that connects mass and weight.

3. Convert to weight.

(a) 1 gram

(b) 85 g

(c) 0.55 kg

(d) 246 kg

(e) 2012 kg

(f) 1.6 tonnes

4. Convert to mass.

(a) 0.4 N

(b) 8 N

(c) 245 N

(d) 490 N

(e) 9800 N

(f) 65 kN

5. (a) Calculate the mass of a 60 kg person on the Moon. (Moon acceleration due to gravity is about 1.6 m s^{-2}.)

(b) Calculate this person's weight on the Moon.

(c) Predict his weight on Earth.

(d) Calculate the weight of a 70 kg astronaut on Earth.

(e) Calculate his mass on the Moon.

(f) Calculate his weight on the Moon.

6. List three essential differences between mass and weight.

7. Jupiter has a gravitational acceleration that is 2.53 times greater than that of Earth. An object on Jupiter weighs 506 N.

(a) What is the mass of the object?

(b) What will be its weight on Earth?

8. A 60 kg astronaut travels to a planet where acceleration due to gravity is 0.38 times that of Earth.

(a) What is the astronaut's weight on Earth?

(b) What is his mass on the planet?

(c) What is his weight on the planet?

9. On Earth an astronaut steps onto a set of scales. As she does, the surface of the scales moves down and compresses a spring in the workings. Gearing changes this movement into the horizontal swing of the pointer. The pointer becomes stationary at the 60 kg mark.

(a) Explain if the scales are measuring the astronaut's mass or weight.

(b) What would the scales read on the Moon? Justify your answer.

10. Calculate the weight of a 10 kg object at each of the following cities on Earth.

City	Gravitational field at this position (m s^{-2})
Sydney	9.7979
Hong Kong	9.8099
Toronto	9.8054
Helsinki	9.8189
Baghdad	9.7964
Seoul	9.7994
Manila	9.7844
New York	9.8024
Mexico City	9.7799

SET 2 Gravitational Field

For relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

1. (a) What is a field?

(b) More specifically, what is a gravitational field?

(c) Where do we find gravitational fields?

(d) Define gravitational field strength.

(e) Give the mathematical equation for gravitational field strength.

(f) What are the units for gravitational field strength?

(g) What do we commonly refer to gravitational field as?

2. (a) What property of a mass changes if its position in a gravitational field changes?

(b) State the relationship between the work done on an object and its changing position in a gravitational field.

3. (a) Explain why the gravitational field strength at the surface of the Earth is not constant.

(b) Given that the gravitational force acting on an object gives it its weight force, by equating the two equations for gravitational force and weight force, derive another equation for gravitational field strength.

(c) Use this equation to explain why the gravitational field strength due to the Earth decreases as we go further away from the Earth.

(d) What does the '*r*' (or perhaps you used *R*, or *d*) in your equation stand for?

4. (a) Recall the equation we used in earlier work to calculate the changes in the gravitational potential energy of masses close to the surface of the Earth.

(b) When calculating these changes, what does the '*h*' or Δh stand for?

(c) Explain why we cannot use this equation to calculate the gravitational potential energy changes for objects which are not close to the Earth's surface.

5. (a) What does the work to cause the energy change when an object moves from a higher gravitational potential energy position to a lower gravitational potential energy position?

(b) What does the work to cause the energy change when an object moves from a lower gravitational potential energy position to a higher gravitational potential energy position?

6. An astronaut in his space suit weighs 882 N on Earth where the gravitational field strength is 9.8 N kg^{-1}.

(a) What would he weigh on a planet where the gravitational field strength was 2.2 N kg^{-1}?

(b) What would he weigh on a planet where the gravitational field strength was 12.2 N kg^{-1}?

7. Compare the gravitational field strength at sea level on Earth with the field strength on top of Mount Everest. Explain your answer.

8. (a) An 80 kg man moves from a place on Earth where the gravitational field strength is 9.82 N kg^{-1} to another place where the field strength is 9.77 N kg^{-1}. By how much does his mass change?

(b) Find his change in weight.

9. (a) Explain, in terms of gravitational field strength, why all objects, regardless of mass, will fall towards the Earth at the same rate when they are dropped from the same height.

(b) Will the rate at which they fall change if they are dropped from a significantly higher height? Justify your answer.

10. Planet X has a mass 6 times greater than that of the Earth and twice the Earth's diameter. What will be the strength of the gravitational field on its surface?

11. (a) Given the mass of our Sun as 2.00×10^{30} kg, and its diameter as 1 392 530 km, find the strength of its gravitational field on its surface.

(b) Estimate the gravitational field strength on the surface of a neutron star that has five times the mass of our Sun, and a radius of 10 km.

12. What is the acceleration due to gravity at an altitude of 2×10^5 m above the Earth's surface?

13. A hypothetical planet has a radius three times that of Earth, but has the same mass. What is the acceleration due to gravity near its surface?

14. Find the altitude above the Earth's surface where Earth's gravitational field strength would be two-thirds of its value at the surface.

15. The gravitational field intensity at the surface of a planet is 3.4 N kg^{-1}. If a planet's mass is 6.8×10^{22} kg, what is its radius?

16. Calculate the acceleration of an 85 kg astronaut on the International Space Station (ISS) when the ISS is at a height of 350 km above Earth's surface.

17. Compare the gravitational field strength if you were in a satellite in orbit 1000 km above the Earth to what it would be if you were in orbit 1000 km above Mars (take the mass of Mars as 6.42×10^{23} kg, and its radius as 3400 km).

18. At what distance from the Earth's surface is the gravitational field strength 7.33 N kg^{-1}?

19. On the surface of Planet X an object has a weight of 63.5 N and a mass of 22.5 kg. What is the gravitational field strength on the surface of Planet X?

20. On the surface of Planet Y, which has a mass of 4.83×10^{24} kg, an object has a weight of 50.0 N and a mass of 30.0 kg. What is the radius of this planet?

21. What is the gravitational field strength 1.27×10^7 m above the Earth's surface?

22. Planet B has a mass of 4.0×10^{22} kg and a radius of 5.0×10^5 m. What is the gravitational field strength on the surface of planet B?

23. (a) A receiver at sea level detects a signal from a satellite in a circular orbit when it is passing directly overhead. The microwave signal is received 75 ms after it was transmitted from the satellite. Calculate the altitude of the satellite. (Velocity of electromagnetic radiation = 3×10^8 m s^{-1}.)

(b) Determine the gravitational field strength of the Earth at the position of the satellite.

24. A 100 kg astronaut feels a gravitational force of 750 N when placed in the gravitational field of a planet.

(a) What is the gravitational field strength at the location of the astronaut?

(b) What is the mass of the planet if the astronaut is 2×10^6 m from its centre?

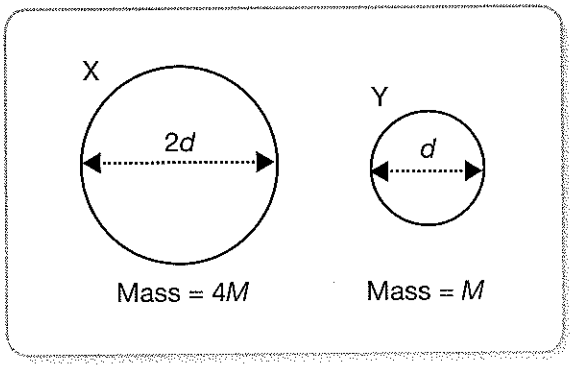
25. The gravitational field strength at the surface of a planet, X, is 18 N kg^{-1}.

(a) Calculate the minimum amount of energy required to lift a 7.0 kg rock a vertical distance of 20 m from the surface of X if the gravitational field can be considered to be uniform over such a small distance.

(b) State whether your answer to (a) would be different if the 7.0 kg mass were lifted a vertical distance of 20 m from a point near the top of the highest mountain on planet X. Explain your answer.

(c) Calculate the gravitational field strength at the surface of another planet, Y, that has the same mass as planet X, but twice the diameter of X.

26. Consider the two planets shown below. Their masses and diameters are given.



If the gravitational field on the surface of X is 8 m s^{-2}, what is it on the surface of Y?

SET 3 Newton's Law of Universal Gravitation

For relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

1. Determine the gravitational force between two 10 kg masses 6 m apart.

2. Determine the gravitational force acting on a 500 kg satellite which is orbiting the Earth at an altitude of 300 km. The radius of Earth is 6380 km and its mass is 6×10^{24} kg.

3. Two objects have a gravitational force of F newtons acting between them. Predict what happens to this force if:

(a) The distance between the objects is doubled.

(b) The distance between the objects is halved.

(c) One of the objects is replaced by an object with twice the mass of the original object.

(d) Both objects are replaced by objects which each have three times the original mass.

(e) The distance between the objects is tripled *and* one of the objects is replaced by an object with four times the initial mass.

4. Two planets have masses M and $3M$, and radii R and $2R$ respectively.

(a) Compare the weight (as a ratio) of a 10 kg mass on the surface of each.

(b) How would this ratio change if the masses were 50 kg masses instead of 10 kg masses?

5. Given that the mass of the Moon is 7.34×10^{22} kg and that its average diameter is 3.48×10^{3} km, find:

(a) The gravitational force acting on a 50 kg mass on its surface.

(b) The gravitational force on a 500 kg satellite which orbits the Moon at an altitude of 300 km.

(c) The gravitational force on a 500 kg satellite orbiting the Moon with an orbital radius of 2000 km.

(d) The strength of the gravitational field at an altitude of 300 km above the surface of the Moon.

(e) The strength of the gravitational field in the orbital position of the satellite in (c).

6. Use the data in the table to answer this question.

Planet	Diameter (km)	Mass (kg)
Mars	6.79×10^{3}	6.42×10^{23}
Jupiter	1.43×10^{5}	1.90×10^{27}
Saturn	1.20×10^{5}	5.69×10^{26}
Uranus	5.18×10^{4}	8.68×10^{25}

(a) Find the gravitational force between Mars and a 50 kg mass on its surface.

(b) Find the gravitational force between Jupiter and a 50 kg mass on its surface.

(c) Find the weight of a 50 kg mass on the surface of Saturn.

(d) Find the weight of a 50 kg mass on the surface of Uranus.

(e) Find the weight of a 50 kg mass which is orbiting Mars at an altitude of 600 km.

(f) Find the weight of a 50 kg mass which is orbiting Jupiter at an altitude of 600 km.

(g) Find the gravitational force between Saturn and a 50 kg mass in orbit at an altitude of 400 km.

(h) Find the gravitational force between Uranus and a 50 kg mass in an orbit with an orbital radius of 2.70×10^{7} km.

(i) Find the gravitational field strength on the surface of Mars.

(j) Find the gravitational field strength 300 km above the surface of Jupiter.

(k) Compare the gravitational field strengths on the surfaces of Saturn and Uranus.

7. If the mass of Titan, one of Jupiter's moons, is 1.35×10^{23} kg and its radius is 2570 km determine:

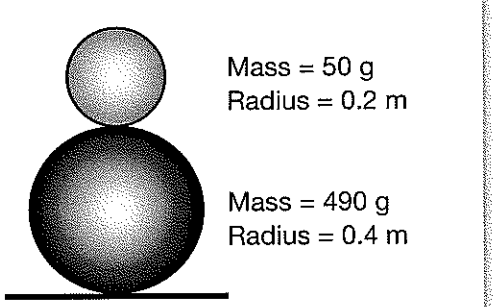
(a) The acceleration due to gravity at its surface.

(b) The force on a 750 kg rock which is on the surface of Titan.

8. (a) What force does Earth exert on an 80.0 kg astronaut at an altitude equivalent to 2.5 times Earth's radius?

(b) What is the gravitational field strength of the Earth at this position?

9. A 490 kg solid sphere with a radius of 0.40 m is on the surface of the Earth. Resting on top of this sphere is a second sphere, radius 0.2 m and mass 50 g as shown in the diagram.



(a) Find the net force the top sphere exerts on the bottom sphere.

(b) Find the magnitude of the gravitational force exerted by the bottom sphere on the top sphere.

10. If the mass of the Moon is 7.35×10^{22} kg and the orbital radius of the Moon about the Earth is 3.85×10^{5} km, determine the force of gravitational attraction between the Earth and the Moon.

11. A planet has two moons. The first is 3 times as far from the centre of the planet as the second and has a mass that is 4 times the mass of the second. What is the ratio of the gravitational force on the first planet to the gravitational force on the second?

12. Two spherical balls are placed so their centres are 3.72 m apart. The force between them is 2.45×10^{-8} N. If the mass of the smaller ball is 54.3 kg, what is the mass of the other ball?

13. How far (in km) from the centre of the Earth would a person with a weight of 470 N on the surface of the Earth have to be to have a weight of 56.2 N?

14. A mass of 20.0 kg is located 4.0 m to the right of a mass of 30.0 kg.

(a) What is the force on the 20.0 kg mass?

(b) What is the force on the 30.0 kg mass?

15. Christian and Mia have masses of 55 kg and 49 kg respectively. They sit 1.5 m apart in the front centre of Mr S's physics class. Determine the magnitude of the force of gravitational attraction between them.

16. The centres of two masses, X and Y, are separated by a distance R. The force of attraction between the two masses is F. How does this force change if the following changes are made?

(a) The mass of X is doubled.

(b) The distance between the masses was doubled.

(c) The mass of X was doubled and the mass of Y was tripled.

(d) Both the masses were halved, as was the distance between them.

(e) Mass X was halved, mass Y was tripled, and the distance between them increased by 50%.

17. Calculate the distance between two 150 kg masses that have a gravitational attraction of 8.34×10^{-7} N.

18. How far above the Earth's surface would you have to be for your weight to be only one fifth (0.2) times your weight on the surface of the Earth?

19. Compare your weight in a satellite in orbit 1000 km above the Earth to what it would be if you were in orbit 1000 km above Mars (take the mass of Mars as 6.42×10^{23} kg, and its radius as 3400 km).

20. The mass of Pluto is 1.3×10^{22} kg. Pluto is 5.76×10^{9} km from Earth. What is the force of gravity between Pluto and Earth?

SET 4 Gravitational Potential Energy

For relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

1. Where would an object be if its gravitational potential energy was zero? Explain your answer.

2. Define gravitational potential energy for an object which is a significant distance above the surface of the Earth (or any other planet), giving the equation we use to find it, and the units we use to measure it.

3. Explain why we do not use the equation $E_p = mgh$ to calculate the gravitational potential energy of an object in space.

4. Explain, using appropriate laws of physics and mathematical equations if appropriate, why the gravitational potential energy of an object at a particular point in space is always a negative quantity.

5. Calculate the gravitational potential energy between two satellites, each with a mass of 500 kg, 5 km apart in space.

6. A 7 tonne satellite orbiting Earth has an orbital radius of 7100 km. What is its gravitational potential energy? Take the mass of the Earth as 6×10^{24} kg.

7. The Hubble telescope, mass 11 340 kg orbits Earth at an altitude of 595 km. What is its gravitational potential energy? Take the mass of the Earth as 5.974×10^{24} kg, and its diameter as 12 760 km.

8. (a) Find the gravitational potential energy of the Moon with respect to the Earth. The mass of the Moon is 7.34×10^{22} kilograms and the mass of the Earth is 5.974×10^{24} kilograms. The orbital radius of the Moon about the Earth is 384 400 kilometres.

(b) The orbital radius data in (a) includes the radius of the Earth and the radius of the Moon. However, in questions like 5, and 6 and 7 above, we did not include the size of the satellite in our deliberations. Why could the satellite's size be ignored?

9. Use the data in the table to answer this question.

Planet	Diameter (km)	Mass (kg)
Mercury	4.88×10^3	3.58×10^{23}
Venus	1.21×10^4	4.87×10^{24}
Earth	1.28×10^4	5.97×10^{24}
Neptune	4.93×10^4	1.02×10^{26}

(a) Find the gravitational potential energy of a 75 kg mass on the surface of Mercury.

(b) Find the gravitational potential energy of a 75 kg mass on the surface of Venus.

(c) Find the gravitational potential energy of a 150 kg mass 400 km above the surface of Earth.

(d) Find the gravitational potential energy of a 150 kg mass 800 km above the surface of Neptune.

(e) Find the gravitational potential energy of a 250 kg satellite in orbit at an altitude of 3000 km above the surface of Mercury.

(f) Find the gravitational potential energy of a 250 kg satellite orbiting Venus with an orbital radius of 6.5×10^6 km.

SET 5 Changes in Gravitational Potential Energy

For relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

1. (a) What is the gravitational potential energy between a 100 kg object and a 200 kg object 25 m apart?

(b) If the two objects are now moved until they are 50 m apart, will the gravitational potential energy between them increase or decrease?

(c) What will happen to the gravitational force between them?

(d) Explain why your answers to (b) and (c) are different. (If they aren't, they should be!)

2. A 500 kg satellite is in orbit around the Earth at an altitude of 10 000 km. Calculate the work needed to be done to put this satellite into a new orbit of altitude 20 000 km.

3. A 250 kg satellite has an orbital radius of 6680 km. By how much will its gravitational potential energy change if this orbit is changed to one with an orbital radius of 7380 km?

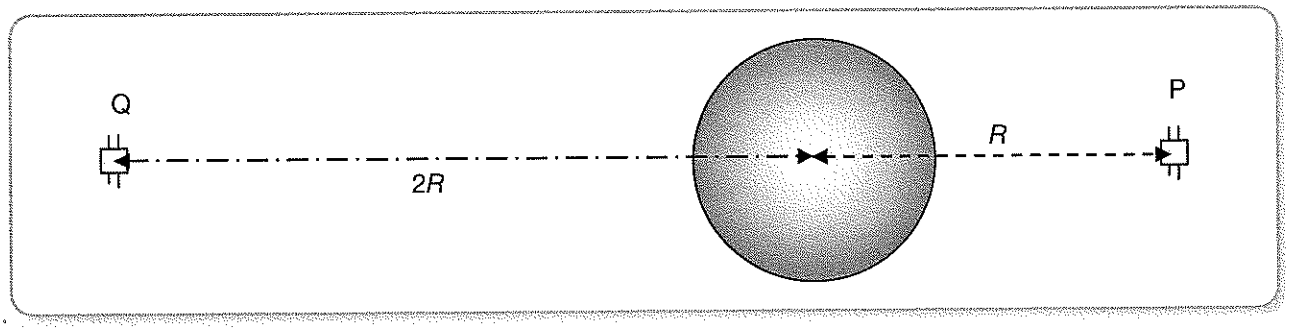
4. Satellite P, mass 250 kg, in orbit around the Earth, has gravitational potential energy equal to -1.37×10^{10} J.

(a) Calculate the orbital radius of this satellite.

(b) Calculate its altitude.

(c) Calculate the change in its gravitational potential energy if this altitude was doubled.

5. A satellite, X, is in orbit around the Earth with an orbital radius of R km, in position P as shown in the diagram. In this position it has $-E$ joules of gravitational potential energy.



(a) What would be the gravitational potential energy of this satellite if it was moved to position Q, with an orbital radius of $2R$ km?

(b) A new satellite with triple the mass of X is placed into orbit in position P. What will be its gravitational potential energy at P?

(c) By how much will its gravitational potential energy change if it is moved to position Q?

6. Consider the information in the table.

Planet	Diameter (km)	Mass (kg)
Mercury	4.88×10^3	3.58×10^{23}
Venus	1.21×10^4	4.87×10^{24}
Earth	1.28×10^4	5.97×10^{24}
Neptune	4.93×10^4	1.02×10^{26}

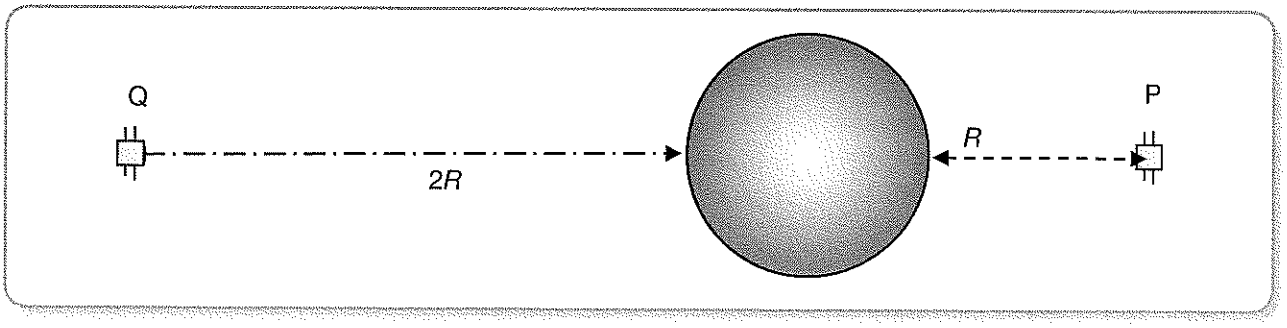
Imagine identical satellites in orbit above each of these planets with the same orbital radius.

(a) Which satellite would have the largest gravitational potential energy? Justify your answer.

(b) Which satellite would have the smallest gravitational potential energy? Justify your answer.

(c) A satellite is in orbit around Mercury with an orbital radius of R. What will be the orbital radius of this satellite around Venus in order to have the same gravitational potential energy?

7. E joules of work are done to put satellite P into orbit at an altitude of R km above the Earth. A second, identical satellite, Q, is put into an orbit at twice the altitude of the first satellite as shown in the diagram.



(a) Would twice as much work need to be done to put this second satellite into orbit? Justify you answer.

(b) What would be the value of the altitude of the second satellite if twice the amount of work was done on it in putting it into orbit compared to the first satellite?

8. Satellite S, mass 500 kg is in orbit around a planet of mass 3×10^{23} kg, and diameter 8000 km, at an altitude of 12 000 km.

(a) What is the gravitational potential energy of S at this altitude?

(b) By how much will its gravitational potential energy change if its orbital radius is doubled?

(c) By how much will its gravitational potential energy change if its orbital radius is halved?

(d) By how much will its gravitational potential energy change if its altitude is doubled?

(e) By how much will its gravitational potential energy change if its altitude is halved?

9. (a) Determine the change in the gravitational potential energy of a 100 kg object if it was moved from the surface of the Earth to the top of a 250 m high building by two different equations.

(b) Use your answers to justify our use of the equation $\Delta E_P = m \times 9.8 \times \Delta h$ for objects close to the Earth's surface.

10. Jupiter has a mass of 1.90×10^{27} kg and a diameter of 1.43×10^5 km.

(a) What is the strength of the gravitational field on the surface of Jupiter?

(b) What work is done in putting a 1.5 tonne satellite into an orbit with an altitude of 15 000 km?

(c) What additional work would be needed to put the satellite into orbit at an altitude of 25 000 km?

11. A 250 kg satellite orbits a planet at an orbital radius of 8731 km where the field strength is 35.9 N kg^{-1}.

(a) Show that the gravitational potential energy of an object of mass m_o in orbit with an orbital radius of R is given by the equation $E_P = gm_oR$.

(b) Use the equation from (a) to find its gravitational potential energy at this position.

(c) Use your answer to (a) to calculate the mass of the planet.

(d) Its orbit changes to a position where the gravitational field strength is 37.5 N kg^{-1}. Is this new position at a lower or higher altitude? Justify your answer.

(e) Calculate the altitude of the new position.

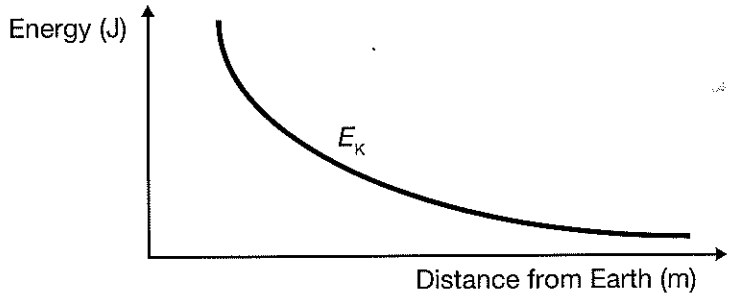
12. (a) What would be the altitude of the orbit of a 300 kg satellite orbiting Earth if its gravitational potential energy is -5×10^9 J?

(b) What would be the gravitational field strength at the orbital altitude?

(c) What would be the orbital radius of the satellite if it was placed into a different orbit where the gravitational field strength is double the amount you calculated in (b)?

(d) By how much would its gravitational potential energy change in moving it to this new position?

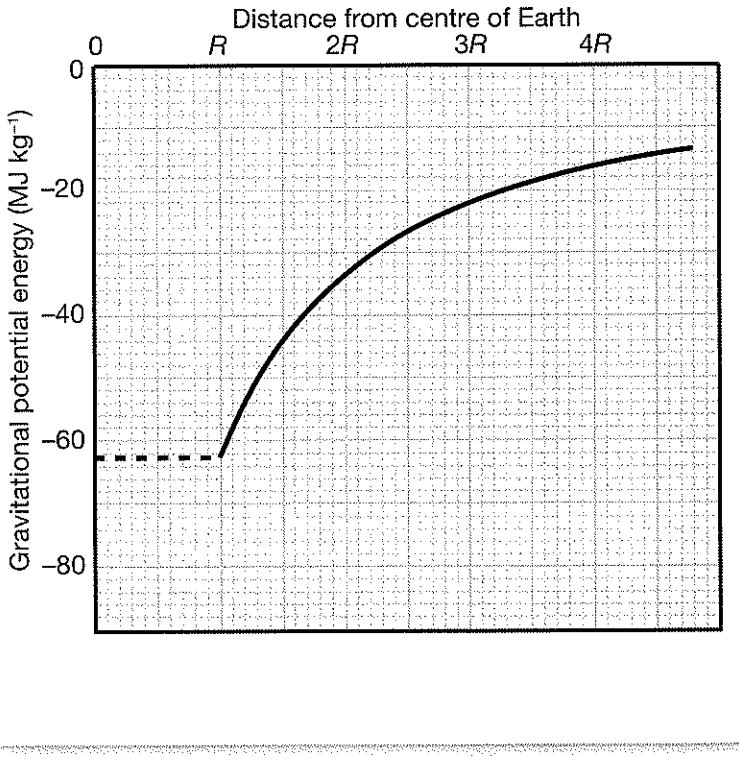
13. The graph shows how the kinetic energy (E_K) of a spaceship changes as it moves away from the Earth. The engines of the ship are turned off, and it is coasting through space with constant speed.



(a) Draw another graph, or perhaps another line on the same graph, to show how the gravitational potential energy, E_P of this spaceship changes as it moves away from the Earth.

(b) Justify your answer to (a) in terms of the principles of physics you used to answer it.

14. The graph shows how the gravitational potential energy varies with distance in the region above the surface of the Earth. R is the radius of the Earth, which is 6380 km.



(a) Calculate the gravitational potential energy at the surface of the Earth (this would be equal to the gravitational potential energy possessed by a 1 kg mass at the surface).

(b) Use the graph to calculate:

(i) The gravitational potential energy at a distance $2R$ from the centre of the Earth.

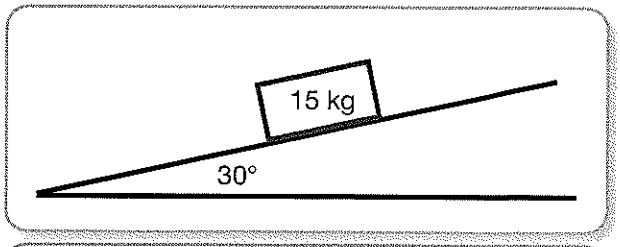
(ii) The change in the potential energy of a 1200 kg satellite when it is raised from the surface of the Earth into a circular orbit of radius $3R$.

(c) Use the data in the graph to determine the gravitational field strength at a distance $2R$ from the centre of the Earth.

SET 6 Motion on an Inclined Plane

1. Show, by drawing a labelled vector diagram, that the acceleration of an object on an incline is given by $a = g \sin \theta$, where θ is the angle of inclination of the incline.

2. A 15 kg block sits on a smooth ramp which is inclined at 30° as shown in the diagram.

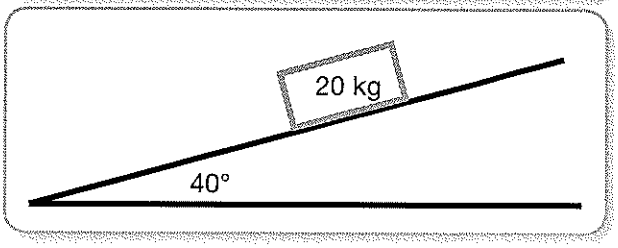


(a) Calculate the component of the weight force acting parallel to the ramp.

(b) Calculate the net force acting on the block.

(c) What will be the block's acceleration?

3. A 20 kg block sits on a ramp which is inclined at 40° as shown in the diagram. The ramp applies a frictional force of 0.15 N kg^{-1} on the block.



(a) Copy the diagram and label all the forces acting on it.

(b) By drawing a vector diagram, show that there is a net force acting on the block.

(c) Calculate the component of the weight force acting parallel to the ramp.

(d) Calculate the total frictional force acting on the block.

(e) Calculate the net force acting on the block.

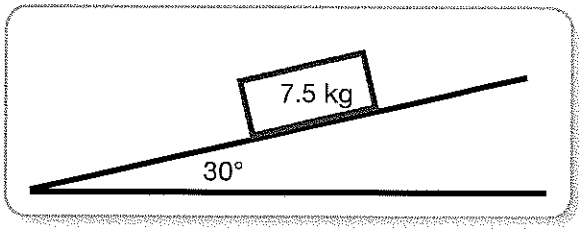
(f) What will be the block's acceleration?

4. A barrel of mass 150 kg is held at rest on a smooth ramp inclined at 45° by a string parallel to the ramp.

(a) What is the tension in the string?

(b) What is the force the ramp puts on the barrel?

5. A 7.55 kg block sits on a ramp which is inclined at 30° as shown in the diagram. The block is sliding down the ramp with an acceleration of 3.4 m s^{-2}.



(a) Calculate the component of the weight force acting parallel to the ramp.

(b) Calculate the net force acting on the block.

(c) Calculate the frictional force in N kg^{-1} acting on the block.

6. A mass of 200 kg is being pulled up a 20° incline by a rope which is parallel to the incline with an acceleration of 0.25 m s^{-2}.

(a) Calculate the net force acting on the mass.

(b) If the friction between the mass and the incline is 250 N, find the tension in the rope.

(c) Calculate the force the incline places on the mass.

7. A 54 kg cyclist is riding his 16 kg bike down a 15° incline and is accelerating at 0.8 m s^{-2}.

(a) Calculate the total net force acting on the cyclist and the bike.

(b) Determine the downhill component of the weight of the cyclist and the bike.

(c) What total frictional force is acting on the bike?

8. A 2.5 tonne truck moves up a smooth 30° incline with a constant speed of 3.0 m s^{-1}.

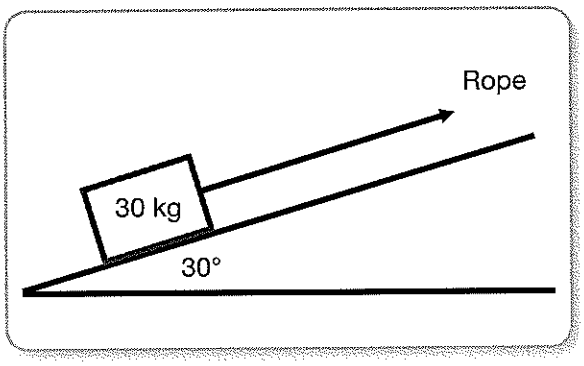
(a) What is the net force on the truck?

(b) What is the component of the weight force of the truck down the incline?

(c) What force must be applied to the truck so that it moves up the incline at constant speed?

(d) If the truck's engine suddenly cut out and the brakes failed to work, what would the truck do?

9. A 30 kg block sits on a ramp inclined at 30° as shown in the diagram. The block is pulled up the ramp with an acceleration of 0.75 m s^{-2} by a rope which is parallel to the surface of the incline.



(a) Calculate the component of the weight force acting parallel to the ramp.

(b) Find the net force acting on the block.

(c) Calculate the tensional force in the rope if the acceleration of the block remains the same, but there is a total frictional force of 5.0 N opposing the motion of the block up the ramp.

(d) Find the new net force acting on the block.

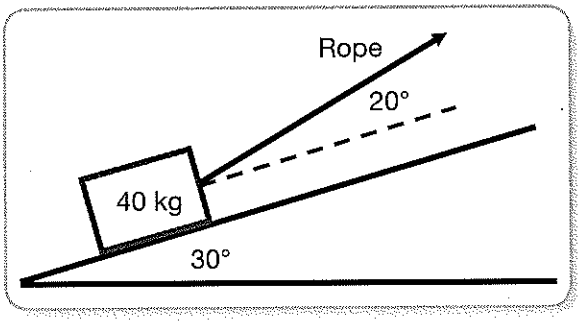
(e) Calculate the tensional force in the rope.

10. A block of mass 300 kg is held at rest on a smooth ramp inclined at 15° by a rope which is inclined at 20° to the ramp.

(a) What is the tension in the rope?

(b) What is the force the ramp puts on the block?

11. A 40 kg block sits on a smooth ramp which is inclined at 30° as shown in the diagram. The block is pulled up the ramp with an acceleration of 0.75 m s^{-2} by a rope which is inclined at 20° to the ramp.

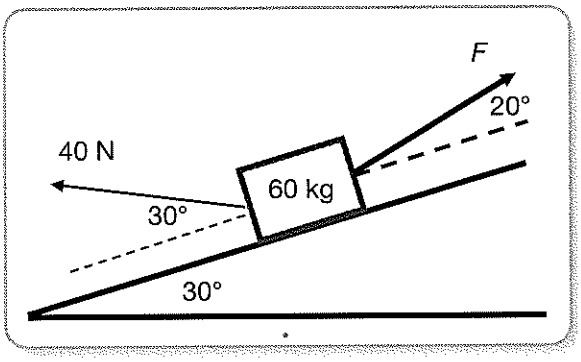


(a) What is the net force acting on the block?

(b) Find the component of the weight force of the block acting down the incline.

(c) What is the tension in the rope?

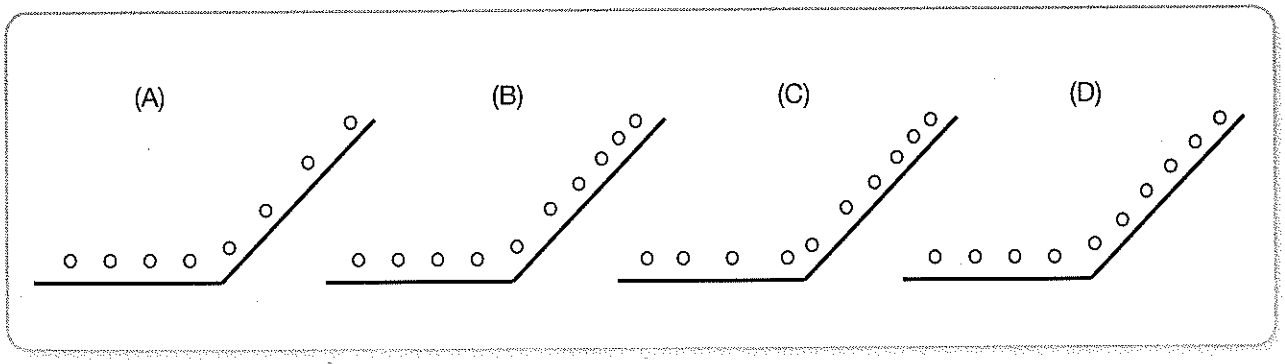
12. A block is at rest on an inclined plane as shown in the diagram.



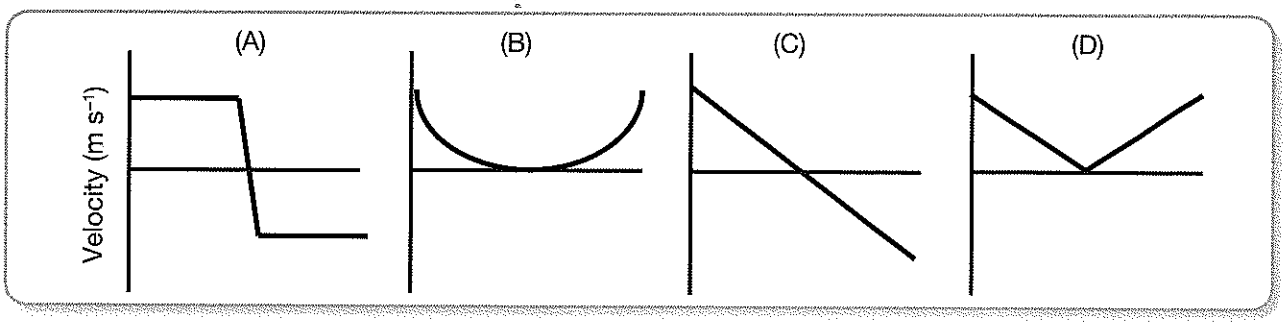
(a) What is the value of *F*?

(b) What will be the value of the normal reaction of the surface on the block?

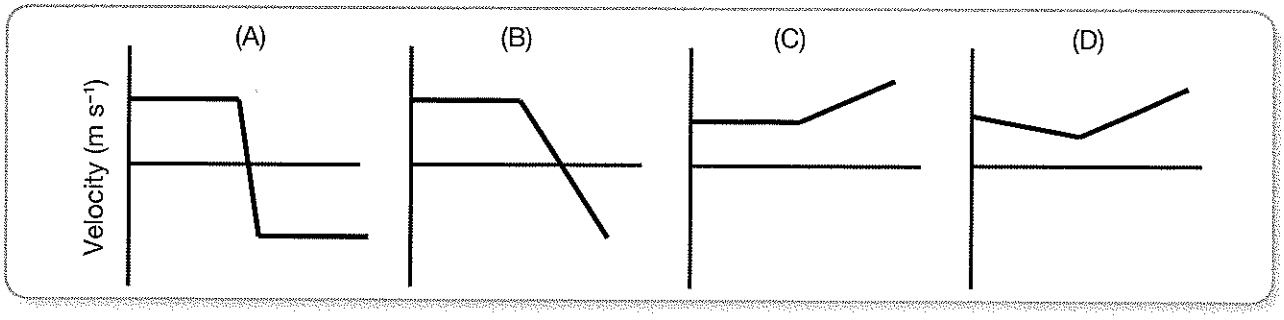
13. Two boys are playing ice hockey on a street in the middle of winter when a thin layer of ice covers the road. A stray puck travels across the frictionless ice and then up the friction-free incline of a driveway. Which of the following diagrams best represents the motion of the puck as it travels across the level street and then up the driveway?



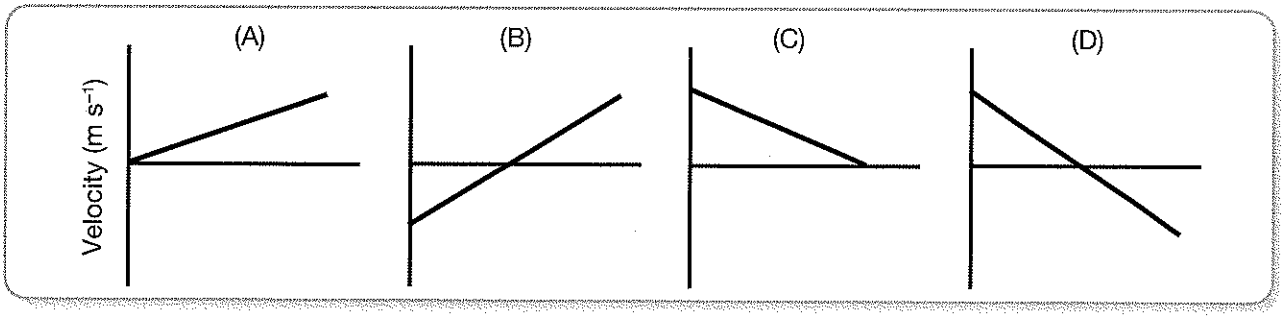
14. Jack stands at the bottom of the driveway and kicks a soccer ball. The ball rolls northward up the driveway and then rolls back to Johnny. Which one of the following velocity-time graphs best represents the velocity of the ball as it rolls up the driveway and back down to Jack?



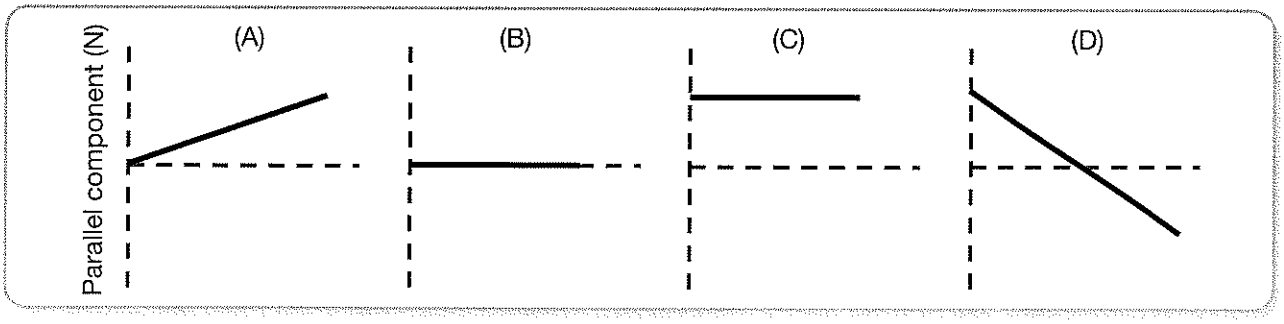
15. A bowling ball was bowled across the rough concrete floor of a warehouse, out the door and down the driveway. Which of the velocity-time graphs below would be an appropriate representation of the ball's motion as it rolls across the horizontal surface and then down the incline? Consider frictional forces.



16. Which of the velocity-time graphs would represent the motion of a car tyre released from rest as it rolls down a hill?



17. A block is pulled up a smooth incline at a constant speed by a force inclined at 30° to the incline. Which graph best represents how the magnitude of the component of the force parallel to the incline changes over time?



SET 7 Projectile Motion 1: Projectile Launched Vertically

1. An aeroplane is flying horizontally at 240 m s^{-1} at an altitude of 1500 m. A small parcel is dropped from the aeroplane.

(a) What is the initial horizontal velocity of the parcel?

(b) What is the initial vertical velocity of the parcel?

(c) How long does it take the parcel to reach the ground?

(d) What horizontal distance will it travel in this time?

(e) What is the final horizontal velocity of the parcel (just as it hits the ground)?

(f) What its final vertical velocity?

(g) Find the final velocity of the parcel.

2. A car is driving at 12 m s^{-1} on a flat road towards the edge of a cliff. Its brakes fail just as it approaches the edge of the cliff. It goes over the edge and hits the ground at the bottom 3.5 s later.

(a) What was its initial vertical velocity?

(b) How far out from the edge of the cliff does it hit the bottom?

(c) What is the height of the cliff?

(d) With what velocity does it hit the ground at the bottom of the cliff?

3. A ball moving horizontally at 6 m s^{-1} rolls off the edge of a 125 m cliff.

(a) How long will it take the ball to hit the water at the bottom of the cliff?

(b) How far out from the edge does the ball hit the water?

(c) Find its velocity on hitting the water.

4. A ball is thrown horizontally with a speed of 14.0 m s^{-1} from a height of 2.0 m above the ground.

(a) Find its initial horizontal velocity.

(b) Find its initial vertical velocity.

(c) How long does it take to reach the ground?

(d) What horizontal distance will it travel in this time?

(e) Find the final horizontal velocity of the ball just as it hits the ground.

(f) Find its final vertical velocity.

(g) What is the final velocity of the ball?

5. A gun is aimed directly at the centre of the bullseye in a target 60 m away. It fires a bullet horizontally at 300 m s^{-1} directly at the bullseye. Where does the bullet hit the target?

6. Two balls are thrown at the same time horizontally outwards from the top of a 147 m high building. Ball A is thrown at 15 m s^{-1} and ball B at 17.5 m s^{-1}.

(a) Which ball will hit the ground first, and by how much?

(b) How far apart will the balls be when they hit the ground?

(c) What will be the vertical velocity of ball A just as it hits the ground?

(d) What will be the vertical velocity of ball B just as it hits the ground?

7. An arrow is fired horizontally at 30 m s^{-1} at an apple in a tree. Just as the arrow starts its flight, the apple falls from the tree. If the arrow and the apple are 15 m above the ground at the start, and the apple is 45 m from the archer, find:

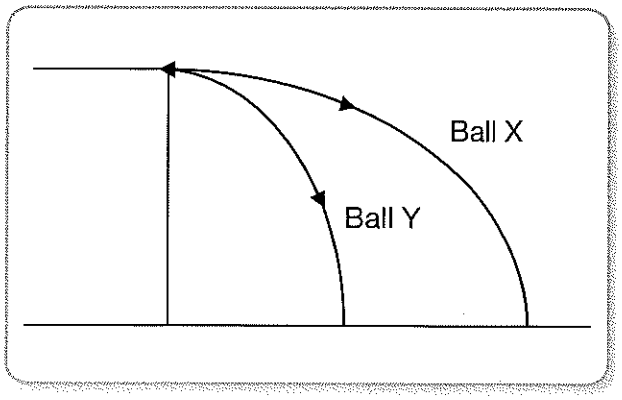
(a) How long it takes the arrow to reach the apple.

(b) How far above the ground the arrow is after travelling the 45 m.

(c) How far the apple has fallen by the time the arrow has travelled 45 m.

(d) Whether or not the arrow hits the apple. Justify your answer.

8. Ball X is fired horizontally from a 1.2 m high table at 2.6 m s^{-1}. Ball Y is projected vertically out from the edge of the table at 1.5 m s^{-1}.



(a) How much further out from the edge of the table does ball Y land? About:

(A) 0.5 m

(B) 0.7 m

(C) 1.0 m

(D) 1.25 m

(b) With what velocity will X hit the floor?

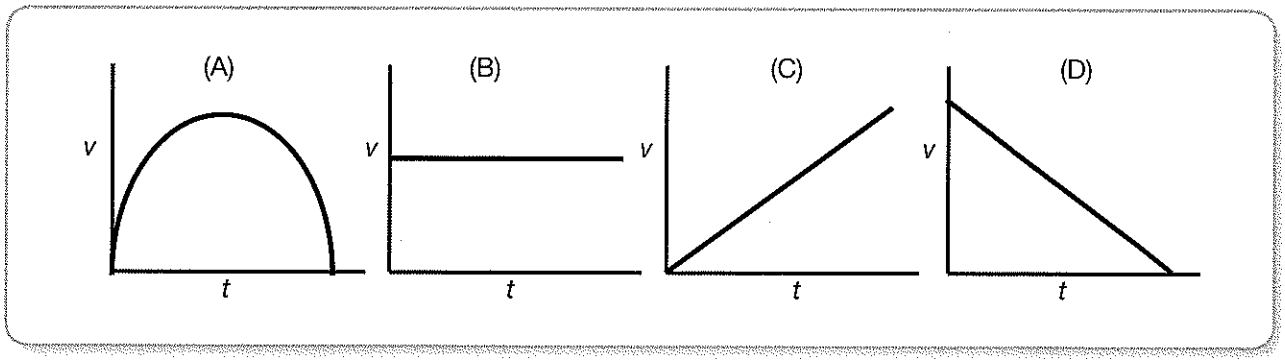
(A) 4.85 m s^{-1}

(B) 5.08 m s^{-1}

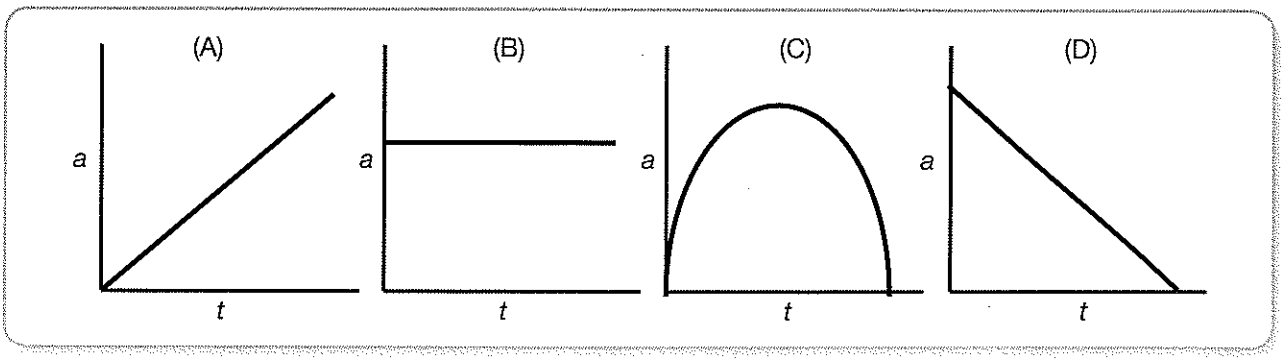
(C) 5.50 m s^{-1}

(D) 23.52 m s^{-1}

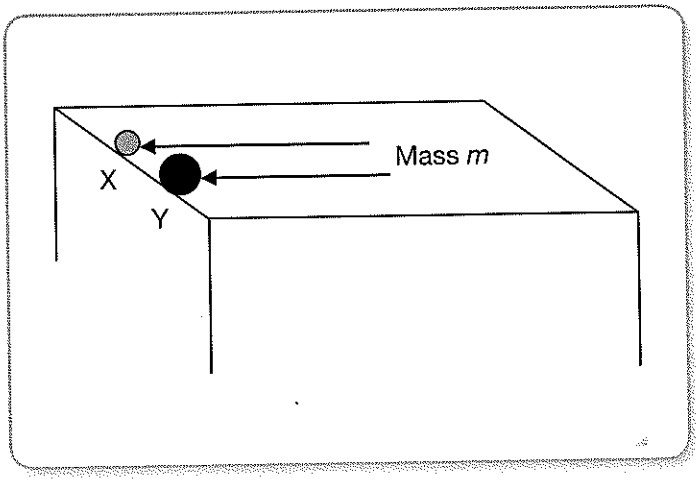
9. A ball is rolled at different speeds along a horizontal benchtop until it falls over the edge towards the floor. Which graph best shows the velocity of the ball as it falls to the floor?



10. A ball is rolled at constant speed along a horizontal benchtop until it falls over the edge towards the floor. Which graph best shows the acceleration of the ball as it falls to the floor?



11. A student rolls two balls X and Y, X with mass m and Y with mass $2m$ across a benchtop so that they leave the edge at the same time and with the same speed.



Four students made statements about the flight of the two balls:

Jacinta: Y hits the ground before X.

Chin: X and Y hit the ground at the same time.

Mario: X hits the ground twice as far away from the table compared to Y.

Pasqual: X hits the ground the same distance from the table as Y.

Whose statement about the two balls is correct?

(A) Mario and Jacinta.

(B) Mario and Chin.

(C) Pasqual and Jacinta.

(D) Pasqual and Chin.

12. Several balls are rolled at different speeds along a benchtop until they fall over the edge towards the floor. Which statement about these balls is correct?

(A) All four balls will hit the floor at the same time.

(B) The slowest ball will hit the floor first; the fastest will hit it last.

(C) The fastest ball will hit the floor first; the slowest will hit it last.

(D) All balls will land in the same position at the same time.

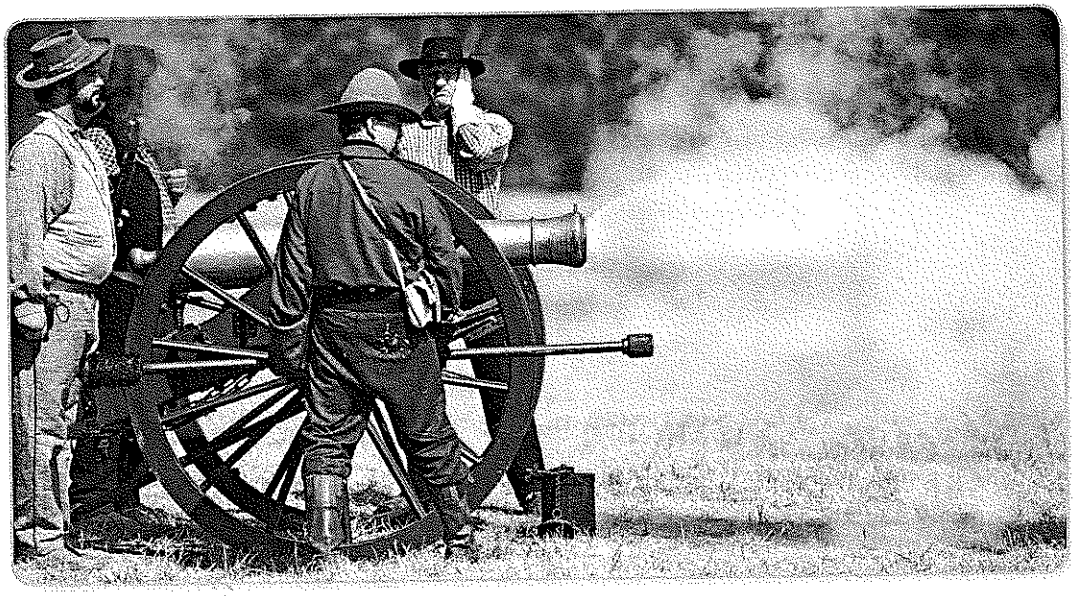
13. A cannon was fired at an elevation of 40°. It was then loaded with an identical charge and ball and fired again at an elevation of 50°. Which statement about the two cannonballs is correct?

(A) The first cannonball will have a shorter time of flight.

(B) The first cannonball will have a longer time of flight.

(C) The first cannonball will have the longer range.

(D) The first cannonball will have the shorter range.



SET 8 Projectile Motion 2: Projectile Landing at Same Level as Launched

1. Joe and Blow are playing catch. Joe throws a ball to Blow, 100 m away at an angle of 60° at 30 m s^{-1}. Blow catches the ball at the same level as it was thrown from.

(a) What was its time of flight?

(b) What was its initial vertical velocity?

(c) What was its maximum height above the throw position?

2. A footballer wants to kick an 'up and under' towards the goal. He kicks it at 20 m s^{-1} at 70° to the horizontal.

(a) How long will it be in the air?

(b) What horizontal distance will the ball travel (assume it is caught at the same height from which it is kicked)?

(c) What will be its maximum height above its position of being kicked?

3. A 55 kg high jumper leaves the ground with 1800 J of kinetic energy. As she 'clears the bar' she has 650 J of kinetic energy.

(a) Assuming that she clears the bar by 15 cm, how high was the bar set?

(b) What is her speed as she arches over the bar?

(c) What is the vertical component of her jump velocity?

(d) How long does the jump take from take-off to ground landing?

4. A cricket player hits a 'six', the ball landing 105 m away from the wicket over the boundary 'on the full'. The maximum height reached by the ball was 30 m above the field.

(a) What was the velocity of the ball as it left the bat?

(b) What was its time of flight?

5. The largest horizontal distance a particular person can throw a ball is 98 m.

(a) At what angle to horizontal must it be thrown?

(b) What will be the speed at which it is thrown?

(c) What will be its time of flight?

Now suppose the person throws the ball vertically into the air at the same speed.

(d) How high will it rise?

(e) What will be its time of flight?

6. A boy, standing on the tray of a flat top truck throws a ball vertically upwards at 8.2 m s^{-1} and catches it again at the same level. The truck was moving at 20 km h^{-1}.

(a) What was the time of flight of the ball?

(b) How high did it rise?

(c) How far did the truck travel while the ball was in the air?

7. A BMX bike rider rides up a ramp inclined at 22° and jumps a 60 m gap to land on an identical ramp on the other side of the gap.

(a) How long will it take to make the jump?

(b) What will be the horizontal component of his velocity?

(c) How fast does he need to 'take off' from the first ramp in order to make the jump safely?

(d) What will be the vertical component of his velocity?

(e) What will be the maximum height of the bike above the top of the ramps?

(f) If his take-off speed was the same, but the ramps were inclined at 30°, what would be the maximum safe gap?

8. A golfer hits a ball at 60 m s^{-1} at an angle of 30° to the horizontal.

(a) What will be its time of flight?

(b) How far will it travel down the level fairway before it hits the ground?

(c) What will be its maximum height above the tee?

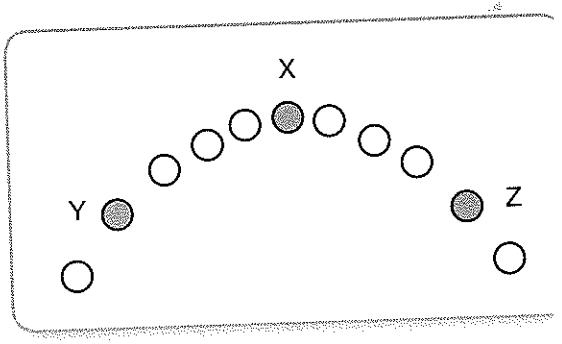
(d) How much further (or less) will the ball travel down the fairway if the golfer hits it from the tee at 45° above horizontal at the same speed?

9. (a) A projectile is launched into the air at 60° to the horizontal. Which statement about the projectile is true when it is at its maximum height?

(A) $a_x = 0$ (B) $a_y = 0$ (C) $v_x = 0$ (D) $\Delta x = 0$

(b) At what angle should a projectile be launched so that its range is maximum?

10. Consider three positions, X, Y and Z, in the path of a projectile as shown by the darkened circles in the diagram.



(a) Which choice about the acceleration of the projectile at X, Y and Z is correct?

(A) Acceleration at Y is less than the acceleration at X.

(B) Acceleration at X is the same as the acceleration at Z.

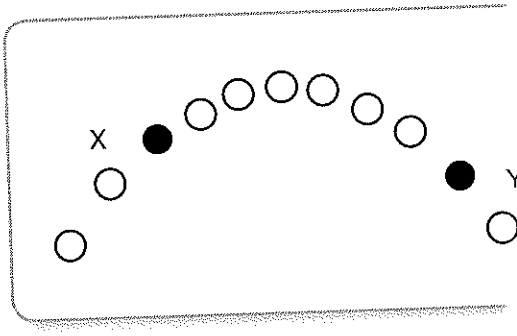
(C) Acceleration at X is greater than the acceleration at Y.

(D) Acceleration at Y is in the opposite direction to the acceleration at Z.

(b) Three projectiles are launched at 20°, 40° and 60° to the horizontal at 20, 30 and 40 m s^{-1} (in that order). Calculate the ratio of their accelerations.

11. The diagram shows the path of a projectile.

Which choice correctly shows the directions of the velocity and acceleration of the projectile at points X and Y?



	Velocity at X	Acceleration at X	Velocity at Y	Acceleration at Y
(A)	→	↓	→	↓
(B)	↗	↑	↘	↓
(C)	↗	↓	↘	↓
(D)	↑	→	↓	→

12. A ball is thrown upwards from point P and follows a parabolic path. Its highest point is point X. Which statement about the acceleration of the ball is true?

(A) It is zero at X.

(B) It is maximum at X.

(C) It is maximum at P and minimum at X.

(D) It is the same at X as it is at P.

13. The diagram shows the paths of the flights of two projectiles, X and Y.

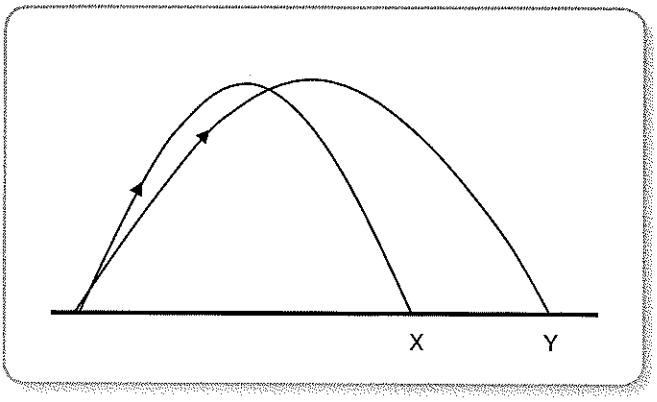
Which of the following is different for X and Y?

(A) Their accelerations.

(B) Their times of flight.

(C) Their maximum *y*-displacements.

(D) Their initial velocities.



14. The diagram shows a stroboscopic photograph of a projectile which has a time of flight of 10.0 s

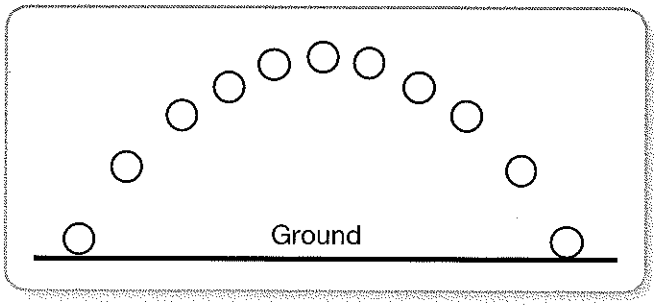
What was its initial vertical speed?

(A) 49 m s^{-1}

(B) 98 m s^{-1}

(C) 147 m s^{-1}

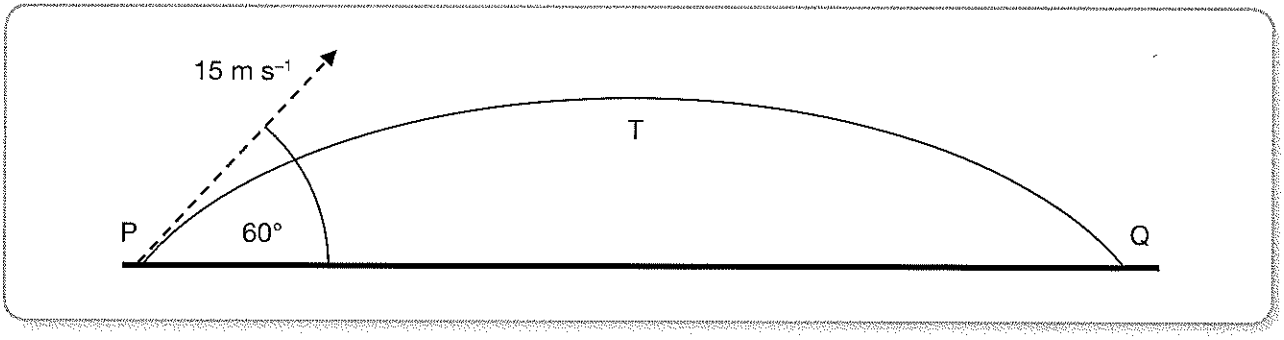
(D) Additional information is required to determine this.



15. A ball is hit into the air at 45 m s^{-1} and 30° to the horizontal. What is its speed 2 s later?

(A) 2.9 m s^{-1} (B) 19.6 m s^{-1} (C) 25.4 m s^{-1} (D) 31.9 m s^{-1}

16. A projectile is fired from a cannon at 15 m s^{-1} at an angle of 60° from position P, landing at position Q as shown in the diagram. Point T marks the maximum height of the projectile.



(a) What is the horizontal velocity of the projectile at P compared to at Q?

(b) What is the vertical velocity of the projectile at P compared to at Q?

(c) What is its acceleration at P compared to Q?

(d) What is the horizontal velocity of the projectile at P compared to at Q?

(e) What is its horizontal velocity at T?

(f) What is its vertical velocity at T?

(g) What is its acceleration at T?

(h) How long does it take the projectile to travel from P to T?

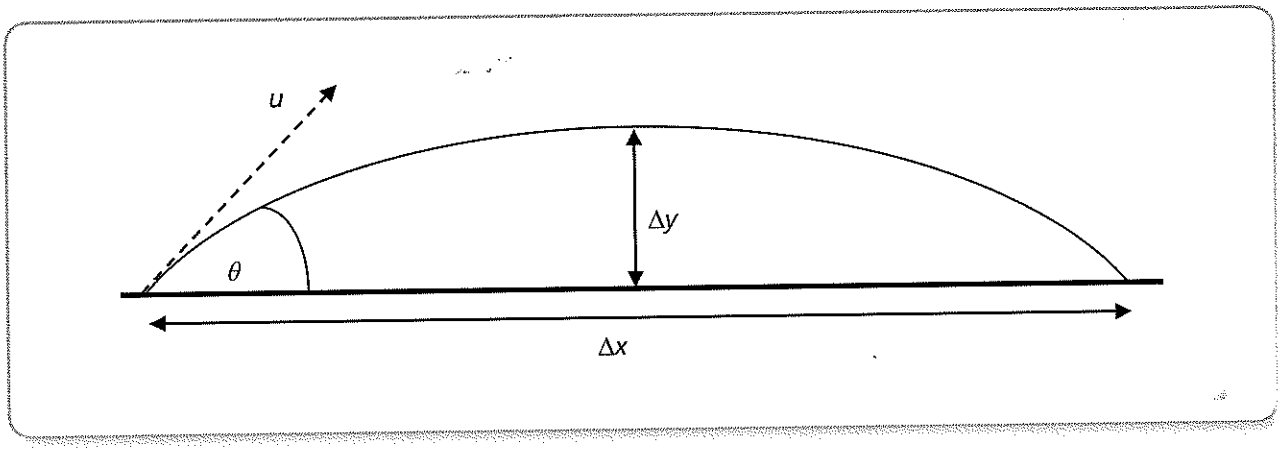
(i) How long does it take the projectile to travel from T to Q?

(j) What is its time of flight?

(k) What is the range of the projectile?

(l) What is the velocity *f* the projectile just before it hits the ground at Q?

17. The diagram shows the path taken by a 600 g projectile.

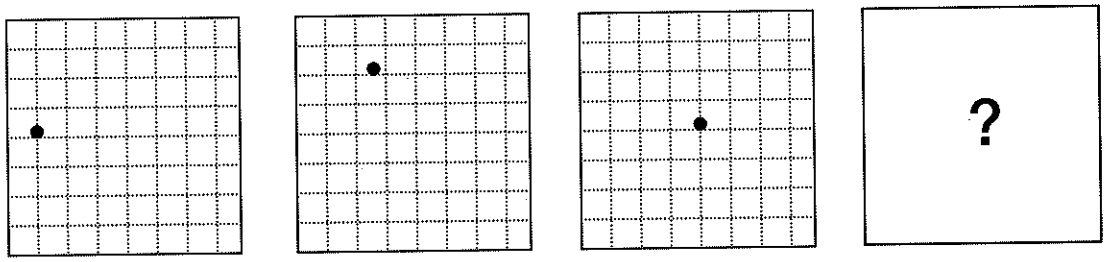


(a) If the total energy of the projectile at time zero is 120 J, what is its initial speed?

(b) If the maximum height of the ball, Δy, is 8.0 m, what will be the speed of the projectile at this point?

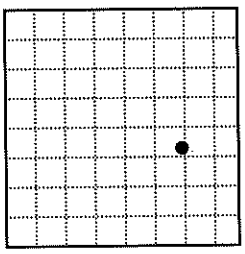
(c) Determine the value of θ.

18. Some photographs are taken of a ball moving in a parabolic path in front of a grid. The time interval between photographs is identical. The diagrams show the first three photographs of the ball's flight.

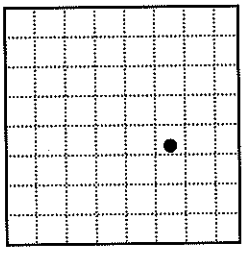


Which choice correctly shows the next photograph in the series?

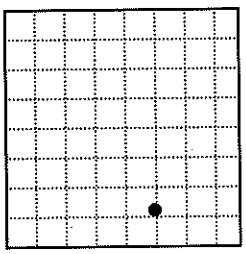
(A)



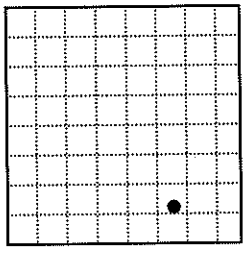
(B)



(C)



(D)



SET 9 Projectile Motion 3: Projectile Landing at Different Level to Launch

1. A shot put is thrown from shoulder height, 1.23 m above the ground at an angle of 30° to the horizontal with a speed of 7.35 m s^{-1}.

(a) How long before it hits the ground?

(b) How far will it be thrown?

(c) What will be its maximum height above the thrower's shoulder?

(d) Calculate the velocity with which it hits the ground.

2. A golfer hits a ball and it lands directly into the hole on the green, a distance of 160 m away, and 50 m below the position where it was hit. The golf ball is in flight for 5 s.

(a) What is the golf ball's initial horizontal velocity?

(b) What is its initial vertical velocity?

(c) Calculate the velocity with which the ball leaves the tee after being hit.

(d) How long does it take to reach its maximum height?

(e) What is its maximum height?

3. A basketball player shoots for a basket from a distance of 4.0 m. The hoop is 3.5 m above court level, and the player releases the ball from a height of 2.0 m. The ball goes through the hoop 1.25 s after shooting it.

(a) What is the horizontal component of the ball's initial velocity?

(b) What is the vertical component of the ball's initial velocity?

(c) With what velocity does the player shoot the ball?

(d) For how long does the ball rise?

(e) What is its maximum height of the ball above its release point?

4. A projectile is fired from the top of a 147 m high cliff at 60 m s^{-1} at an angle of 30° to the horizontal.

(a) What is its initial horizontal velocity?

(b) What is its initial vertical velocity?

(c) What is its maximum height above the launch position?

(d) What is its time of flight?

(e) What is its range?

(f) What is its final horizontal velocity (just before it hits the ground)?

(g) What is its final vertical velocity?

(h) What is its final velocity?

5. A projectile is fired from the top of a 1102.5 m high cliff at 40 m s^{-1} at an elevation of 50°.

(a) For how long is the projectile above its launch position?

(b) What is its total time of flight?

(c) What is its maximum height above the launch position?

(d) What is its range?

(e) What is its velocity just before it hits the ground at the bottom of the cliff?

6. A cannonball is fired at 150 m s^{-1} at an elevation of 60° at a cliff which is 300 m away.

(a) At what height above the ground does the projectile hit the cliff?

(b) What will be its time of flight?

(c) What will be its vertical velocity as it hits the cliff?

(d) With what velocity will it hit the cliff?

7. A cannon on top of a 150 m high cliff fires a cannonball at 40 m s^{-1} at an elevation of 30° at a sailing ship in the water directly below but 250 m out from the bottom of the cliff. The mast on the ship is 30 m high.

(a) How long will it take the ball to cover the distance to the ship?

(b) Assume that if the ball hits the ship, it will hit somewhere on the mast. Does the cannonball hit the mast?

(c) If yes, where does it hit it? If no, by how much does it miss?

(d) What will be the maximum height of the ball above its firing position?

(e) For how long will it be above its firing position?

(f) What is the time of flight of the ball?

(g) With what velocity does it hit the mast (if it hits) or the water (if it misses)?

8. A projectile fired from the top of a castle 75 m high hits a trebuchet in a valley 120 m horizontally away from the castle and 50 m below its ground level 6.0 s after being fired.

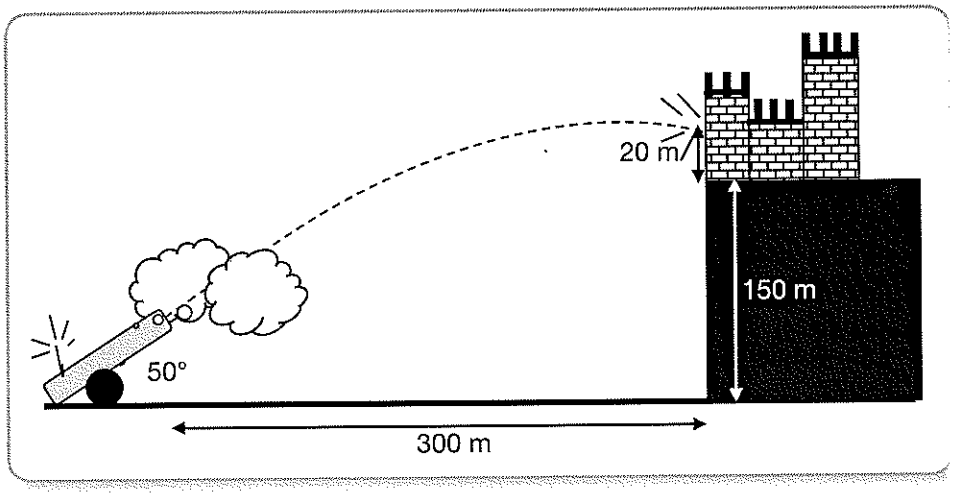
(a) What was its initial horizontal velocity?

(b) What was the velocity with which the projectile was fired?

(c) What was its maximum height above the castle?

(d) With what velocity does it hit the trebuchet?

9. The diagram (not drawn accurately) shows a cannonball being fired at an elevation of 50° at a castle on top of a 150 m high cliff 300 m away. The cannonball hits the castle 20 m up its front wall. The height of the cannon is negligible compared to the other heights.



(a) What will be its time of flight?

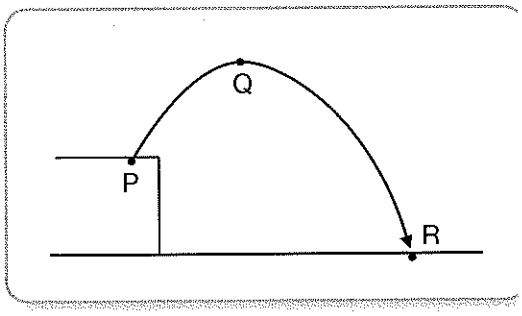
(b) What is the speed of the cannonball as it leaves the cannon?

(c) What is its maximum height above the ground?

(d) Despite the drawing, during which part of its flight does it hit the castle, on the way up or on the way down?

(e) With what velocity does the cannonball hit the castle wall?

10. A projectile follows the pathway shown.



Which statement about this projectile is correct?

(A) The time of flight from P to R is twice the time from P to Q.

(B) The acceleration of the projectile is independent of its launch velocity.

(C) The vertical component of the velocity is the same at P, Q and R.

(D) The horizontal component of the velocity is greatest at Q.

11. A projectile is fired from the top of the wall of a castle and rises 47 m into the air during its flight to hit a gunpowder storage hut 250 m away and 40 m below the level of the cannon.

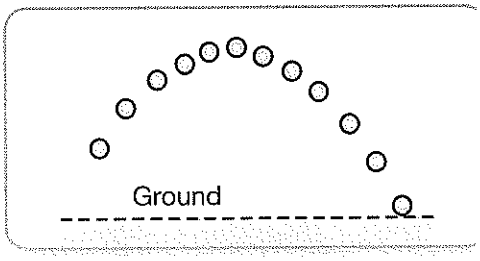
(a) What was its time of flight?

(b) What was the velocity with which the projectile was fired?

(c) How long is the projectile above its firing position?

(d) With what velocity does it hit the gunpowder hut?

12. A tennis ball is hit into the air and follows the path shown.



(a) Which statement regarding the flight of the ball is correct?

(A) The velocity of the ball at its highest point is zero.

(B) The velocity of the ball is always changing.

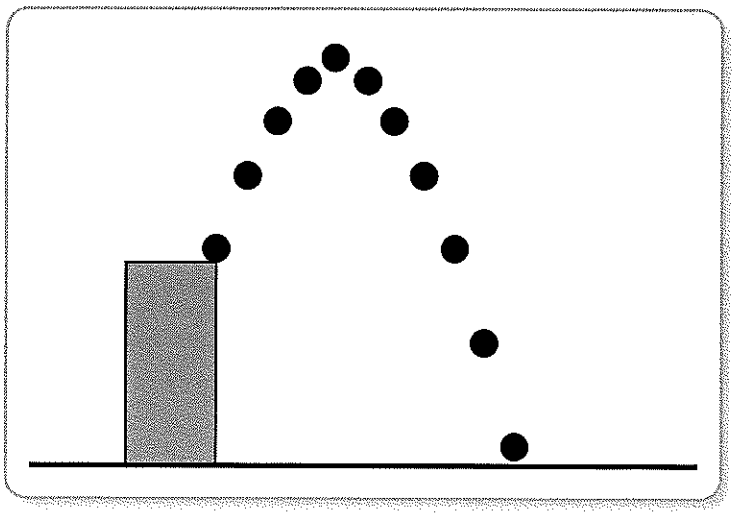
(C) The direction of the ball's acceleration changes when it reaches the highest point.

(D) The acceleration of the ball at its highest point is zero.

(b) The final displacement of this projectile could be described as being negative. Explain this.

SET 10 Analysing Projectile Motion Diagrams

1. The diagram shows a film clip of a ball projected into the air. The camera took the clip at 2.5 frames per second.



(a) What was the time of flight of the ball?

(b) How long did it take to reach its maximum height?

(c) What was its initial vertical velocity?

(d) How high did it rise?

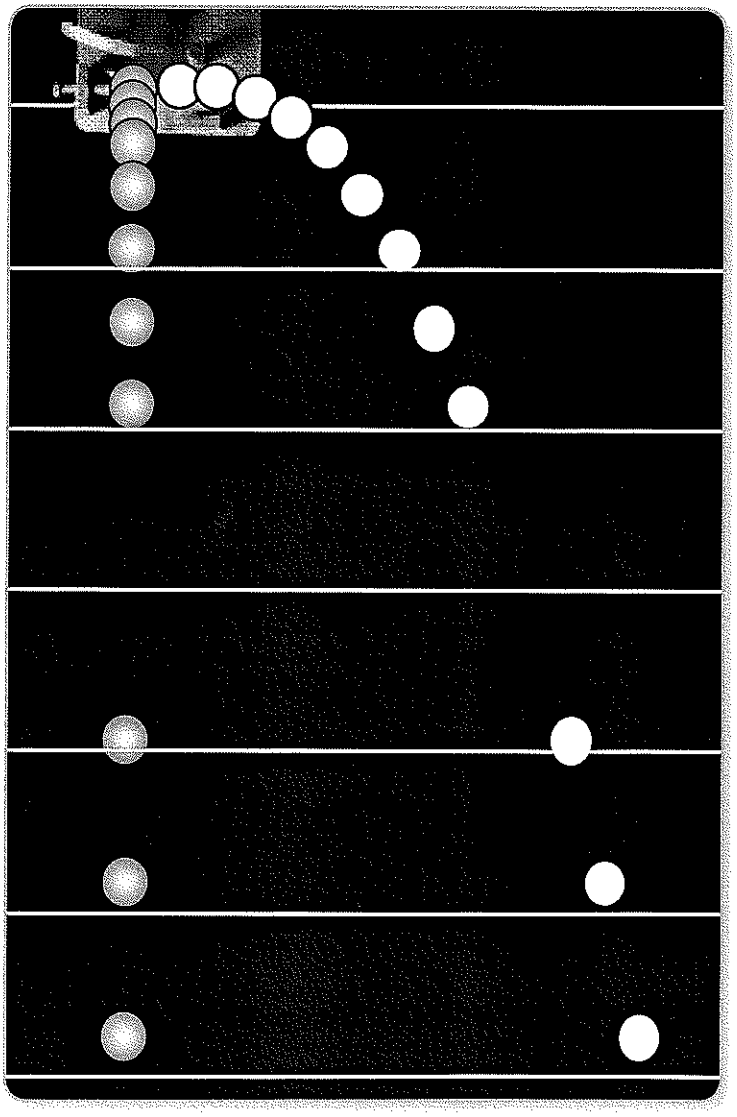
(e) What was its range?

(f) What was its initial horizontal velocity?

(g) What was its initial velocity?

(h) From what height was the ball projected?

2. The diagram shows a photographic record of a grey ball falling vertically and a white ball projected horizontally. The distance between the horizontal white lines on the diagram is 10 cm. Some images of both balls are missing.



(a) What is the time interval between the pictures?

(b) How many images of the grey ball are missing?

(c) Draw in these missing images.

(d) How many images of the white ball are missing?

(e) Draw in these missing images.

(f) Find the time of flight of the balls.

(g) Find the instantaneous vertical velocity of the grey ball as the last image was taken.

(h) What was the initial velocity of the white ball?

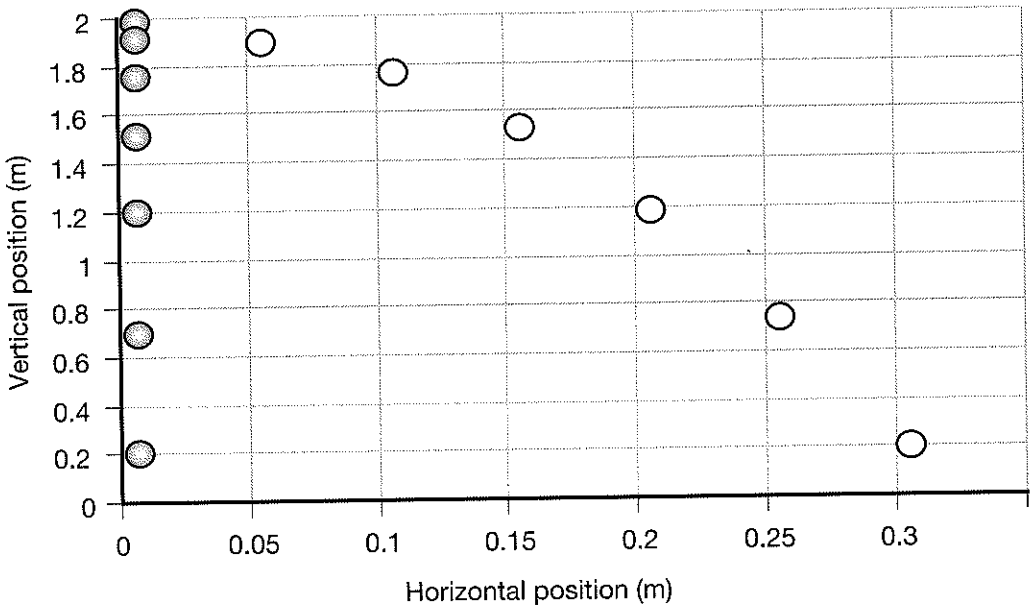
(i) Find the instantaneous vertical velocity of the white ball as the last image was taken?

(j) What was the range of the white ball?

(k) What was the horizontal velocity of the white ball?

(l) What was the instantaneous velocity of the white ball as the last image was taken?

3. The graph shows the positions of two balls at regular intervals of time. The grey ball was dropped vertically, the white ball projected horizontally outwards.



Situation 1 Balls falling close to Earth's surface

(a) Determine the initial horizontal velocity of the white ball.

(b) Analyse the information in the graphs to determine the total time of flight for each ball.

(c) Find the time interval between each image of the balls.

(d) Calculate the vertical velocity of the white ball when it is at the position represented by the last image on the graph.

(e) Calculate the velocity of the white ball when it is at the position represented by the last image on the graph.

Situation 2 Balls falling close to surface of planet X

On planet X, the time between successive images of the balls is 0.28 s.

(f) Analyse the information to determine the total time of flight for each ball on planet X.

(g) Determine the initial horizontal velocity of the white ball on planet X.

(h) Calculate the strength of the gravitational field on planet X.

(i) Calculate the vertical velocity of the white ball when it is at the position represented by the last image on the graph on planet X.

(j) Calculate the velocity of the white ball when it is at the position represented by the last image on the graph on planet X.

Situation 3 Balls falling close to surface of planet Y

The gravitational field strength on planet Y is 15.3 m s^{-2}.

(k) Calculate the vertical velocity of the white ball when it is at the position represented by the last image on the graph on planet Y.

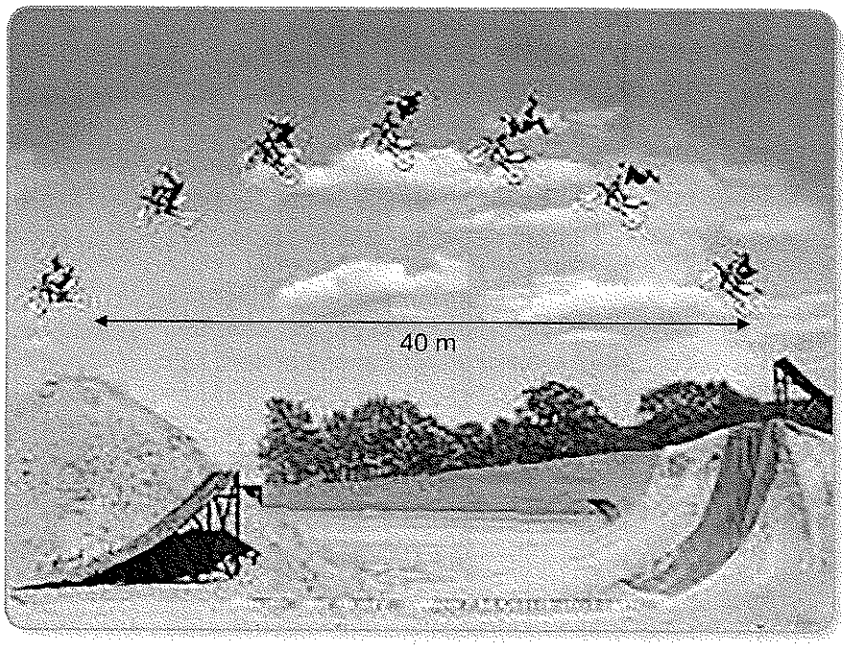
(l) Analyse the information to determine the total time of flight for each ball on planet Y.

(m) Find the time interval between each image of the balls on planet Y.

(n) Determine the initial horizontal velocity of the white ball on planet Y.

(o) Calculate the velocity of the white ball when it is at the position represented by the last image on the graph on planet Y.

4. The diagram represents a stroboscopic photograph of the motion of a motorbike stunt rider jumping across a gap from a ramp inclined at 40°. The distance between the back of the first bike in the photo and the last bike is 40 m. The camera used to take the pictures took one frame every 0.43 s.



(a) How long did it take the bike to travel the 40 m shown?

(b) Estimate the horizontal velocity of the bike and rider.

(c) Calculate his initial velocity.

(d) What was his maximum height above the top of the launch ramp?

5. The picture shows a photograph taken as a person jumps horizontally out from a vertical cliff. He hits the water 2.0 s after jumping.

(a) Calculate, or estimate the height of the cliff.

(b) At what rate (frames per second) did the camera take the photo?

(c) Estimate how far out from the base of the cliff does he hit the water.

(d) Use your answer to (c) to calculate his horizontal velocity.

SET 11 Uniform Circular Motion

For relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

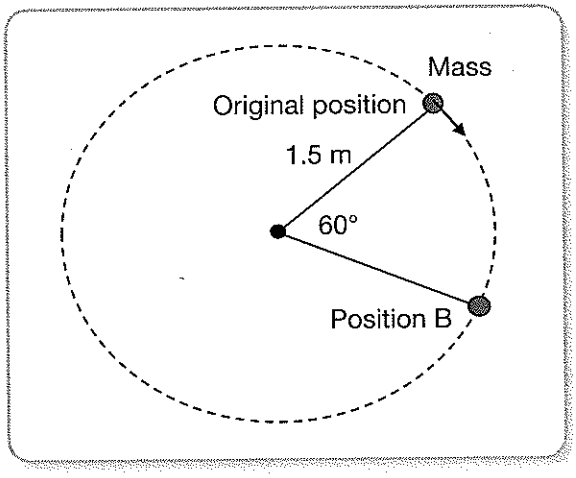
1. What is the essential characteristic in physics for uniform circular motion?

2. (a) Calculate the rotational speed (also called the linear or tangential speed) of a point on the surface of Earth at the equator.

(b) Calculate the magnitude of the centripetal acceleration of a point on the surface of the Earth at the equator.

(c) If the Earth was a perfect sphere, how would this affect the gravitational field of the Earth on the surface at the equator compared to the value at the poles? Justify your answer.

3. A mass of 2.5 kg lies on a perfectly smooth horizontal surface and is connected to a light string which is 1.5 m long as shown in the diagram. The mass completes 4 rotations per second.



(a) Calculate the linear speed of the mass.

(b) Calculate the magnitude of the acceleration of the mass.

(c) Calculate the magnitude of the tension in the string.

(d) Calculate how long it takes the mass to move from its original position to position B.

(e) Draw a vector diagram to determine the magnitude and direction of its acceleration as it moves from its original position to B.

(f) Compare your answers to (b) and (e).

4. A 400 kg satellite orbits 300 km above the surface of the Earth at 7740 m s^{-1}.

(a) Using the formula for gravitational field strength, calculate the strength of the gravitational field at the position of this satellite.

(b) Using the formula for centripetal acceleration, calculate the acceleration of the satellite.

(c) Compare your answers to (a) and (b) and account for any similarity or difference.

(d) Using Newton's formula for gravitational force, calculate the gravitational acting on the satellite.

(e) Using the formula for centripetal force, calculate the centripetal force acting on the satellite.

(f) Compare your answers to (e) and (f) and account for any similarity or difference.

5. (a) Taking the orbital radius of the Earth about the Sun as 1.5×10^8 km, calculate the orbital speed of the Earth through space.

(b) Use your answer to (a) to calculate the centripetal acceleration of the Earth towards the Sun.

(c) Use your answer to (b) to calculate the gravitational force between the Earth and the Sun.

(d) Use the gravitational field strength equation to find the strength of the gravitational field due to the Sun at the position of Earth.

(e) Use Newton's equation to find the gravitational force between the Earth and the Sun.

(f) Account for similarities or differences in your answers to (b) and (d).

(g) Account for similarities or differences in your answers to (c) and (e).

6. A hammer thrower swings a 5 kg hammer around at 3.5 m s^{-1}. The distance from his shoulder to the tip of the hammer is 1.75 m. What force does he need to apply to stop the hammer from flying off too soon?

7. In an amusement park, the Volcano ride consists of a train which travels in a horizontal circle at relatively high speeds. The students estimated that the radius of the track was 35 m and they timed it at one revolution every 6.5 s at its highest speed.

(a) What is the highest speed of the Volcano ride?

(b) What is the acceleration of the students riding the train when it is travelling at its highest speed?

8. The Singapore Flyer Ferris wheel has a diameter of 165 m and is as tall as a 42 storey building. It takes about 30 minutes to carry its 780 riders once around the circle of travel.

(a) What is the highest speed of the Ferris wheel?

(b) What is the acceleration of the passengers riding it?

9. During the spin cycle of a washing machine, the clothes move to the outer wall of the spinning drum as it spins at 1800 revolutions per minute. The radius of the drum in the washing machine is 30 cm.

(a) Explain why the clothes end up flattened against the walls of the spinning barrel.

(b) Determine the speed of the clothes on the wall of the spin barrel.

(c) Find the acceleration of the clothes.

10. Elmira, New York, boasts of having the fastest carousel ride in the world. The merry-go-round at Eldridge Park, radius 7.4 m, takes riders on a spin at 8.0 m s^{-1}.

(a) Determine the time for outside riders to make one complete circle.

(b) Find the acceleration of the riders.

11. A CD-ROM drive rotates the disc at 1200 revolutions per minute.

(a) What is the speed of the outer edge of the disc 5.6 cm from the centre?

(b) What is the acceleration of the edge of the disc?

12. A merry-go-round in a park has a diameter of 6 m. Three children sit on it. Mary, mass 30 kg sits on the edge. Julie, mass 40 kg sits 1.5 m in from the edge, and Tommy, mass 50 kg, sits on the hub of the axle right at the centre. What is the ratio of the centripetal forces acting on them?

13. In the window of a toy store a battery-powered plane hangs from a string and flies in a horizontal circle. The 650 g plane makes a complete circle every 2.2 seconds. The radius of the circle is 0.950 m.

(a) Determine the orbital speed of the plane.

(b) Calculate its centripetal acceleration.

(c) What is the centripetal force acting on the plane?

14. A discus thrower whirled the 1.6 kg discus in a circle with a radius of 1.1 m, so that it was moving at a speed of 52 m s^{-1} when he let it go. Determine the net force acting upon the discus just before it was let go.

15. In a pairs figure skating competition, the skaters hold hands and the male swings the female around in a circle, his body acting as the axis of rotation. In this manoeuvre, the female skater's body stretches out virtually horizontal to the ice floor. Calculate the force between their hands if the female skater has a mass of 50 kg and her centre of mass rotates in a circle with a radius of 65 cm once every 2.0 s.

16. A driver makes a turn off a road at 18.0 m s^{-1}. The turning radius of the level curve is 36.0 m. If the car has a mass of 1250 kg, determine:

(a) The acceleration of the car as it goes through the corner.

(b) The net force acting on the car.

(c) The minimum frictional force between the tyres and the road to ensure that the car does not slide out of the corner.

(d) The frictional force per unit mass acting on the car as it turns the corner.

17. A 1500 kg car takes a horizontal curve on a level roadway at a speed of 23 m s^{-1}. The radius of the curve is 65 m. Determine the total frictional force required to keep the car on the curve.

18. A student is swinging a rubber stopper of mass 200 g around his head on a 1.2 m long string. The stopper makes 1 revolution each second.

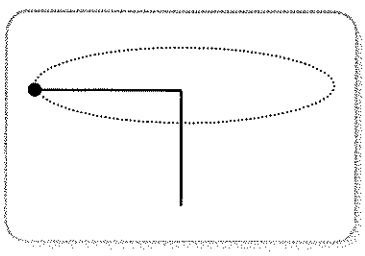
What is the tension in the string?

(A) 0.52 N

(B) 7.90 N

(C) 9.48 N

(D) 94.8 N



SET 12 Circular Motion on an Inclined Track

1. What is the smallest radius of an unbanked (flat) track around which a 60 kg motorcyclist riding a 100 kg bike can travel if her speed is 30 km h^{-1} and the friction between the tyres and the road is 0.28 N kg^{-1}?

2. A 1050 kg car travels around a turn of radius 70 m on a flat road. The frictional force between the tyres and the road is 0.80 N kg^{-1}.

(a) What is the maximum speed the car can travel without slipping?

(b) Does this result depend on the mass of the car? Explain.

3. A train travels at a constant speed around a curve of radius 225 m. A lamp hanging on a light cord swings out to an angle of 20.0° while the train turns. Find the speed of the train.

4. Determine the minimum angle at which a road should be banked so that a car traveling at 20.0 m s^{-1} can safely negotiate the curve if the radius of the curve is 200.0 m.

5. If a curve with a radius of 65 m is properly banked for a car travelling 75 km h^{-1}, what must be the coefficient of static friction (the frictional force per kilogram of mass of the car) for a car not to skid when travelling at 90 km h^{-1}?

6. A car is driven around a circle with a radius of 200 m and bank angle 10 degrees. Calculate the maximum safe velocity of the car if friction is negligible.

7. An airplane is flying in a horizontal circle at a speed of 460 km h^{-1}. Its wings are tilted 40° to the horizontal and force is provided by lift that is perpendicular to the wing surface. What is the radius of the circle?

8. A curve has a radius of 50 metres and a banking angle of 15°. What is the maximum safe speed for a car on this curve?

9. A turn of radius 100 m is being designed for a maximum speed of 25 m s^{-1}. At what angle should the turn be banked?

10. A circular speedway track has a radius of 300 m. The track is banked at 30°. What is the maximum speed for a car to safely travel on this track?

11. A curve with a radius of 90 m is properly banked for a car travelling at 60 km h^{-1}. At what angle is the curve banked?

12. An airplane is circling an airport by travelling a horizontal circular path at a speed of 400 km h^{-1}. The bank angle of the wings is 20°. What is the radius of the circular path the plane is flying?

13. A jet fighter is flying around the airport with a speed of 800 km h^{-1} along a circular path with a radius of 2 km.

(a) What acceleration is the pilot experiencing?

(b) At what angle must the wings of the plane be banked?

14. A curved road has a radius of curvature of 200 m and a bank angle of 10°. What is the highest speed at which a car can round the curve safely?

15. The curves at either end of the track at a velodrome have a radius of curvature of 50 m banked at 15°.

(a) What is the maximum speed a cyclist could take these curves?

(b) If the maximum safe speed for a different velodrome track, same radius of curvature was 85 km h^{-1}, at what angle would the end curves need to be banked?

16. At what angle should a curve of 200 m radius be banked so that no friction is required when a car travels at 60 km h^{-1} around the curve?

17. The Daytona 500 is a major event of NASCAR (National Association for Stock Car Auto Racing) season. It is held at the Daytona International Speedway in Daytona, Florida. The turns in this track have a maximum radius of 316 m and are banked at 31°. Assuming no friction, what is the maximum safe speed for the race cars to travel around them?

18. A turn of radius 250 m is being designed for a speed of 40 m s^{-1}. At what angle should the turn be banked?

19. An exit ramp on a certain highway has a radiu of curvature of 75 m. If the ramp is banked at 20 degrees, what is the maximum speed that can safely be executed?

SET 13 Orbital Velocity

For all relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

1. (a) Knowing that the gravitational force acting on a satellite in orbit is a centripetal force, use the equations for these forces to derive a formula for calculating orbital velocity.

(b) Use this formula to determine the orbital velocity of a satellite in orbit above Earth at an altitude of 300 km.

2. A 270 kg communication satellite is placed in a geostationary orbit 35 780 km above a ground level relay station. What is the speed of the satellite in orbit?

3. A 500 kg satellite is orbiting Earth with an orbital speed of 5000 m s^{-1}.

(a) What is its orbital radius?

(b) What is its altitude?

(c) Repeat these calculations for a satellite with a mass of 250 kg with the same orbital velocity.

4. Three satellites, masses M, $2M$ and $3M$ orbit the Earth. Their orbital radii are R, $4R$ and $25R$ respectively.

(a) Determine the ratio of their orbital velocities.

(b) Determine the ratio of the gravitational field strengths at the positions of the satellites.

(c) Determine the ratio of the forces of gravity acting on the satellites.

5. Three satellites, masses M, $2M$ and $3M$ orbit the Earth. They each have the same orbital velocity.

(a) Determine the ratio of their orbital radii.

(b) Determine the ratio of the gravitational field strengths at the positions of the satellites.

(c) Determine the ratio of the forces of gravity acting on the satellites.

6. (a) Three satellites of masses M, $4M$ and $16M$ orbit the same planet. If their orbital velocities are the same, what is the ratio of their orbital radii?

(b) Three identical satellites orbit planets with masses of M, $4M$ and $16M$. If their orbital velocities are the same, what is the ratio of their orbital radii?

(c) Three identical satellites orbit planets with masses of M, $4M$ and $16M$. If their orbital radii are the same, what is the ratio of their orbital velocities?

7. (a) What will be the orbital velocity of a 400 kg satellite in orbit at an altitude of 300 km around planet X which has a mass of 3.5×10^{25} kg and a diameter of 5000 km?

(b) At what altitude above planet X would a 300 kg satellite have an orbital speed of 8000 m s^{-1}?

(c) A 500 kg satellite orbits planet X's only moon with an orbital radius of 2000 km and a velocity of 2750 m s^{-1}. What is the mass of the moon?

8. (a) Calculate the orbital speed of the Moon in its orbit around Earth given the distance between the Earth and Moon as 3.85×10^8 m, and its orbital period as 27.3 days.

(b) Determine the acceleration of the Moon about the Earth.

9. Taking the orbital radius of the Earth about the Sun as 1.5×10^8 km, and the mass of the Sun as 2.0×10^{30} kg, determine:

(a) The orbital speed of the Earth around the Sun.

(b) The acceleration of the Earth about the Sun.

10. The Incredible Hulk is hoping to put a baseball in orbit by throwing it horizontally from the top of Mount Newton. The summit of Mount Newton is approximately 10 km above Earth's surface. With what speed must he throw the ball in order to put it into orbit?

11. A satellite of mass 1000 kg is launched with a speed of 10 km s^{-1}. It settles into a circular orbit of radius 8.68 × 10^3 km above the centre of the Earth. What is its orbital speed?

12. In 2009, NASA's Messenger spacecraft became the second spacecraft to orbit the planet Mercury. Messenger orbited Mercury about 200 km above its surface. Taking the radius of Mercury as 2.44 × 10^3 km, and its mass as 3.30 × 10^{23} kg, determine:

(a) The orbital speed of Messenger.

(b) The orbital period of Messenger.

13. A satellite of mass 200 kg is launched from a site on Earth's equator into an orbit 200 km above the surface of Earth. Assuming a circular orbit and no air friction:

(a) Find the orbital speed of the satellite.

(b) Find the circumference of the orbit.

(c) What is the period of the orbit?

14. (a) What will be the linear speed of an Earth satellite in a circular orbit at an altitude of 300 km?

(b) What is its period of revolution?

15. Determine the orbital speed of the International Space Station which is orbiting at 350 km above the surface of the Earth.

16. A rocket of mass 4.0 × 10^3 kg is in orbit around Earth with an orbital radius of 7.6 × 10^3 km.

(a) What is its orbital speed?

(b) Find its gravitational potential energy.

(c) What is the strength of the Earth's gravitational field at the position of the satellite?

(d) Determine the weight of the satellite while it is in orbit.

(e) Find the gravitational force acting on the satellite.

17. At what distance from the surface of the Earth is the orbital velocity 5000 m s^{-1}?

18. Compare the orbital velocities of two identical satellites in orbit around the Earth and Mars, both at altitudes of 1000 km. The mass of Mars is 6.42 × 10^{23} kg and its diameter is 6.792 × 10^3 km.

19. (a) A 5000 kg spacecraft is approaching the Moon, radius 1738 km, mass 7.35 × 10^{22} at 6840 m s^{-1}, and wants to orbit 510 km above its surface. What is the speed of orbit at this altitude?

(b) They aim the spacecraft to a point 510 km to the side of the Moon, turn the spacecraft around and use the main engines which generate a thrust of 6 × 10^4 N to slow it down. What will be the deceleration of the spacecraft while the engines are firing?

(c) For what time do they burn their main engine to reduce their speed to the proper orbital speed?

20. Which of the following does *not* affect the orbital velocity of a satellite?

(A) Its altitude.

(B) Its mass.

(C) The mass of the planet it orbits.

(D) The gravitational force of the planet.

21. Which statement about the orbits of satellites is correct?

(A) The greater the mass and altitude, the slower the satellite travels.

(B) The greater the mass and altitude, the faster the satellite travels.

(C) The higher the altitude of the orbit, the faster the satellite travels.

(D) The higher the altitude of the orbit, the slower the satellite travels.

22. The orbit of the Earth about the Sun is almost circular. The closest and farthest distances are 1.47 × 10^8 and 1.52 × 10^8 km, respectively. Calculate the missing values in the table. The mass of the Sun is 2.0 × 10^{30} kg.

(a)	Orbital speed at closest distance (m s^{-1})		Orbital speed at furthest distance (m s^{-1})	
(b)	Gravitational field at closest distance (N kg^{-1})		Gravitational field at furthest distance (N kg^{-1})	
(c)	Gravitational force at closest distance (N)		Gravitational force at furthest distance (N)	
(d)	Gravitational potential energy at closest distance (J)		Gravitational potential energy at furthest distance (J)	

SET 14 Kepler's Laws of Planetary Motion

1. Which of the following outlines Kepler's first law?

(A) Planets move in circles with the Sun at the centre.

(B) Planets move along an elliptical path with the Sun at the centre.

(C) Planets move along an elliptical path with the Sun at one of the foci.

(D) Planets move along an elliptical path with the Earth at one of the foci.

2. In simple language, Kepler's second law means that:

(A) Planets move more rapidly when near the Sun than when farther away.

(B) Planets close to the Sun have shorter periods than those farther away.

(C) The Sun is the centre of planetary orbits.

(D) Slower planets are closer to the Sun.

3. Kepler's third law says that:

(A) A planet moves more rapidly when near the Sun than when farther away.

(B) Planets closer to the Sun have shorter periods than those farther away.

(C) The Sun is the centre of planetary orbits.

(D) Slower planets are closer to the Sun.

4. Kepler's theory was based upon his analysis of the accurate observations made by whom?

(A) Nicolaus Copernicus.

(B) Tycho Brahe.

(C) Galilei Galileo.

(D) Sir Isaac Newton.

5. If Mars was orbiting the Sun in an orbit four times as big as its current orbit, what would its orbital period be compared to its period now?

(A) 4 (B) 8

(C) 16 (D) 24

6. A satellite orbiting Earth at twice the radius of the Earth orbits in about 4.0 hours. How many hours would it take a satellite orbiting at 6.6 times the Earth's radius to complete an orbit?

(A) 4 (B) 8

(C) 16 (D) 24

7. The law of planetary motion that relates the speed of a planet to its position in its orbit is:

(A) Kepler's first law.

(B) Kepler's third law.

(C) The law of equal areas.

(D) The law of inertia.

8. To change the period of an artificial satellite that is orbiting the Earth, we need to change its:

(A) Orbital radius. (B) Mass.

(C) Density. (D) Size and shape.

9. Asteroid X orbits the Sun at a distance of 2 AU while asteroid Y, which is one half as large as X, orbits at 4 AU.

Which asteroid will take longer to go around the Sun according to Kepler's third law?

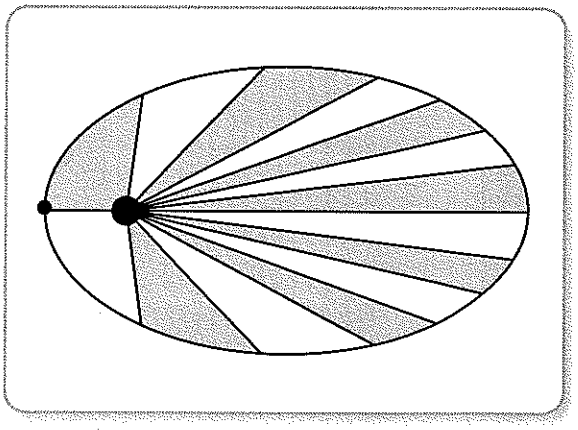
(A) Asteroid X.

(B) Neither, both will take the same time.

(C) Asteroid Y.

(D) More information is required.

10. Consider the diagram below which shows the path of a planet orbiting a star.



According to Kepler, if the arc length of each of the sectors shown represents the same period of time for the orbiting body, then:

(A) The planet is orbiting with constant speed.

(B) The planet travels faster further from the star.

(C) The grey and white sectors are all equal area.

(D) The grey and white sector areas represent the total distance travelled by the planet in each period of time.

SET 15 Kepler's Law of Periods

For relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

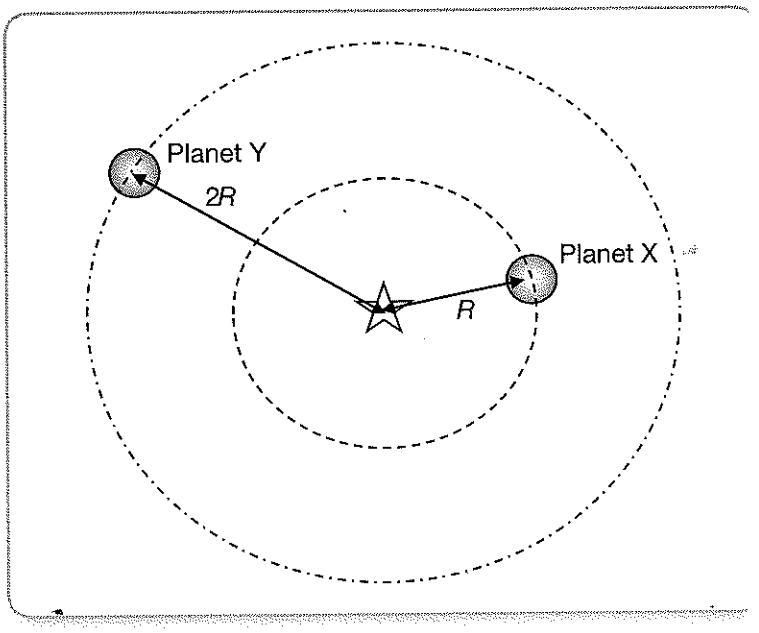
1. Two identical planets X and Y orbit the same star at distances of R and $2R$ as shown in the diagram. The orbital velocity of X is v, and its period of revolution is T.

(a) Determine the period of the orbit of Planet Y in terms of T.

(b) Determine the orbital velocity of planet Y in terms of v.

(c) How would your answers to (a) and (b) vary if planet Y had twice the mass of planet X? Justify your answer.

(d) How would your answers to (a) and (b) vary if the mass of the star was double its present value? Justify your answer.



2. Consider the information in the table which refers to five moons orbiting planet Y.

(a) Find values for the missing quantities P, Q, R and S and then

(b) Calculate the mass of planet Y.

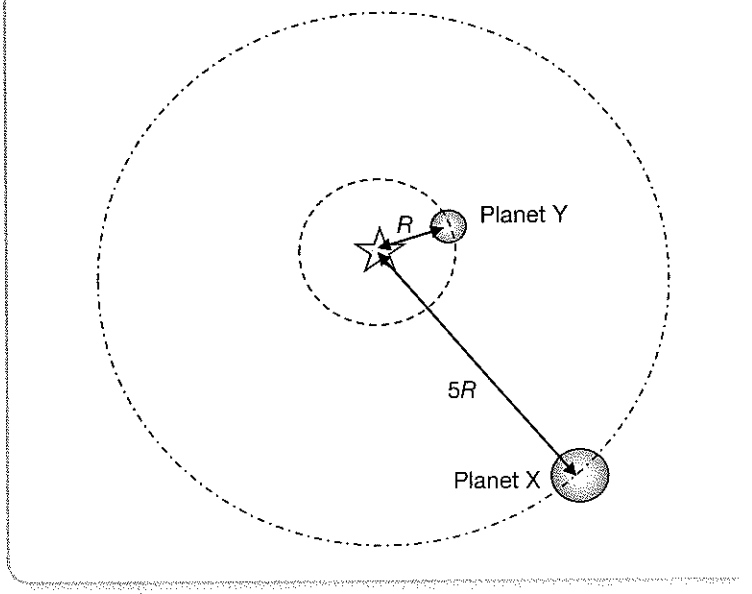
Moon	Period (Earth days)	Orbital radius (km)
Kryptos	30	240 000
Arktor	90	P
Linadox	Q	50 000
Pallidon	9.95	R
Lunrock	S	20 000

3. Planets X and Y orbit the same star. The mass of planet X is 4 times the mass of planet Y. The orbital radius of planet X is 5 times that of planet Y.

(a) What is the ratio of the orbital period of planet X to that of planet Y?

(b) What is the ratio of the orbital velocity of planet X to that of planet Y?

(c) What is the ratio of the gravitational force between the star and planet X to that of planet Y?



4. The Moon orbits Earth in about 27.3 days and its orbital radius is about 385 000 km. Find the mass of the Earth according to this data.

5. The orbital radius of the Earth about the Sun is 150 000 000 km. Taking the time it takes the Earth to orbit the Sun as 365.25 days, find the mass of the Sun.

6. A geostationary satellite remains at the same position above the Earth's surface at all times. What is its altitude above the Earth's surface?

7. Jupiter, mass 1.94×10^{27} kg, orbits the Sun with an orbital radius of 7.5×10^{8} km and a period of 11.86 Earth years. Use this data to find the mass of the Sun.

8. The Moon has an orbital period of 27.3 days and an average orbital radius of 3.85×10^{5} km. Use this data to find the altitude of an artificial satellite of Earth if its period is 90 minutes.

9. The period of Earth's Moon is 27.3 days and it has a mean orbital radius of 3.8×10^{5} km. What is the altitude of a satellite orbiting the Earth once every 14.0 days?

10. The rings of a Saturn-like planet are composed of chunks of ice that orbit the planet. The inner radius of the rings is 78 000 km, while the outer radius is 190 000 km. The mass of this planet is 5.69×10^{26} kg. Find the period of an orbiting chunk of ice at the inner radius.

11. Neptune is an average distance of 4.5×10^{9} km from the Sun. Estimate the length of the Neptunian year given that the Earth is 1.50×10^{8} km from the Sun on the average.

12. Phobos, a moon of Mars, was discovered in 1877. Its orbital radius is 9380 km and its orbital period is 0.318 days. Determine the mass of Mars based on this data.

13. A satellite has a mass of 5850 kg and is in a circular orbit 4.1×10^{5} m above the surface of a planet. The period of the orbit is two hours. The radius of the planet is 4.15×10^{6} m.

(a) Find the mass of the planet.

(b) Find the gravitational field strength on the surface of the planet

(c) What is the weight of the satellite when it is on the planet's surface?

14. Geostationary satellites orbit the Earth above the equator completing one orbit in 24 hours.

(a) Calculate the orbital radius of a geostationary satellite.

(b) Calculate the orbital speed of a geostationary satellite.

(c) Calculate the acceleration of a geostationary satellite.

15. Given that Earth is 1.5×10^{11} m away from the Sun and a certain comet orbits the Sun in 925 Earth days, find the average distance from the Sun to the comet.

16. Planet X, mass $3M$ orbits star S with an orbital radius of R. Planet Y, mass $2M$ orbits the same star with an orbital radius of $4R$.

(a) What is the ratio of the orbital period of planet X to that of planet Y?

(b) What is the ratio of the orbital velocity of planet X to that of planet Y?

(c) What is the ratio of the gravitational force between the star and planet X to that of planet Y?

17. Mercury has a mass of 3.58×10^{23} kg. Pluto has a mass of 1.27×10^{22} kg. What is the Kepler's law ratio of r^3/T^2 for Mercury to Pluto?

(A) 1 : 1

(B) About 3 : 1

(C) About 27 : 1

(D) Need period data to determine this ratio.

18. Planet P has a mass of 4×10^{24} kg and a diameter of 10 000 km. A satellite orbits P at an altitude of 2000 km. What is the period of the satellite?

(A) 1.41 hours (B) 1.61 days

(C) 1.98 days (D) 1.61 years

19. The table gives information about two of the moons orbiting planet R.

Moon	Radius of orbit (km)	Orbital period (days)
X	100 000	2.0
Y	400 000	T

What is the orbital period of moon Y?

(A) 1.4 days (B) 4.0 days

(C) 16 days (D) 32 days

SET 16 Escape Velocity

For relevant questions in this set, take the radius of the Earth as 6380 km, and its mass as 6×10^{24} kg.

1. (a) Clarify the concept of escape velocity.

(b) How will the surface escape velocity (the escape velocity from the surface of a planet) compare with the orbital escape velocity of a satellite (the escape velocity needed from the orbital altitude)?

(c) Explain how a rocket can escape Earth's gravitational field at a speed less than the escape velocity.

2. (a) If an object is given kinetic energy equal to its gravitational potential energy on the surface of the Moon, then it will escape the gravitational field of the Moon and reach infinity. Use this concept to derive an equation for the escape velocity, v_e, of an object, mass m, from the Moon.

(b) A particular neutron star has a radius of 1.6×10^4 m and a mass of 3.1×10^{21} kg. Determine the escape speed, v_e, from the surface of the star.

(c) Determine the gravitational field strength on the surface of this star.

(d) The period, T, of rotation of the star is 2 minutes. Use your answer to (d) to prove that matter is not lost from the surface of the star as a result of its high speed of rotation.

3. (a) Find the escape velocity for a 500 kg projectile from the surface of the Earth.

(b) Find the escape velocity for a 250 kg projectile from the surface of the Earth.

(c) Explain any similarity or difference in your answers to (a) and (b).

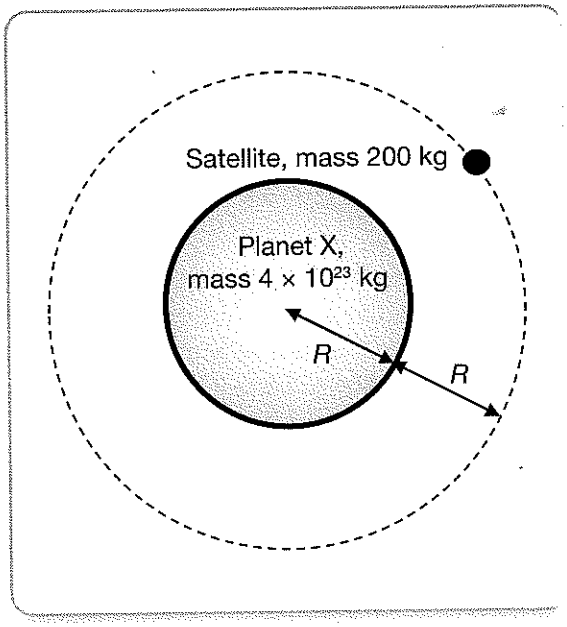
4. The escape velocity of Mars is 5×10^3 m s^{-1} and its radius is 3.397×10^3 km. According to this data, what is the mass of Mars?

5. (a) What is the escape speed on a spherical asteroid whose radius is 525 km and whose gravitational acceleration at the surface is 2.7 m s^{-2}?

(b) What is the mass of the asteroid?

(c) What will happen to a particle leaving the asteroid's surface with a vertical speed of 1000 m s^{-1}?

6. A satellite of mass 200 kg is in orbit around a planet at an altitude of R m which is equal to the radius of the planet. The planet has a mass of 4×10^{23} kg and radius R equal to 5000 km.



(a) Find the surface escape velocity for X.

(b) Determine the escape velocity from the orbital position.

(c) Explain the difference in the two values.

7. (a) Consider two planets A and B. They are of the same radius but planet A is more massive than planet B. Which has the larger escape velocity?

(b) Suppose that the mass of A is actually 9 times the mass of B. How does the escape velocity from A compare to B?

8. Uranus is about 15 times more massive than Earth and its radius is about 4 times Earth's. Given the Earth's escape velocity from Earth's surface is 11 200 m s^{-1}, determine the escape velocity from the surface of Uranus.

9. What is the escape velocity from Jupiter given that its mass is 300 times that of the Earth and its radius is 10 times larger? Take the Earth's escape velocity as 11 200 m s^{-1}.

10. The radius of the Sun is about 6.96×10^5 km and its surface gravitational field is 274 N kg^{-1}. Calculate the escape velocity from the surface of the Sun.

11. The radius of the Moon is 1737 km and the acceleration due to the Moon's gravity is 1.62 m s^{-2}. Calculate the Moon's escape velocity.

12. Calculate the escape velocity of a spacecraft in an Earth orbit with an altitude of 200 km.

13. (a) Imagine an alien race with a planet shrinking ray that reduced the Earth until it was a black hole. The escape velocity for a black hole is equal to the speed of light, 3×10^8 m s^{-1}. What would be the new size of the Earth?

(b) The aliens repeated the shrinking process with the Sun, which has a radius approximately 110 times that of Earth, and a mass of 2.0×10^{30} kg.

(i) What was the escape velocity from the Sun before that aliens shrunk it?

(ii) What would be its new size after it was also shrunk to black hole size?

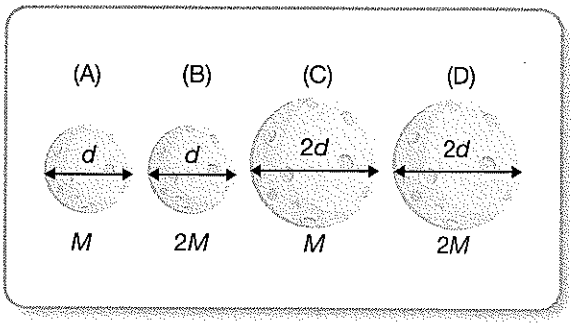
14. A 15.6 km diameter, spherical asteroid with a mass of 3.6×10^9 kg is 400 million kilometres from the Sun.

(a) What is the orbital speed of the asteroid?

(b) What is its orbital period?

(c) What is its surface escape velocity?

15. The diagrams show the sizes and masses of four planets, A, B, C and D. For which planet would the escape velocity be greatest?



16. The escape velocity for planet X is greater than the escape velocity for planet Y. Which statement about planets X and Y is correct?

(A) If the mass of X < the mass of Y, then the diameter of X must be > the diameter of Y.

(B) If the mass of X > the mass of Y, then the diameter of X must be > the diameter of Y.

(C) If the mass of X < the mass of Y, then the diameter of X must be < the diameter of Y.

(D) If the mass of X > the mass of Y, then the diameter of X must be < the diameter of Y.

17. Which of the following factors does *not* affect the magnitude of escape velocity?

(A) The diameter of the planet.

(B) The mass of the planet.

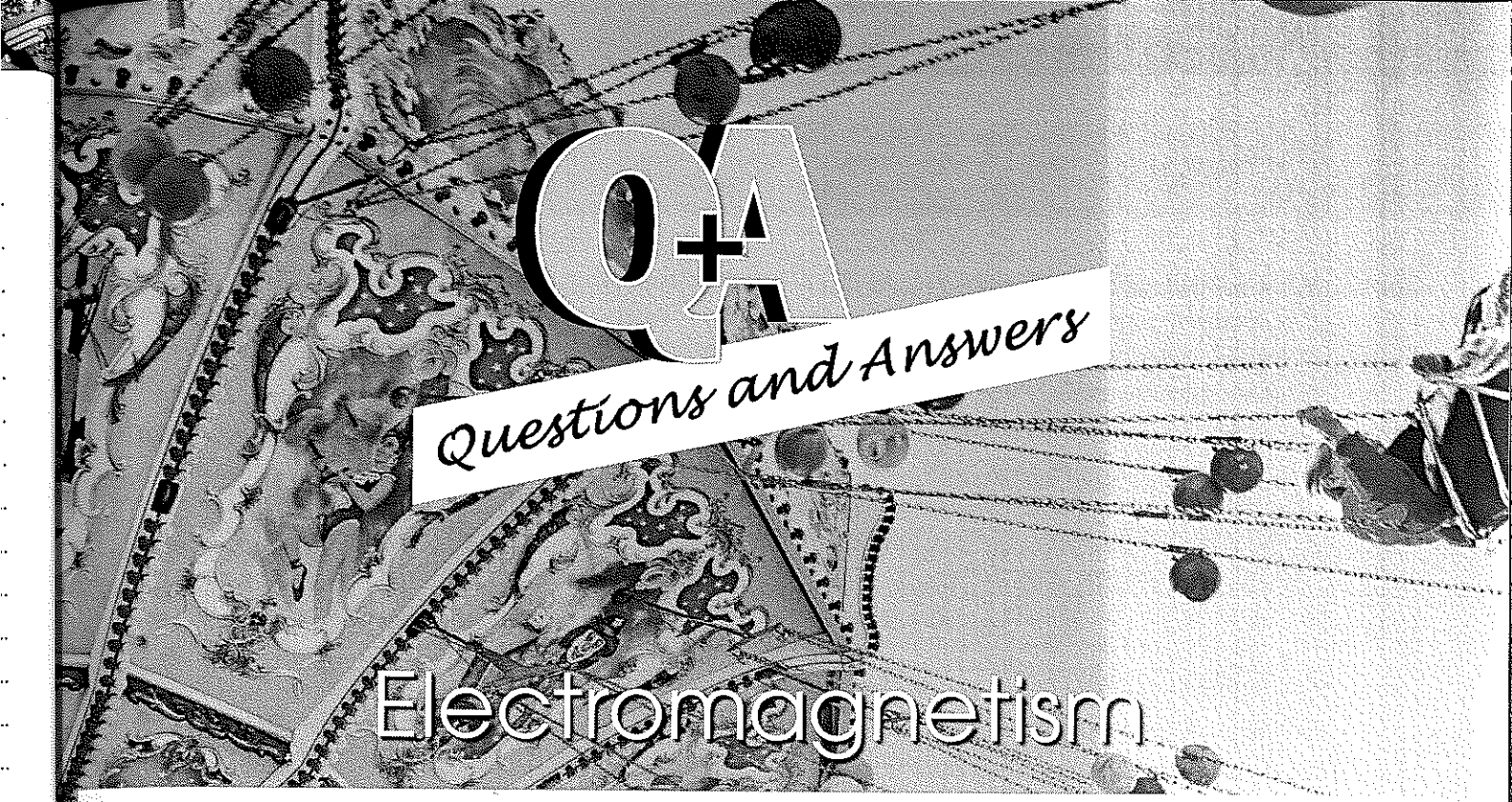
(C) The mass of the rocket.

(D) The universal gravitational constant.

18. Using the information in the table, calculate the missing values in the table.

Object	Mass (kg)	Diameter (km)	Escape velocity (m s^{-1})
Mercury	(a)	4879	4300
Venus	4.9×10^{24}	12 104	(b)
Earth	5.97×10^{24}	(c)	11 1274
Mars	(d)	6794	5021
Jupiter	1.90×10^{27}	142 984	(e)
Saturn	5.69×10^{26}	(f)	35 568
Uranus	(g)	51 800	21 144
Neptune	1.03×10^{26}	49 250	(h)
Pluto	1.27×10^{22}	(i)	1209
The Sun	(j)	1 392 530	619 021
The moon	7.35×10^{22}	3575	(k)

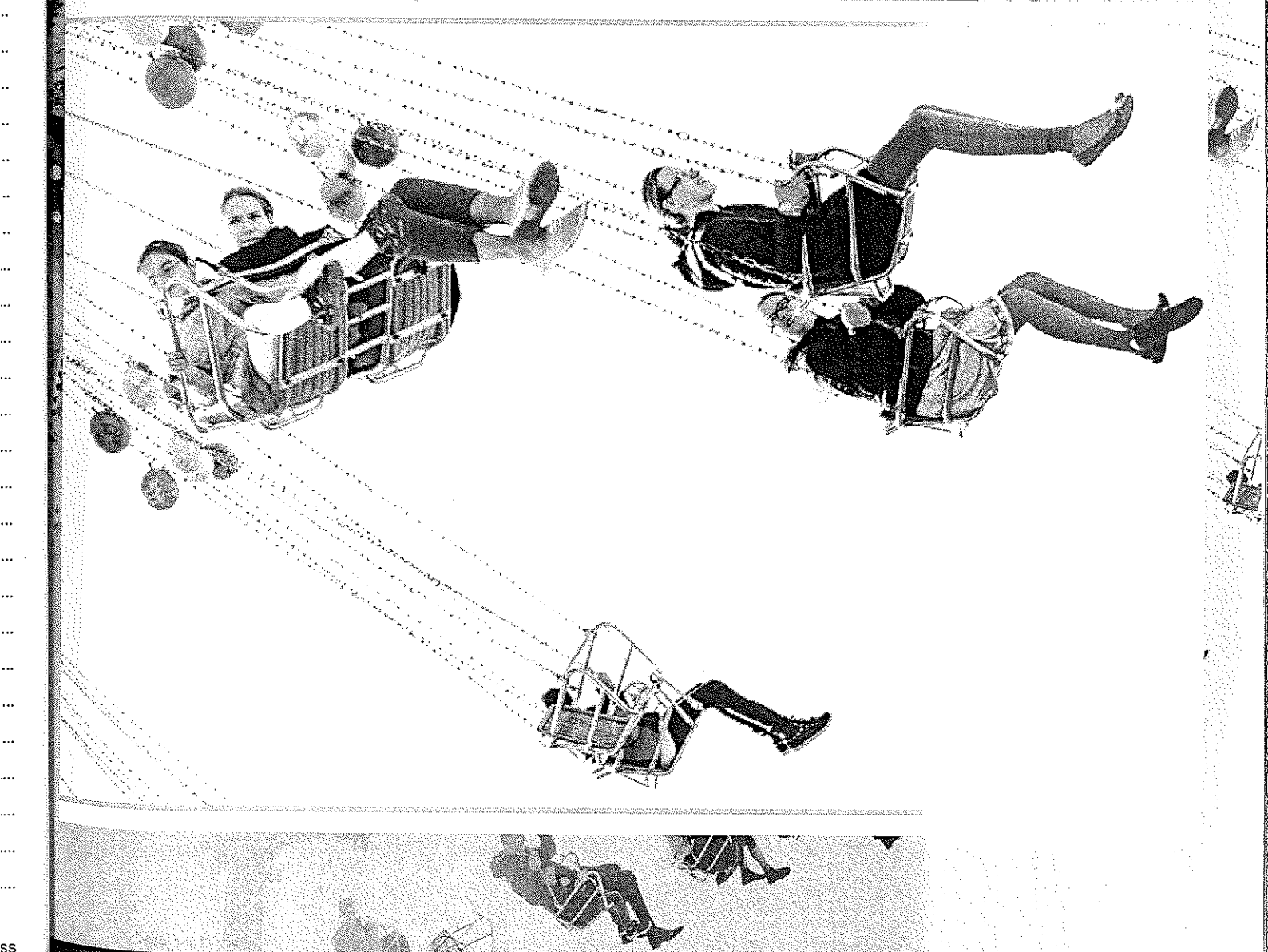
Notes



Q+A

Questions and Answers

Electromagnetism



SET 17 Electrostatic Charge

1. (a) What charge does X have if it has equal numbers of protons and electrons?

(b) What charge does Y carry if it has an excess of electrons?

(c) What charge does Z carry if it has a deficiency of electrons?

(d) Why, in (c) above did the question not ask – What charge does an object carry if it has an excess of protons?

(e) What is the unit, and its symbol, used to measure electrostatic charge?

(f) Define this unit.

(g) What is the charge on an electron?

(h) What is the charge on a proton?

(i) How does electrostatic charge connect to current electricity?

2. How many electrons are lost or gained in producing the following charged objects?

(a) Object P with a charge of –0.6 C.

(b) Object Q with a charge of + 0.75 C.

(c) Object R with charge +4 mC.

(d) Object S with charge –6 mC.

(e) Object T with charge –0.3 μC.

3. What charge does each of the following carry?

(a) E with a deficiency of 4.0×10^{6} electrons.

(b) F with an excess of 2.5×10^{20} electrons.

(c) G with an excess of 1.6×10^{10} electrons.

(d) H with a deficiency of 6.4×10^{9} electrons.

(e) I with an excess of 7.5×10^{16} electrons.

4. When a motor is running, 16 C of charge passes through it in 0.5 s. What current is flowing?

5. 1.6 A flows through a conductor for 15 s. How many electrons pass each point in the conductor in this time?

6. 8.0×10^{16} electrons flow through a conductor every 0.25 s. What current is flowing?

7. If 32 mA of current flows in a wire, how long would it take for one mole of electrons (6.023×10^{23} electrons) to pass a point in the wire?

8. A 60 W light globe circuit carries 0.25 A when it is turned on. If the light is left on for 6 hours each night for a month (30 days), calculate:

(a) The charge passing through the circuit each second.

(b) The number of electrons passing through the circuit each second.

(c) The total charge passing through the circuit during the 6 hours.

(d) The total number of electrons flowing through the circuit in the 6 hours.

(e) The total charge passing through the circuit during the month.

(f) The total number of electrons flowing through the circuit in the month.

9. (a) 6.4×10^{19} electrons flow in a circuit each second. What current flows?

(b) 1.6×10^{17} electrons flow in a circuit each second. What current flows?

(c) 3.2×10^{15} electrons flow in a circuit each minute. What current flows?

(d) 8.0×10^{19} electrons flow in a circuit each minute. What current flows?

(e) 4.0×10^{12} electrons flow in a circuit each 0.1 s. What current flows?

10. (a) How many electrons flow in a circuit if the current in the circuit is 0.3 A?

(b) How many electrons flow in a circuit if the current in the circuit is 2.0 A?

(c) How many electrons flow in a circuit if the current in the circuit is 4.5 mA?

(d) How many electrons flow in a circuit if the current in the circuit is 3.6 μA?

(e) How many electrons flow in a circuit if the current in the circuit is 8.0 μA?

SET 18 Coulomb's Law

1. State Coulomb's law of electrostatics and give the formula used to find the force between static charges. Identify each symbol in the equation.

2. Two charges, A, 3.0×10^{-8} C, and B, 4.0×10^{-8} C are 5 mm apart. What is the force between them?

3. Two charges, X, 3.0×10^{-6} C, and Y, -5.0×10^{-6} C are 10 mm apart. Find the force between them.

4. Two charges, P, 4.0 μC, and Q, 9.0×10^{-6} C are 6 cm apart. What is the force between them?

5. Two charges, E, –3.5 mC, and F, 8.0 μC are 14 cm apart. What is the force between them?

6. Two charges, A, 4.0 nC, and B, 12.0 nC are 4 mm apart. What is the force between them?

7. (a) The force between two identical charges P and Q is 4.0×10^{-8} N attraction. If the charges are 6 mm apart, what is their magnitude?

(b) Are the charges positive or negative? Explain your answer.

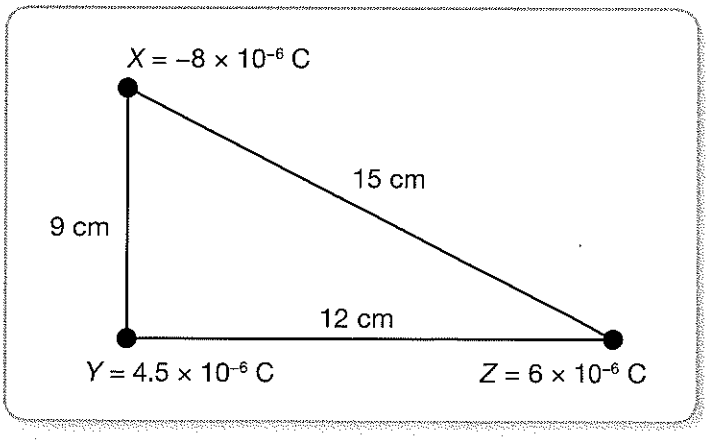
8. Charge X is three times the magnitude of charge Y. X and Y are 5 mm apart and repel each other with a force of 4.5×10^{-1} N. What is the magnitude and sign of each charge?

9. Charges X, Y and Z are in the corners of a triangle as shown in the diagram.

(a) Calculate the force on X due to the other two charges.

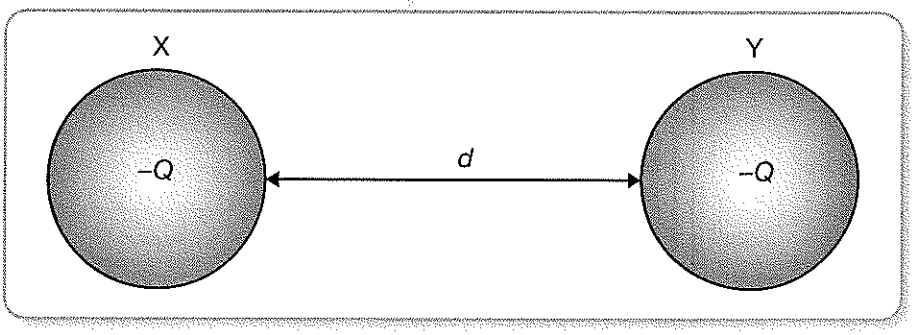
(b) Calculate the force on Y due to the other two charges.

(c) Calculate the force on Z due to the other two charges.



10. X and Y are two charged spheres distance *d* metres apart. Each sphere carries a charge of –*Q*. The force between the spheres is 2.0×10^{-5} N repulsion.

What will be the force between the charges in each of the following cases?



(a) +2*Q* is added to sphere X.

(b) Instead, the distance between X and Y is doubled.

(c) Instead, an additional charge of +3*Q* is added to sphere Y.

(d) Instead, the distance between X and Y is reduced to one third its original value.

(e) Instead, the charge on X is reduced to –0.5*Q*, the charge on Y is increased to –2.5*Q* and the distance between them is halved.

11. Consider a point charge of 2.50×10^{-6} C. What is the magnitude of the force on a charge of 3.00×10^{-8} C placed 20 cm from this charge?

(A) 1.69×10^{2} N

(B) 3.375×10^{-3} N

(C) 3.375×10^{3} N

(D) 1.69×10^{8} N

12. Two isolated particles, X and Y are 4 m apart. Particle X has a net charge of 2Q, and Y has a net charge of Q. The ratio of the magnitude of the electric force on X to that on Y is:

(A) 1 : 1

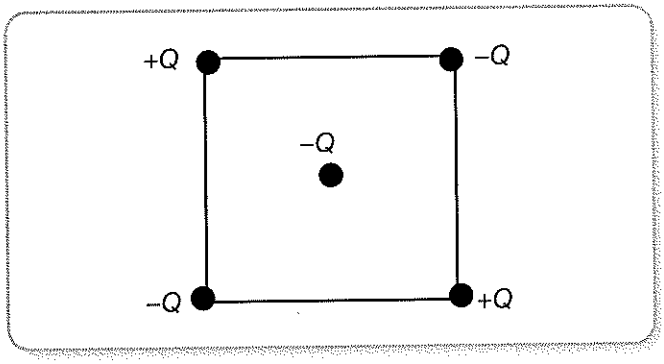
(B) 2 : 1

(C) 1 : 2

(D) 1 : 4

Use the following information to answer the next TWO questions.

Four charges identical in magnitude are placed at the corners of a square. A charge of –Q is placed in the centre of the square.



13. Which statement about the force acting on the charge at the centre is correct?

(A) It is zero.

(B) It is directed to the right.

(C) It is directed down the page.

(D) It is directed into the page.

14. The charge at the centre is now changed to +Q. What effect does this have on the force acting at the centre?

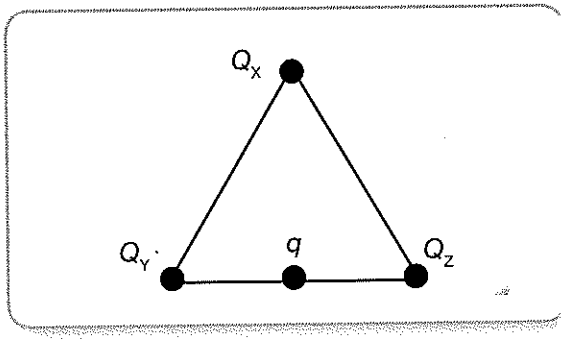
(A) It reverses its net direction.

(B) It doubles the magnitude of the force.

(C) It halves the magnitude of the force.

(D) It has no effect.

15. Three identical charges are placed at the corners of an equilateral triangle. A fourth charge, q, is placed midway between two of these charges as shown.



Is it possible to choose a value for q so that the force on it due to the three charges is zero?

(A) Yes, because it is midway between Q_Y and Q_Z.

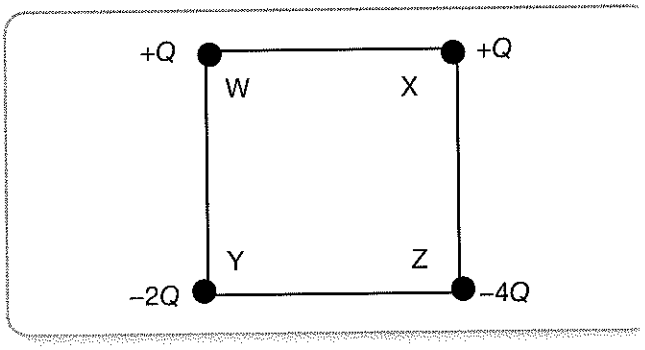
(B) Yes, if the charge on q is opposite the charge on Q_X.

(C) No, because the force vectors on it can never add to zero.

(D) No, because the force vectors due to the three charges all act in the same direction.

Use the following information to answer the next TWO questions.

Consider the system below that consists of four charges placed at the corners of a square. The force charge X puts on charge W is F newtons.



16. In terms of F, what is the force Y places on W?

(A) F (B) $-F$

(C) $2F$ (D) $-2F$

17. What is the force Y places on Z in terms of F?

(A) $2F$ (B) $4F$

(C) $-8F$ (D) $+8F$

SET 19 Electric Field

1.
(a) Define an electric field.
(b) State the units used to measure electric field.
(c) Recall two equations used to calculate electric field.
(d) In which direction relative to the field will a positive charge move when placed in the field?
(e) In which direction relative to the field will a negative charge move when placed in the field?
(f) Why do we use arrows to represent an electric field?
(g) How can we compare the relative strengths of electric fields by looking at diagrams which represent them?
(h) What is the direction of electric field relative to a positive charge?
(i) What is the direction of electric field relative to a negative charge?
(j) What is the direction of electric field relative to a high potential position?
(k) What is the direction of electric field relative to a low potential position?

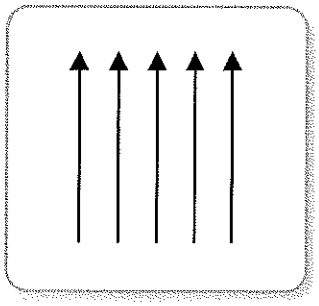
2. A uniform electric field acts to the left. In which direction will each of these particles accelerate?
(a) A proton.
(b) A positron (same mass as electron, but opposite charge).
(c) A neutron.
(d) An antiproton (same mass as proton, but opposite charge).
(e) A helium nucleus (He^{2+}).

3. It is technically incorrect to say that a charge in an electric field will move in a particular direction because of the action of the field. What is the more correct statement and why should we say this in preference?

4. The magnitude of the electric field at a point in space is equal to:
(A) The force that would be felt by a proton if placed at that point.
(B) The force that would be felt by a electron if placed at that point.
(C) The force that would be felt by a test charge if placed at that point.
(D) The force per unit charge at that point.

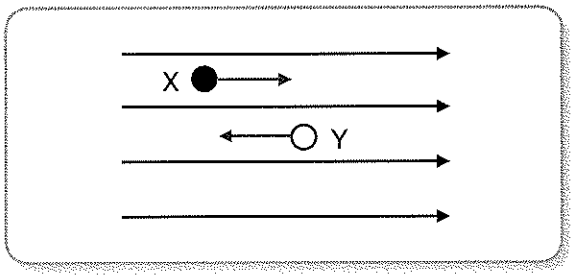
5. Which of the following properties of electric field lines is *incorrect*?
(A) A tangent to a field line gives the direction of the field at that point.
(B) The number of field lines per unit cross-sectional area indicates the magnitude of the field.
(C) Field lines can cross only at right angles.
(D) Field lines can never cross.

6. Suppose an electric field is constant in a certain region of space and directed upward as shown. An electron is fired to the right into the field. What is the direction of the force that the field exerts on the electron once it has entered the field?
(A) To the left.
(S) To the right.
(C) Upward.
(D) Downward.



7. The definition of the direction of electric field refers to the direction of the force acting on a 'test charge' in the field. What is this test charge?
(A) It is a zero magnitude charge.
(B) It has a charge of +1 C.
(C) It has a charge of $+1.6 \times 10^{-19}$ C.
(D) It can be +1 C or –1 C depending on the field.

8. The diagram shows two particles in an electric field directed from left to right. The arrows attached to the particles show the direction each particle will move because of the field.



(a) Which end of the field is the high potential end?

(b) Which end of the field would be 'positive'?

(c) Identify the sign of the charge on X.

(d) Identify the sign of the charge on Y.

9. Which statement about electric field lines is correct?

(A) They are vectors in the direction of the electric field.

(B) They cross each other midway between like charges.

(C) They are directed away from negative towards positive.

(D) They are always parallel to each other.

10. In a uniform electric field directed from left to right, which way will a negatively charged particle move?

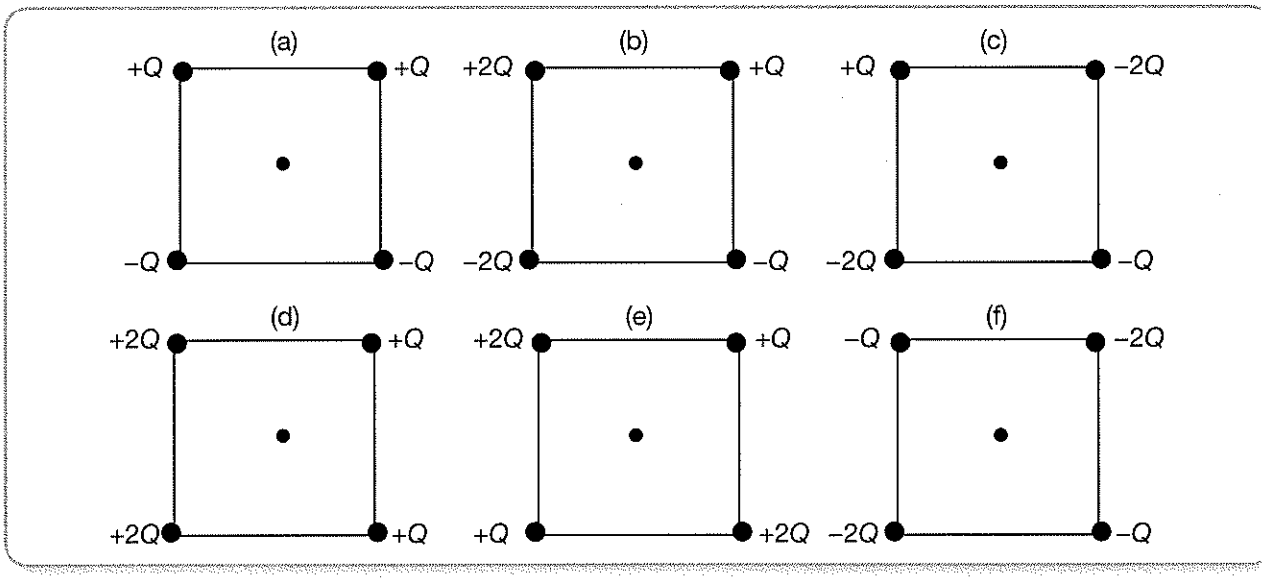
(A) Left.

(B) Right.

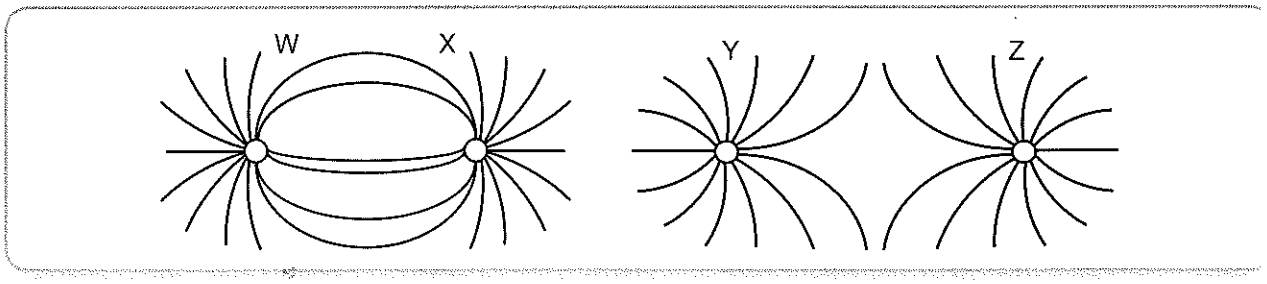
(C) Up at 90° to the field.

(D) Down at 90° to the field.

11. Four charges are arranged on the corners of six different rectangles in the situations shown below. Copy each rectangle separately, and place a vector with its tail at the centre of each rectangle showing the direction of the electric field at that point. No indication of magnitude is required, although you will have to consider the magnitudes of the forces involved in order to get the correct direction.

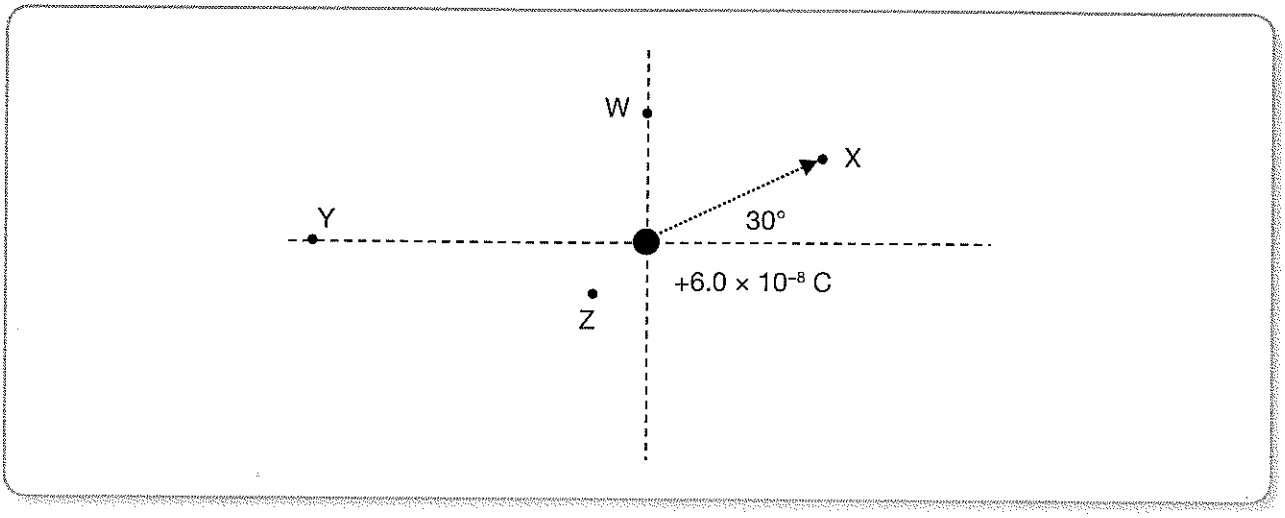


12. Suggest possible signs for the charges W. X, Y and Z in the diagrams below.



SET 20 Electric Field Strength

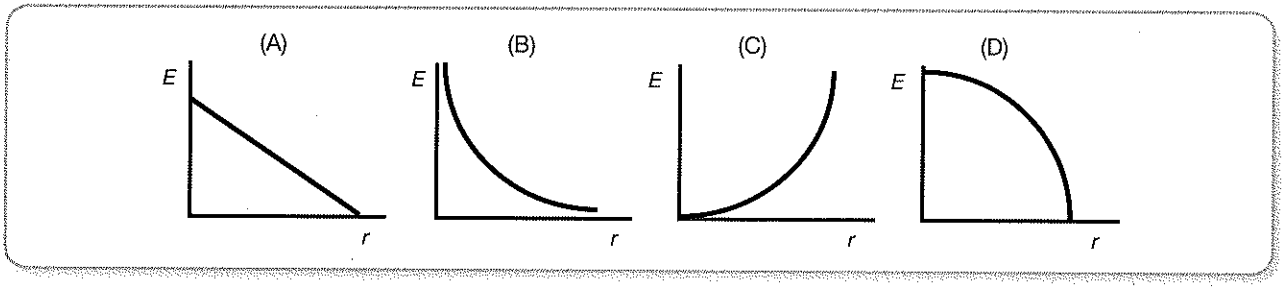
1. Consider an isolated point charge of $+6.0 \times 10^{-8}$ C as shown. Point W is 15 cm from the charge, X is 20 cm from the charge, Y is 45 cm from it, and Z is 7.5 cm away.



(a) Find the magnitude and direction of the electric field at each point W, X, Y and Z.

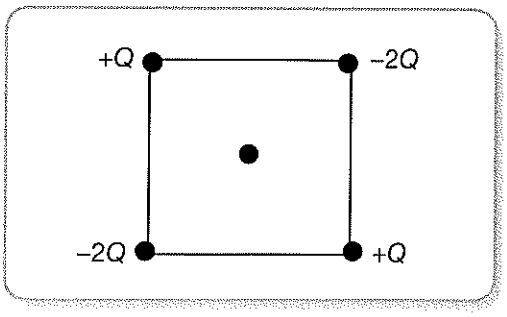
(b) Find the new magnitude and direction of the field at each point W, X, Y and Z if the 6.0×10^{-8} C charge is removed and replaced with a charge of -3.0×10^{-8} C.

2. A solid, small point sphere has radius R and carries positive charge Q. Which of the following graphs represents the electric field E as a function of the distance r from the centre of the sphere?



Use the following information to answer the next TWO questions.

Four charges identical in magnitude are placed at the corners of a square. A charge of $-Q$ is placed in the centre of the square. The magnitude of the electric field at the centre of the square due to the $+Q$ charge at the top left corner of the square is E.



3. What is the magnitude of the field at the centre due to the charge at the bottom left of the square?

(A) 0 (B) E (C) $-E$ (D) $2E$

4. What is the field at the centre due to all four corner charges?

(A) 0 (B) $\sqrt{(2E)}$ towards the top of the square.

(C) $\sqrt{(2E)}$ towards the bottom of the square. (D) $4E$

5. Calculate the electric field strength 12 cm from a charge of -3.0×10^{-9} C.

6. The field strength 2 cm from a charge is 5×10^{3} N C^{-1} directed towards the charge. What is the value of the charge?

7. At point P, some distance from a charge of magnitude 4.8×10^{-6} C, the field strength is 1.2×10^{4} N C^{-1} directed away from the charge.

(a) How far is point P from the charge?

(b) What sign is the charge? Justify your answer.

8. Calculate the electric field strength 15 cm from a charge of 5.0×10^{-8} C.

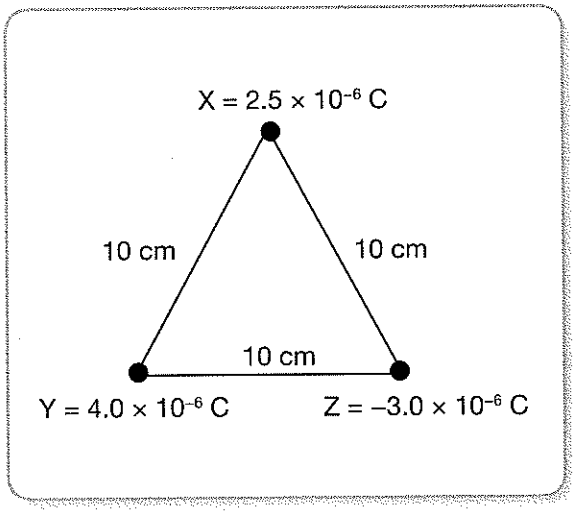
9. The field strength 14 cm from a charge is 5×10^{-6} N C^{-1} directed towards the charge. What is the value of the charge?

10. At point P, some distance from a charge of magnitude 6.4×10^{-6} C, the field strength is 5.6×10^{3} N C^{-1} directed towards the charge.

(a) How far is point P from the charge?

(b) What sign is the charge? Justify your answer.

11. Three charges, X, Y and Z are at the corners of a 10 cm equilateral triangle as shown.

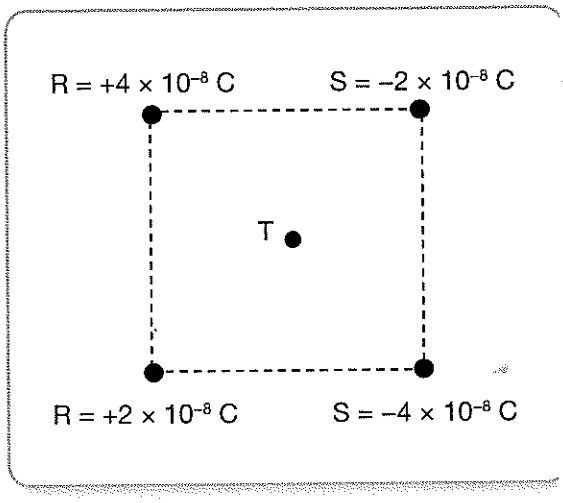


(a) Find the electric field at X due to the other two charges.

(b) Find the electric field at Y due to the other two charges.

(c) Find the electric field at Z due to the other two charges.

12. Four charges, P, Q, R and S are at the corners of a 10 cm square as shown.

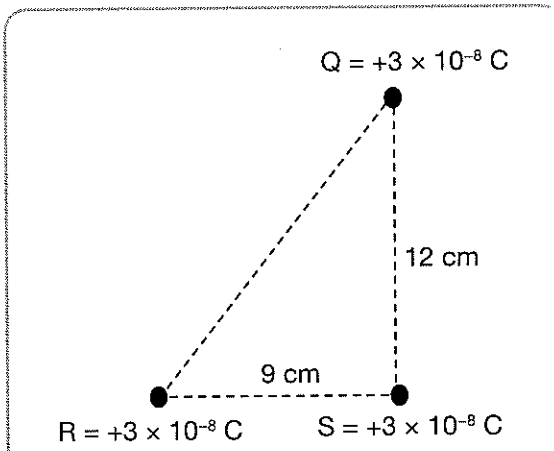


(a) Determine the electric field at charge P due to the other three charges.

(b) Determine the electric field at charge Q due to the other three charges.

(c) Find the electric field at T, in the centre of the square due to the four charges.

13. Three point charges, Q, R and S are placed at the corners of a right-angled triangle as shown in the diagram.



(a) Determine the strength of the electric field at charge Q due to the other two charges.

(b) Determine the *E* strength of the electric field at charge R due to the other two charges.

(c) Determine the strength of the electric field at charge S due to the other two charges.

SET 21 Force on a Charge in an Electric Field

1. A helium nucleus, charge + 3.2×10^{-19} C experiences a force of 6.4×10^{-15} N towards the south due to the action of an electric field. What is the magnitude and direction of the electric field at that point?

(A) 5×10^{-5} N C^{-1} north.

(B) 5×10^{-5} N C^{-1} south.

(C) 2×10^{5} N C^{-1} north.

(D) 2×10^{4} N C^{-1} south.

2. What is the force on $-3.6\ \mu$C in an electric field of strength 5.0×10^{-4} N C^{-1} west?

(A) 1.8×10^{-9} N west.

(B) 1.8×10^{-9} N east.

(C) 139 N west.

(D) 139 N east.

3. What is the acceleration of a proton on an electric field of strength 4.0×10^{-6} N C^{-1}? The charge on a proton is $+1.6 \times 10^{-19}$ C and its mass is 1.7×10^{-27} kg.

(A) 4.25×10^{-14} m s^{-2}

(B) 2.66×10^{-3} m s^{-2}

(C) 376.5 m s^{-2}

(D) 2.35×10^{13} m s^{-2}

4. A charge of -6.0×10^{-8} C experiences a force of 1.5×10^{-4} N east at point P. What is the strength of the electric field at P?

(A) 2.5×10^{3} N C^{-1} east.

(B) 2.5×10^{3} N C^{-1} west.

(C) 4×10^{-4} N C^{-1} east.

(D) 4×10^{-4} N C^{-1} west.

5. What is the force on a charge of +8.0 nC in an electric field of strength 7.5×10^{-3} N C^{-1} west?

(A) 6×10^{-11} N west.

(B) 6×10^{-11} N east.

(C) 1.1×10^{-6} N west.

(D) 1.1×10^{-6} N east.

6. A force of 2.4×10^{-4} N east acts on a charge at point P in an electric field of strength 6.0×10^{-4} N C^{-1} east. What is the magnitude and sign of the charge?

(A) +0.4 C

(B) $+1.44 \times 10^{-7}$ C

(C) −0.4 C

(D) -1.44×10^{-7} C

7. A charge of $-8.0\ \mu$C experiences a force of 2.1×10^{-5} N south at point P. What is the strength of the electric field at P?

(A) 0.38^{5} N C^{-1} south.

(B) 0.38 N C^{-1} north.

(C) 2.6 N C^{-1} south.

(D) 2. 6 N C^{-1} north.

8. A force of 2.5×10^{-10} N south acts on a charge in an electric field of strength 1.8×10^{-3} N C^{-1} south. What is the charge?

(A) 7.2×10^{6} C south.

(B) 7.2×10^{6} C north.

(C) $+1.4 \times 10^{-7}$ C south.

(D) -1.4×10^{-7} C north.

9. What is the acceleration of an electron in an electric field of strength 2.5×10^{-6} N C^{-1} east? The mass of an electron is 9.1×10^{-31} kg.

(A) 2.3×10^{-6} m s^{-2} west.

(B) 2.3×10^{-6} m s^{-2} east.

(C) 4.4×10^{5} m s^{-2} west.

(D) 4.4×10^{5} m s^{-2} east.

10. An electron experiences a force of 6.0×10^{-14} N south. What is the magnitude and direction of the electric field at that point?

(A) 3.75×10^{5} N C^{-1} south.

(B) 3.75×10^{5} N C^{-1} north.

(C) 2.67×10^{-6} N C^{-1} south.

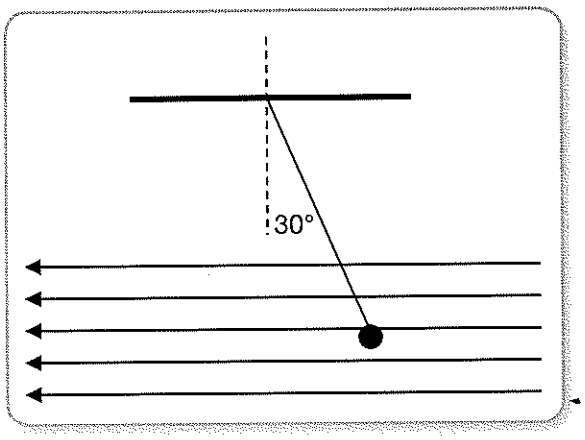
(D) 2.67×10^{-6} N C^{-1} north.

11. The charge on a proton is $+1.6 \times 10^{-19}$ C and its mass is 1.67×10^{-27} kg. What is the acceleration of a proton on an electric field of strength 2.4×10^{-9} N C^{-1} west?

(A) 0.23 m s^{-2} west. (B) 0.23 m s^{-2} east.

(C) 4.35 m s^{-2} west. (D) 4.35 m s^{-2} east.

12. A uniform electric field points to the left. A small metal ball charged to –2.0 mC hangs at a 30° angle from a string of negligible mass attached to the ceiling as shown.



The tension in the string is measured to be 0.1 N. What is the magnitude of the electric field?

(A) 0.05 N C^{-1} (B) 2.5 N C^{-1}

(C) 25 N C^{-1} (D) 50 N C^{-1}

Use the following information to answer the next TWO questions.

X, carrying a charge of $+5.4 \times 10^{-10}$ C and Y, with a charge of $+4.8 \times 10^{-10}$ C are separated by 36 cm.

13. What is the electric field midway between X and Y?

(A) 3.0 N C^{-1} towards X.

(B) 3.0 N C^{-1} towards Y.

(C) 16.7 N C^{-1} towards X.

(D) 16.7 N C^{-1} towards Y.

14. What would be the force on an electron placed midway between these charges?

(A) 4.8×10^{-19} N towards X.

(B) 4.8×10^{-19} N towards Y.

(C) 2.7×10^{-18} N towards X.

(D) 2.7×10^{-18} N towards Y.

Questions 16 and 17 use information from Question 15.

15. What will be the electric field 0.5 cm from a charge of 3.8×10^{-10} C?

(A) 3.42 N C^{-1} (B) 6.84 N C^{-1}

(C) 13.68 N C^{-1} (D) 68.4 N C^{-1}

16. What is the force on an electron in this field?

(A) 5.5×10^{-19} N

(B) 1.1×10^{-18} N

(C) 2.2×10^{-18} N

(D) 1.1×10^{-17} N

17. What would be the acceleration of the electron?

(A) 6.0×10^{11} m s^{-2}

(B) 1.1×10^{12} m s^{-2}

(C) 2.4×10^{12} m s^{-2}

(D) 1.2×10^{13} m s^{-2}

Use the following information to answer the next TWO questions.

A force of 1.75×10^{-14} N in an easterly direction acts on a charge in an electric field of 3.6×10^{-5} N C^{-1} west.

18. What is the magnitude of the charge?

(A) 4.86×10^{-10} C (B) 6.3×10^{-9} C

(C) 6.3×10^{-19} C (D) 2.06×10^{9} C

19. What is the sign of the charge?

(A) Positive because the force and field are in opposite directions.

(B) Negative because the force and field are in opposite directions.

(C) Positive because the force and field are in the same direction.

(D) Negative because the force and field are in the same direction.

20. Two charges M, 3×10^{-6} C and N, 2×10^{-6} C are 54.5 cm apart. Where will the electric field be zero?

(A) 24.5 cm from M.

(B) 27.5 cm from M.

(C) 30 cm from M.

(D) 36 cm from M.

Use the following information to answer the next TWO questions.

An electron is in an electric field of strength 2.5 N C^{-1}.

21. What force acts on the electron?

(A) 4×10^{-19} N in the direction of the field.

(B) 4×10^{-19} N in the opposite direction to the field.

(C) 4×10^{-19} N at 90° to the field.

(D) 4×10^{-19} N parallel to the field.

22. How would this force vary if a magnesium ion, Mg^{2+} was placed in the field?

(A) Force would be half the magnitude and in the opposite direction to the field.

(B) Force would be twice the magnitude and in the opposite direction to the field.

(C) Force would be half the magnitude and in the direction of the field.

(D) Force would be twice the magnitude and in the direction of the field.

23. Protons experience a 4.26×10^{-19} N force to the right when placed in an electric field.

What is the strength and direction of the electric field?

(A) 6.82 V m^{-1} to the right.

(B) 6.82 V m^{-1} to the left.

(C) 2.66 V m^{-1} to the right.

(D) 2.66 V m^{-1} to the left.

24. A force of 1.2×10^{-12} N south acts on a charge at point P in an electric field of strength 2.4×10^{-4} N C^{-1} north.

(a) Calculate the magnitude of the charge.

(b) Identify the sign of the charge. Justify your answer.

25. An electron, charge -1.6×10^{-19} C, in a cathode ray tube experiences a force of 3.0×10^{-15} N towards the south.

Calculate the magnitude and direction of the electric field at that point.

26. Calculate the acceleration of a proton on an electric field of strength 4.0×10^{-5} N C^{-1}. The charge on a proton is $+1.6 \times 10^{-19}$ C and its mass is 1.67×10^{-27} kg.

27. A charge of +6.0 μC experiences a force of 1.5×10^{-3} N east at point P. Calculate the strength of the electric field at P.

28. A charge of –3.0 μC experiences a force of 4.2×10^{-3} N south at point P. Calculate the strength of the electric field at P.

29. A charge of +4.0 μC is in an electric field of strength 1.5×10^{-3} N C^{-1} north at point P. Calculate the force on the charge.

30. A charge of –4.8 μC is in an electric field of strength 4.0×10^{-3} N C^{-1} west at point P. Calculate the force on the charge.

31. A force of 1.5×10^{-3} N east acts on a charge at point P in an electric field of strength 1.5×10^{-3} N C^{-1} west. Calculate the magnitude and sign of the charge.

32. A force of 1.2×10^{-12} N south acts on a charge at point P in an electric field of strength 3.6×10^{-3} N C^{-1} south. Calculate the magnitude and sign of the charge.

33. An electron, charge -1.6×10^{-19} C, in a cathode ray tube experiences a force of 6.0×10^{-15} N towards the south. Calculate the magnitude and direction of the electric field at that point.

34. Calculate the acceleration of a proton on an electric field of strength 2.55×10^{-5} N C^{-1}. The charge on a proton is $+1.6 \times 10^{-19}$ C and its mass is 1.7×10^{-27} kg.

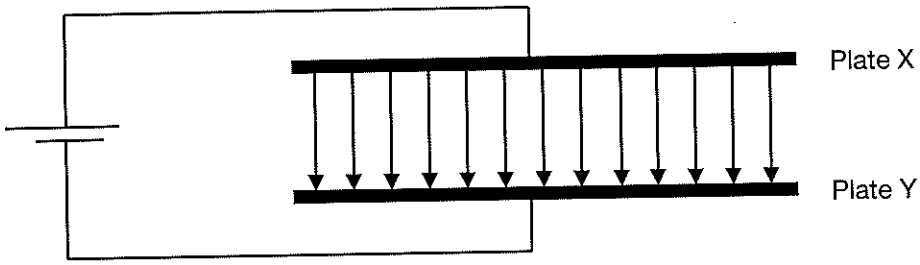
35. An electron flows through a 20 cm long conductor which has a potential difference of 30 V applied across its ends. What is the force on the electron?

36. (a) Compare the force acting on a proton and an electron fired at equal speeds, in the same direction, into a electric field. Justify your answer.

(b) Compare the forces if the direction of travel of the particles was opposite. Justify your answer.

SET 22 Electric Field Strength Between Parallel Plates

1. The diagram shows electric field lines representing the electric field between two parallel plates X and Y.

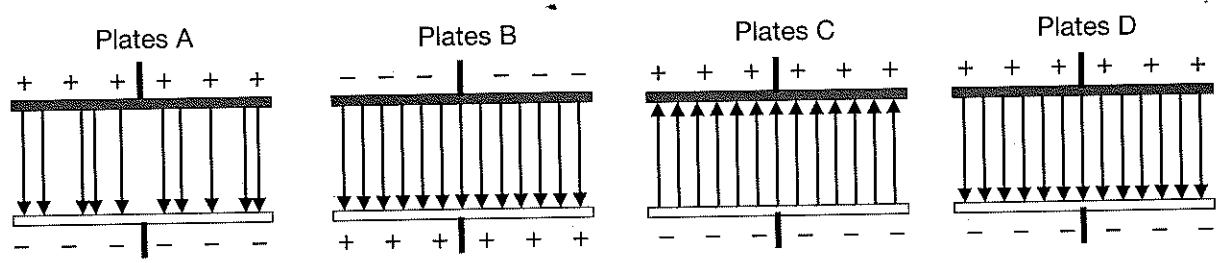


(a) Describe the nature of the field between the plates.

(b) Which plate is the high potential plate?

(c) Which plate is the positively charged plate?

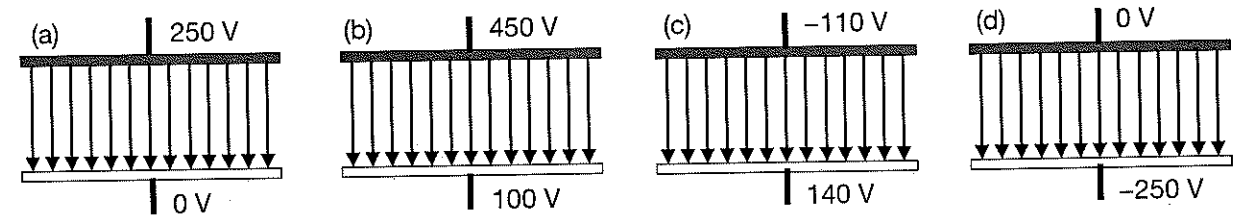
2. The diagrams show the electric field lines between four pairs of charged parallel plates.



(a) Which fields have been drawn correctly?

(b) Explain why the others are incorrect.

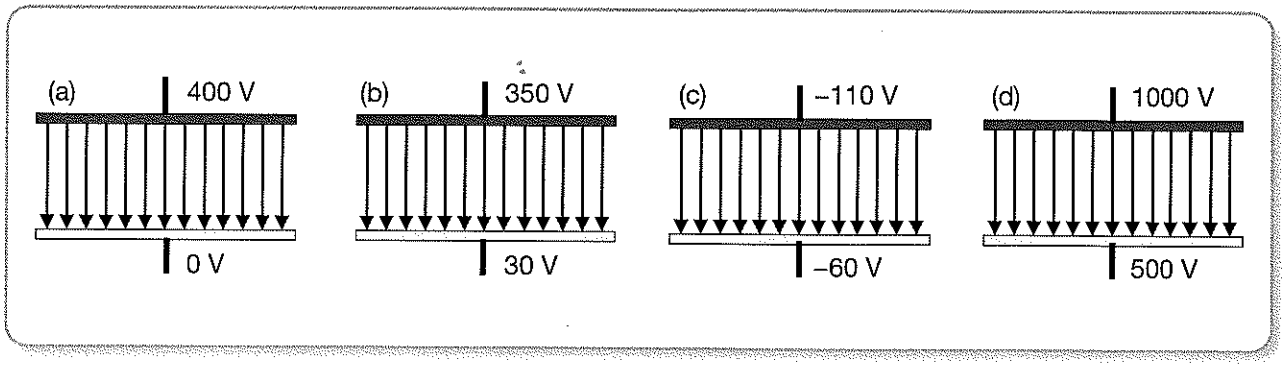
3. Consider the four pairs of charged parallel plates below. Each pair of plates is separated by 10 mm of air.



(a) Calculate the strength of the electric field between each pair of parallel plates.

(b) Explain the similarity of your answers.

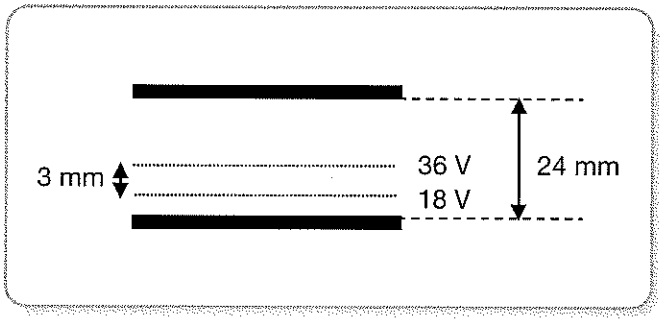
4. Consider the four pairs of charged parallel plates below. Each pair of plates is separated by 10 mm of air.



Calculate the strength of the electric field between each pair of parallel plates.

Use the following information to answer the next TWO questions.

The diagram shows two equipotential lines 3.0 mm apart between two charged, parallel plates. The plates are 24 mm apart.



5. What is the potential difference between the plates?

(A) 144 V
(B) 288 V
(C) 1500 V
(D) Unable to be determined with this data.

6. What is the strength of the electric field between the plates?

(A) 6 V m^{-1}
(B) 600 V m^{-1}
(C) 6000 V m^{-1}
(D) Unable to be determined with this data.

7. A charge of 4.0×10^{-8} C experiences a force of 2.0×10^{-3} N when it is placed in between two parallel charged plates which have a potential difference of 250 V across then.

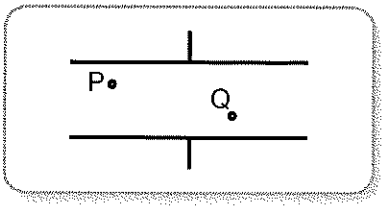
What is the distance between the plates?

(A) 8×10^{-6} m (B) 5 mm
(C) 1.25 m (D) 200 m

8. The potential difference across two parallel plates is 40 V. This produces an electric field of strength 1.6×10^{2} V m^{-1}. What is the separation between the plates?

(A) 0.4 mm (B) 40 mm
(C) 0.25 m (D) 2.5 m

9. Consider points P and Q between two parallel, charged plates as shown. Which statement about the electric field at P and Q is correct?



(A) $E_P > E_Q$
(B) $E_P < E_Q$
(C) $E_P = E_Q$
(D) Cannot be determined without knowing the charges on the plates.

10. A positively charged particle with a mass of 5.0×10^{-4} kg is placed between two parallel plates, 4.5 mm apart and the voltage across the plates adjusted until the particle is suspended between them. The potential difference across the plates is 850 V at this time.

(a) What is the magnitude of the charge on the particle?

(b) Which plate of the two is the positive plate? Justify your answer.

11. Two parallel plates have a potential of V volts across them. They are 4.0 cm apart. At point P, midway between the two plates, the strength of the electric field is E N C^{-1}.

(a) What will be the strength of the electric field at point Q, 1.0 cm from the positive plate?

(b) What will be the strength of the electric field at point Q, 1.0 cm from the negative plate?

(c) What will be the strength of the electric field at point P, if the voltage across the plates is halved?

(d) What will be the strength of the electric field at point P, if the distance between the plates is doubled?

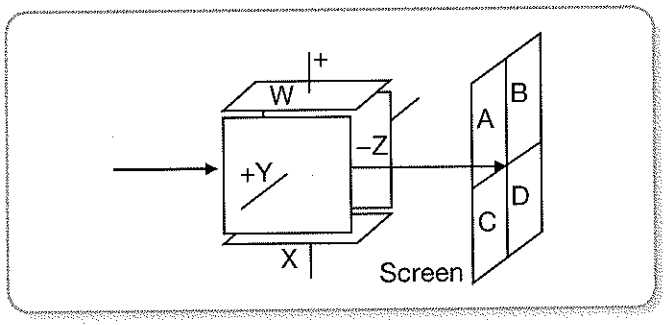
(e) What will be the strength of the electric field at point P if a charge of 6.0×10^{-6} C is placed at P?

(f) What will be the ratio of the forces acting on charges of 6.0×10^{-6} C placed at P and at Q?

Use the following information to answer the next TWO questions.

A beam of cathode rays is fired through the centre of the gap between two pairs of charged parallel plates as shown. If undeflected, they would hit the screen dead centre as shown by the arrow.

A potential of 10 V is applied across plates W and X. A potential difference of 20 V is applied across plates Y and Z. The polarities of the plates are shown on the diagram.



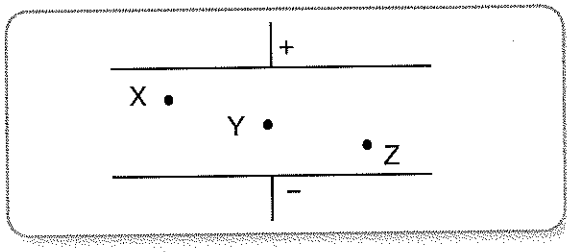
12. In which region does the beam hit the screen?

(A) A (B) B
(C) C (D) D

13. Which plate polarities would result in the cathode ray beam hitting section C of the screen?

(A) X positive, Y positive.
(B) W positive, Y positive.
(C) X positive, Z positive.
(D) W negative, Y negative.

14. The diagram shows three identically charged particles in the electric field between two plates. Which statement about the forces acting on these charges due to the electric field is correct?



(A) $F_X > F_Y > F_Z$ (B) $F_X > F_Z$, and $F_Z = 0$
(C) $F_Z > F_Y > F_X$ (D) $F_X = F_Y = F_Z$

Use the following information to answer the next TWO questions.

Two parallel plates have a potential difference of 300 V between them. They are 2.0 cm apart.

15. What is the strength of the electric field between the plates?

(A) 6 V m^{-1} (B) 150 V m^{-1}
(C) 600 V m^{-1} (D) 15 000 V m^{-1}

16. What would be the new strength if the potential difference between the plates was halved, and the distance between them doubled?

(A) 150 V m^{-1} (B) 3750 V m^{-1}
(C) 1200 V m^{-1} (D) 60 000 V m^{-1}

17. The electric field between two parallel plates is E V m^{-1}. Predict the new field if:

(a) The distance between the plates is tripled.

(b) The voltage across the plates is doubled.

(c) The distance between the plates is tripled and the voltage across them doubled.

(d) The charge on a particle placed between the plates is doubled.

(e) The area of each plate is doubled.

(f) The distance between the plates is doubled and the potential difference across them is doubled.

18. An electron moving at 2.0×10^3 m s^{-1} enters an electric field between two parallel plates separated by 10 cm. The potential difference between the plates is 75 V.

(a) What is the force on the electron?

(b) Describe the motion and the path taken by the electron as it moves into the field.

SET 23 Work Done By a Field

1. A test charge moves from point A to point B in an electric field. The work done by the field during this process is equal to:

(A) The potential difference between points A and B.

(B) The difference in the potential energy of the charge at point A and at point B.

(C) The difference between the value of the electric field at point A and at point B.

(D) Zero if the electric field is uniform.

2. Consider a uniform electric field directed from left to right. In which case would an external force do work in moving a positive test charge in this field?

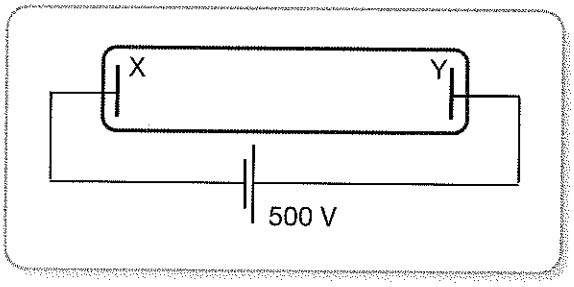
(A) If it was moved from left to right.

(B) If it was moved from right to left.

(C) If it was moved sideways at 90° to the field.

(D) If it was moved vertically upwards through the field.

3. The diagram represents a cathode ray tube. Each second, 2.5×10^{15} electrons cross the evacuated gap between X and Y.

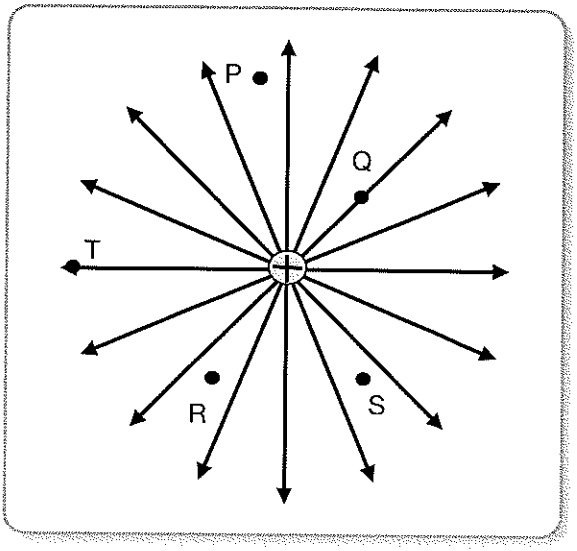


(a) What is the direction of the electric field inside the cathode ray tube? Justify your answer.

(b) What current flows through the circuit which includes the cathode ray tube?

(c) How much work is done on each electron as it moves between X and Y?

4. Consider the following diagram which shows several points (P, Q, R, S and T) in the electric field around a positive point charge.



(a) At which point would the electric field be strongest? Justify your answer.

(b) Compare the work needed to move identical positive charges from P to R and from P to T. Justify your answer.

(c) Imagine a positive test charge brought from infinity to each point near this charge. Which point would require the most work done on the test charge?

In talking to students about this situation, a teacher said that no work needs to be done to move a positive test charge from R to S.

(d) Assess this statement.

(e) Would the same statement hold if the test charge was negative? Justify your answer.

5. A negative point charge, Q at point X midway between two oppositely charged parallel plates moves at constant speed to point Y also midway between the plates. Which statement is correct?

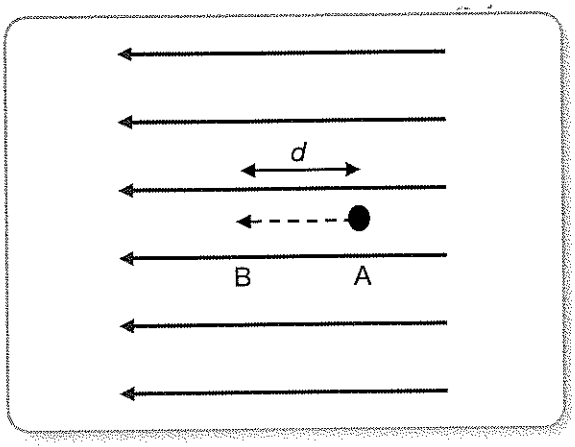
(A) The work done in moving Q is zero.

(B) Work needs to be done on Q to move it.

(C) The field does work on Q when it moves.

(D) Work will have to be done on Q by an external force.

6. A positive charge is moved from point A to point B, a distance of d metres, in an electric field as shown in the diagram.



(a) Write an expression for the work done on the charge moving from X to Y in terms of the force and the distance it moves.

(b) Write an expression for the work done on the charge in terms of the charge and the potential through which it moves (V).

(c) Write an expression for the force applied to the charge to do this work in terms of the charge and the electric field strength.

(d) Combine your equations to derive an expression for the strength of the electric field in terms of the potential and distance through which the charge moves.

(e) What units are used to measure electric field?

7. A lightning strike transfers 250 C of charge 1.6 km to the ground through a potential difference of 1.5×10^6 V. How much energy was carried by the lightning?

(A) 1.67×10^{-4} J

(B) 6000 J

(C) 2.34×10^5 J

(D) 3.75×10^8 J

8. The potential across a 25 cm long discharge tube is 30 000 V. What is the kinetic energy of an electron, just before it hits the anode?

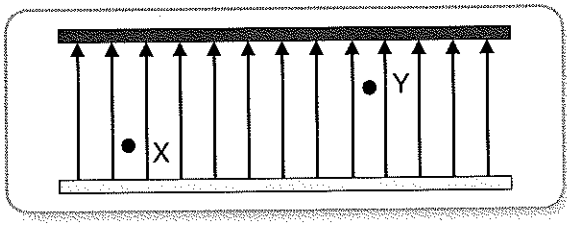
(A) 1.92×10^{-14} J

(B) 1.92×10^{-16} J

(C) 1.20×10^{-15} J

(D) 4.80×10^{-15} J

9. The diagram shows the electric field between two parallel plates and two positions X and Y, between them. A charge of -8.0×10^{-18} C is placed at point X.



(a) Identify the positive plate. Justify your answer.

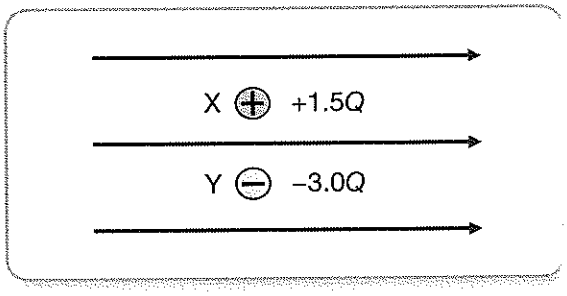
(b) Predict the direction of the force on the charge. Justify your answer.

(c) The charge is moved from X to Y. Identify any changes to the force on it. Justify your answer.

(d) In moving from X to Y, is any work done on the charge? If so, identify the source of the work.

(e) If a similar, but positive charge was moved from X to Y, identify the change in the force acting on it. Explain your answer.

10. The diagram shows two charges, X equal to 1.5Q, and Y, equal to –3.0Q, in an electric field which is directed from the left to the right.



(a) If left undisturbed, what will happen to each of these charges?

(b) Identify if work needs to be done, and by what, if X is moved to the left.

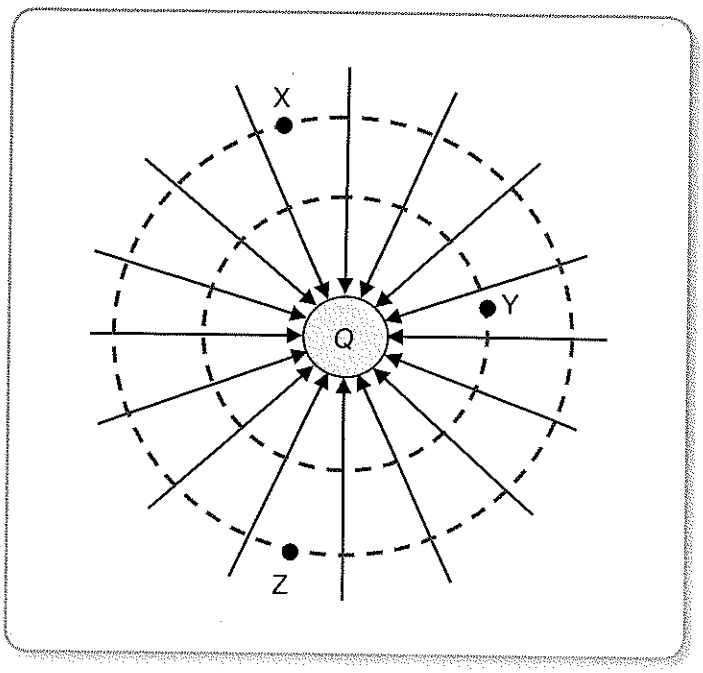
(c) Identify if work needs to be done, and by what, if X is moved to the right.

(d) Identify if work needs to be done, and by what, if Y is moved to the left.

(e) Identify if work needs to be done, and by what, if Y is moved to the right.

(f) In each case in (b) to (d) above, compare the amounts of work needed to be done on X compared to Y.

11. The diagram shows the electric field around a charge Q, and two equipotential lines (the dotted circles) drawn around the charge.



(a) A charge is moved along the equipotential line from point X to point Z on this exponential line. Which of the following statements is true?

(A) Work is done whenever a charge is moved along an equipotential line.

(B) No work is required to move a charge along an equipotential line.

(C) Equipotential lines are parallel to field lines.

(D) Work is done whenever a charge is moved from one equipotential line to another.

(b) What would be the true statement if the charge was moved from X to Y?

12. The electric field plates of an oscilliscope are 5.0 mm apart and have a potential difference of 20 V across them.

(a) What is the strength of the electric field between the plates?

(b) Describe the motion of an electron released from the negative plate and travelling to the positive plate.

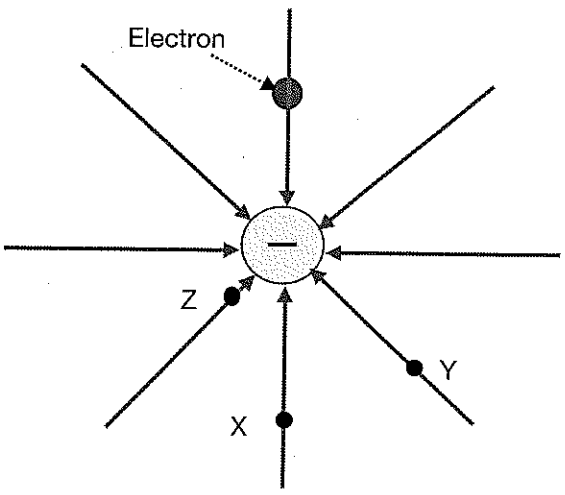
(c) What will be the force on this electron?

(d) What will be its acceleration?

(e) What work will the field do on this electron?

(f) With what speed will the electron reach the positive plate?

13. An electron is at point X in the electric field near a point negative charge as shown in the diagram.



Points Y and Z are other points in the field. Y is the same distance from the charge as X, but Z is much closer to the charge.

As the electron moves from X to Z, 4.8×10^{-15} J of work is done on it.

(a) What is the source of the work done on the electron, the field or an external force? Justify your answer.

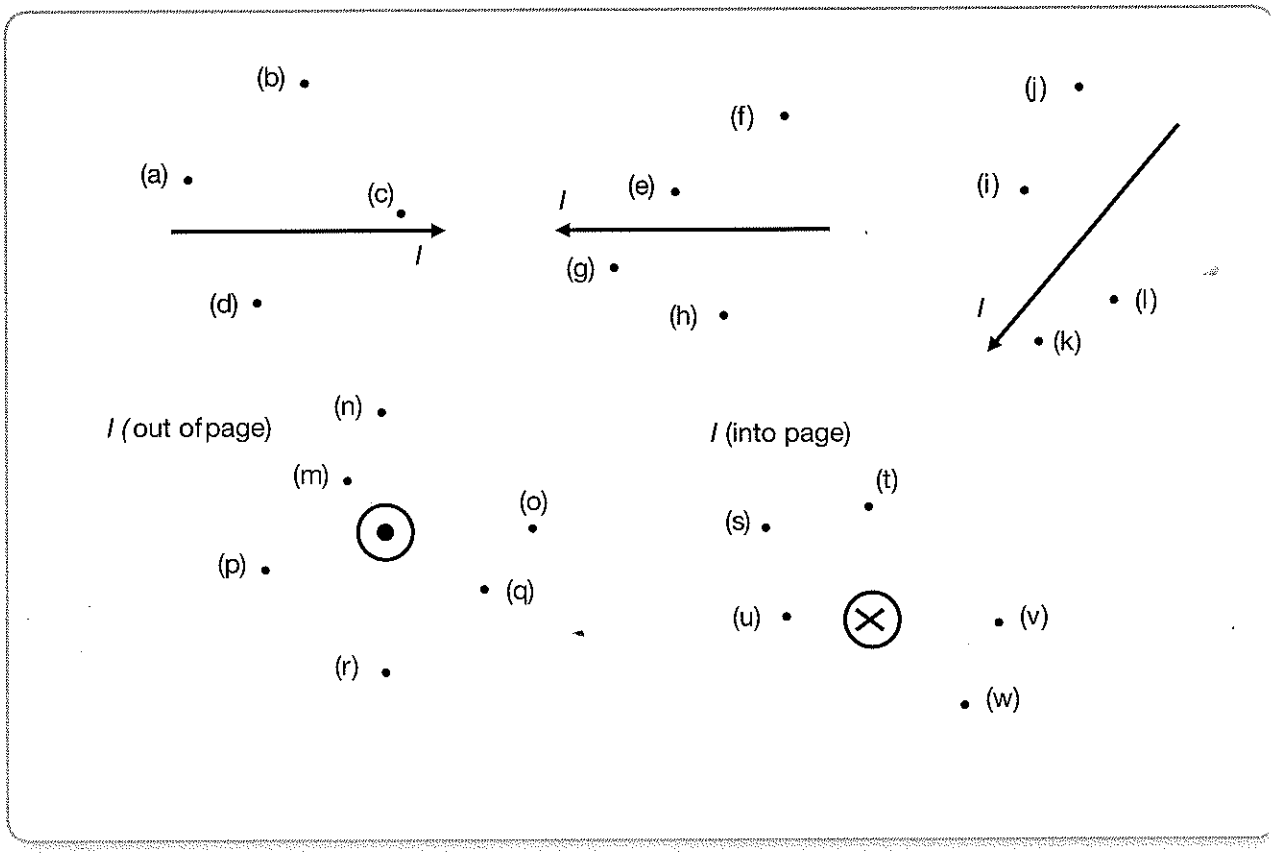
(b) Calculate the potential difference between X and Z.

(c) Determine the potential difference between Y and Z.

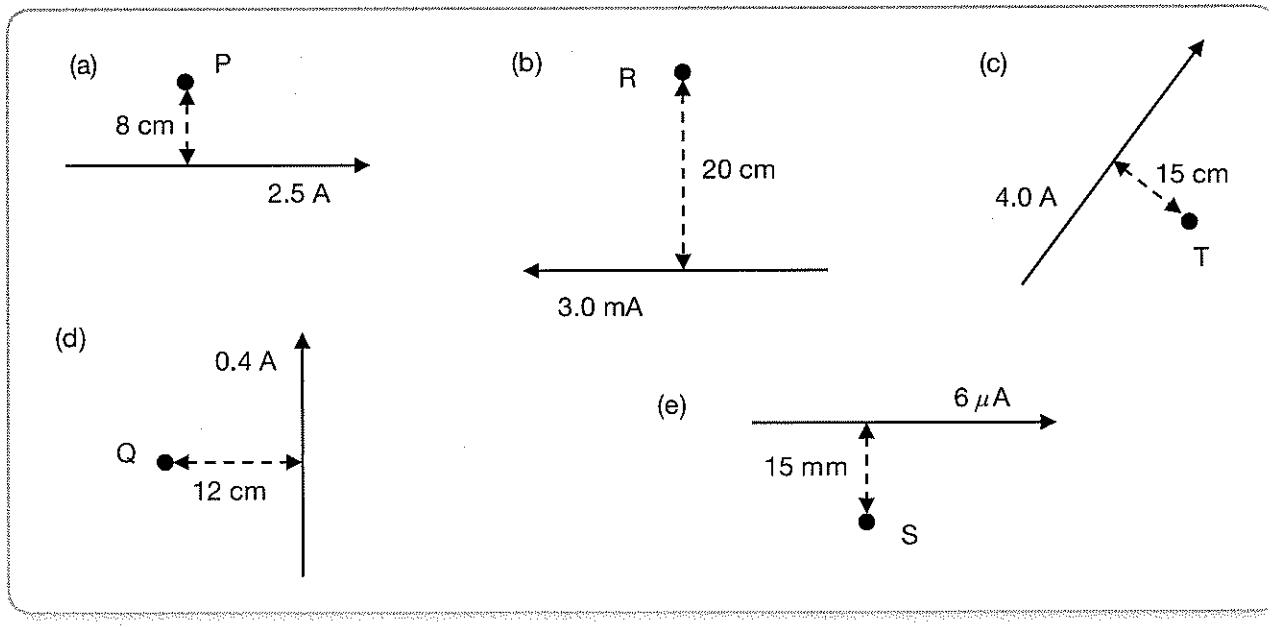
(d) Determine the potential difference between X and Y.

SET 24 Magnetic Field Around Straight Conductors

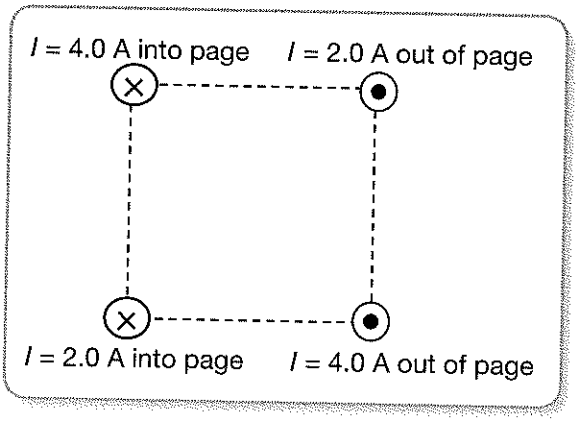
1. Determine the direction of the magnetic field at each of the points shown in the diagrams below.



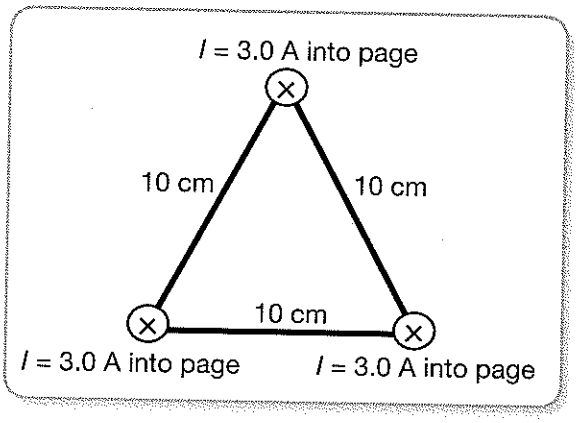
2. Find the magnitude and direction of the magnetic field at each of the labelled points in the diagrams below.



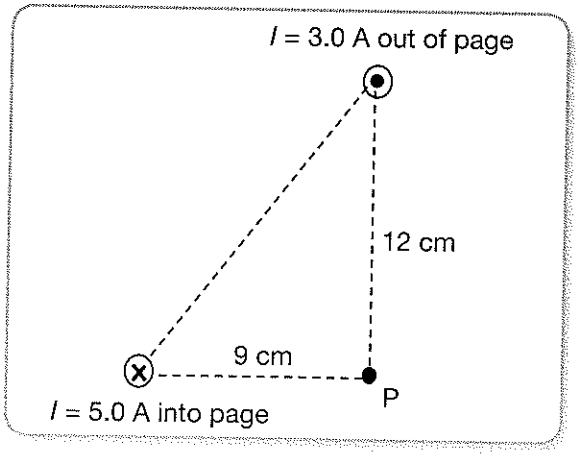
3. Four current carrying conductors are at the corners of a 10 cm square as shown. Determine the magnitude and direction of the magnetic field at the centre of the square.



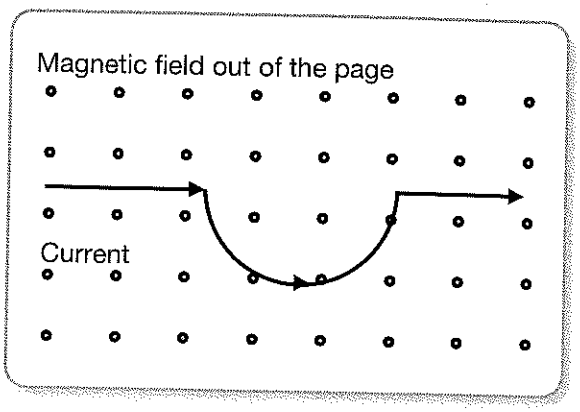
4. Three current carrying conductors are at the corners of a 10 cm equilateral triangle as shown. Determine the magnitude and direction of the magnetic field at the centre of the triangle



5. Two current carrying conductors are at the corners of a right-angled triangle as shown. Determine the magnitude and direction of the magnetic field at point P, the third corner of the triangle.



6. The diagram shows a current carrying coil in a magnetic field directed into the page.



What is the direction of the resultant force on the semicircular section of the conductor?

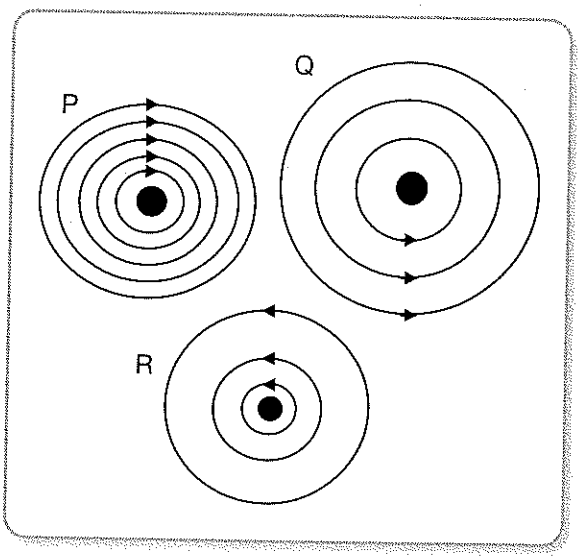
(A) Up the page.

(B) Out of the page.

(C) Down the page.

(D) There is zero resultant force.

7. The diagrams show the magnetic fields around three straight conductors P, Q and R. The black dots represent the conductors. Note that the direction of the current flowing in them is not given.



(a) Determine the direction of the current in each of the conductors.

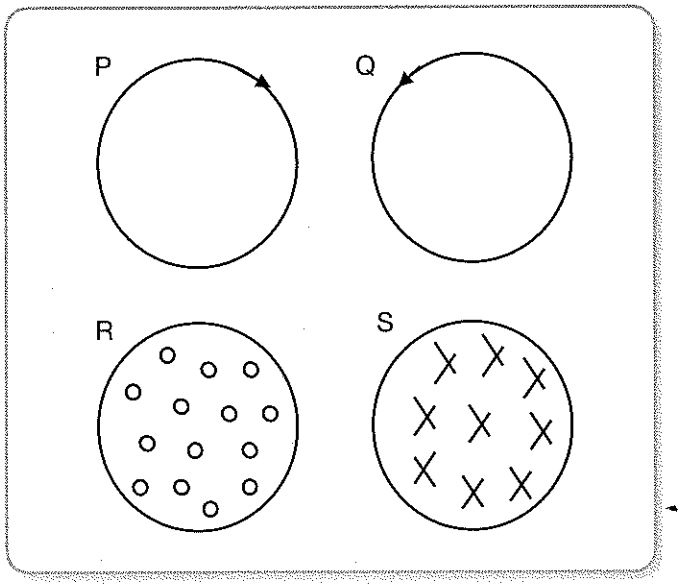
(b) Compare the magnitudes of the three currents in the conductors. Justify your answer.

(c) Comment on the field lines around each of the conductors.

SET 25 Magnetic Fields and Solenoids

Use the information below to answer the next THREE questions.

Consider the current carrying coils P, Q, R and S below.



1. Which statement about these coils is correct?

(A) The magnetic field in coil P is out of the page.

(B) The magnetic field in coil Q is out of the page.

(C) The current in coil R is clockwise.

(D) The current in coil R is anticlockwise.

2. As you look at the coils, which represent a north magnetic pole facing you?

(A) P and S.

(B) Q and R.

(C) P and R.

(D) Q and S.

3. If the coils were replaced by bar magnets in which one(s) would we be looking at a south pole?

(A) P only.

(B) R only.

(C) P and S only.

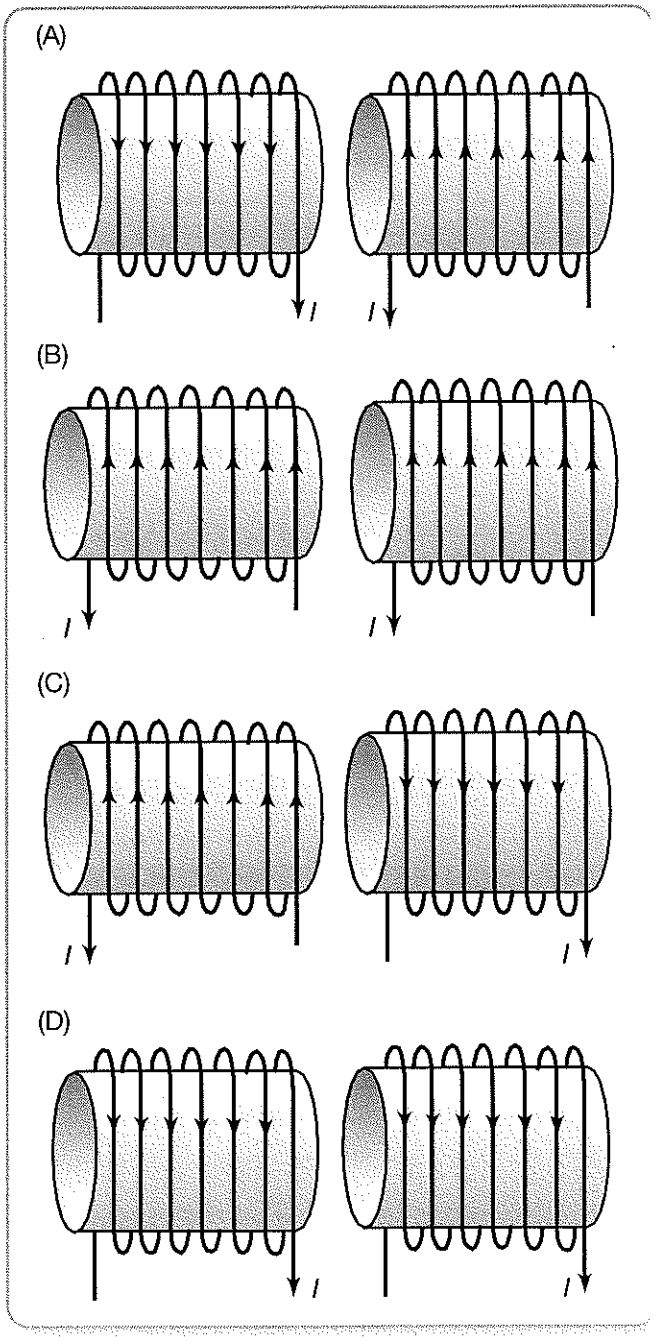
(D) R and Q only.

4. (a) What is a solenoid?

(b) A solenoid induces a stronger magnetic field if it has a soft iron core. Why?

(c) Would a soft copper core work in the same way? Explain your answer.

5. Which of the following pairs of solenoids will have an attractive magnetic field between them?



SET 26 Magnetic Flux and Flux Density

1.
(a) What is magnetic flux?
(b) What is flux density?
(c) What is an alternate term we use for flux density?
(d) What is the unit we use to measure magnetic flux?
(e) What is the unit we use to measure flux density?

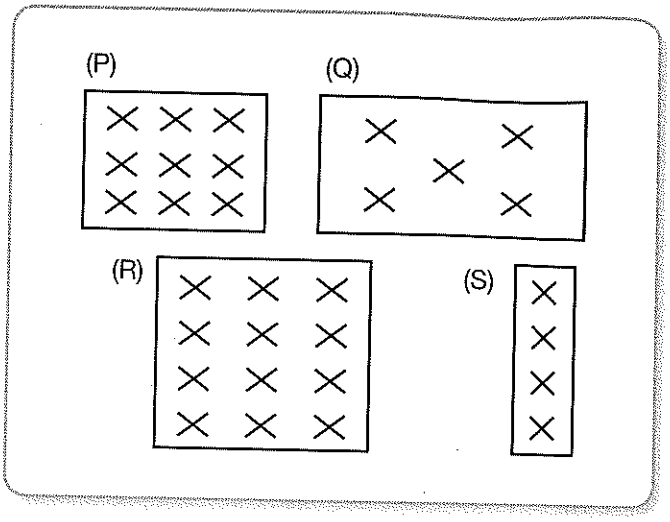
2. A square coil with 12 cm sides has 4.0 Wb of flux through it.
(a) Calculate the flux density in the coil.
(b) Predict the magnetic flux and flux density through the coil if its area was doubled.
(c) Predict the magnetic flux and flux density through the coil if the length of its sides was doubled.
(d) Predict the magnetic flux and flux density through the coil if the area was reduced to a 6 cm square.
(e) Predict the magnetic flux and flux density through the coil if the area was increased to an 18 cm square.

3. Which statement gives the correct relationship between magnetic flux and magnetic flux density?
(A) $\Phi = B \times A$
(B) $\Phi = B/A$
(C) $\Phi = B \times A^2$
(D) $\Phi = B/A^2$

4. A square has a magnetic flux of intensity B through it. What is the new intensity if the area of the square is doubled?
(A) $0.25B$
(B) $0.5B$
(C) $2B$
(D) $4B$

Use the following information to answer the next TWO questions.

The diagrams show the magnetic flux through four areas.



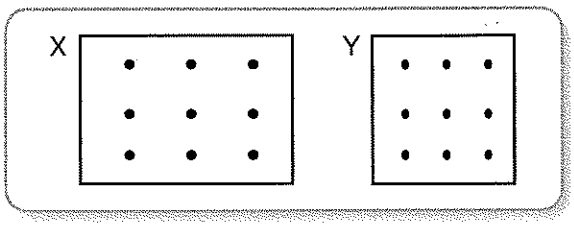
5. Which area contains the largest magnetic flux?
(A) P
(B) Q
(C) R
(D) S

6. Which area contains the smallest magnetic field intensity?
(A) P
(B) Q
(C) R
(D) S

7. An area is permeated by a magnetic flux of Φ which is equivalent to a flux density of B.
Which choice shows the new flux and flux density through this area if the area is halved?

	Magnetic flux	Flux density
(A)	Φ	B
(B)	Φ	$2B$
(C)	2Φ	B
(D)	2Φ	2B

8. Consider the magnetic fields in areas X and Y below.



Which statement about these magnetic fields is correct?

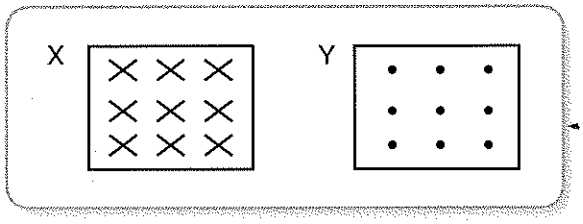
(A) $B_X = B_Y$

(B) $B_X > B_Y$

(C) $B_X < B_Y$

(D) $B_X = -B_Y$

9. Consider areas X and Y, both permeated by magnetic fields.



Which statement about these magnetic fields is correct?

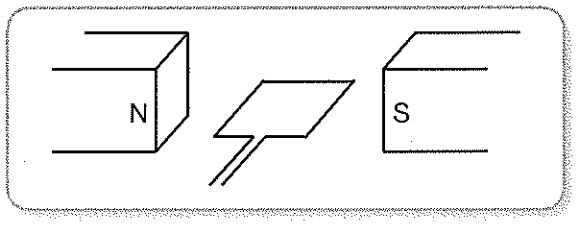
(A) Flux through X = flux through Y.

(B) Flux through X > flux through Y.

(C) Flux through X < flux through Y.

(D) Flux through X = –flux through Y.

10. The diagram shows a coil between the poles of two magnets.



Which action will increase the magnetic flux through this coil?

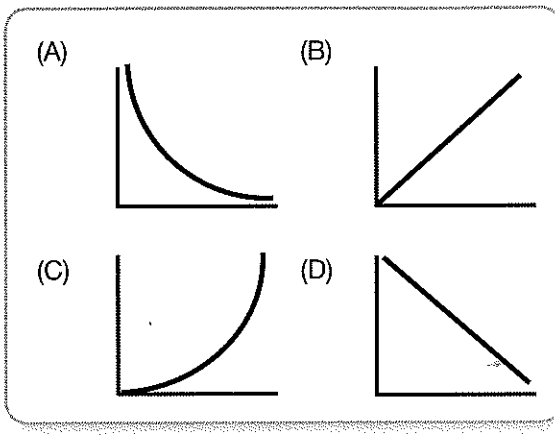
(A) Increase the strength of the magnets.

(B) Increase the area of the coil.

(C) Increase the number of turns in the coil.

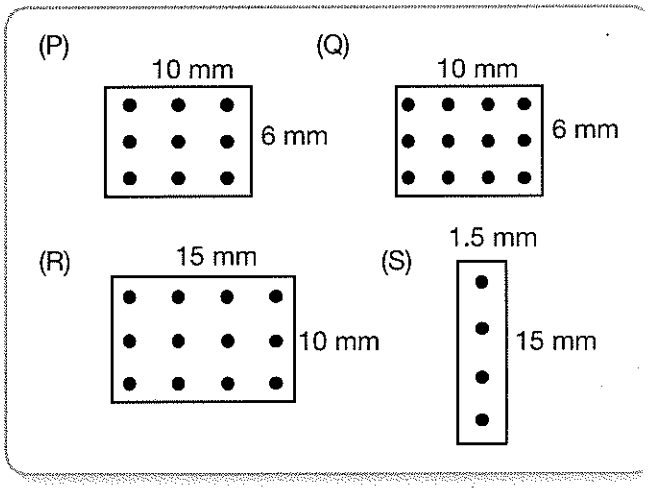
(D) Rotate the coil through 20°.

11. Which graph best shows the relationship between magnetic flux density and area?



Use the following information to answer the next TWO questions.

The diagrams show the magnetic flux through four areas.



12. What is the ratio of the magnetic flux in area R to that in S?

(A) 1 : 1

(B) 3 : 1

(C) 4 : 1

(D) 6.7 : 1

13. What is the ratio of the magnetic flux density in area P to that in Q?

(A) 1 : 1

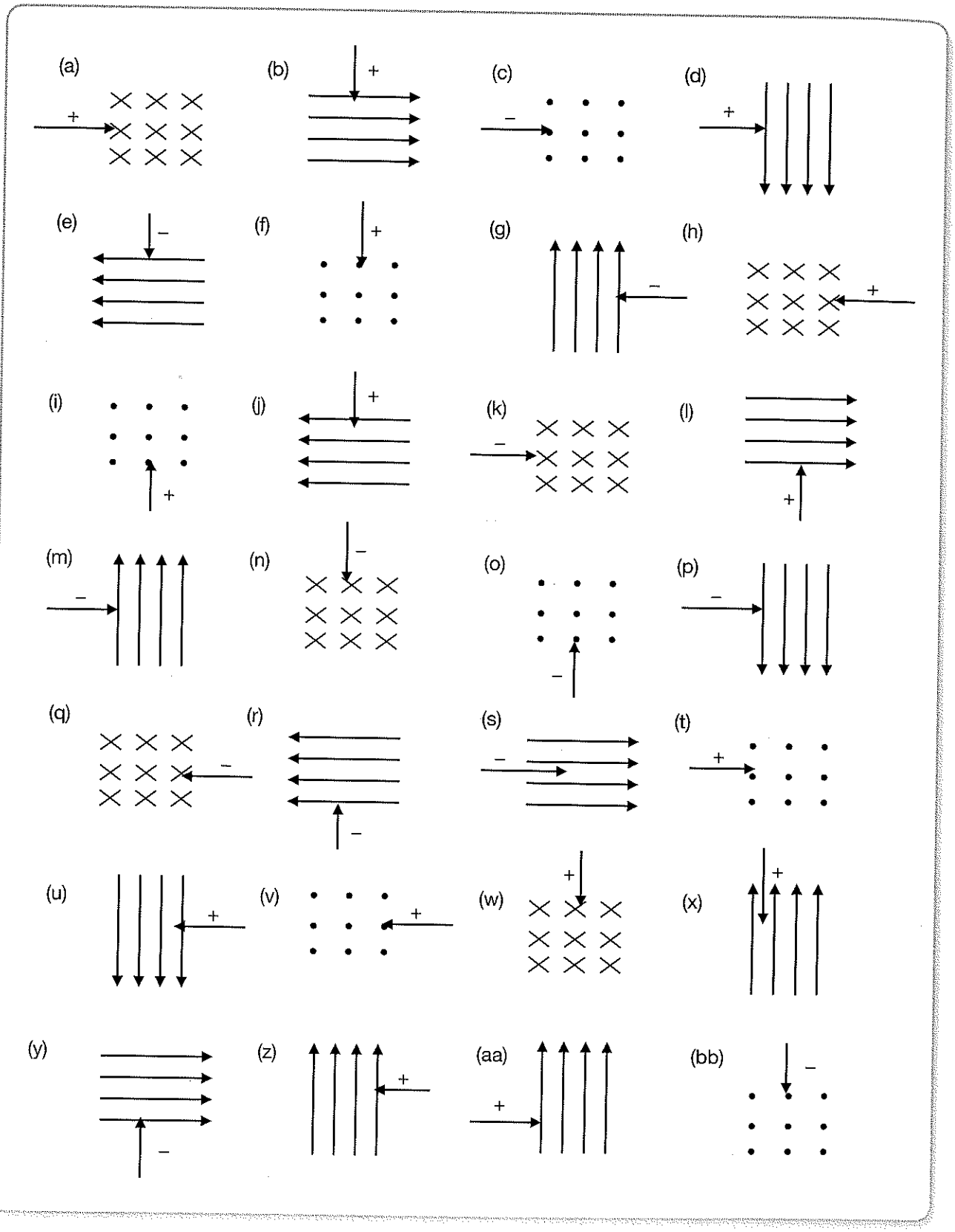
(B) 1.3 : 1

(C) 3 : 4

(D) 4 : 3

SET 27 Moving Charges in a Magnetic Field 1

1. Predict the direction of the force acting on each charge as it enters each magnetic field shown below.



SET 28 Moving Charges in a Magnetic Field 2

1. A beam of electrons, accelerated by a 2.4 kV electron gun passes into a magnetic field of strength 9.7×10^{-3} T at 90° to the field.

(a) Find the speed of the electrons as they leave the electron gun.

(b) Find the force on each electron while it is in the magnetic field.

(c) Find the radius of curvature of the path taken by the electrons.

2. A charge of 4.8×10^{-19} C is travelling north at 2.5×10^{7} m s^{-1}, perpendicular to a magnetic field of 0.25 T directed into the page.

(a) Calculate the force on the charge.

(b) Deduce the direction the charge will move while it is in the field.

3. A 3.2×10^{-19} C charge travels south at 4.0×10^{7} m s^{-1}, at 90° to a magnetic field directed into the page. It experiences a force of 6.4×10^{-12} N.

(a) Find the strength of the magnetic field.

(b) Deduce the direction the charge will move while it is in the field.

4. A charge moves at 1.25×10^{6} m s^{-1} at 90° to a magnetic field of strength 0.4 T. The charge experiences a force of 8.0×10^{-13} N.

(a) Calculate the magnitude of the charge.

(b) What other information do we need to determine the sign of the charge?

5. A 8.0×10^{-19} C charge moves at 5×10^{5} m s^{-1} from north to south into a magnetic field into the page. A force of 4.4×10^{-13} N acts on it.

(a) Find the magnitude of the field.

(b) Deduce the direction of movement of the charge in the field.

6. A charge of 4.8×10^{-19} C is travelling north at 2.5×10^{7} m s^{-1}, perpendicular to a magnetic field of 0.25 T directed vertically down.

(a) Calculate the force on the charge.

(b) Deduce the direction the charge will move while it is in the field.

7. A charge of 3.2×10^{-19} C is travelling south at 4.0×10^{7} m s^{-1}, perpendicular to a magnetic field directed vertically down. It experiences a force of 6.4×10^{-12} N.

(a) Calculate the strength of the magnetic field.

(b) Deduce the direction the charge will move while it is in the field.

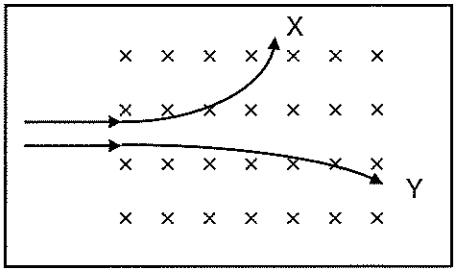
8. Three electrons are fired at 2.5×10^{5} m s^{-1} from the east, west and south into a magnetic field of strength 0.02 T north.

(a) Calculate the magnitude of the force on each electron.

(b) Deduce the direction of the force on each electron.

(c) Describe the path each electron will take while it is in the field.

9. The diagram shows two particles entering and travelling through a magnetic field.



(a) Predict the sign of each charge. Justify your answer.

(b) Give reasons for the difference in the radius of curvature of each particle.

10. A negatively charged particle travelling from left to right enters a magnetic field and is deflected out of the page. What is the direction of the magnetic field?

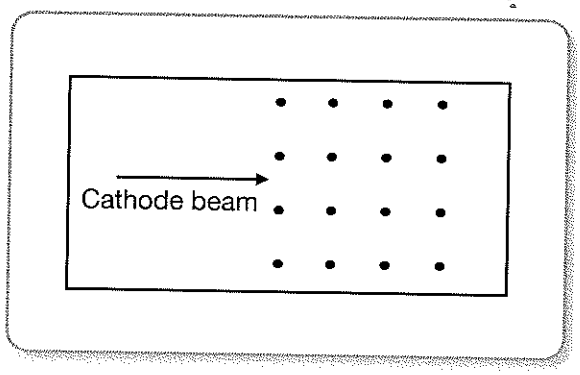
(A) Out of the page.

(B) Into the page.

(C) Towards the top of the page.

(D) Towards the bottom of the page.

11. A beam of cathode rays enters a magnetic field directed vertically out of the page as shown.



Which way will the cathode rays be deflected as they enter the magnetic field?

(A) Towards the bottom of the page.

(B) Towards the top of the page.

(C) Out of the page.

(D) Into the page.

Use the following information to answer the next TWO questions.

Two charged particles X and Y are fired into the same magnetic field. The charge on X is twice that of Y, but Y is moving twice as fast as X.

12. Which choice correctly states the ratio of the force acting on X due to the magnetic field to that acting on Y?

(A) $F_X : F_Y = 4 : 1$

(B) $F_X : F_Y = 2 : 1$

(C) $F_X : F_Y = 1 : 1$

(D) $F_X : F_Y = 1 : 2$

13. How would the sign of the charge on X and Y affect this force?

(A) If the signs are both negative, the deflections would be in opposite directions.

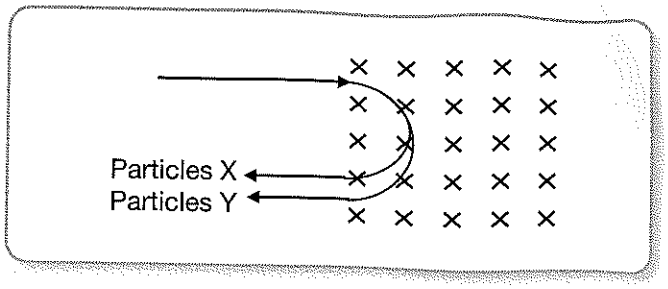
(B) If the signs are both positive, the deflections would be in the same directions.

(C) If the signs are opposite, the deflections would be in the same directions.

(D) The sign of the charge will have no influence on the deflection.

Use the following information to answer the next TWO questions.

A beam of charged particles moves through a magnetic field directed into the page as shown.



14. Which statement about the particles is correct?

(A) Both are negatively charged and X has more mass than Y.

(B) Both are positively charged and X has more mass than Y.

(C) Both are negatively charged and Y has more mass than X.

(D) Both are positively charged and Y has more mass than X.

15. Which change would increase the radius of curvature of the beams through the field?

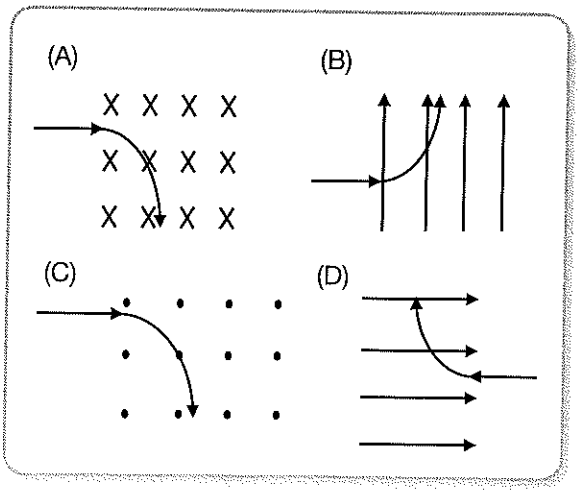
(A) Increase the strength of the magnetic field.

(B) Use a particle with a greater charge but the same mass.

(C) Use a particle with identical charge but less mass.

(D) Fire the particle into the field with a greater speed.

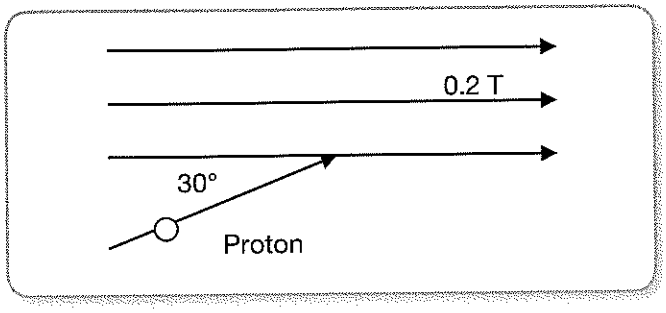
16. Which diagram shows the path taken by a positively charged particle in a magnetic field?



SET 29 Moving Charges in a Magnetic Field 3

Use the following information to answer the next TWO questions.

A proton moving at 4.5×10^3 m s^{-1} enters a magnetic field as shown.



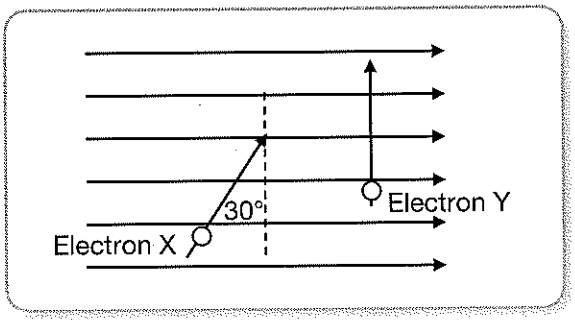
1. What is direction of the magnetic force on the proton as it enters the field?

(A) Up the page.
(B) Down the page.
(C) Into the page.
(D) Out of the page.

2. What would be the magnitude of the force on the proton?

(A) 1.25×10^{-16} N
(B) 1.25×10^{-17} N
(C) 7.2×10^{-17} N
(D) 1.44×10^{-16} N

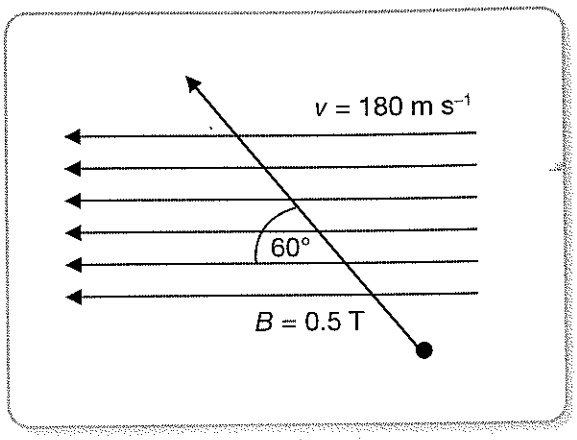
3. Two electrons are moving at the same speed in a magnetic field as shown.



What is the ratio of the force on electron X to that on electron Y?

(A) 0.5 : 1 (B) 0.866 : 1
(C) 1 : 1 (D) 2 : 1

4. The diagram shows a particle of mass 3.6×10^{-25} kg and charge 6.4×10^{-19} C entering a magnetic field of strength 0.6 T at 180 m s^{-1} at an angle of 60°.



(a) Find the force on the particle while it is in the field

(b) Determine the radius of curvature of the path it takes.

5. A single particle carrying a charge of 1.6×10^{-19} C and with a mass of 4.6×10^{-27} kg, enters a 4.0 T magnetic field into the page at 2.5×10^6 m s^{-1}.

The field covers an area of 0.1 m × 0.1 m. It enters at 90°, 0.05 m up from the bottom left corner.

(a) Calculate the force on the particle while it is in the field.

(b) Calculate the radius of the path the particle takes in the field.

(c) Calculate the speed with which the particle exits the field.

(d) Where will the particle exit the field?

6. Two charged particles, X and Y, of equal mass 4.0×10^{-5} kg, are fired from left to right, at 250 m s^{-1} into a uniform magnetic field of flux density 0.3 T directed into the page.

X has a charge of +3.0 mC and Y −1.5 mC.

(a) Draw a diagram to show the path of each particle through the field.

(b) Calculate the radius of curvature of each path.

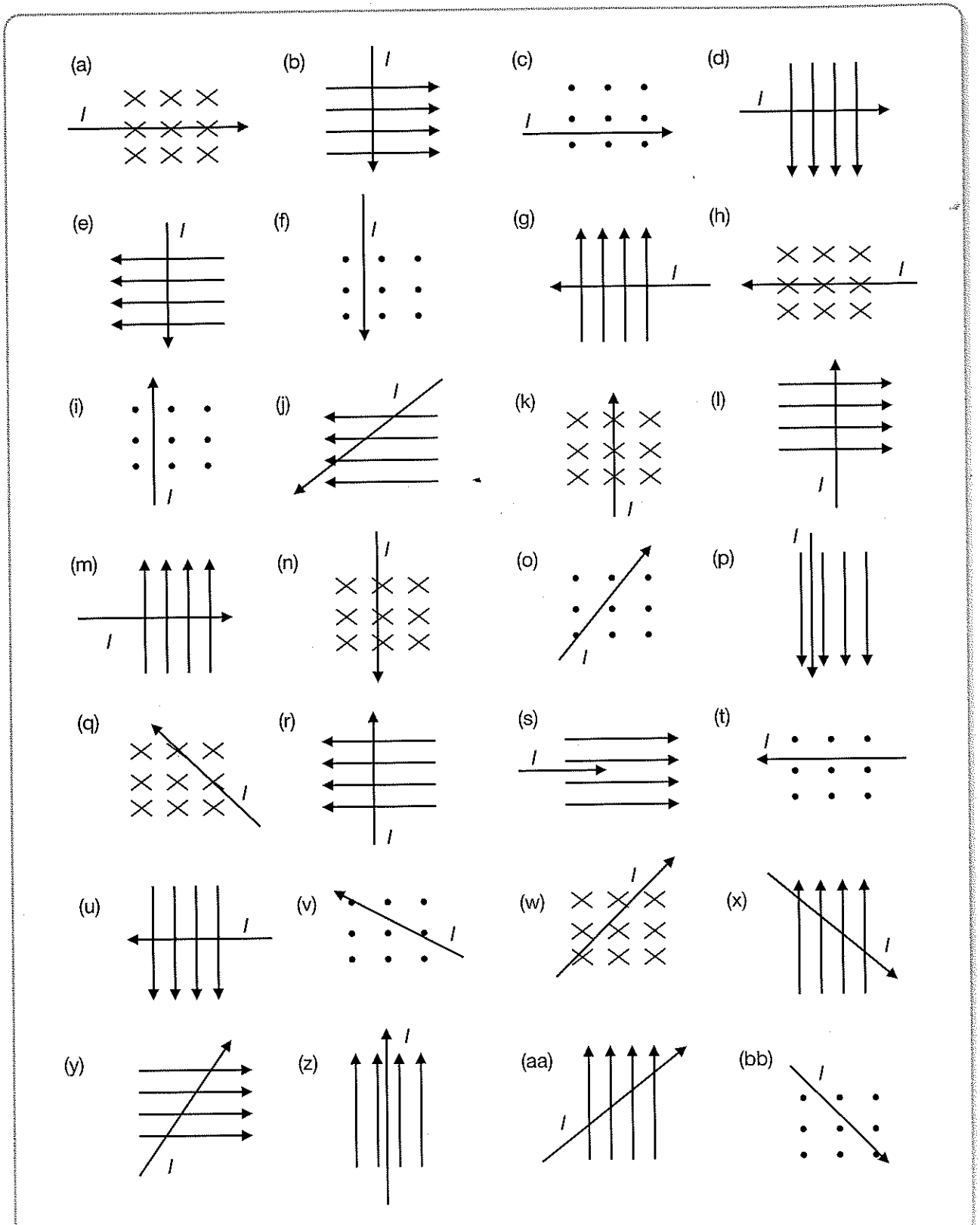
7. The table relates to the force on charged particles moving perpendicularly to magnetic fields. N, S, E and W refer to compass directions, IP means into the page, OOP out of the page. Calculate the missing quantities.

	Force on charge (N)	Magnetic field (T)	Charge (C)	Velocity ($m\ s^{-1}$)
a	4.32×10^{-17} IP	0.25 E		50 N
b		0.15 IP	-4.8×10^{-19}	30 W
c	8.0×10^{-17} E	0.6 N	1.2×10^{-18}	
d	1.5×10^{-17} E	0.02 IP		120 S
e	1.2×10^{-18} OOP		-1.8×10^{-18}	40 E
f		0.4 E	6.0×10^{-18}	250 IP
g	8.0×10^{-16} OOP		1.5×10^{-17}	75 W
h	3.2×10^{-18} S		-6.4×10^{-19}	15 OOP
i	2.4×10^{-17} W	2.5 IP		60 N
j		3.0 OOP	3.6×10^{-19}	90 N
k	9.0×10^{-16} OOP	0.1 E	1.5×10^{-17}	
l	3.0×10^{-17} IP	0.5 S	-4.5×10^{-18}	

8. The information in the table relates charges moving into magnetic fields. N, S, E and W refer to traditional compass directions, IP means into the page, OOP out of the page. Calculate the missing quantities.

	Force on charge (N)	Magnetic field (T)	Charge (C)	Velocity of charge ($m\ s^{-1}$)	Angle between field and motion
a	4.32×10^{-18} IP	0.25 S		50 E	30
b		0.15 N	-4.8×10^{-19}	30 IP	60
c	0.8×10^{-18} N	0.6 E	1.2×10^{-18}		45
d	1.5×10^{-17} OOP	0.02 E		120 S	20
e	1.2×10^{-18} OOP		-1.8×10^{-18}	40 N	120
f		0.4 OOP	-6.0×10^{-18}	250 E	160
g	4.2×10^{-17} E	0.7 S	1.6×10^{-17}	75 OOP	
h	3.3×10^{-18} OOP	0.45 E	6.4×10^{-19}	15 S	
i	2.4×10^{-17} W	2.5 N	2.1×10^{-19}		50
j	8.2×10^{-17} IP		3.5×10^{-19}	90 N	60
k	9.0×10^{-16} OOP	1.2 N	-1.4×10^{-17}	157 W	
l	3.0×10^{-17} IP	0.5 W		24 S	34

1. Predict the direction of the force acting on each current carrying conductor in each magnetic field shown below.



SET 31 The Motor Effect 2

1. (a) Define the motor effect.

(b) What factors influence the magnitude of the motor effect?

(c) Express your answer to (b) above mathematically.

(d) What factors influence the direction of the motor effect?

(e) What causes the production of all electromagnetic forces?

2. Chose the correct option for the following.

(a) Electric currents do/do not interact with each other.

(b) Electric currents do/do not interact with magnetic fields.

(c) Stationary charges do/do not interact with magnetic fields.

(d) Moving charges do/do not interact with magnetic fields.

(e) Magnetic fields do/do not interact with each other.

(f) A magnetic field is/is not induced around all current carrying conductors.

(g) A magnetic field is/is not induced around all moving charges.

3. A current carrying conductor in a magnetic field experiences a force of F newtons. Find the force on it if its length is halved and the current flowing in it is doubled.

(A) $0.5F$

(B) F

(C) $2F$

(D) $4F$

4. A current carrying conductor in a magnetic field experiences a force of F newtons. Find the force on the conductor if the magnetic field is doubled in intensity.

(A) $0.5F$

(B) F

(C) $2F$

(D) $4F$

5. A current carrying conductor makes an angle of θ with a magnetic field and experiences a force of F newtons. What would be the effect on the force on the conductor if θ is increased?

(A) Force remains at F.

(B) Force will increase.

(C) Force will decrease.

(D) Unable to be answered unless values known.

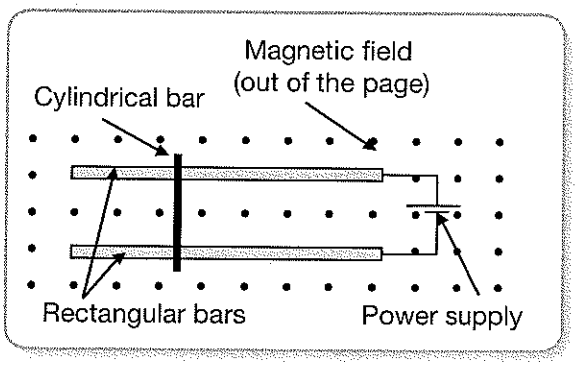
6. Which choice correctly identifies the way we use the right hand palm rule?

	Fingers	Thumb	Direction palm faces
(A)	B	I	F
(B)	B	F	I
(C)	F	B	I
(D)	I	B	F

7. A cylindrical conducting bar (black) lies at rest across two rectangular bars (grey) as shown in the diagram.

The system is within a vertical magnetic field directed out of the page (black circles).

The power supply is switched on, sending a current through the bars.



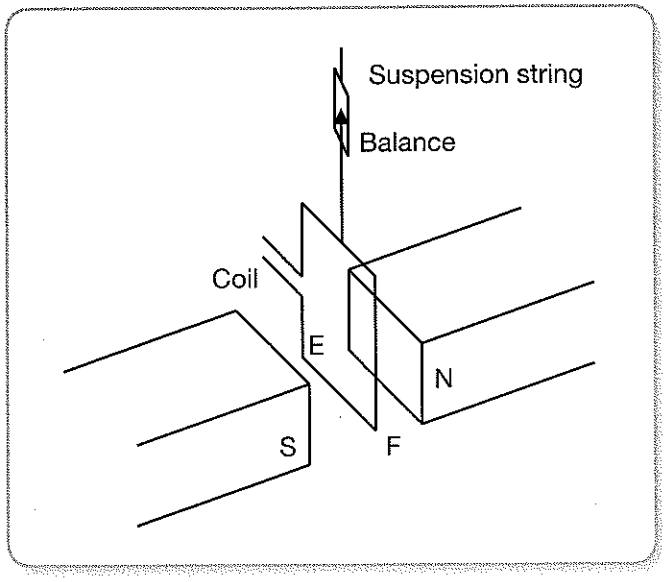
(a) What will the black bar do?

(b) Explain your answer.

(c) The students then reversed the polarity of the power source and the magnetic field and then repeated the experiment. What would the black bar do now? Justify your answer.

Use the following information to answer the next FIVE questions.

A coil, containing 500 turns of wire, was suspended so that its lower side, EF, 5 cm long, was between the jaws of a powerful magnet.



Varying currents were passed through the coil and the reading on the balance was recorded. The results are shown in the table.

Current in coil (A)	Reading on balance (N)
0	4.3
1	4.1
2	3.9
3	3.7
4	3.5
5	3.3

8. What would be the reading on the balance when 7.5 A of current flowed through the coil?

(A) 3.0 N (B) 2.9 N
(C) 2.8 N (D) 2.7 N

9. What was the mass of the coil?

(A) 20.4 grams (B) 43.9 grams
(C) 1.8 kg (D) 42.14 kg

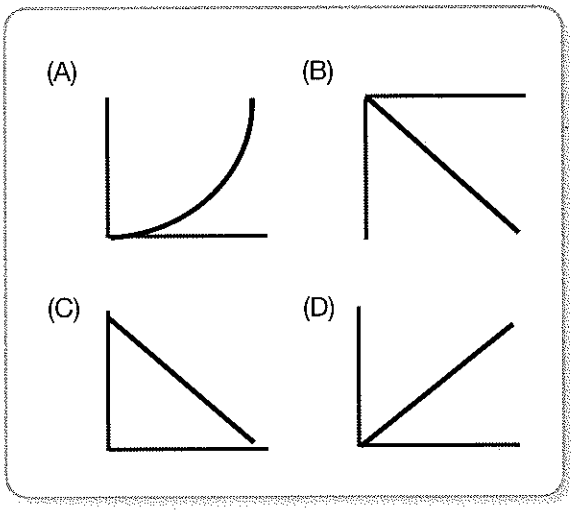
10. In which direction does the current flow through the coil?

(A) From E to F.
(B) From F to E.
(C) Alternately from E to F and then from F to E.
(D) Unable to be determined without additional data.

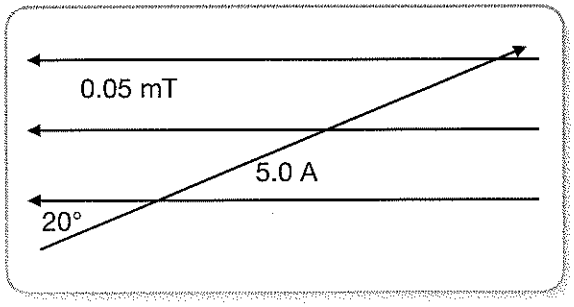
11. What is the strength of the magnetic field?

(A) 4.0 T
(B) 2.6×10^{-2} T
(C) 8.0×10^{-3} T
(D) 8×10^{-5} T

12. Which graph correctly shows the relationship between the force on a conductor due to the motor effect and the current flowing in the coil?



13. A 15 cm conductor makes an angle of 20° with a 0.05 mT magnetic field directed towards the west as shown.



Find the magnitude and direction of the force on the conductor due to the magnetic field.

SET 32 Forces on Straight Conductors in Magnetic Fields

1. A current carrying conductor is at an angle of 30° to a magnetic field. It experiences a force of *F*. Predict the new force on the conductor if:

(a) The field is doubled in strength.

(b) The current flowing in the conductor is halved.

(c) The length of the conductor in the field is halved.

(d) The angle the conductor makes with the field is increased to 60°.

(e) Predict the effect on the force of the conductor if all four changes above are made at the same time.

2. A conductor carrying 2.5 A experiences a force of 0.15 N when 0.4 m of its length is placed in a magnetic field of intensity 0.5 T.

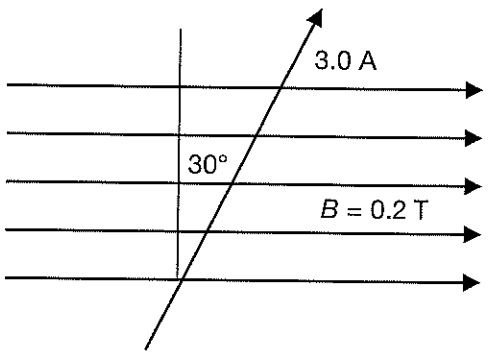
(a) Calculate the angle the conductor makes with the field.

(b) Predict the angle where the conductor will experience maximum force. Justify your answer.

(c) At what angle will the conductor experience zero force? Justify your answer.

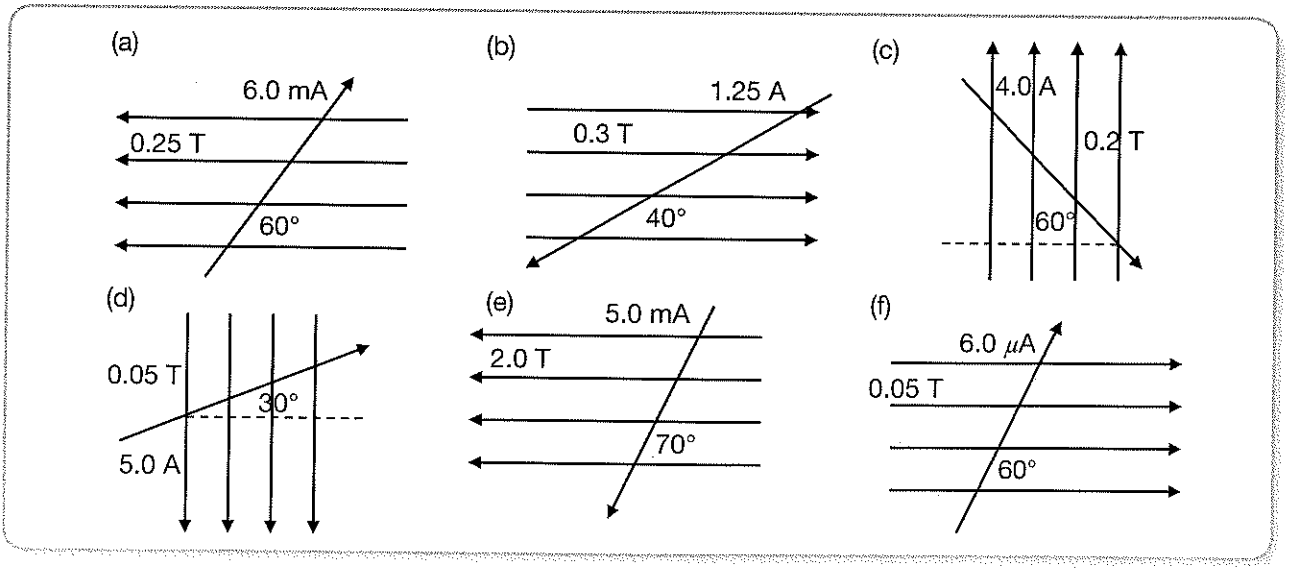
(d) Predict the force on a new conductor with half the electrical resistance of the initial conductor placed in an identical position in the same field. Justify your answer.

3. Calculate the force on the conductor in a magnetic field of strength 0.2T shown in the diagram. It carries a current of 3.0 A and the length of the conductor in the magnetic field is 0.25 m. The conductor makes an angle of 30° with the field as shown.



4. Two long, parallel, current carrying conductors attract each other with a force of *F* newtons. The current in each, and the distance between them is tripled. Predict the new force between the wires.

5. Each conductor shown below has 0.2 m of its length in the magnetic field. Calculate the force on each conductor due to the magnetic field.



SET 33 Torque on a Coil 1

1. (a) What is a torque?

(b) Write two equations for the torque on a current carrying coil in a magnetic field and identify the symbols used.

2. A coil in a magnetic field carries a current. Predict the effect on the torque on the coil of:

(a) Halving the current.

(b) Reversing the current direction.

(c) Doubling the turns in the coil.

(d) Increasing the strength of the magnetic field four times.

(e) Replacing the coil with an identical one of lower resistivity.

3. A rectangular coil 4 cm × 6 cm is part of a motor. The coil has 500 turns, has its long side between the jaws of a magnet of strength 0.05 T and carries a current of 3.0 A.

(a) Calculate the force on each long side of the coil.

(b) Calculate the maximum torque experienced by the coil.

(c) Predict the minimum torque experienced by the coil.

(d) Calculate the torque when the coil is at 30° to the field.

(e) Calculate the force on each long side when it is in this position.

4. A square coil of side 5.0 cm with 1500 turns lies horizontally in the plane of a magnetic field of intensity 0.6 T directed left to right. A current of 2.5 mA into the page flows in PQ, the side closest to the north pole of the magnet.

(a) Calculate the force on side PQ.

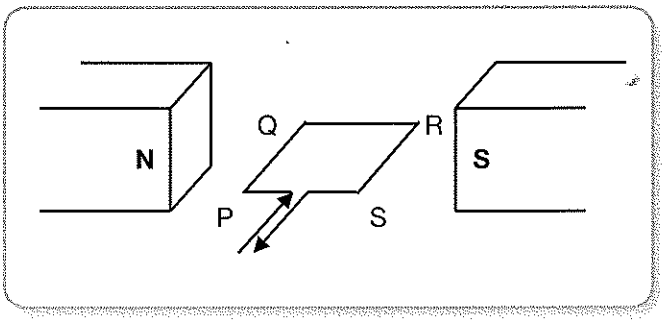
(b) Find the maximum torque on the coil.

(c) Calculate the torque when the coil has rotated through 120°.

(d) Calculate the change in flux through the coil when it moves from its initial position through 90°.

Use the following information to answer the next FOUR questions.

Consider a 200 turn coil in a 0.02 T magnetic field carrying a current of 6 A as shown. The length of side PQ is 10 cm and QR is 6 cm.



5. What is the magnitude of the torque on the coil in its present position?

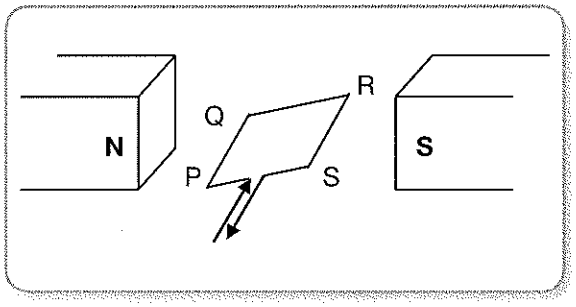
(A) 0 N m

(B) 0.144 N m

(C) 14.4 N m

(D) 144 N m

6. The coil is rotated so that it makes an angle of 30° with the magnetic field. What is the magnitude of the new torque on the coil?



What is the magnitude of the torque on the coil in this new position?

(A) 0 N m

(B) 0.125 N m

(C) 12.5 N m

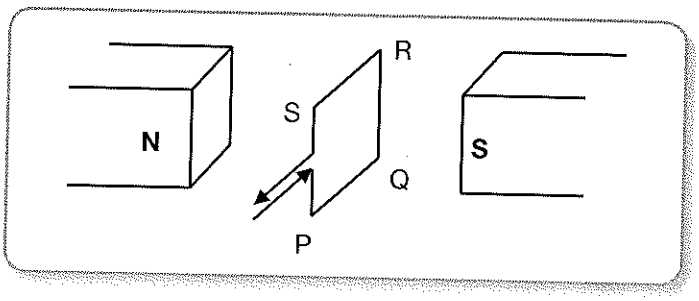
(D) 125 N m

7. The coil is rotated so that it is perpendicular to the magnetic field.

What is the magnitude of the torque on the coil in this new position?

(A) 0 N m (B) 0.144 N m

(C) 14.4 N m (D) 144 N m



8. Consider the same coil in its original position Both the magnetic field and the current are reversed.

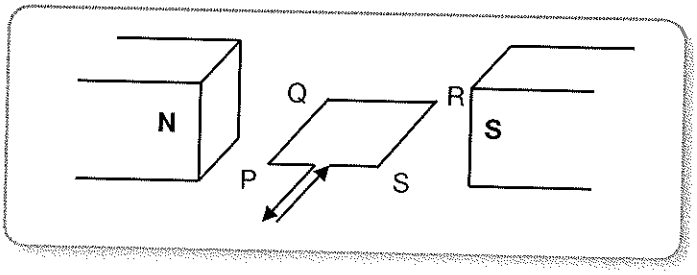
How do these changes affect the torque?

(A) The magnitude will be unchanged but the direction will be reversed.

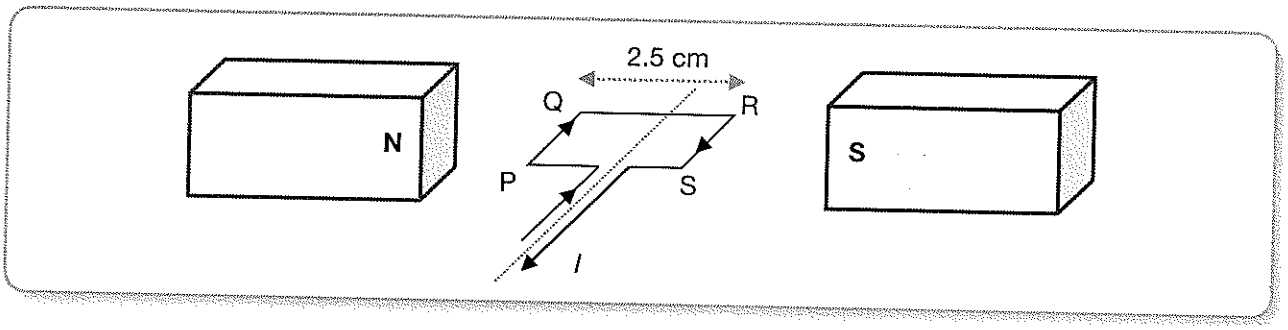
(B) The magnitude will double but the direction will be unaffected.

(C) The magnitude will be halved but the direction will be unaffected.

(D) The magnitude and direction will be unchanged.



9. A square coil of side 2.5 cm lies horizontally in the plane of a uniform magnetic field of intensity 0.5 T as shown in the diagram. The coil has 250 turns and carries 2.5 A.



(a) Calculate the force on side PQ.

(b) Calculate the maximum torque acting on the coil.

(c) Calculate the change in flux through the coil when it moves from its initial position through 90°.

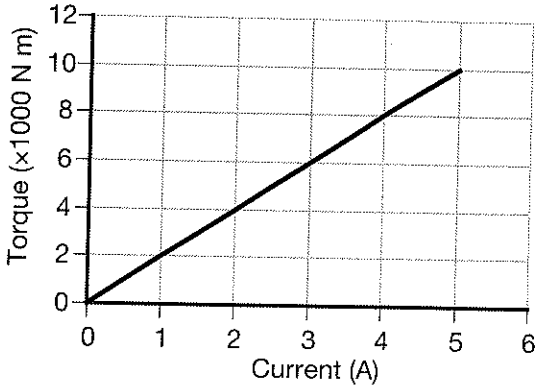
(d) Predict what will happen to this coil after it rotates 90° if current keeps flowing through it.

10. The graph shows the relationship between the torque acting on an 8.0 cm square coil in a magnetic field of intensity 0.3 T, and the current flowing through it. The field is parallel to the plane of the coil.

(a) Calculate the number of turns in the coil.

(b) As this coil turns in the field, the torque it experiences varies. Explain why this occurs.

(c) Describe, with the aid of an appropriate graph, how the torque on the coil changes as it rotates through 360°.



SET 34 Torque on a Coil 2

1. The information in the table refers to a coil in, and parallel to, a magnetic field. Magnitudes only are shown. Find the missing quantities.

	Torque on coil (N m)	Number of turns (n)	Field strength (T)	Current in coil (A)	Coil area ($m^2 \times 10^{-4}$)
a		250	0.5	3.6	16
b		400	0.25	0.4	9
c	6.72×10^{-2}	300	1.4	0.2	
d	0.96	75	2.0	4.0	
e	1.08	80	3.6		25
f	1.80	750	0.2		50
g	2.16	1000		2.4	60
h	2.88×10^{-2}	800		1.5	12
i	9.00×10^{-2}		0.3	5.0	30
j	1.44		2.5	3.0	64

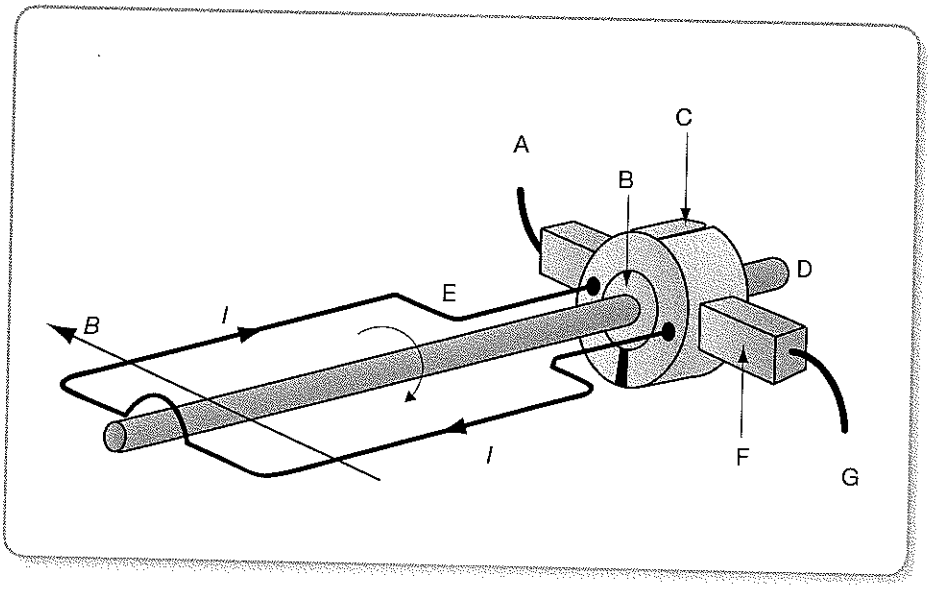
2. The information (magnitudes only) below refers to coils in magnetic fields. Calculate the missing values.

	Torque on coil (N m)	Turns (n)	Field strength (T)	Current (A)	Coil area (cm^2)	Angle between field and coil
a		400	0.2	3.5	16	30
b		125	0.4	0.6	9	60
c	0.063	275	1.8	0.1		45
d	0.698	45	4.0	2.8		120
e	3.016	750	3.5		25	50
f	0.194	300	0.3		50	35
g	0.85	1200		1.2	60	55
h	0.30	600		1.8	12	70
i	4.56		6.0	4.0	30	25
j	6.28		0.8	2.0	64	130
k	0.17	35	3.5	0.4	40	
l	0.47	50	5.0	1.5	25	

SET 35 Simple DC Motors

1. Outline the principles by which a simple DC motor operates.

2. The diagram shows a simple DC motor. The directions of the magnetic field and the current through the coil are shown.



(a) Draw up a table to identify each of labelled components of this motor and the function of each part.

(b) Which way will the coil shown in the diagram rotate?

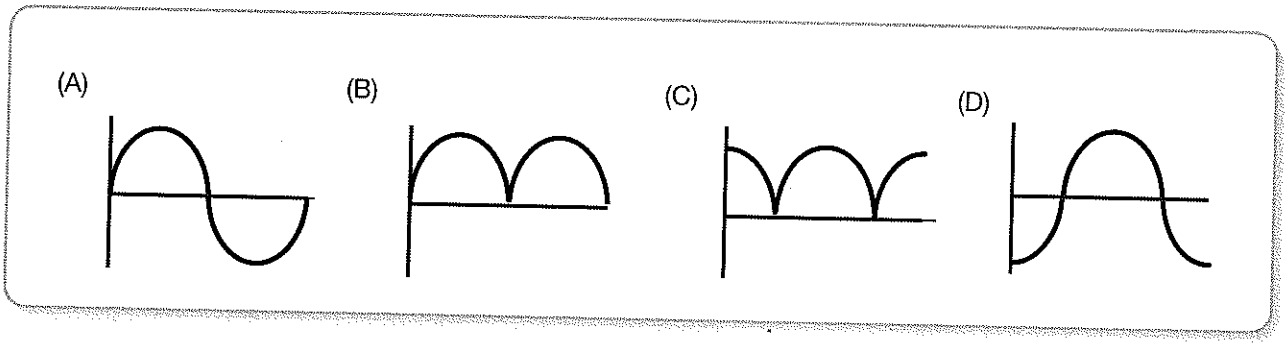
(c) Explain the function of the split rings in a simple DC motor.

(d) Explain two ways the efficiency of the motor above could be improved.

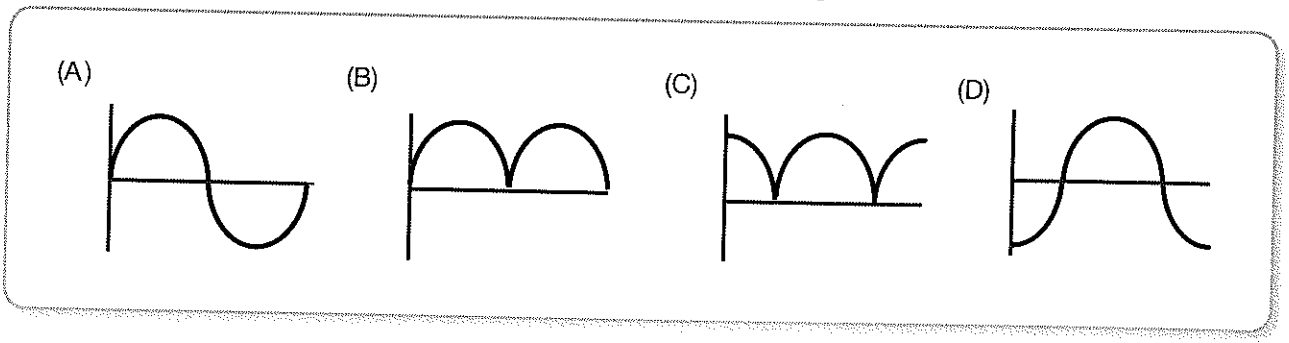
(e) Motors usually use electromagnets to produce magnetic fields rather than permanent magnets. Why?

(f) How could we increase the strength of the electromagnetic field used in a motor?

3. A coil in a DC motor rotates through 360° within a magnetic field. Initially the plane of the coil is parallel to the magnetic field. Which graph correctly shows the torque acting on the coil?



4. A coil in a DC motor rotates through 360° within a magnetic field. Initially the plane of the coil is parallel to the magnetic field. Which graph best shows the magnetic flux through the coil?



SET 36 Simple AC motors

1. Compare the structure of a simple AC motor and a simple DC motor.

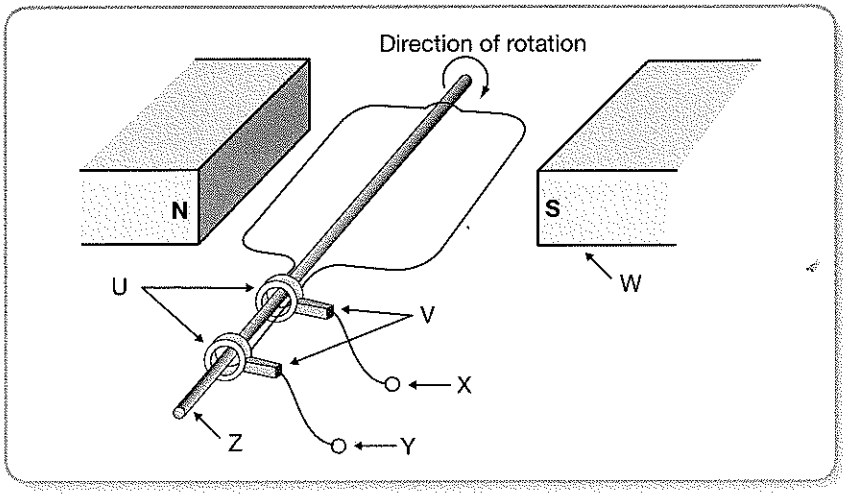
2. The diagram shows some of the parts in a simple AC motor.

(a) Identify each labelled part of the simple AC motor and state its function.

(b) What is the function of the slip rings on an AC motor?

(c) Which way does current flow through the coil of the motor?

(d) Which wire is connected to the positive terminal of the power supply?



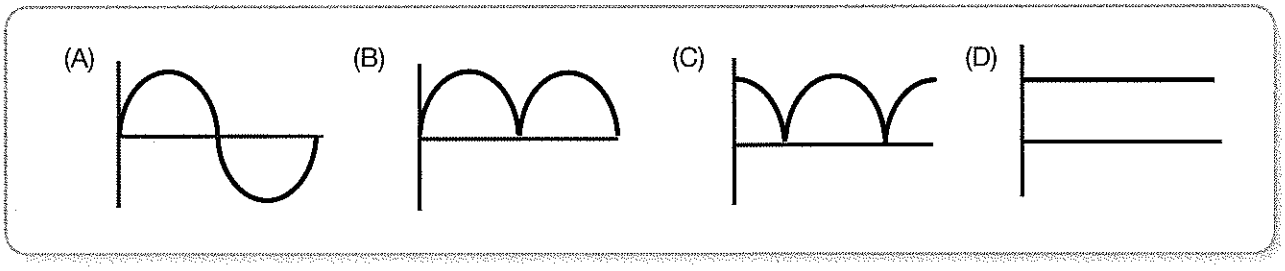
3. Three students gave the following answers to a question which asked for the function of the slip rings in a simple AC motor. Assess whether or not each of these three answers are correct.

Student X: The function of the slip rings in a simple AC motor is to change the direction of the flow of current through the coil every 180° of rotation so that the torque on the coil is always in the same direction.

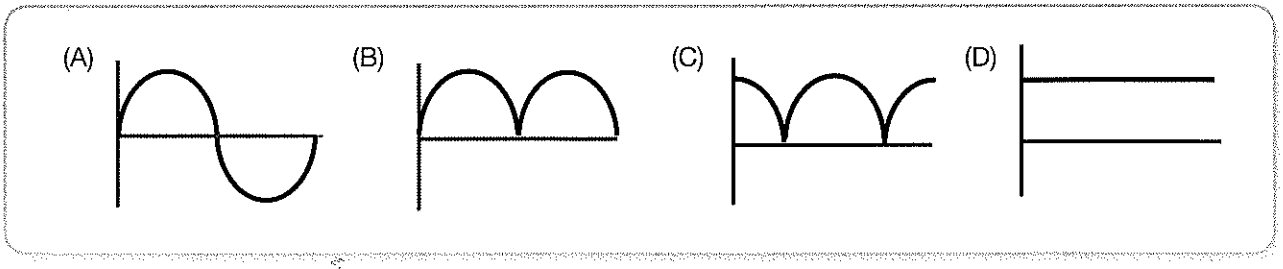
Student Y: The function of the slip rings in a simple AC motor is to ensure that the direction of the current flow though the coil is in the same direction relative to the magnetic field so that the torque on the coil is always in the same direction.

Student Z: The function of the slip rings in a simple AC motor is simply to act as electrical contact points between the power supply and the rotor coil.

4. A simple AC motor has radial magnets. Which graph correctly shows how the flux through the motor coil changes as it rotates?



5. A simple AC motor has radial magnets. Which graph correctly shows how the torque on the motor coil changes as it rotates from its starting position parallel to the field?



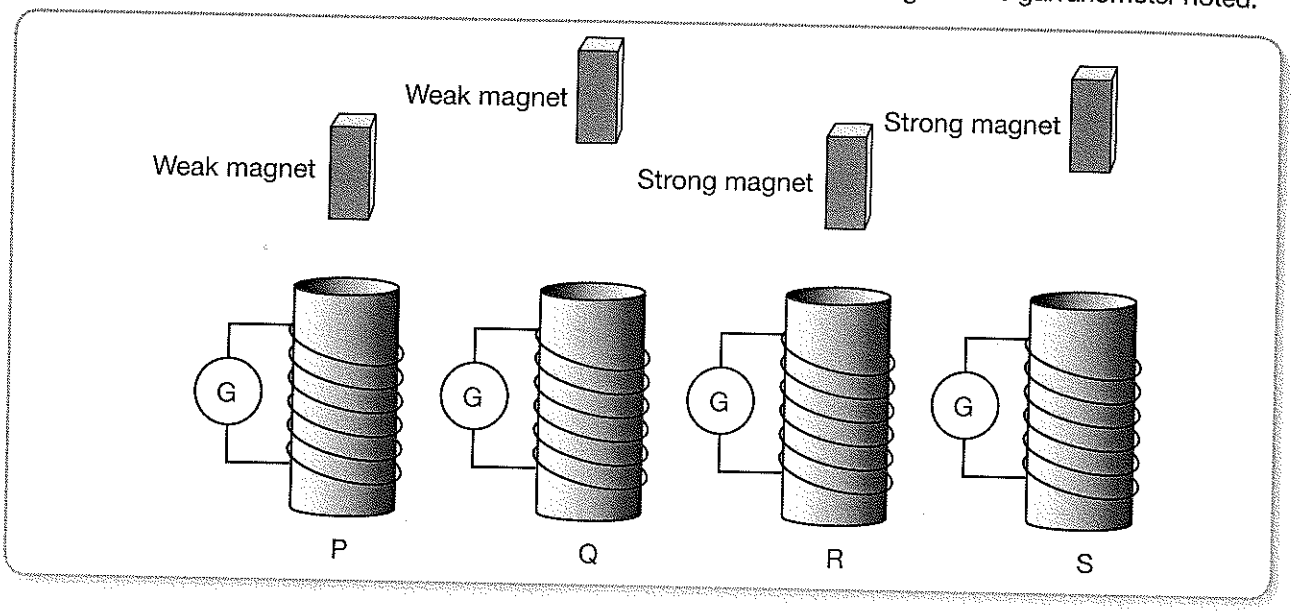
SET 37 Faraday and Induction

1. (a) State Faraday's laws of electromagnetic induction.

(b) Briefly describe the experiments done by Faraday in his work.

2. A magnet is continuously moved towards and away from a coil which is connected to an ammeter. The magnet does not ever move inside the coil. Explain what will be observed on the ammeter.

3. The diagram below shows how the equipment was set up for a Faraday type experiment. A coil of wire was wound on a thin cardboard tube and connected to a galvanometer. In different runs, a weak and a strong magnet were dropped from different heights through the tube and the readings on the galvanometer noted.



(a) There are two possible relationships that could be studied using this apparatus. What are these two possible aims for the experiment?

(b) For each of your two purposes (your answer to (a) above), identify which set-ups (P, Q, R and S) would be used to study each one.

(c) A student predicted that the emf's detected in the galvanometers would be, in increasing order, P < Q < R < S. Assess this prediction.

(d) What would the result of P and Q alone show? Explain why.

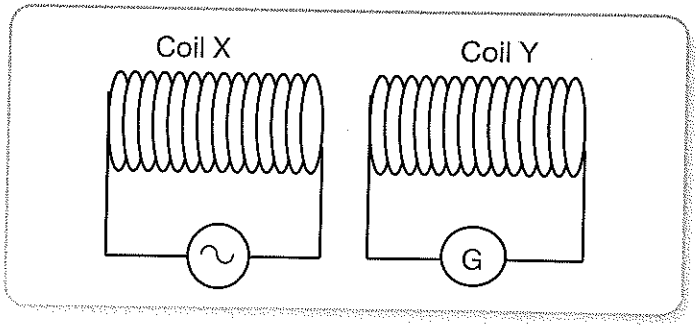
(e) State three factors which have been controlled in this experiment.

(f) Imagine that a strong magnet had been used in run Q. How would this change the purpose of the experiment?

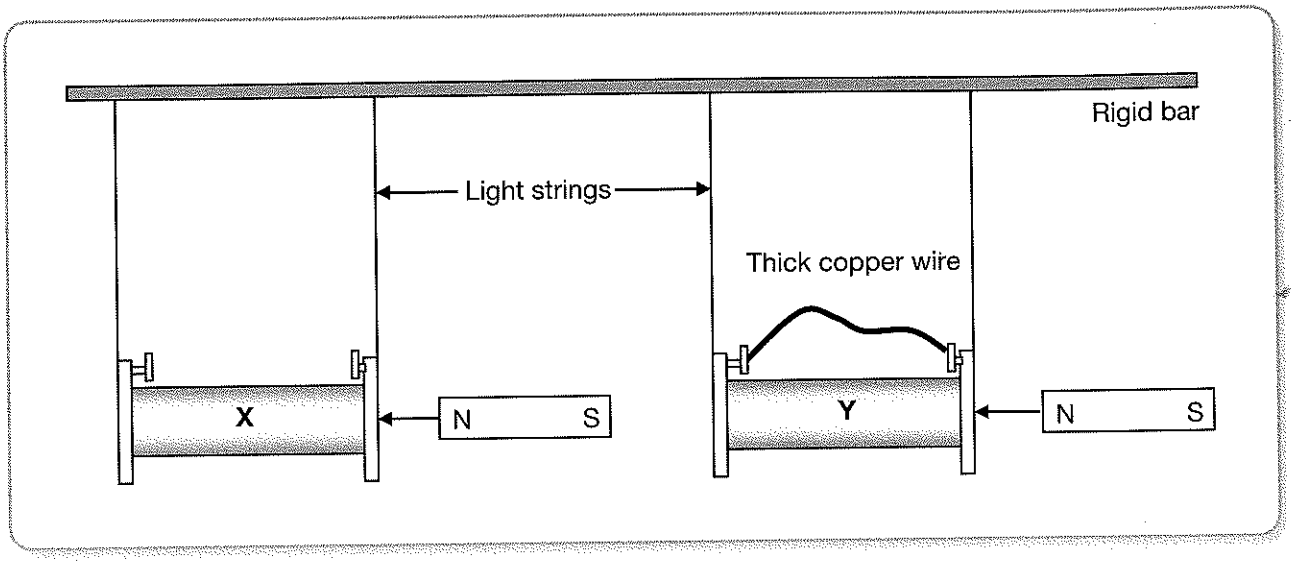
(g) List the induced emf's in increasing order if the weak magnet in Q is replaced by a strong magnet.

4. A student set up two coils as shown. She noticed that when the AC supply to circuit X was turned on, the galvanometer in circuit Y showed a current flowing in coil Y.

Explain this observation.



5. Two light solenoids are suspended by light strings from a rigid bar so that they can swing like pendulums. Solenoid Y differs from X in that the two ends of its coil are joined by a thick copper wire. The ends of the wire in solenoid X are not connected. A magnetic north pole is moved towards each solenoid. The magnets are stopped just as they enter the solenoids.



(a) Describe whether or not an emf will be induced in each coil. Justify your answer.

(b) Describe whether or not a current will be produced in each coil. Justify your answer.

(c) Describe the movement of the coils as the magnets approach them

6. A magnet is moved towards a coil which is part of a closed circuit. As a result, a current of 1 ampere is induced in the circuit. What will be the induced current if:

(a) The magnet is moved away from the coil at the same speed?

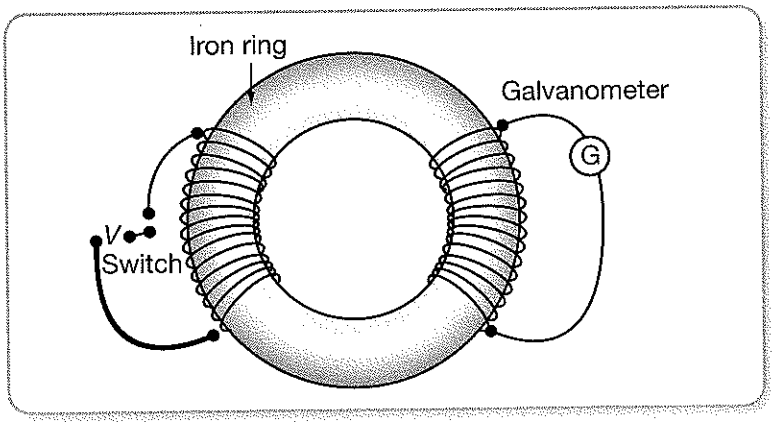
(b) The magnet is moved towards the coil at twice the speed?

(c) The magnet is moved away from the coil at one third the speed?

(d) Two magnets are moved towards the coil at the original speed?

7. The diagram shows the coil used by Faraday in his experiments on electromagnetic induction.

Describe and explain what happens in the galvanometer in each of the following situations.



(a) The switch is closed and left closed.

(b) The switch, initially closed, is opened.

(c) The switch is continuously closed and opened.

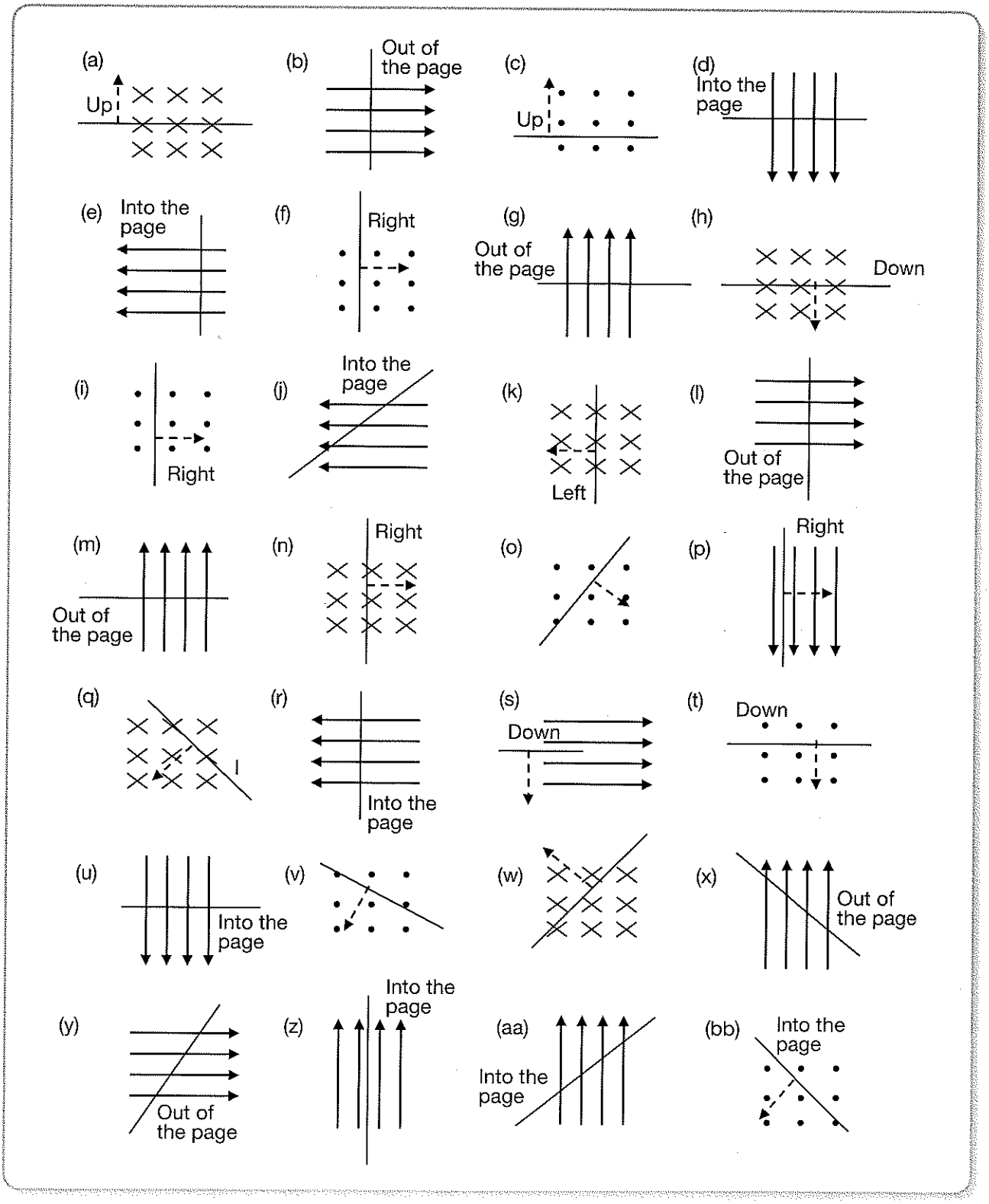
(d) How did Faraday interpret these observations?

(e) The equation Faraday derived to quantify his observations is as follows.

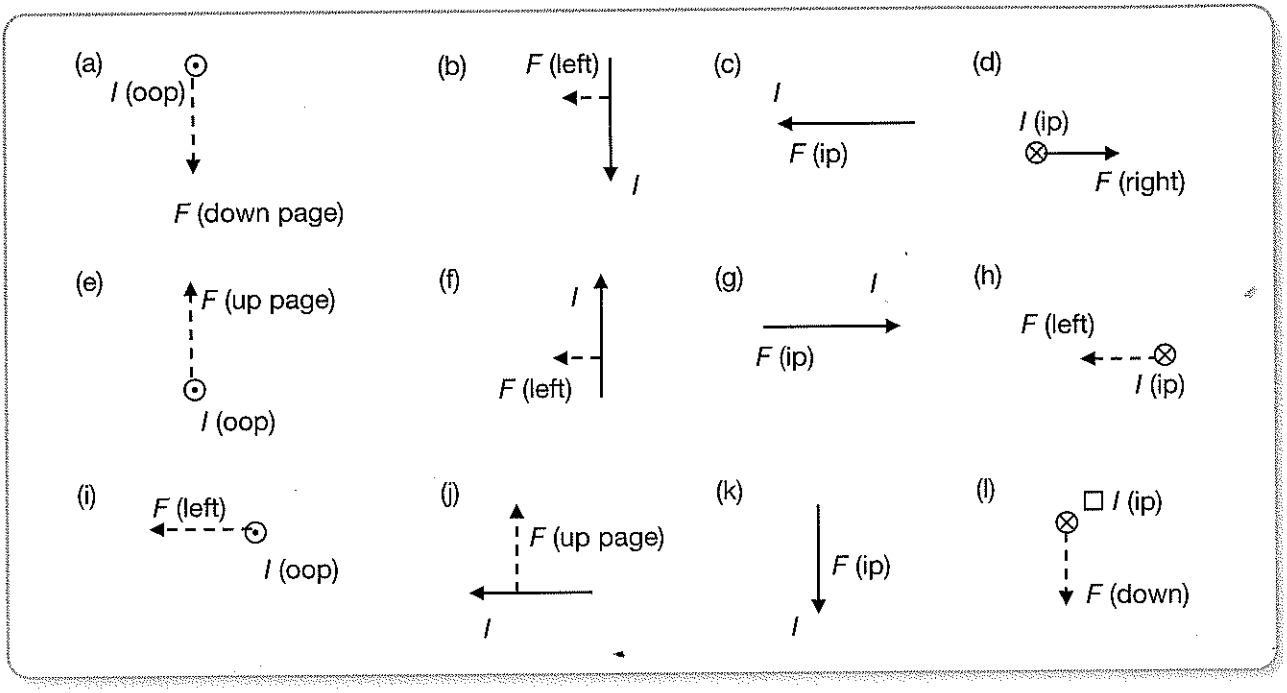
Identify each of the symbols used in this equation, taking particular care to explain the minus sign.

$$\varepsilon = \frac{-n\Delta\Phi}{\Delta t}$$

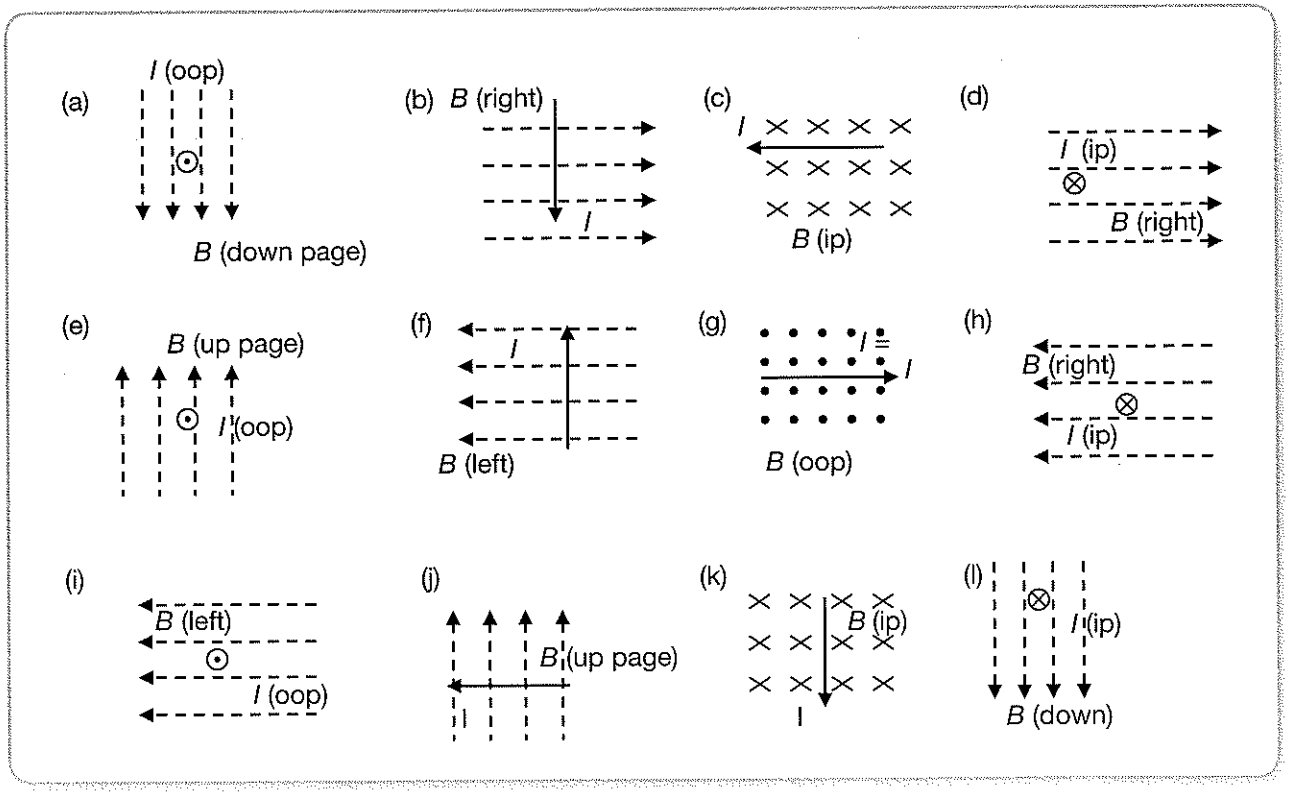
1. Predict the direction of the induced current in each of the conductors moving in the directions indicated through each magnetic field shown in the diagrams below. The dashed arrows show the direction of movement of each conductor.



2. Use Lenz's law to determine the direction in which the magnetic field must be directed in order to produce the induced emf shown in each wire. *F*, represented by the dashed lines when in the plane of the page, refers to the direction of the force moving the conductor through the magnetic field, as do the terms ip, oop, left, right, up and down.

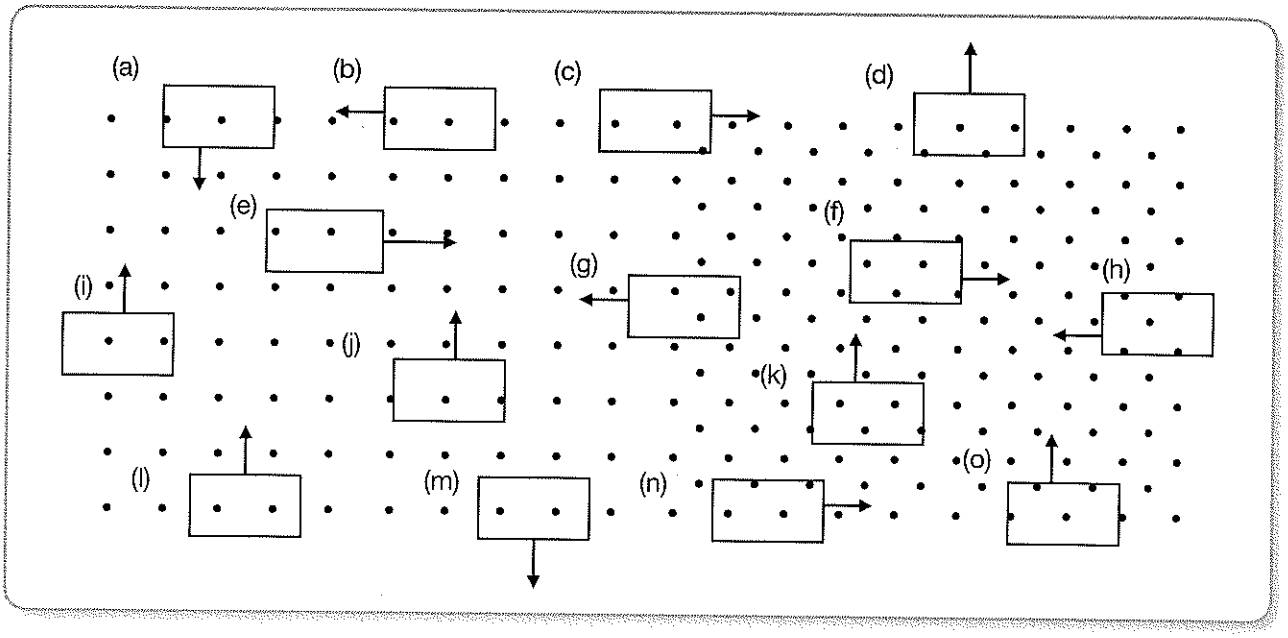


3. Use Lenz's law to determine the direction in which each conductor must be moved in order to produce the induced emf shown in each wire. *B*, represented by the dashed lines when in the plane of the page, refers to the direction of the magnetic field in the vicinity of the conductor.

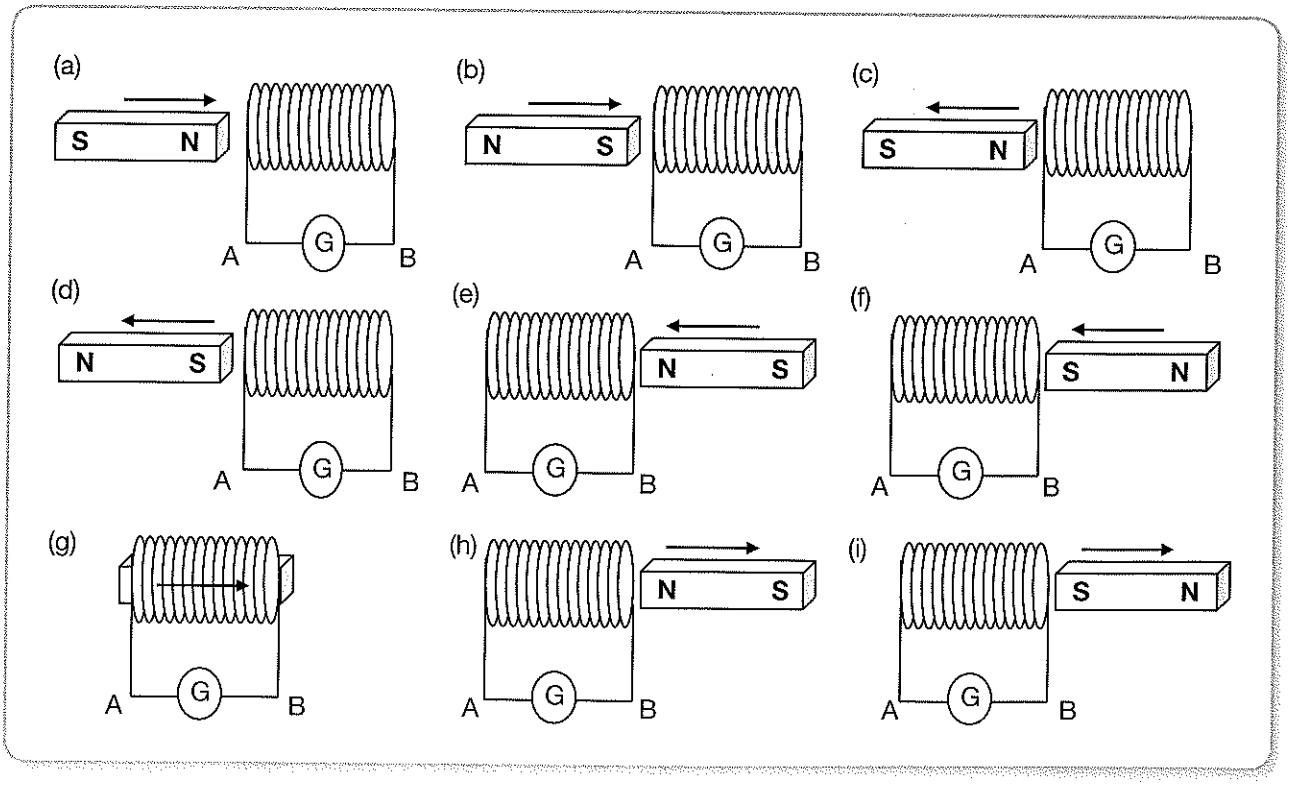


SET 39 Lenz's Law and Coils

1. The diagram shows several rectangular coils moving through a magnetic field directed out of the page. Predict the direction of any induced emf in each coil.

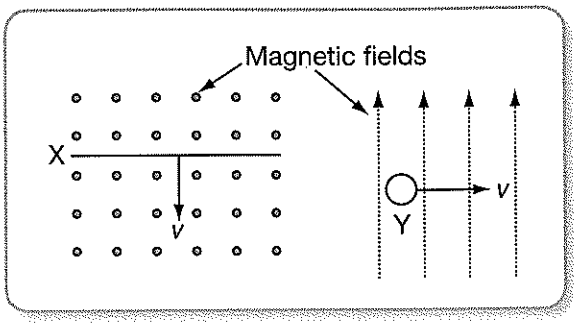


2. Predict the direction of the induced emf in each of the coils shown below. Express your answers as AGB or BGA depending on which way you predict the current produced will flow through the galvanometer. The arrows above each magnet indicate which way the magnet is moving relative to the coil. Note that in (g), the magnet is completely inside the coil.



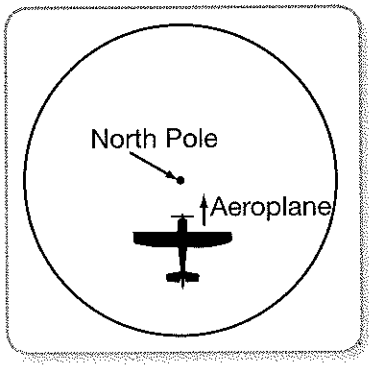
3. Consider two conductors X and Y moving through magnetic fields as shown.

Which choice correctly gives the directions of the induced emf's in X and Y?



	Emf in X	Emf in Y
(A)	To the left	Into the page
(B)	To the left	Out of the page
(C)	To the right	Into the page
(D)	To the right	Out of the page

4. An aeroplane flies towards the North Pole (where there is a considerable vertical magnetic field) as shown in the diagram.



(a) Which statement is correct?

(A) There will be a build-up of electrons on the left wing.

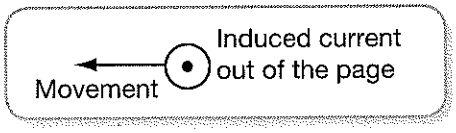
(B) There will be a build-up of electrons on the right wing.

(C) There will be a build-up of electrons at the nose of the plane.

(D) There will be no effect on the electrons, they will be evenly distributed.

(b) Justify your answer.

5. As a result of moving a vertical conductor to the left through a magnetic field, a current is induced in it out of the page as shown.

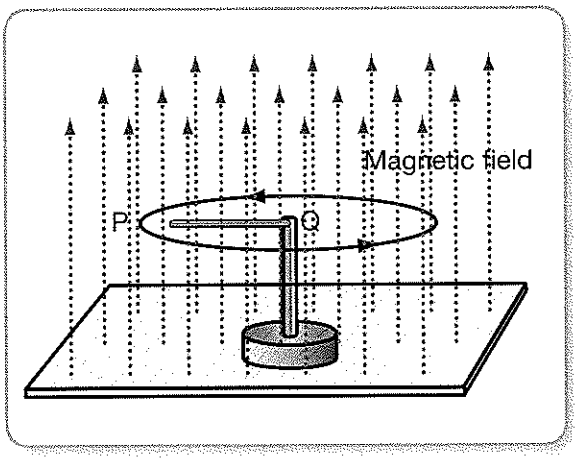


What is the direction of the magnetic field?

(A) Down the page. (B) Up the page. (C) Into the page. (D) Out of the page.

Use the following information to answer the next TWO questions.

A metal rod PQ is attached to a frictionless pivot on a heavy base sitting on a bench. The rod rotates clockwise with constant speed. A uniform magnetic field acts vertically upwards out of the bench.



6. Which choice predicts the direction of the induced emf in the rod PQ as it rotates?

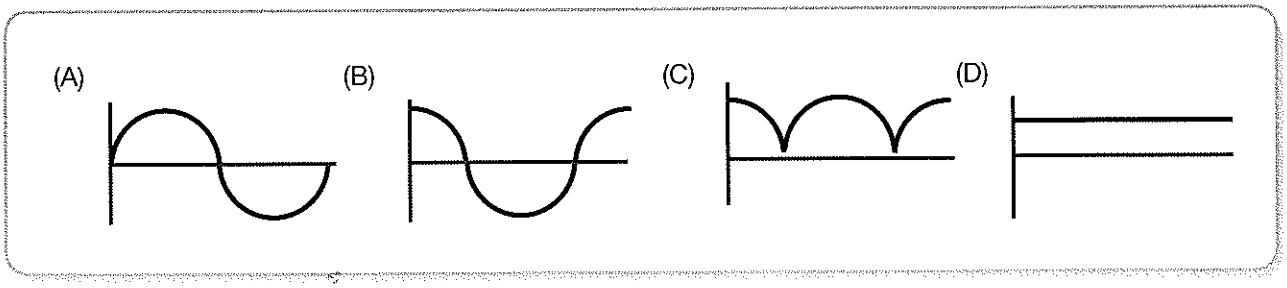
(A) From P to Q.

(B) From Q to P.

(C) Alternating between P and Q.

(D) From the top of the rod to the bottom of the rod.

7. Which graph best shows the emf induced in the rod?

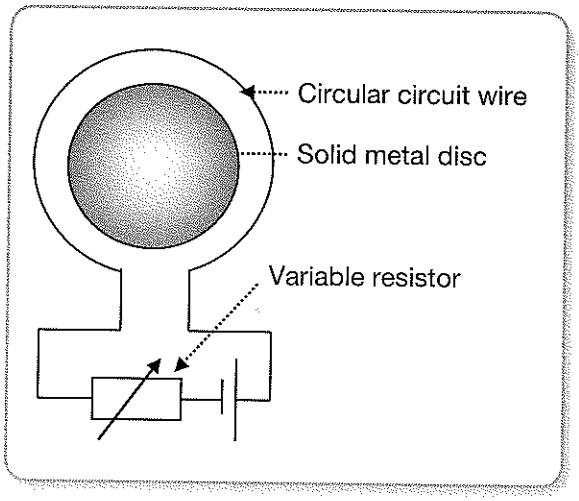


A In general

1. (a) What is an eddy current?

(b) Recall when eddy currents are produced.

2. Consider a circular circuit which loops around a solid metal disc as shown.



Explain what will happen in the metal disc if the power is turned on in the circuit and:

(a) The variable resistor is set at a fixed value.

(b) The magnitude of the variable resistor is decreased continuously.

(c) The magnitude of the variable resistor is increased continuously.

(d) The power supply is changed to an AC supply and the resistor value is kept constant.

B Electromagnetic brakes

3. Compare the advantages and disadvantages of electromagnetic brakes and conventional brakes.

4. Explain why electromagnetic brakes will work regardless of the speed of the object they stop.

5. Outline, with reference to Lenz's law, how magnetic brakes work.

6. Explain why electromagnetic brakes work more smoothly than conventional brakes.

7. Compare the safety of electromagnetic and conventional brakes.

8. Explain what happens when a solid, thin metal plate is dropped vertically downwards:

(a) Into a horizontal magnetic field.

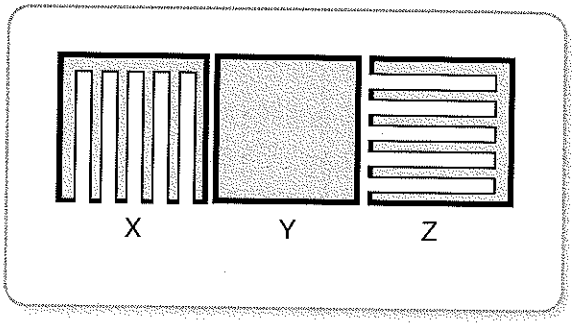
(b) Through, and totally enclosed within a horizontal magnetic field.

(c) Out of a horizontal magnetic field.

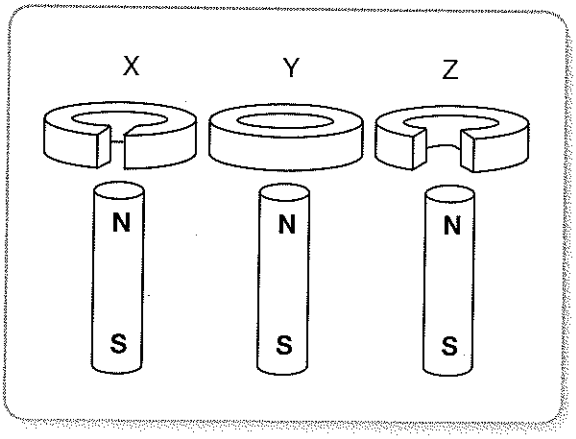
9. (a) Which principles in physics govern the production of eddy currents?

(b) Why must eddy currents follow Lenz's law?

10. Consider three metal plates shown below. Predict and explain what would happen to them if they were dropped into and moved through the same magnetic field directed into the page.



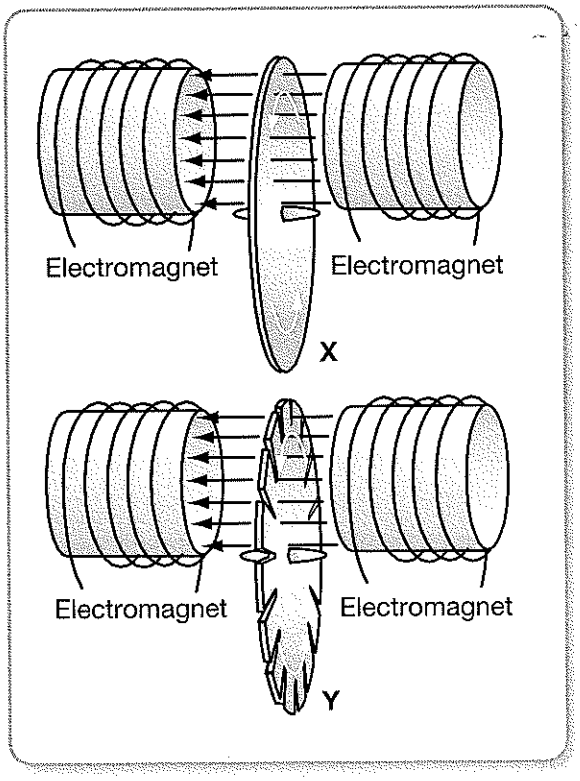
11. Three rings are dropped over three identical magnets as shown.



(a) In which order do the rings reach the bottom of the magnets?

(b) Justify your answer to (a).

12. Two aluminium discs are rotated between electromagnets as shown in the diagrams.



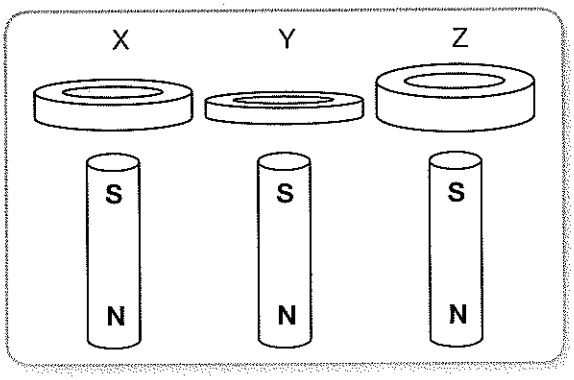
(a) Predict, and account for, what happens to disc X as it spins on its axle.

(b) Compare the rotation of disc Y with that of disc X.

(c) Deduce the directions of any eddy currents set up in disc X.

(d) Deduce the directions of any eddy currents set up in disc Y.

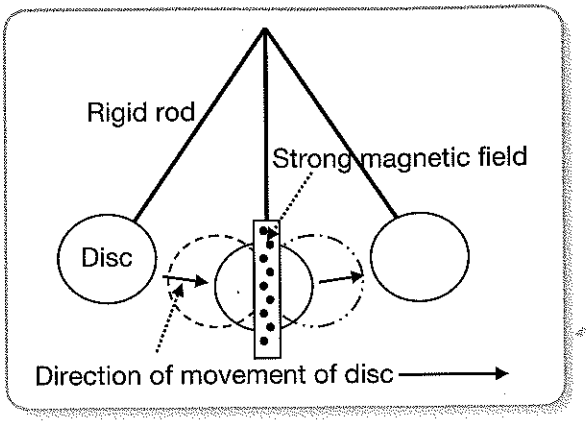
13. Three rings are dropped over identical magnets. The thickness of the rings varies.



(a) In which order do the rings reach the bottom of the magnets?

(b) Justify your answer to (a).

14. A metal disc is attached to a rigid rod and swings from left to right through a magnetic field directed out of the page as shown.



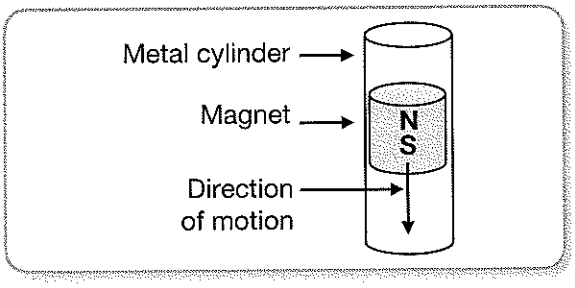
Predict the direction of eddy currents as the disc:

(a) Enters the magnetic field (dashed disc).

(b) Is in the position shown in the diagram.

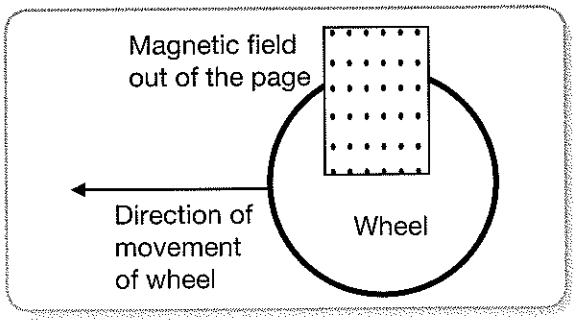
(c) Exits the field (dash/dotted disc).

15. Consider a magnet falling through a metal cylinder as shown.



Identify where, and in what direction(s) any eddy currents are formed in the metal cylinder.

16. A wheel moving from right to left passes into a magnetic field directed out of the page as shown.



(a) In what direction are eddy currents induced in the wheel?

(b) What will happen to the wheel as a result of this?

C Induction cooktops

17. Explain how electromagnetic induction is used in an induction cooktop.

18. Predict the efficiencies of three induction cooktop saucepans, X made of iron, Y made of copper, and Z made of aluminium. Explain your answer.

19. Induction cooktops are safer than conventional cooktops. Give reasons why.

20. Induction cooking is much more efficient than conventional cooktops. Suggest reasons for this.

21. Would an induction cooktop work on DC current? Explain.

22. Which choice correctly contrasts the efficiency of conventional coil cooktops and induction cooktops?

	Conventional coils	Induction cooktops
(A)	About 20%	About 60%
(B)	About 40%	About 80%
(C)	About 60%	About 20%
(D)	About 80%	About 40%

23. What is directly responsible for producing the heat for cooking food on an induction cooktop?

(A) The induced AC current in the base of the saucepan.

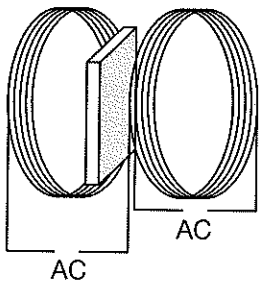
(B) The changing magnetic fields induced by the AC current in the coils under the benchtop.

(C) Interaction between the induced AC current in the saucepan and the resistance of the metal in the base.

(D) The interaction between the induced magnetic fields around the coil under the bench and the resistance of the metal in the base.

Use the following information to answer the next TWO questions.

In an experiment, a square sheet of aluminium was placed between two coils each carrying 20 amperes of AC current for 5 minutes as shown in the diagram. The experiment was repeated using an identically sized sheet of iron.



24. Which choice correctly predicts the results of this experiment?

(A) Both sheets will remain at room temperature.

(B) The iron sheet will get hotter than the aluminium sheet.

(C) The aluminium sheet will get hotter than the iron sheet.

(D) Both sheets will get equally warm.

25. The AC supply is replaced with a DC supply with double the voltage. How will the results of the experiment change?

(A) Both sheets will remain at room temperature.

(B) The iron sheet will get hotter than during the original experiment.

(C) The aluminium sheet will get hotter than the iron sheet, both hotter than in the original experiment.

(D) Both sheets will get warmer than the original experiment.

26. Where does electromagnetic induction occur in an induction cooktop?

(A) In the base of the saucepan.

(B) Around the coils carrying the current under the benchtop.

(C) Between the coils under the benchtop and the base of the saucepan.

(D) In the saucepan base and around the coils under the benchtop.

SET 41 Back Emf in Motors

1. (a) What is a back emf?

(b) Where do back emf's occur?

(c) How is back emf produced?

(d) Why does back emf always oppose supply emf?

(e) What is the effect of back emf on a motor?

(f) How can back emf be reduced? Explain your answer.

(g) How does the inclusion of a soft iron core in a motor affect the back emf?

(h) Explain what will happen to the back emf in a motor if it slows down?

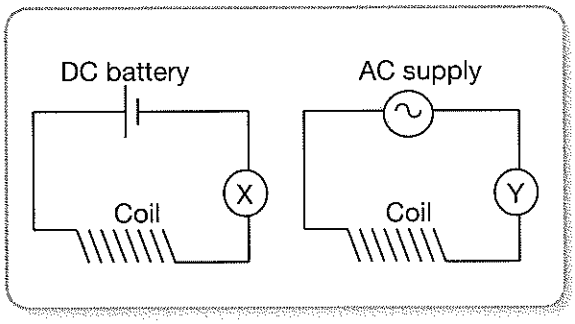
2. (a) Why is a motor more likely to burn out if it jams rather than when it is running normally?

(b) How is a motor protected against this?

3. (a) When the load on a DC motor is increased, its speed decreases. What other quantity decreases?

(b) Explain your answer.

4. Identical globes, X and Y, are connected into circuits as shown. They have an identical glow. Note that the circuits do not affect each other.



Explain what happens to the globes when a soft iron core is inserted into each coil.

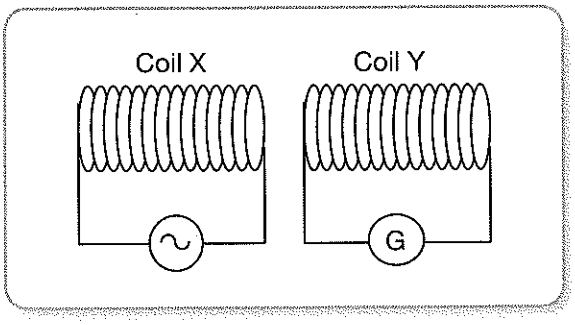
5. A car mechanic noticed that when his 240 V DC motor was operating at maximum speed, the voltage across its coil was significantly less than 240 V. Why?

6. (a) A motor is connected in series to an ammeter and a 12 V DC supply. The resistance of the coil is 2.0 Ω. When the motor is operating at full speed, the ammeter reads 3.5 A. What is the best explanation for this?

(b) What is the magnitude of the back emf in this motor?

7. As a builder drills into a brick wall, the friction on the drill bit causes the armature of the drill to slow down. Describe what happens to the current and the back emf in the drill's coils as the drill slows down.

8. A student set up two coils as shown below. She noticed that when the AC supply to circuit X was turned on, the galvanometer in circuit Y showed a current flowing in coil Y.



(a) Explain why this happens.

(b) The student placed a soft iron core inside the coils X and Y. What will she observe? Explain your answer.

(c) What is another observation the student will notice when the soft iron cores are in the coils?

9. Back emf is an example of which law or principle in physics?

(A) Newton's third law.

(B) The motor effect.

(C) Faraday's law of induction.

(D) The law of conservation of energy.

SET 42 Transformers

1.
(a) What is the function of transformers?
(b) Describe how the structure of transformers relates to their function.
(c) Explain why a transformer will not work with DC input.
(d) Identify the two different types of transformers.
(e) State where each type of transformer is used and why it is used there.

2. A transformer has a 5000 turn primary coil. It is to be used to convert 240 V AC to 450 V AC.
(a) Identify this as a step-up or step-down transformer.
(b) Calculate the number of coils in the secondary coil.

3. The output coil of a transformer has four times as many coils as the input coil.
(a) Identify this as a step-up or step-down transformer.
(b) If the input voltage was 100 V, state the output voltage.
(c) If the input current was 20 A, state the output current.

4. A battery charger is used to charge 1.5 V batteries through a 12 V transformer. It has 2400 turns in the input coil. Assume 100% efficiency.
(a) Calculate the number of turns in the output coil.
(b) If the input current is 10 mA, calculate the output current.

5. A transformer changes 240 V to 5000 V.
(a) Calculate the ratio of the number of turns in each coil.
(b) Calculate the ratio of the currents in each coil.
(c) Identify the output coil.
(d) Predict where in your home this transformer might be used.

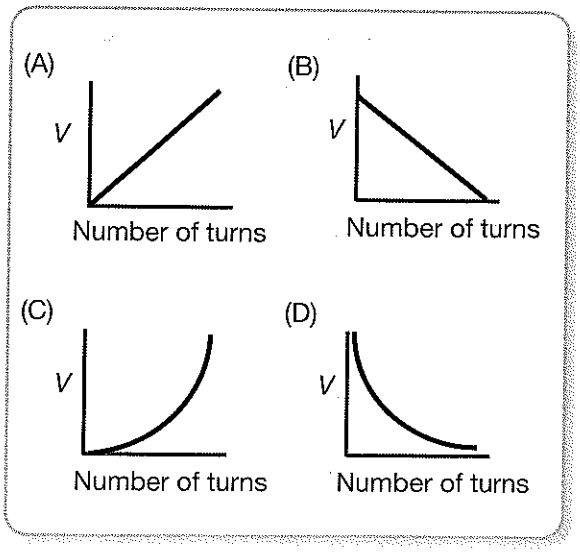
6. Assess each of the following statements as they apply to our use of transformers in household appliances.
(a) Transformers are used to reduce currents to minimise energy losses during transmission of electricity through wires.
(b) Transformers are used to increase currents so that some appliances can produce more power.
(c) Transformers are used to change voltages so that appliances can operate on correct voltages.
(d) Transformers are used to change voltages so that less power is used when smaller appliances are used.

7. Which choice shows the relationship between the number of coils in the primary and secondary coils of a transformer and the primary and secondary voltages?

(A) $\frac{n_P}{n_S} = \frac{(VI)_P}{(VI)_S}$ (B) $\frac{n_P}{n_S} = \frac{V_S}{V_P}$

(C) $\frac{n_P}{n_S} = \frac{V_P}{V_S}$ (D) $\frac{n_P}{n_S} = \frac{I_P}{I_S}$

8. Which graph shows the relationship between the number of turns in the secondary coil of a transformer, and the voltage it produces.



9. Which appliance uses a step-up transformer?

(A) Hot water system.
(B) Microwave oven.
(C) Mobile phone charger.
(D) Refrigerator.

10. Which appliance uses a step-down transformer?

(A) Air conditioning unit.
(B) LCD TV.
(C) Microwave oven.
(D) Toaster.

11. Which choice describing the function of transformers is correct?

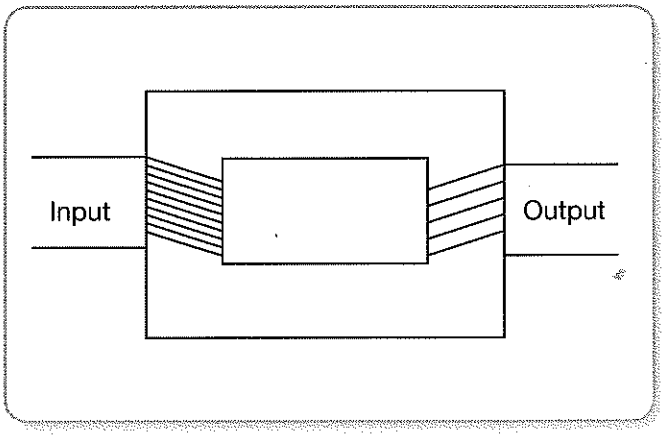
	Step-up transformer	Step-down transformer
(A)	Increases voltage and current.	Decreases voltage and current.
(B)	Increases voltage, decreases current.	Decreases voltage, increases current.
(C)	Decreases voltage and current.	Increases voltage and current.
(D)	Decreases voltage, increases current.	Increases voltage, decreases current.

12. Which choice describing transformers is correct?

	Step-up transformer		Step-down transformer	
	Input coil	Output coil	Input coil	Output coil
(A)	*n* turns	*n* turns	*n* turns	*n* turns
(B)	*n* turns	More than *n* turns	*n* turns	More than *n* turns
(C)	*n* turns	Less than *n* turns	*n* turns	Less than *n* turns
(D)	*n* turns	More than *n* turns	*n* turns	Less than *n* turns

Use the following information to answer the next TWO questions.

Consider the following accurate diagram of a transformer.



12. Which statement best describes this transformer?

(A) It is a step-up transformer and its input voltage will be greater than its output voltage.
(B) It is a step-up transformer and its input voltage will be less than its output voltage.
(C) It is a step-down transformer and its input voltage will be greater than its output voltage.
(D) It is a step-down transformer and its input voltage will be less than its output voltage.

13. If the input voltage was 120 V, what would be the output voltage?

(A) 60 V
(B) 80 V
(C) 180 V
(D) 240 V

Use the following information to answer the next TWO questions.

A transformer has an input voltage of 100 V. It outputs 10 A at 20 V.

14. What is the current in the primary coil?

(A) 1 A
(B) 2 A
(C) 5 A
(D) 50 A

15. What law in physics do you apply to get this answer?

(A) Faraday's law of induction.

(B) Ohm's law.

(C) Power input = power output.

(D) The law of conservation of energy.

Use the following information to answer the next TWO questions.

A neon shop sign operates at maximum efficiency if a voltage of 6000 V is applied.

16. What would be the ratio of the number of coils in the secondary coil to the number in the primary coil of the transformer needed to run it efficiently from a 240 V supply line?

(A) 1 : 25

(B) 1 : 40

(C) 25 : 1

(D) 40 : 1

17. Why is such a high voltage needed?

(A) Gases are not good conductors of electricity so to get the electrical discharges needed to light up the tube a high voltage is needed.

(B) Gases are not good conductors of electricity so to get the electrical discharges a very low gas pressure is required inside the tubes.

(C) Neon signs are large and a large voltage is required to enable electrons to travel the length.

(D) Neon signs use a lot of energy so a large input of energy is required.

18. A transformer is to be used to run a 45 V system from a 240 V supply line. Which choice shows a possible number of turns in each of its primary and secondary coils?

	Primary coil	Secondary coil
(A)	500	94
(B)	7500	1250
(C)	40 000	7500
(D)	10 000 000	1 500 000

19. Some students built a transformer and connected it to the DC terminals of a power supply. It failed to work. Which choice best accounts for this?

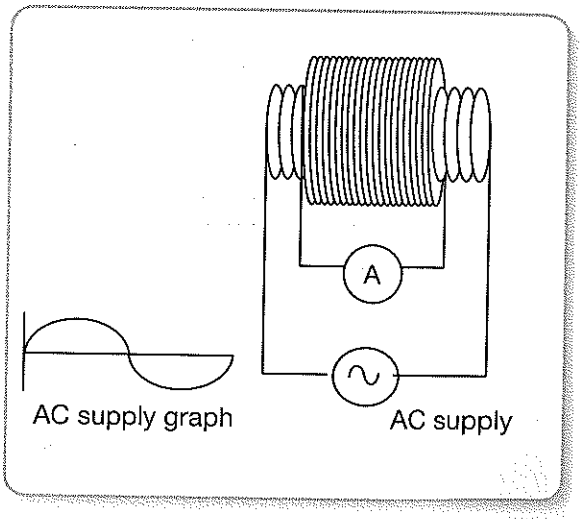
(A) Electromagnetic induction only occurs in the presence of changing magnetic fields.

(B) The DC current was probably not large enough to cause induction of strong magnetic fields.

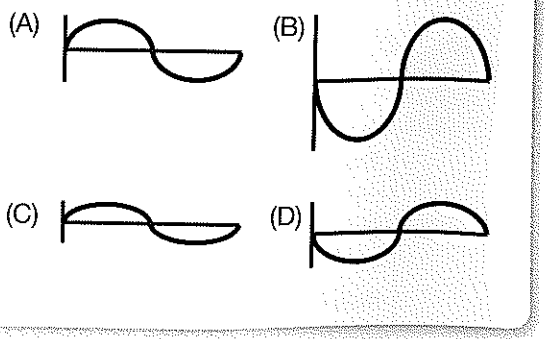
(C) Alternating current is needed for the electrons in the transformer coils to oscillate back and forth.

(D) They probably connected the transformer coils into the circuit with reverse polarities

20. Some students built a transformer by wrapping one coil of wire around another coil and connecting these to an AC supply and a galvanometer as shown in the diagram. The graph of the AC supply is also shown.



Which graph best represents the output from this transformer?



SET 43 Transformers and Electricity Transmission

1. Outline the functions of step-up and step-down transformers in electrical transmission systems.

2. What is the main purpose of step-up transformers in electrical transmission?

(A) To increase voltage so that power can be delivered more efficiently to the consumer.

(B) To increase current so that more energy gets to the consumer.

(C) To decrease voltage so that less energy is lost as heat energy in the transmission wires.

(D) To decrease current so that more power is delivered to consumers.

3. Current is transmitted to homes, shops, factories and schools as AC. Which choice best explains the reason for this?

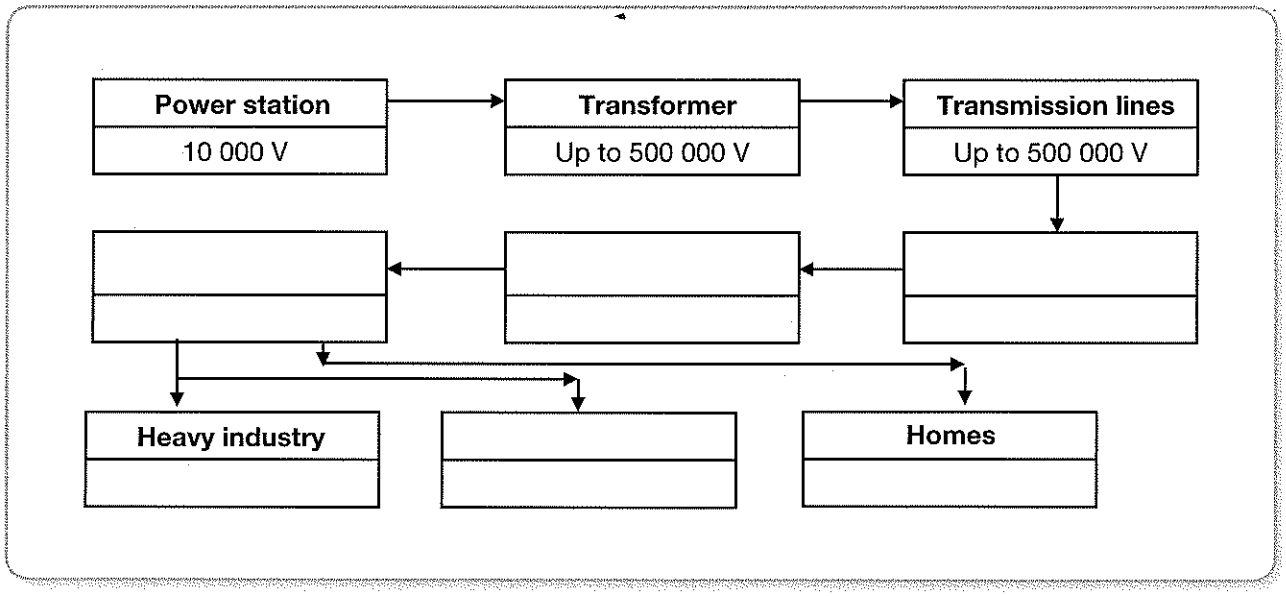
(A) More energy is available if transmission from power stations is at low voltages and low currents.

(B) Less energy is available if transmission from power stations is at high voltages and high currents.

(C) More energy is lost as heat if transmission from power stations is at high voltages and low currents.

(D) Less energy is lost as heat if transmission from power stations is at high voltages and low currents.

4. Electrical energy is generated at about 10 000 V at power stations, but is not transmitted, nor used by consumers at this voltage. Copy and complete the diagram below to show the changes in voltage that occur as the energy is transmitted to consumers. Also identify where each voltage change occurs.



5. Copy and complete the table for the transmission of electricity from a 40 MW power station over 50 km of 2.5 ohm km^{-1} lines.

Transmission voltage	400 kV	300 kV	200 kV	100 kV
Current in lines ($I = P/V$)				
Power loss ($P = I^2R$)				
Loss (%)				

6. Which of the following would help reduce heat loss in electrical transmission lines?

(A) Use thinner wire in the transmission lines.

(B) Increase the current flowing through the lines.

(C) Increase the voltage at which the current is transmitted.

(D) Use conductors with a higher resistivity.

7. A power station generates 250 kW of electricity and transmits it at 3.0×10^5 V. It sends this power to a town 40 km away through transmission lines which have a resistance of 0.3 $\Omega\ m^{-1}$.

(a) Calculate the current in the transmission lines.

(b) Calculate the voltage drop between the power station and the town.

(c) From this deduce the voltage available at the town.

(d) Calculate the power loss in the lines.

8. The diagram shows a variable output transformer. Depending on the output connections chosen, the output voltage can be varied.

(a) How many different output voltages are possible?

(b) Suppose the input voltage is 300 V. Estimate the output voltage (round off to the nearest 5 V) when the connection used is:

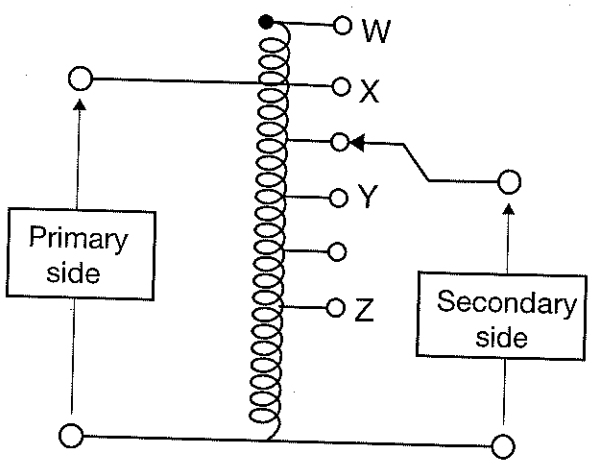
(i) W

(ii) X

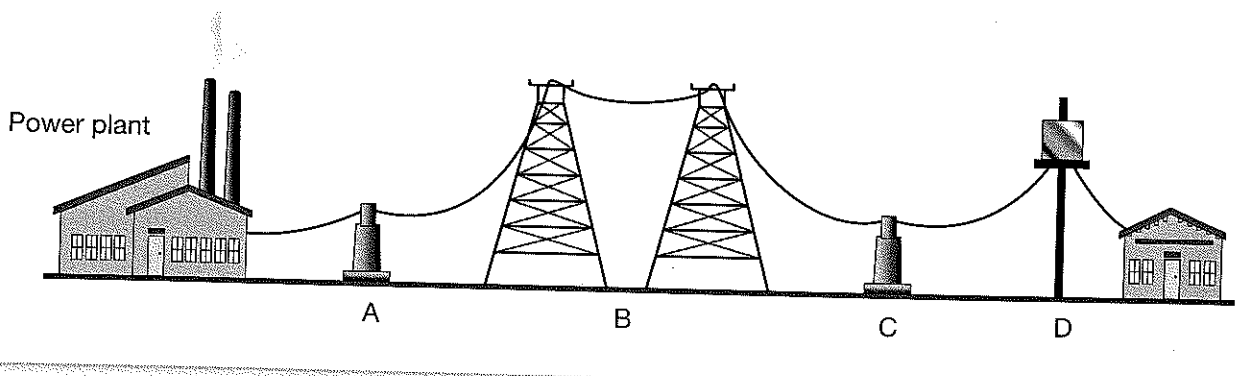
(iii) Y

(iv) Z

(c) What is the output voltage (nearest 5 V) at the connection shown in the circuit?



9. Consider the following diagram which shows an electrical transmission system. At which position in this system would you find a step-up transformer?



10. Summarise three major impacts of the production and use of electricity on the environment.

11. Summarise three major impacts of the production and use of electricity on our society (i.e. people).

SET 44 The DC Generator

1. (a) What is the function of a generator?

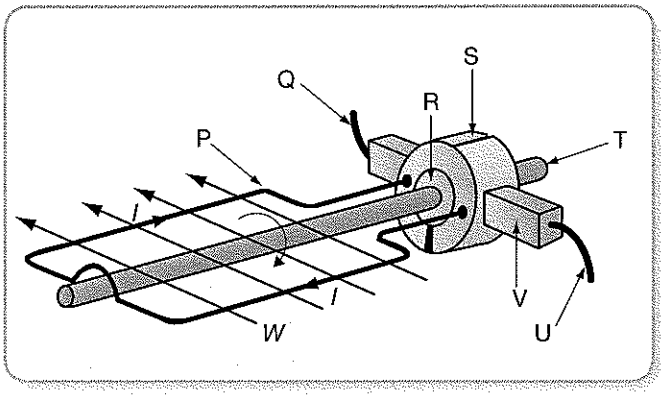
(b) What is the main energy change occurring in a generator?

(c) On what scientific principle does a generator work?

(d) What variables in a generator affect the emf produced by it?

(e) What is the unit used for rate of change in flux?

2. The diagram shows a DC generator. Use it to answer this question.



(a) Identify each labelled part of the generator shown in the diagram.

(b) Clarify the function of the split ring commutator.

(c) How would the operation of this generator be different if the magnetic field was supplied by radial magnets rather than by rectangular magnets?

3. A generator produces an emf of *V* volts. Predict and explain the effect on the emf if all other factors remain constant but:

(a) The number of turns in the coil is doubled.

(b) The original coil is rotated at twice the speed.

(c) The magnetic field is halved.

(d) The size of the coil (area) is increased.

(e) Two coils at 90° to each other are used.

4. A DC generator coil lies with its area parallel to a magnetic field.

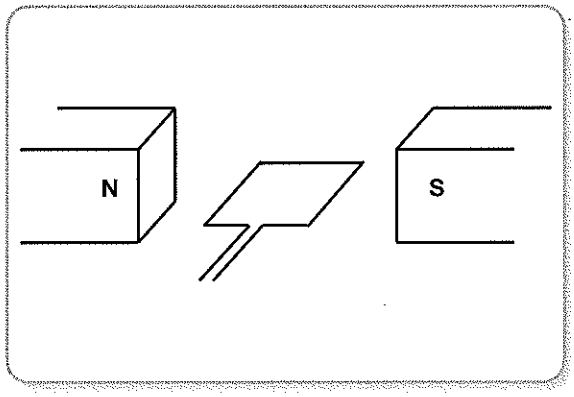
(a) Sketch a graph to show how the flux through the coil changes as it rotates through 360°.

(b) Sketch a graph to show how the rate at which flux is cut changes as it rotates through 360°.

(c) Sketch a graph to show how the emf produced in the coil changes as it rotates through 360°.

(d) Sketch a graph to show the emf delivered to the external circuit as the coil rotates through 360°.

5. The diagram represents a DC generator which produces an output voltage as the coil rotates in the magnetic field.



If the voltage produced as it passes the position shown is 8 V, what will it be as it rotates, at the same speed, at 90° to this position?

(A) 0 V

(B) Less than 8 V but > 0 V

(C) 8 V

(D) Unable to determine with this data.

6. Who discovered the generation of an electric current by a magnet moving near a coil?

(A) Faraday.

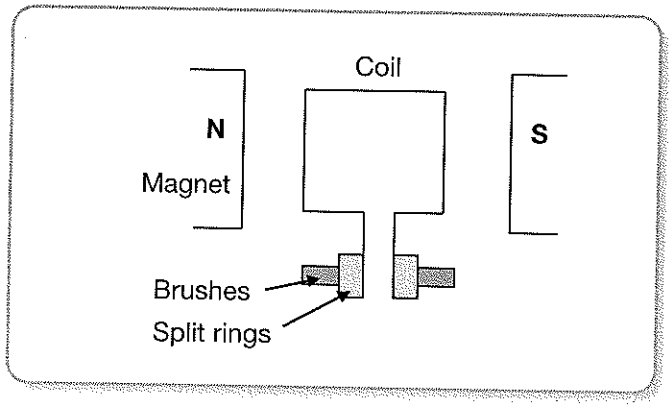
(B) Hertz.

(C) Oersted.

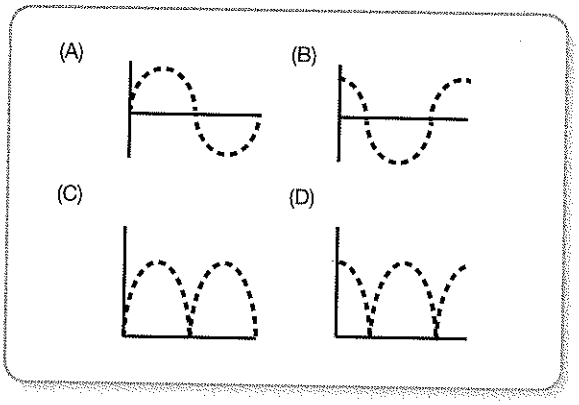
(D) Tesla.

Use the following information to answer the next TWO questions.

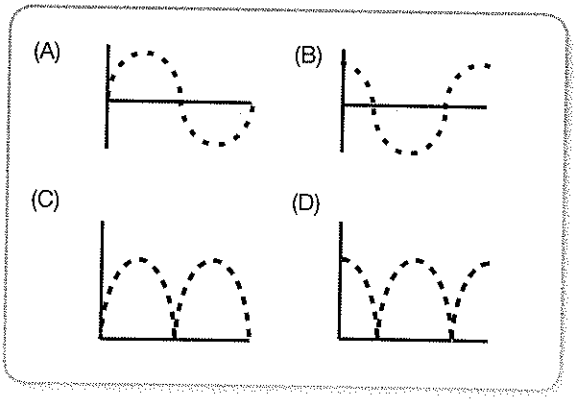
The diagram shows equipment used to generate a current. At the instant of time shown in the diagram, the coil is in full rotation.



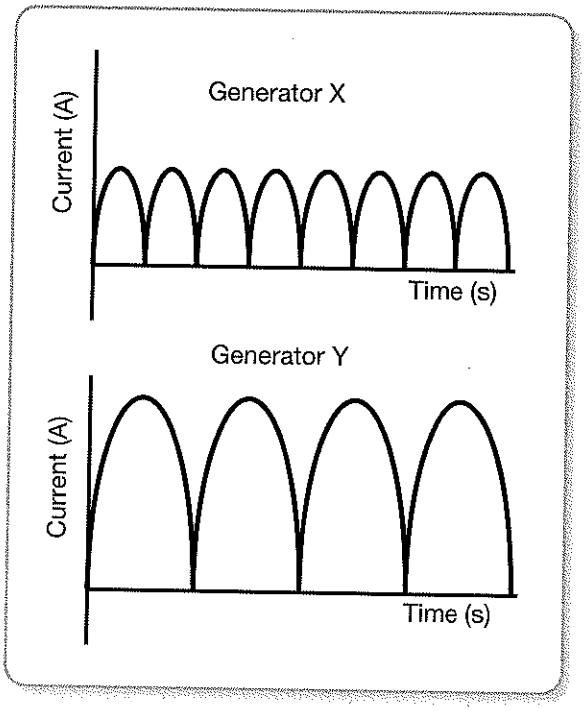
7. Assuming that the coil starts from the position shown, which graph best shows the current produced internally by this device as it rotates through 360°?



8. Assuming that the coil starts from the position shown, which graph best shows the current delivered to the external circuit by this device as it rotates through 360°?



9. The graphs below show the current delivered by two different DC generators.



(a) Explain, in terms of possible structural differences between the two generators, why these graphs are different.

(b) Explain, in terms of possible operational differences between the two generators, why these graphs are different.

10. Why is it necessary to have a soft iron core in a generator?

(A) To reduce the eddy currents produced.

(B) To minimise the back emf produced.

(C) To allow the magnetic field to permeate the coil.

(D) To improve generation of emf by intensifying the magnetic field in the space occupied by the coil.

11. Why should the soft iron core of a DC generator be laminated?

(A) To reduce the back emf in the electromagnet coils.

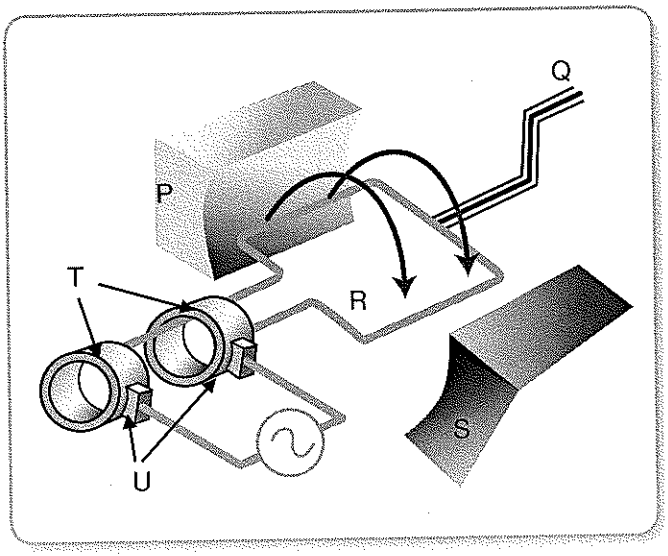
(B) To prevent eddy currents forming in the soft iron core.

(C) To minimise heating effects in the soft iron core.

(D) All of the above.

SET 45 The AC Generator

1. The diagram shows a simple AC generator.



(a) How do we know it is an AC generator?

(b) Identify each of the labelled parts.

(c) What is the functional energy change occurring in this generator?

2. Which of the following can be used to show the production of alternating current?

(A) Move a wire that is part of a closed circuit through a magnetic field at varying speeds.

(B) Move a solenoid that is connected into a circuit up and down.

(C) Rotate a current carrying coil between the poles of a pair of radial magnets.

(D) Move a magnet in and out of a solenoid which is part of a closed circuit.

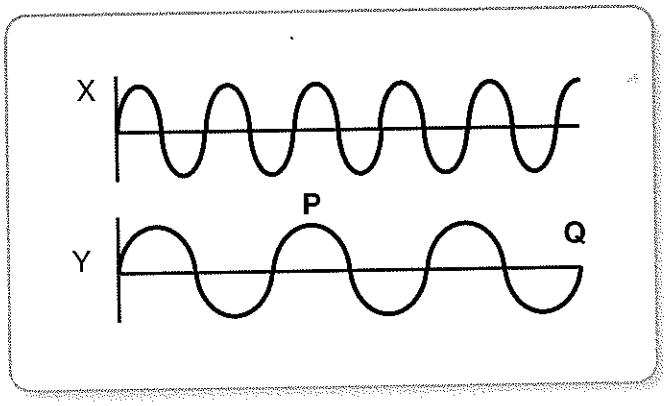
3. A magnet is continuously moved towards and away from a coil which is connected to an ammeter. The magnet does not ever move inside the coil.

(a) What will be observed on the ammeter?

(b) What would be observed if the ends of the coil were connected to a voltmeter?

Use the following information to answer the next SIX questions.

The graphs show the outputs from two generators X and Y over a period of time. They are drawn to the same scale and represent the same period of time.



4. What two things about these generators could be true?

(A) X is turning faster and has fewer turns in its coil than Y.

(B) X is turning faster and has more turns in its coil than Y.

(C) X is turning slower and has fewer turns in its coil than Y.

(D) X is turning slower and has more turns in its coil than Y.

5. Justify your answer to Question 4 in terms of the structure and/or operation of the generator.

6. If the total time represented by each graph is 0.01 s, determine the frequency of the emf generated by each generator.

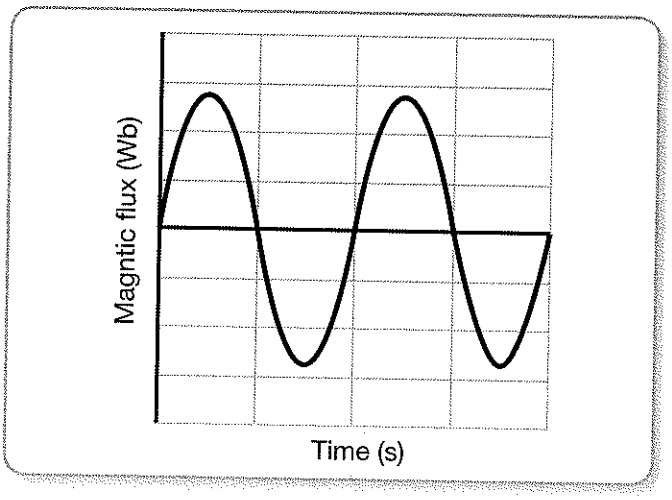
7. Points P and Q on graph Y represent a maximum emf output and zero emf output respectively. Describe the position of the coil in the generator at times P and Q.

8. What is the rate of rotation of the coil in X compared to that in Y?

9. Given that X is rotating faster than Y, explain why the magnitudes of the output emf from each can be identical.

Use the following information to answer the next TWO questions.

The graph shows how the flux through a coil in a generator changes over time.



10. When does this generator supply zero voltage?

(A) All the time because change of flux is constant.

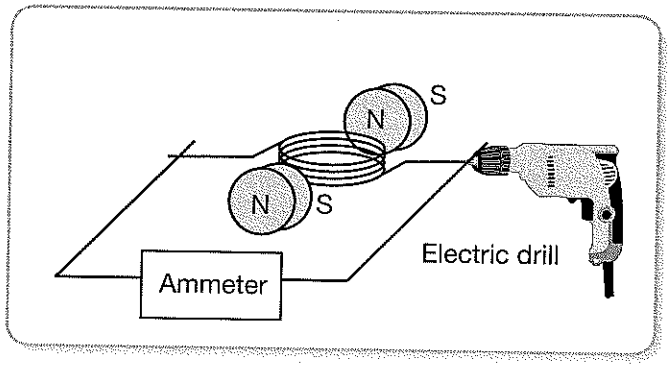
(B) At no time. A voltage is constantly generated.

(C) At times when the graph passes through zero on the flux axis.

(D) At times when the graph is at maximum or minimum amplitude.

11. Justify your answer.

12. Consider the diagram.

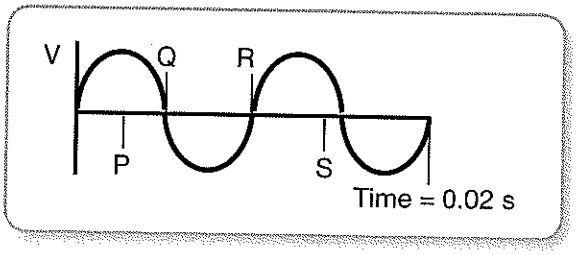


(a) What is the diagram showing?

(b) Describe how it will work.

(c) Outline, in terms of principles in physics, why it works.

13. The graph shows how the voltage output from a generator varied with time.



(a) When does the coil of the generator cut flux at the greatest rate?

(A) At time P.

(B) At time Q.

(C) At time R.

(D) At time S.

(b) Justify your answer.

(c) Which statement about this generator is correct?

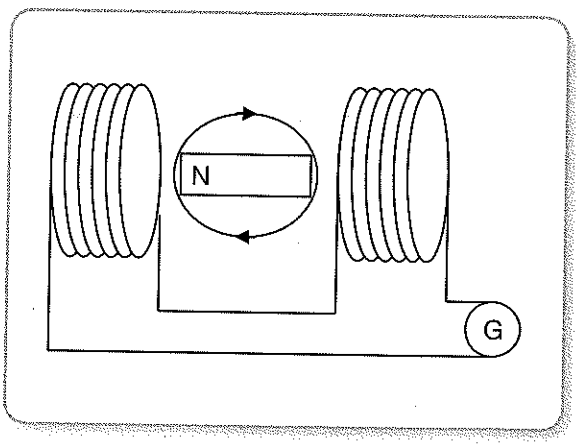
(A) It is a 50 Hz DC generator.

(B) It is a 100 Hz DC generator.

(C) It is a 50 Hz AC generator.

(D) It is a 100 Hz AC generator.

14. A student set up two coils either side of a strong magnet as shown in the diagram. He then spun the magnet in a circle and observed the reading on the galvanometer.



What would he observe?

(A) AC output.

(B) A constant DC output.

(C) A varying DC output.

(D) No change in reading.

1. (a) What is the functional difference between a motor and a generator?

(b) What difference in energy changes occur in a motor and a generator?

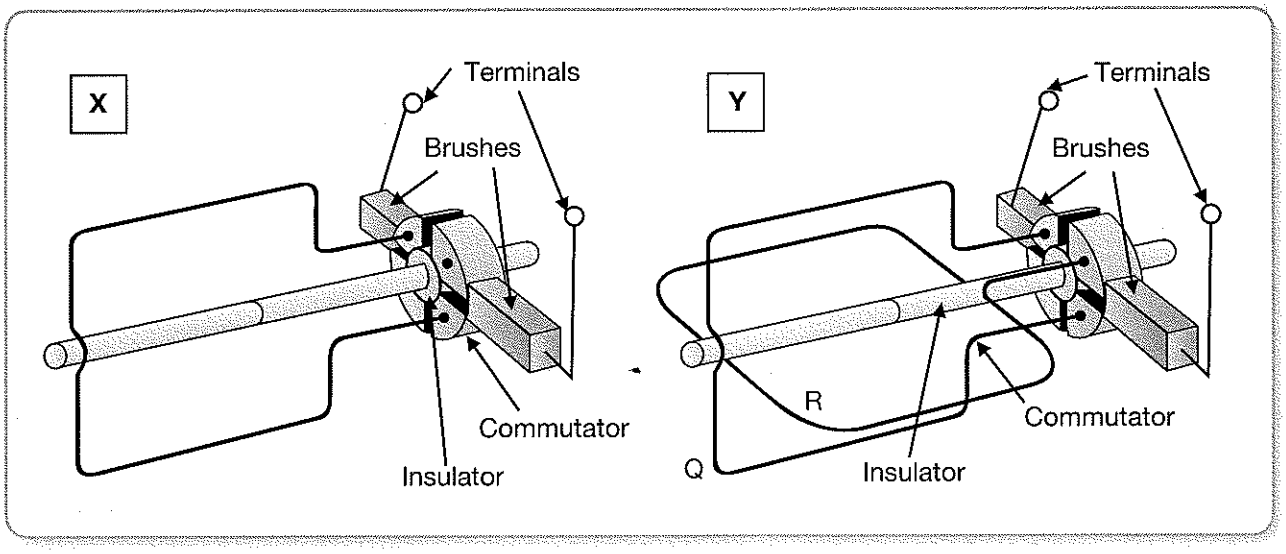
(c) What structural differences are there between a motor and a generator?

(d) Why is a motor subject to a back emf while a generator is not?

(e) Compare the principles in physics involved in a motor and a generator.

(f) What is the structural difference between simple AC and DC generators?

2. The diagrams below show two electrical devices, X and Y.



(a) Discuss whether these devices are motors or generators. Justify your answer.

(b) Are these devices DC or AC devices? Justify your answer.

(c) Describe any physical difference between the two devices.

(d) Justify the need for the two different types of commutators.

(e) Taking the two devices shown as generators, sketch a graph to show the emf generated internally for two complete rotations of:

(i) The coil in X.

(ii) Coil Q in Y on its own.

(iii) Coil R in Y on its own.

(iv) The net emf generated in device Y.

(f) Taking the two devices shown as generators, sketch a graph to show the emf output (delivered to the external circuit) for two complete rotations of:

(i) The coil in X.

(ii) Coil Q in Y on its own.

(iii) Coil R in Y on its own.

(iv) The net emf generated in device Y.

(g) Although some resource materials may refer to slip rings as commutators they are actually not commutators. Explain this.

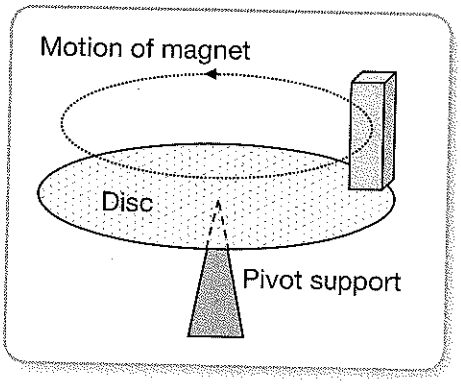
(h) Explain the function of the insulating disc between the commutator and the axles in these devices.

SET 47 The AC Induction Motor

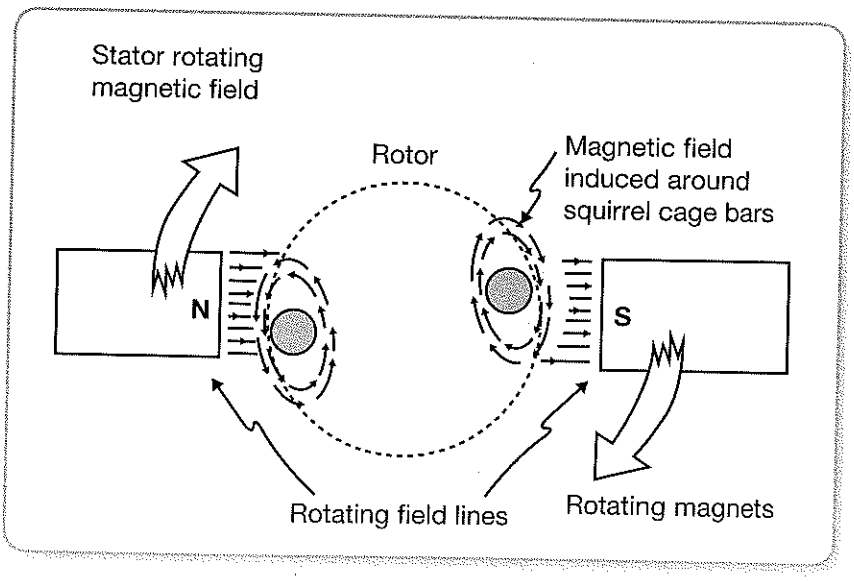
1. What is an induction motor?
2. Briefly describe the structure of a squirrel cage motor.
3. Outline how an AC induction motor works. You should be able to answer this question in five dot point statements.
4. What are the main structural differences between AC motors and AC induction motors?
5. What are the advantages of AC induction motors?
6. What are the disadvantages of AC induction motors?
7. Where are AC induction motors used?

Use the following information to answer the next FOUR questions.

An aluminium disc which is free to move is pivoted horizontally as shown below. A magnet is moved in a circular path above the disc.



8. What will the disc do as a result of the motion of the magnet?
   (A) Stay stationary.
   (B) Rotate in the same direction as the magnet.
   (C) Rotate in the opposite direction of the magnet.
   (D) Either (B) or (C) depending on the pole of the magnet closest to the disc.
9. What are, and who discovered the three principles in physics being applied here?
10. What is this apparatus modelling?
    (A) A DC motor. (B) An AC generator. (C) A galvanometer. (D) An AC induction motor.
11. Why is the output from an AC induction motor lower than other motors?
    (A) The squirrel cage is relatively heavy and difficult to move.
    (B) The magnetic fields lose significant energy as they rotate.
    (C) Significant energy is lost in the induction of currents in the squirrel cage.
    (D) The squirrel cage becomes magnetised and opposes attempts to move it.
12. The diagram shows a simplified induction motor.
    The directions of the magnetic fields around the two squirrel cage bars shown.
    Justify these directions.



SET 48 Electromagnetic Waves

1. (a) What are electromagnetic waves?

(b) How are electromagnetic waves produced?

(c) What properties of electromagnetic waves distinguish them from other types of waves?

(d) What name do we give to the entire range of electromagnetic waves?

(e) What name is given to the energy carried by electromagnetic waves?

(f) Which type of electromagnetic waves carry the highest amount of energy?

(g) Describe the relationship between the wavelength of electromagnetic waves and their frequency.

(h) Describe the relationship between the wavelength of electromagnetic waves and their energy.

(i) Describe the relationship between the frequency of electromagnetic waves and their energy.

(j) Electromagnetic waves are said to have a dual nature. What does this mean?

(k) What is the name of an electromagnetic particle?

2. The diagram shows the types of electromagnetic waves.

Gamma	P	Ultraviolet	Visible light	Infra-red	Q	Radio waves	TV waves

(a) Identify the missing radiations P and Q.

(b) Which end of the diagram has waves with the longest wavelength?

(c) Which waves have the least energy?

(d) Which waves are used mostly in telecommunications?

(e) Which electromagnetic waves are more harmful to humans, P or Q?

(f) Which waves have wavelengths shorter than visible light?

(g) Which of these types of waves has the highest frequency?

(h) Which type of electromagnetic wave do our nerves detect as heat energy?

(i) Which of these types of electromagnetic waves travels the fastest in a vacuum?

(j) What is the speed of electromagnetic waves through a vacuum?

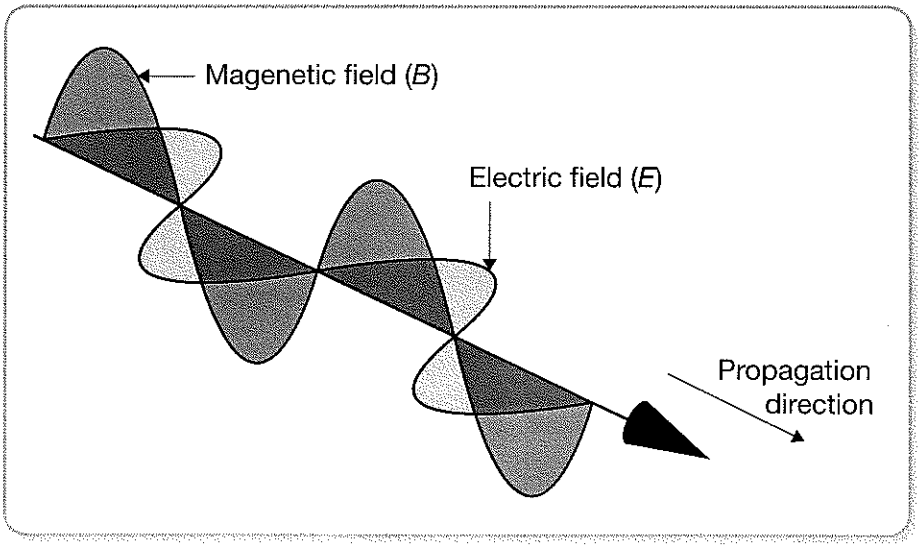
3. On a copy of the diagram of an electromagnetic wave label:

(a) Two wavelengths, one for each of the perpendicular components.

(b) Compare these two wavelengths.

(c) Two amplitudes, one for each of the perpendicular components.

(d) Compare the two amplitudes.



SET 49 Superconductors

1.
(a) What is a superconductor?
(b) Name five types of substances that have been shown to have superconducting properties under the right conditions.
(c) What causes resistance in normal conductors?
(d) Explain what happens to the resistance of a conductor when it is cooled.
(e) What logical conclusion can you draw from your answer to (d)?
(f) This was the same conclusion drawn by scientists at the time superconductivity was discovered. What observation upset their ideas?
(g) What is the resistance of a superconducting material below its critical temperature?
(h) Define or explain the concept of critical temperature.
(i) Considering your answers to (c) and (g), does this mean that no collisions occur between current electrons and lattice particles in superconducting materials? Explain what is happening.
(j) What are the two main characteristic differences between type I and type II superconductors?

2.
(a) Why are researchers hoping to find superconductors which will work at room temperatures?
(b) What implications are there for society if their research is successful?

3. Account for the limited use of superconductors at the present time.

4. Which of the following types of substances can be superconducting materials?
(A) Ceramic compounds.
(B) Metals.
(C) Organic compounds.
(D) All of the above.

5. Which of the following is an advantage of superconducting technology?
(A) Unlimited conductivity in existing transmission lines.
(B) Eliminates the need for transformers.
(C) All transmission lines underground.
(D) Resistance free transmission of electricity.

6. What is the main problem with implementing superconducting technology?
(A) Finding superconducting materials.
(B) The absence of high temperature superconductors.
(C) The cost of developing superconducting materials.
(D) Developing technology to cool superconductors to their critical temperature.

7. One proposed application of superconducting technology is AC current storage. Why would this be an advantage?
(A) We have no method of storing AC current with current technologies.
(B) It would eliminate the need for costly transformers.
(C) It would reduce our reliance on DC batteries.
(D) We would be able to get rid of coal burning power stations.

8. Given the limitations of superconductors, which of the following applications of the technology is most likely?
(A) The generation of AC power in power stations.
(B) The building of superconducting appliances for home use.
(C) Building superconducting components for communications systems.
(D) Magnetic levitation transport systems.

9. Copy the following left hand half sentences into your workbook and then find the correct match from the second column to complete each sentence. Your completed sentences will give you an outline of the BCS theory as to why superconductors become superconductors. (Note that some resources give a much more watered down version of BCS theory. If you wish to check further, Google the original work of Bardeen, Cooper and Schrieffer in this area.)

First part of sentences

(a) In conductors above the critical temperature, valence

(b) When moving through the metal

(c) It is these collisions which give rise

(d) It is thought that valence electrons become detached from atoms and are

(e) Lattice atoms become

(f) At low temperature a lattice

(g) As conduction electrons pass through the lattice there is a Coulomb

(h) This Coulomb

(i) This distortion increases the force

(j) Because electrons move so slowly, they now approach each other within

(k) There is now a Coulomb force of

(l) Electrons moving in opposite directions exchange

(m) The phonons hold them.

(n) The energy of the Cooper pairs is below a critical

(o) When they interact (collide) with other

(p) This holds as long as the temperature

(q) The Cooper pairs are

Second part of sentences

(c) to the resistance of the material.

(g) force of attraction between the lattice nuclei and the electrons.

(p) remains below the critical temperature for that substance.

(a) electrons are relatively free to move when a potential difference is applied.

(f) has little movement.

(q) the carriers of superconductivity.

(i) on the electrons.

(d) free to move when a voltage is applied.

(j) the Coulomb repulsive force range.

(k) attraction between them.

(l) packets of momentum (phonons).

(q) the carriers of superconductivity.

(e) positively charged.

(h) force distorts the lattice.

(n) quantum level (called the Fermi level).

(b) lattice the electrons collide with atoms and lose energy as heat.

(o) electrons or lattice particles, there is no transfer of energy.

(m) together to form Cooper pairs.

10. The graph shows how the temperature of a conductor changes as its temperature is decreased.

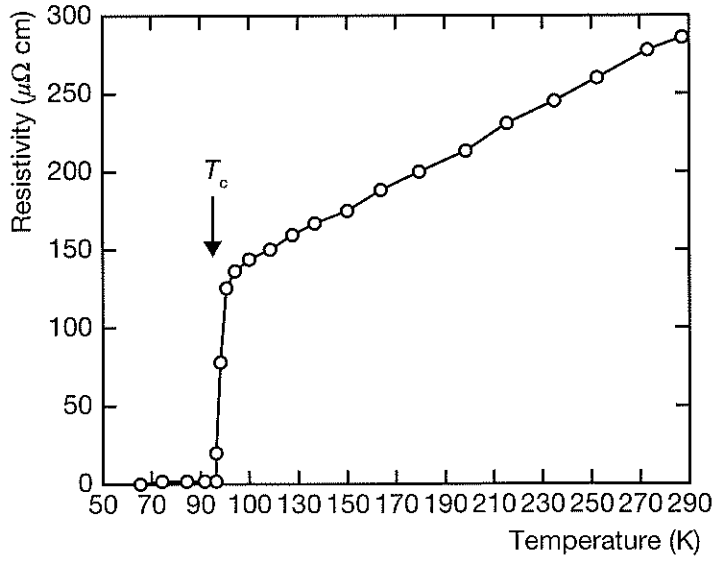
(a) Is this substance a superconducting material? Justify your answer.

(b) What is its critical temperature?

(c) For what range of temperature will its resistivity be zero?

(d) In December 2013, a new superconductor, $TaBa_7Cu_8O_{17+}$ with a critical temperature of 255 K (–18°C) was discovered.

What does this discovery mean to the technology of superconducting materials?

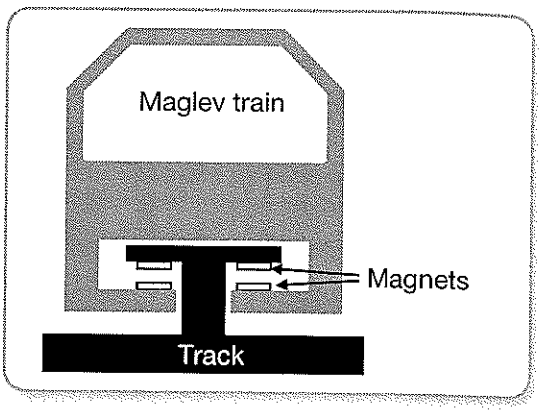


SET 50 Magnetic Levitation

1. (a) What is magnetic levitation (maglev)?

(b) List seven advantages of using magnetic levitation in a transport system such as in maglev trains.

(c) The diagram shows a schematic of a maglev train. Is the maglev caused by magnetic attraction or repulsion? Justify your answer.

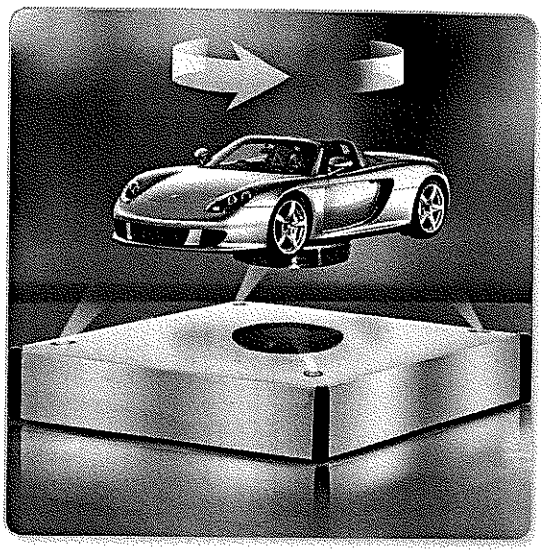


2. The photograph shows a desk novelty item which has a car 'floating' above a square base. When the device is plugged into a power supply and turned on, the car levitates above the base.

(a) Outline how maglev could be used to make this item.

(b) When levitated, the car is quite stable. It does not rock sideways or fall off to one side. If spun around a vertical axis, it spins smoothly. Outline how this stability could be achieved using magnetic fields.

(c) Each time it is activated, the car rises to exactly the same height. Explain why this happens.



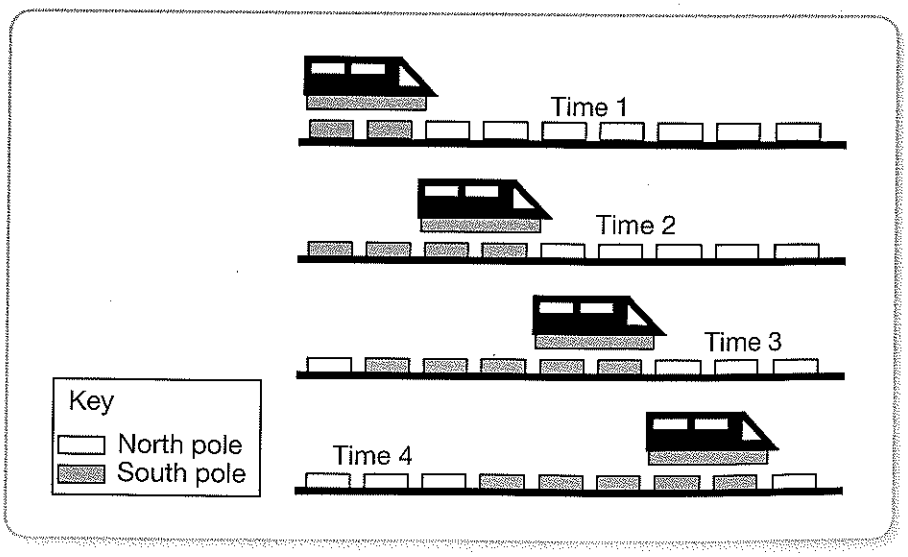
3. Consider the diagram representing the passage of a maglev train over four time intervals. The polarities of the magnets responsible for levitation and assisting in forwards propulsion are shown.

(a) Explain how the levitation of the train is maintained in this design.

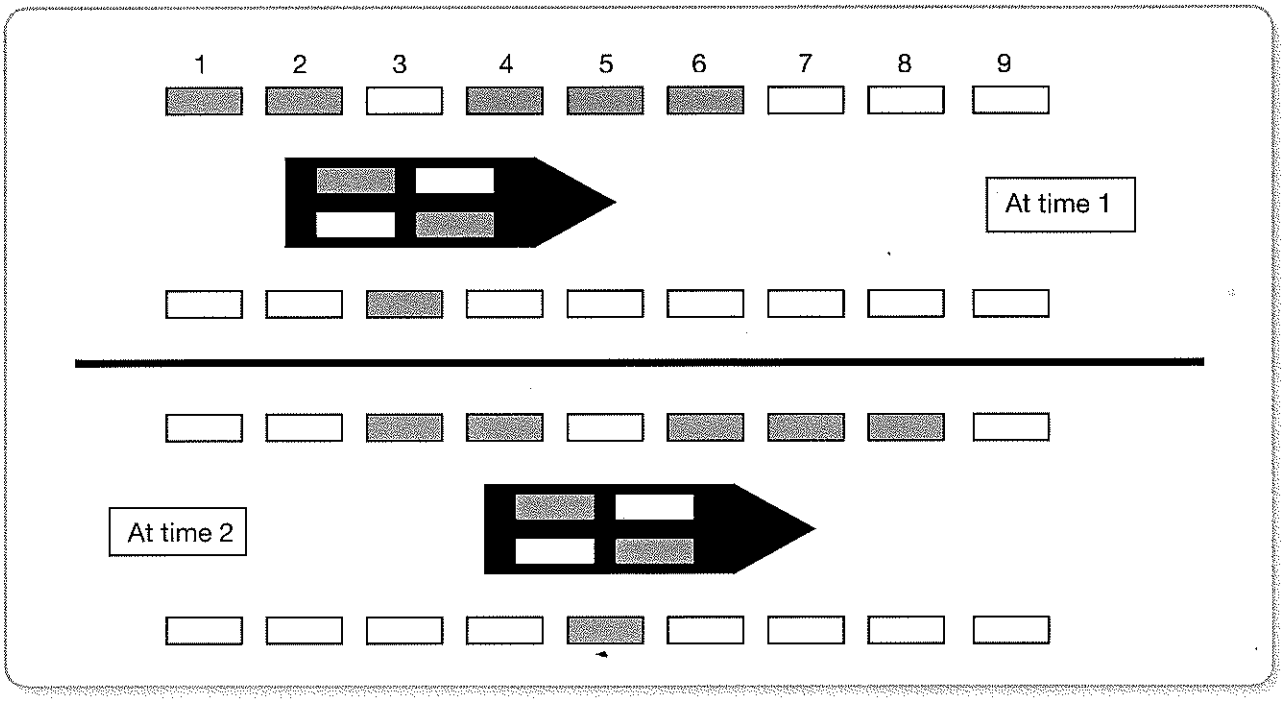
(b) Explain how the track magnets in front of the train assist in forwards propulsion.

(c) Explain how the track magnets behind the train assist in forwards propulsion.

(d) After a short delay, the polarity of the track magnets behind the train are changed. What would be the effect of changing the polarity of these too soon?



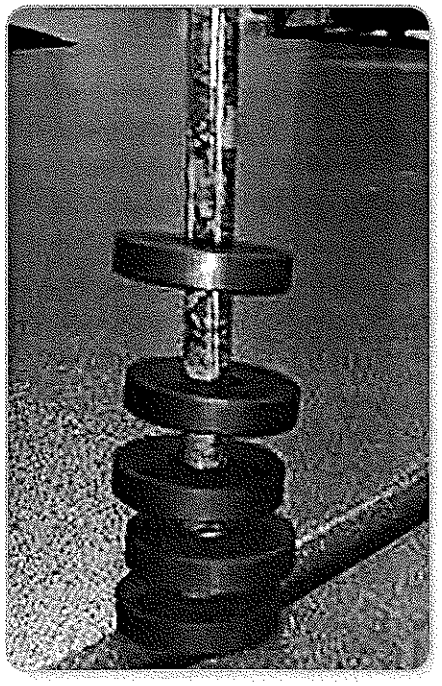
4. Consider the diagram representing the passage of a maglev train over two time intervals. The polarities of the magnets mounted on the side of the guard walls which assist in the propulsion of the train are shown. Those responsible for levitation are not shown. The poles of the magnets are represented by the grey and white shading.



(a) Explain how these diagrams illustrate the forces which result in the movement of the train forwards. It may be useful to get a copy of the diagram and put vectors to represent the forces involved on your copy.

(b) Note that one side of magnets in the time 1 diagram are labelled 1, 2, 3, and so on. Which magnet's polarity change must be timed exactly right for efficient movement forwards. Explain why.

5. The photograph shows a series of donut magnets stacked on a vertical pencil.



Which statement about the magnetic poles of the magnets in this stack is correct?

(A) The top pole of each magnet in the stack must be a north pole.

(B) The top pole of each magnet in the stack must be a south pole.

(C) The top pole of each magnet in the stack could be either a north pole or a south pole.

(D) The top pole of each magnet in the stack will alternate north pole or south pole down the stack.

SET 51 Using Superconducting Technology for Maglev

1. (a) Explain the Meissner effect using the terms paramagnetic and diamagnetic.

(b) What property does a material have if it is paramagnetic?

(c) What property does a material have if it is diamagnetic?

(d) What is a superconducting magnet?

(e) Compare the strength of the electric field developed by superconducting magnets compared to ordinary electromagnets.

(f) Compare the cost of running superconducting electromagnets with that of ordinary electromagnets.

2. Research information to find out how much energy a superconducting maglev (SCMaglev) train uses compared to an ordinary maglev train.

3. Which of the following is a major problem of superconducting technology for maglev trains?

(A) Superconductors require a very low temperature to work.

(B) Superconductors large enough will be too heavy for the system to be efficient.

(C) Superconductor materials are very expensive.

(D) The magnetic force required to elevate the train is huge.

4. Which of the following statements is correct?

(A) Superconducting technology is essential for maglev transport because without it magnetic fields strong enough to do the job cannot be obtained.

(B) Superconducting technology is essential for maglev transportation because without it the maglev would be far less efficient.

(C) Superconducting technology is not essential for maglev but if used it makes the process much more expensive.

(D) Superconducting technology is not essential for maglev but if used, it makes the process less expensive.

5. What is one of the main advantages of a train using magnetic levitation?

(A) It is less expensive to build than conventional trains.

(B) It uses less energy than conventional trains.

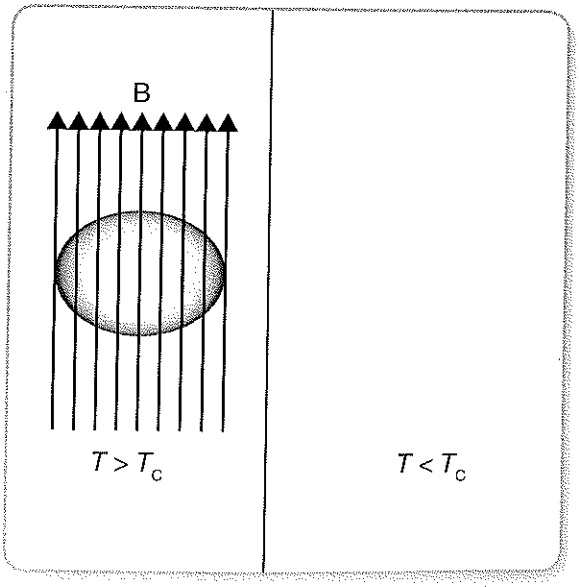
(C) It is more streamlined and so has less air resistance than conventional trains.

(D) There is no rail friction

6. Consider the photograph of a SCMaglev train.

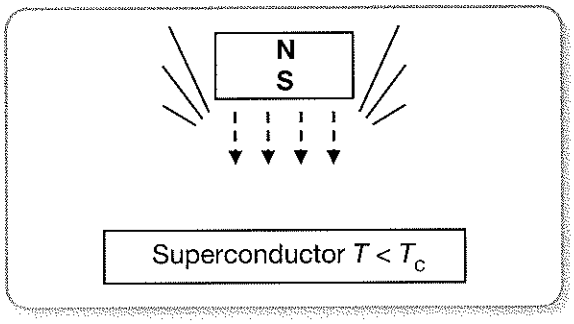
Suggest reasons for the shape of the front of the train.

7. Copy the diagram and complete the second diagram for the superconducting ball when its temperature is below its critical temperature.



Questions 8, 9 and 10 are meant to be done as a set, not alone.

8. Consider a magnet a distance above a superconducting material below its critical temperature. The magnet is lowered towards the superconductor. At an appropriate level, the magnet is let go. It remains levitated above the superconductor. This is shown in the diagram.

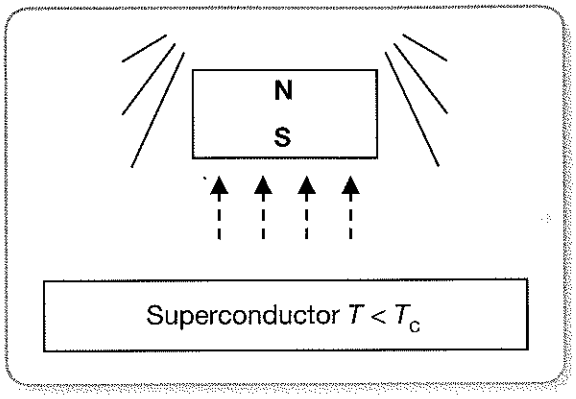


Explain how this levitation occurs in terms of:

(a) The Meissner effect.

(b) The induction of eddy currents in the superconductor.

9. Consider a magnet which is resting on top of a superconducting material which is being cooled. When the temperature of the superconductor falls to its critical temperature, the magnet spontaneously rises and hovers above the superconductor. This is shown in the diagram.



Explain how this levitation occurs in terms of:

(a) The Meissner effect.

(b) The induction of eddy currents in the superconductor.

10. Evaluate the following statements.

(a) Emf is induced in conductors whenever there is relative motion between the conductor and a magnetic field.

(b) Emf is induced in a conductor whenever the magnetic field it is in changes.

(c) Emf is induced in a conductor whenever it moves in a magnetic field.

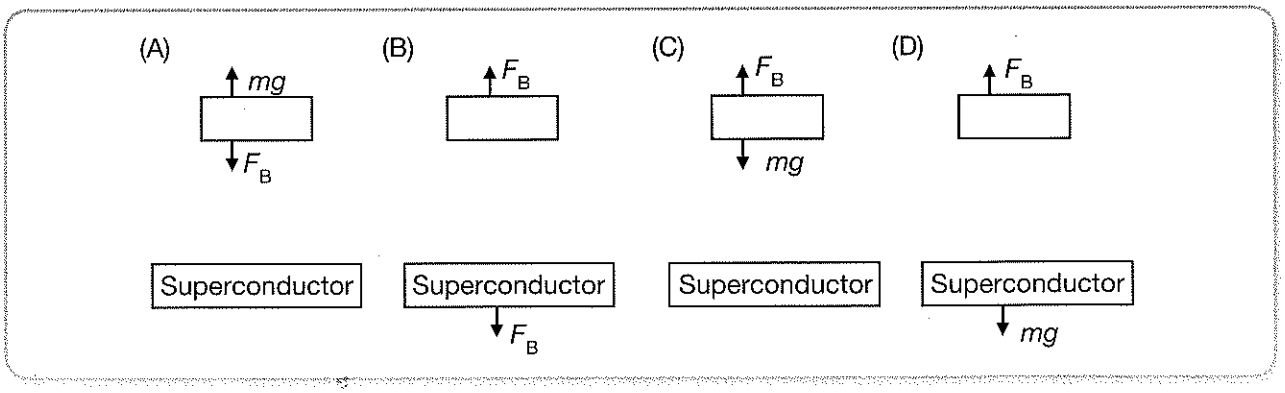
(d) The Meissner effect can explain maglev above a superconductor below its critical temperature.

(e) The induction of eddy currents inside a superconductor can explain maglev above a superconductor below its critical temperature.

(f) Because superconductors are diamagnetic at and below their critical temperature, no eddy currents can be induced in them.

(g) Eddy currents are not the explanation for the levitation of a magnet above a superconducting material at or below its critical temperature.

11. The diagrams represent a magnet levitated above a superconducting material which is below its critical temperature. Which choice best shows the forces involved in this levitation?



SET 52 Magnetic Resonance Imaging (MRI)

1. (a) What is MRI used for?

(b) What particular tissues in the body is MNRI useful for?

(c) Why is MRI used instead of alternate technologies?

(d) What are the two alternative names sometimes used for MRI?

(e) Identify an advantage of MRI over X-rays.

(f) Identify the two main disadvantages of MRI compared to other imaging technologies.

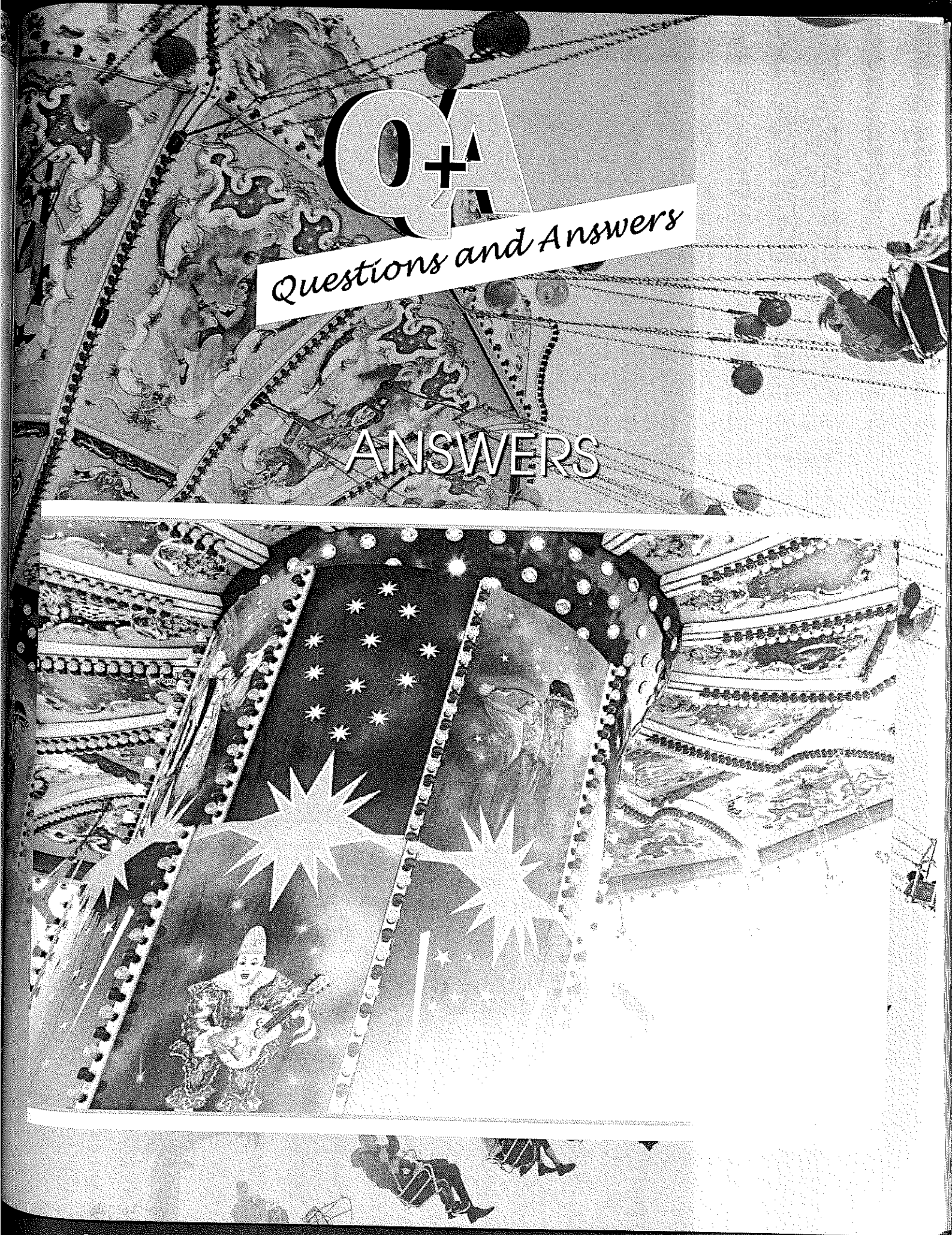
2. There are a number of risks associated with MRI scans. Identify at least three of these risks.

3. Once turned on, MRI machines are seldom turned off, even if there are no patients booked to use them. Why is this?

4. The sentences below, when matched and written out in full, describe the principles underlying the use of MRI. Place an appropriate heading in your book and match them using the order shown in the first half column.

First part of sentences		**Second part of sentences**	
(a)	MRI machines make use of the fact	(a)	field causes many protons to become aligned with its direction.
(b)	Being charged, protons, for example,	(b)	as the Larmor frequency.
(c)	When a person is inside the strong magnetic field of the MRI, the magnetic	(c)	time is shorter for denser tissues.
(d)	A radio frequency electromagnetic pulse is then turned	(d)	that some nuclei behave like small magnetic tops.
(e)	This electromagnetic pulse has the right	(e)	spin of the protons relative to the magnetic field.
(f)	The absorbed pulse changes the direction of the	(f)	to 'relax'.
(g)	This frequency is known	(g)	of the protons returns to their original directions.
(h)	In effect, the protons absorb	(h)	on, producing a varying electromagnetic field.
(i)	After the electromagnetic field is turned off, the spin	(i)	the applied magnetic field.
(j)	The protons then realign with	(j)	so they can be easily identified from the scan.
(k)	They are said	(k)	frequency to be absorbed by the protons.
(l)	During this relaxation they	(l)	energy and move to a higher energy level.
(m)	The rate at which they relax depends	(m)	computers connected to the MRI machine.
(n)	Relaxation times can be measured and plotted by	(n)	can be aligned by a strong magnetic field.
(o)	Because the protons in different tissues in the body relax at different	(o)	rates, the type of tissue can be determined.
(p)	In general the relaxation	(p)	on the type of tissue the protons are in.
(q)	The differences in relaxation times for various tissues are known,	(q)	re-emit the energy they previously absorbed as a radio frequency electromagnetic radiation.

Notes


Q+A
Questions and Answers
ANSWERS

Set 1 Mass and Weight

1. (a) Mass is a measure of the amount of matter in a body.

(b) Mass, does not change regardless of where the object is. Mass is constant, not being changed by the object's position in a gravitational field.

(c) grams (g), kilograms (kg), tonnes (t)

2. (a) Weight is a measure of the force of gravity acting on a mass.

(b) Weight changes according to the strength of the gravitational field the object is in.

(c) Newton (N)

(d) $W_F = mg$

3. Convert mass to kg, and then multiply by 9.8 ($W = mg$).

(a) 9.8×10^{-3} N

(b) 0.83 N

(c) 5.39 N

(d) 2410.8 N

(e) 1.97×10^{4} N

(f) 15 680 N

4. Divide each figure by 9.8 (and convert to grams by dividing by 1000 if needed).

(a) 40.8 g

(b) 0.816 kg (816 g)

(c) 25 kg

(d) 50 kg

(e) 1000 kg

(f) 6632.7 kg

5. (a) 60 kg

(b) 96 N

(c) 588 N

(d) 686 N

(e) 70 kg

(f) 112 N

6. Mass is always constant regardless of its position in the Universe whereas weight changes according to the strength of the gravitational field it is in.

Mass is not a force, weight is a force.

Mass is measured in kilogram, weight in newtons.

(Mass is a scalar quantity – (does not require a direction) while weight is a vector quantity – (requires a direction).

7. (a) 20.4 kg

(b) 200 N

8. (a) 588 N

(b) 60 kg

(c) 223.4 N

9. (a) They work because the astronaut's weight force acts on them, but the scale has been calibrated in mass units rather than weight units, so they report her mass (as do all typical scales we use).

(b) They would read about 10 kg because the astronaut's weight on the Moon is about one sixth of that on Earth, so the spring mechanism inside the scales will only be compressed by one sixth its compression on Earth.

10.

City	Gravitational field at this position ($m\ s^{-2}$)	Weight of 10 kg mass at this position (N)
Sydney	9.7979	97.979
Hong Kong	9.8099	98.099
Toronto	9.8054	98.054
Helsinki	9.8189	98.189
Baghdad	9.7964	97.964
Seoul	9.7994	97.994
Manila	9.7844	97.844
New York	9.8024	98.024
Mexico City	9.7799	97.799

Set 2 Gravitational Field

1. (a) A field is a region in which something experiences a force.

(b) A gravitational field is a region in which masses experience forces.

(c) Gravitational fields are found around all masses.

(d) Gravitational field strength is the force per unit mass an object experiences when placed in the field.

(e) $g = \frac{F}{m}$

(f) N kg^{-1} or m s^{-2}

(g) Acceleration due to gravity.

2. (a) Its gravitational potential energy.

(b) Work done equals the change in the gravitational potential energy.

3. (a) The Earth is not a perfect sphere, and gravitational field strength depends partly on the distance the object is from the centre (of mass) of the Earth.

(b) $mg = \frac{Gm_o m_P}{r^2}$

From which, on cancelling the m_o, and rearranging, $g = \frac{Gm_P}{r^2}$.

(c) This equation clearly shows the dependence of gravitational field on r^{-2}. As we go further from the Earth's centre, the gravitational field becomes exponentially smaller.

(d) r is the distance of the object from the Earth's centre. For an object in orbit, it represents the orbital radius of the orbit (it includes the radius of the Earth, and the altitude of the orbit).

4. (a) $GPE = mgh$ or $GPE = mg\Delta h$

(b) The height of the object above the surface of the Earth, or in the case of Δh, the change in the height of the object above the Earth's surface.

(c) This equation can be used as long as the value for the gravitational field at the positions involved are calculated. It is not used because it is easier to use a more appropriate equation (see later) Reason: The value of 'g' is 9.8 only at the surface so if 'h' is more than a few km the equation is unreliable.

5. (a) The gravitational field.

(b) Some external force, for example, the engine of a spacecraft, the motor of a car hoist.

6. (a) 198 N

(b) 1098 N

7. The gravitational field on top of Mount Everest would be less than that at sea level because the top of Mount Everest is further from the centre of mass of the Earth (taken as being at the centre of the Earth).

8. (a) Mass does not change = 80 kg

(b) 4 N

9. (a) The motion of an object in a gravitational field is described by Newton's three equations of motion, and the mass of an object is not one of the factors affecting that motion. However, in order to fall at the same rate, a larger object will need a larger force acting on it, so the gravitational force will be larger (equal to mass × gravitational field). This ensures that the rate of fall for all masses is the same.

(b) If the height is significant, then the value of the gravitational field (g) will be less, and therefore the initial rate of fall will also be less. As the object falls closer to the Earth, and gravitational field increases, the rate at which its downwards velocity increases will also increase (note that we are ignoring air resistance here).

10. 14.7 N kg^{-1} (or m s^{-2})

11. (a) 275.2 N kg^{-1} (or m s^{-2})

(b) 6.67×10^{12} N kg^{-1} (or m s^{-2})

12. 9.24 N kg^{-1} (or m s^{-2})

13. 1.09 N kg^{-1} (or m s^{-2})

14. 1447 km

15. 1155 km

16. 8.9 m s^{-1}

17. The gravitational field strength in the Earth orbit would 3.3 times greater than that in the Mars orbit.

18. 1.0×10^6 m

19. 2.82 N kg^{-1}

20. 1.4×10^7 m

21. 1.1 N kg^{-1}

22. 10.7 N kg^{-1}

23. (a) 22 500 km

(b) 0.45 N kg^{-1}

24. (a) 7.5 N kg^{-1}

(b) 4.5×10^{23} kg

25. (a) 2520 J

(b) Slightly less because the gravitational field strength would be slightly less at the higher altitude.

(c) 4.5 N kg^{-1}

26. From $g = \frac{GM}{R^2}$, with Y having one quarter the mass and half the radius, g = same as X = 8 m s^{-2}.

Set 3 Newton's Law of Universal Gravitation

1. 1.85×10^{-10} N

2. 4484 N

3. (a) Force will be four time less.

(b) Force will be four times greater.

(c) Force will be doubled.

(d) Force will be nine times greater.

(e) Force will be 9 times smaller because of the distance but 4 times larger because of the mass change, therefore overall it will be $\frac{4}{9}$ times smaller.

4. (a) Weight ratio is 4 : 3.

(b) It would still be the same, 4 : 3.

5. (a) 80.85 N

(b) 588.2 N

(c) 612 N

(d) 1.18 N kg^{-1} (or m s^{-2})

(e) 1.22 N kg^{-1} (or m s^{-2})

6. (a) 185.8 N

(b) 12 39.5 N

(c) 527 N

(d) 432 N

(e) 134.2 N

(f) 1218.9 N

(g) 520.2 N

(h) 3.97×10^{-4} N

(i) 3.72 N kg^{-1} (or m s^{-2})

(j) 24.58 N kg^{-1} (or m s^{-2})

(k) Saturn = 10.54 N kg^{-1} (or m s^{-2}). Uranus = 8.63 N kg^{-1} (or m s^{-2}). Saturn = 1.22 times greater than Uranus.

7. (a) 1.36 m s^{-2}

(b) 1022 N

8. (a) 64.21 N

(b) 0.802 N kg^{-1} (or m s^{-2})

9. (a) 0.49 N downwards

(b) 4.5×10^{-12} N upwards

10. 1.98×10^{20} N (attraction)

11. 4 : 9

12. 93.6 kg

13. 1.85×10^{4} km

14. (a) 2.5×10^{-9} N to the left

(b) 2.5×10^{-9} N to the right

15. 8.0×10^{-8} N

16. (a) Force will double to $2F$
(b) Force will be $0.25F$
(c) Force will become $6F$
(d) Force will be the same, F
(e) Force will be two thirds the original value, $\frac{2F}{3}$

17. 1.34 m

18. 7909 km

19. Your weight in the Earth orbit would be 3.3 times greater than your weight in the Mars orbit.

20. 1.57×10^{11} N

Set 4 Gravitational Potential Energy

1. At infinity. It is only at infinity that an object is outside any gravitational field so that the force acting on it is zero. It therefore has no tendency to 'fall'.

2. The gravitational potential energy of an object at any point in space is equal to the work done in moving it from infinity to the point.
$E_p = -\frac{GMm}{r}$, measured in joules (J).

3. We do not use this equation because the value of g is not constant, and because it refers only to objects close to the surface of Earth (or any other planet or moon) where the zero reference point is the surface of the planet or moon. The equation is more correctly given as $mg\Delta h$, and calculates the *change* in gravitational potential energy close to the surface.

4. The gravitational potential energy at infinity is zero so the only direction it can move is downwards, towards a planet. As it falls its gravitational potential energy will change to kinetic energy. Because the law of conservation of energy must be obeyed, and the object's kinetic energy will be positive, its gravitational potential energy must be negative.

5. -3.335×10^{-9} J

6. -3.95×10^{14} J

7. -6.48×10^{11} J

8. (a) -7.61×10^{28} J
(b) The radius of the satellite is insignificant compared to the other distances and can therefore be ignored.

9. (a) -7.34×10^{8} J
(b) -4.03×10^{9} J
(c) -8.72×10^{9} J
(d) -4.01×10^{10} J
(e) -1.10×10^{9} J
(f) -1.25×10^{7} J

Set 5 Changes in Gravitational Potential Energy

1. (a) -5.34×10^{-8} J
(b) New gravitational potential energy will be -2.67×10^{-8} J. This represents an increase of 2.67×10^{-8} J.
(c) The gravitational force will decrease by a factor of 4 (inversely proportional to distance squared).
(d) If the objects are further apart, then there is a larger energy gap between them, so potential energy will increase whereas gravitational force decreases with increasing distance apart.

2. Gravitational potential energy at 10 000 km = -1.22×10^{10} J. At 20 000 km it will be -7.58×10^{9} J. This represents an increase of 4.62×10^{9} J, which is the amount of work that needs to be done on the satellite.

3. Gravitational potential energy at 6680 km = -1.5×10^{10} J. At 7680 km it will be -1.36×10^{10} J. This represents an increase of 1.4×10^{9} J.

4. (a) 7303 km
(b) 923 km
(c) Gravitational potential energy at the new altitude would be -1.22×10^{10} J which represents a change of $+1.5 \times 10^{10}$ J (using rounded figures).

5. (a) $-0.5E$ J
(b) $-3E$ J
(c) Its gravitational potential energy will increase to $-1.5E$ J (increase by $+1.5E$ J).

6. (a) The satellite above Neptune. With all other variables constant, the determining factor is the mass of the planet.
(b) The satellite above Mercury. With all other variables constant, the determining factor is the mass of the planet.
(c) $13.6R$

7. (a) No. More work would need to be done as a higher orbit involves greater gravitational potential energy, but doubling the altitude does not double the orbital radius, so less than double the work would be involved.

(b) Its altitude would be $2(6380 + R)$ km

8. (a) -6.25×10^8 J.

(b) If the orbital radius is doubled, *GPE* will increase to -3.13×10^8 J which is a change of $+3.13 \times 10^8$ J.

(c) The gravitational potential energy will decrease to -1.25×10^9 J, which is a decrease of 6.25×10^8 J.

(d) The new E_P will be -3.57×10^8 J, which represents an increase of 2.68×10^8 J.

(e) The new E_P will be -1.0×10^9 J which is a decrease of 3.75×10^8 J.

9. (a) Using $\Delta E_P = mg\Delta h$, $\Delta E_P = 245\,000$ J

Using $\Delta E_P = Gm_o M_E\left(\frac{1}{R_{building}} - \frac{1}{R_{ground}}\right) = 245\,787$ J

(b) These two values differ by 0.32%, which given rounding off errors in various values, means that the simpler equation will suffice for changes close to the surface of the Earth.

10. (a) 24.8 m s^{-2}

(b) 4.62×10^{11} J

(c) 2.3×10^{11} J of work must be done to put it into a higher orbit.

11. (a) Given that $E_P = \frac{Gm_o M_P}{R}$, and gravitational field strength $g = \frac{GM_P}{R^2}$, substituting this into the equation for E_P we get $E_P = gm_s R$

(b) -7.83×10^{10} J

(c) 4.1×10^{25} kg

(d) Is a lower altitude because the gravitational field strength has increased, so it must be closer to the planet.

(e) 8538 km

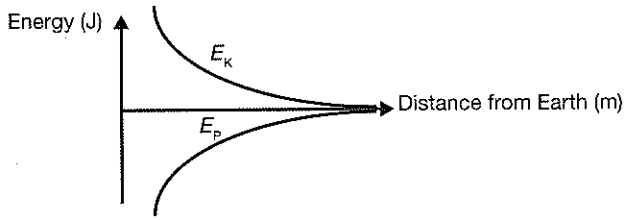
12. (a) 17 632 km

(b) 0.69 N kg^{-1} (or m s^{-2})

(c) 16 979 km

(d) Its E_P would decrease by 2.07×10^9 J to -7.07×10^9 J.

13. (a)



(b) The maximum value for E_P is zero at infinity, so the graph must occupy the negative quadrant of the graph grid and by the law of conservation of energy, the kinetic energy lost will equal the potential energy gained, so the E_P graph will mirror the E_K graph.

14. (a) -62.7 MJ kg^{-1}

(b) (i) Using the graph, go down from the $2R$ on the x-axis to the curve, then across to the intercept on the y-axis, potential energy = about 34 MJ kg^{-1}

(ii) From the graph, potential energy at surface = about 62.5 MJ kg^{-1}

Potential energy at $3R$ = about 22 MJ kg^{-1}

Therefore change = about 40.5 MJ kg^{-1}

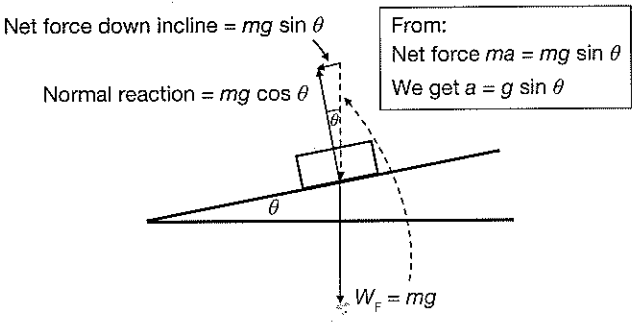
$40.5 \times 1200 = 48\,600$ MJ for the 1200 kg satellite

(c) From the graph, potential energy at $2R$, U = about 33 MJ

Therefore gravitational field = $U \div R = 33 \times 106 \div 6\,380\,000 = 7.17$ m s^{-2}

Set 6 Motion on an Inclined Plane

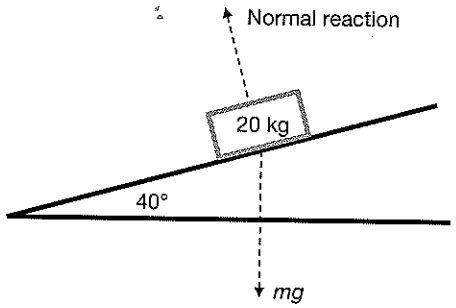
1.



From:
Net force $ma = mg \sin \theta$
We get $a = g \sin \theta$

2. (a) 73.5 N down the incline
(b) 73.5 N down the incline
(c) 4.9 m s^{-2} down the incline

3. (a) and (b) See diagram.
(c) 126 N down the incline
(d) 3 N up the incline
(e) 123 N down the incline
(f) 6.15 m s^{-1} down the incline



4. (a) 1039.4 N (tension in a string acts in both directions in the string)
(b) 1039.4 N perpendicular to the surface of the ramp

5. (a) 36.75 N down the incline
(b) 25.5 N down the incline
(c) 1.5 N kg^{-1} up the incline

6. (a) 50 N
(b) 970.4 N
(c) 1841.8 N upwards at 90° to the surface of the ramp

7. (a) 56 N down the ramp
(b) 177.5 N
(c) 121.5 N up the ramp (against the motion)

8. (a) 0
(b) 12 250 N
(c) 12 250 N up the incline
(d) It would roll back down the incline with an acceleration of 4.9 m s^{-2}

9. (a) 147 N
(b) 22.5 N up the incline
(c) 174.5 N
(d) 22.5 N up the incline

10. (a) If the block is at rest, then forces down the slope = forces up the slope
So, $mg \sin 15° = T \cos 20°$
$300 \times 9.8 \times 0.26 = T \times 0.94$
From which $T = 810.35$ N (no rounding) or 813.2 N (with rounding)
(b) Normal reaction = $mg \cos 15° - T \sin 20°$
$= 300 \times 9.8 \times 0.966 - 813.2 \times 0.34$ (or 810.35×0.34)
$= 2563.6$ N (or 2564.3 N)

11. (a) 30 N
(b) 196 N
(c) 240.5 N

12. (a) If the block is at rest, then forces down the slope = forces up the slope
So, $40 \cos 30° + mg \sin 30° = F \cos 20°$
$40 \times 0.866 + 60 \times 9.8 \times 0.5 = F \times 0.94$
From which $F = 349.8$ N
(b) Normal reaction = $mg \cos 30° - 40 \sin 30° - F \sin 20°$
$= 60 \times 9.8 \times 0.866 - 40 \times 0.5 - 349.7 \times 0.34$
$= 370.3$ N

13. B
14. C
15. D
16. A
17. C

Set 7 Projectile Motion 1: Projectile Launched Horizontally

1. (a) 240 m s^{-1}
(b) 0
(c) 17.5 s
(d) 4199 m
(e) 240 m s^{-1}
(f) 171.5 m s^{-1}
(g) 295 m s^{-1} at 35.5° below the horizontal

2. (a) 0
(b) 42 m
(c) 60 m
(d) 36.3 m s^{-1} at 70.7° down from horizontal (or 19.3° to the vertical)

3. (a) 5.05 s
(b) 30.3 m
(c) 49.85 m s^{-1} at 6.9° to the vertical

4. (a) 14.0 m s^{-1}
(b) 0
(c) 0.64 s
(d) 8.96 m
(e) 14.0 m s^{-1}
(f) 6.26 m s^{-1}
(g) 15.34 m s^{-1} at 24.1° down from horizontal

5. 19.6 cm below the centre of the bullseye

6. (a) Both hit the ground at the same time (both have initial vertical velocity = 0)
(b) 13.7 m apart (they fall for 5.48 s, so (5.48 × 17.5) – (5.48 × 15) = 13.7))
(c) 53.7 m s^{-1}
(d) 53.7 m s^{-1}

7. (a) 1.5 s
(b) 3.975 m (it falls 11.025 m)
(c) 11.025 m
(d) The arrow hits the apple because both have fallen the same vertical distance while the arrow is in flight

8. (a) A
(b) A

9. C

10. B

11. D

12. B

13. A

Set 8 Projectile Motion 2: Projectile Landing at Same Level as Launched

1. (a) 6.67 s
(b) 25.98 m s^{-1}
(c) 34.4 m

2. (a) 3.84 s
(b) 26.2 m
(c) 18 m

3. (a) 1.98 m
(b) 4.86 m s^{-1} (this is her horizontal speed)
(c) 6.47 m s^{-1}
(d) 1.32 s

4. (a) 32.2 m s^{-1} at 48.8° up from horizontal
(b) 4.95 s

5. (a) 45°
(b) 30.99 m s^{-1}
(c) 4.47 s
(d) 49.0 m
(e) 4.47 s

6. (a) 1.67 s
(b) 3.43 m
(c) 9.3 m

7. (a) 2.22 s (write equations for horizontal and vertical time and equate them)
(b) 26.98 m s^{-1}
(c) 29.1 m s^{-1}
(d) 10.9 m s^{-1}
(e) 6.05 m
(f) 74.8 m

8. (a) 6.12 s
(b) 318 m
(c) 45.92 m
(d) 49.23 m further

9. (a) A
(b) 45°

10. (a) B
(b) 1 : 1 : 1 = –9.8 m s^{-2} down

11. C

12. D

13. D

14. A

15. D

16. (a) Same, equal to 15 cos 60° = 7.5 m s^{-1} (horizontal velocity (ignoring air resistance) does not change
(b) Same magnitude (13.0 m s^{-1}) but in opposite directions – up at P, down at Q)
(c) Same = 9.8 m s^{-2} downwards
(d) Same = 7.5 m s^{-1}
(e) 7.5 m s^{-1}
(f) 0
(g) 9.8 m s^{-2} down
(h) 1.33 s
(i) 1.33 s
(j) 2.66 s
(k) 19.95 m (19.9 if time figures not rounded)
(l) 15 m s^{-1} downwards at 60° to horizontal

17. (a) 20 m s^{-1}
(b) 15.6 m s^{-1}
(c) 38.74°

18. D

Set 9 Projectile Motion 3: Projectile Landing at Different Level to Launch

1. (a) 1.0 s
(b) 6.36 m
(c) 0.69 m
(d) 11.69 m s^{-1} down at 33° to the vertical

2. (a) 32 m s^{-1}
(b) 14.5 m s^{-1}
(c) 35.13 m s^{-1} up at 24.4° to the horizontal
(d) 1.48 s
(e) 10.73 m

3. (a) 3.2 m s^{-1}
(b) 7.325 m s^{-1}
(c) 8.0 m s^{-1} at 66.4° up from horizontal
(d) 0.75 s
(e) 2.7 m

4. (a) 51.96 m s^{-1}
(b) 30 m s^{-1}
(c) 45.9 m
(d) 9.33 s
(e) 485 m
(f) 51.96 m s^{-1}
(g) 61.43 m s^{-1} down
(h) 80.5 m s^{-1} at 49.8° down from horizontal

5. (a) 6.25 s
(b) 18.45 s
(c) 47.9 m
(d) 474.4 m
(e) 152.3 m s^{-1} at 80.3° down from the horizontal

6. (a) 441 m
(b) 4 s
(c) 90.7 m s^{-1} upwards
(d) 117.8 m s^{-1} at 50.5° up from the horizontal

7. (a) 7.22 s
(b) No
(c) After 7.22 s its vertical displacement will be 111 m below its firing position, that is, the cannonball will pass 8.97 m above the top of the mast.
(d) 20.42 m
(e) 4.08 s
(f) 7.22 s
(g) 67.4 m s^{-1} at 59° below horizontal

8. (a) 20.0 m s^{-1}
(b) 26.2 m s^{-1} at 40.3° up from horizontal
(c) 14.7 m
(d) 56.0 m s^{-1} at 69.1° down from horizontal

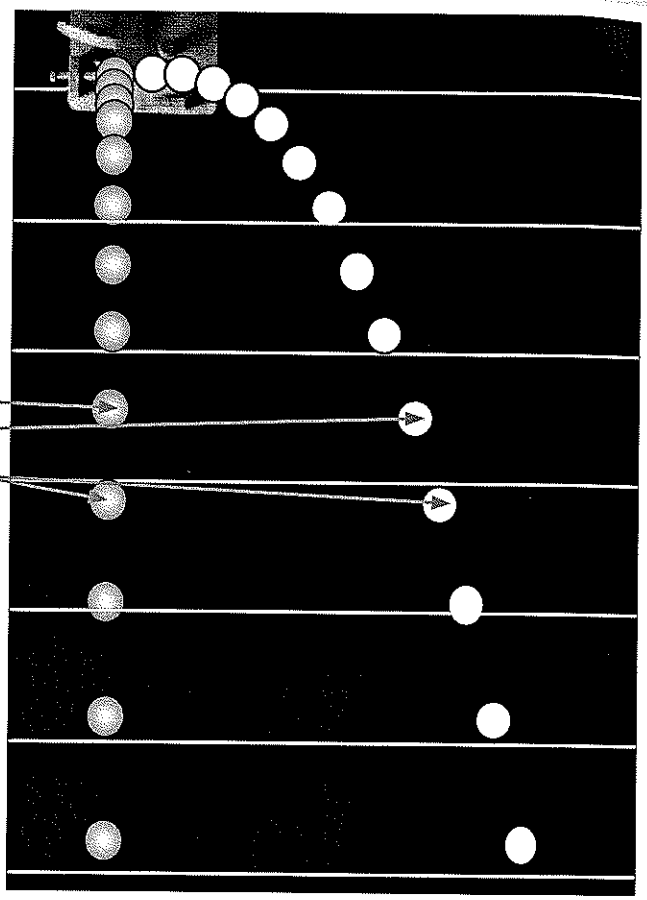
9. (a) 6.1 s
(b) 76.5 m s^{-1}
(c) 175.2 m
(d) On the way down. Its maximum height occurs at time 5.98 s which is before it hits the castle.
(e) 49.2 m s^{-1} at 1.37° below horizontal

10. B

11. (a) 7.3 s
(b) 45.8 m s^{-1} at 41.5° up from the horizontal
(c) 6.2 s
(d) 53.6 m s^{-1} at 50.3° down from horizontal

12. (a) B
(b) In physics, positive and negative values reflect relative directions. In this case, if upwards is taken as the positive direction, then the final displacement, below the point of launch would be a negative value and could therefore be described as negative.

Set 10 Analysing Projectile Motion Diagrams



1.
(a) 4 s
(b) 1.6 s
(c) 15.7 m s^{-1}
(d) 12.5 m
(e) About 19.6 m
(f) 4.9 m s^{-1}
(g) About 16.4 m s^{-1} 72.7° to the horizontal
(h) About 15.74 m

2.
(a) About 0.029 s (assuming 5 time intervals from zero position to first 10 cm line)
(b) 2
(c)
(d) 2
(e)
(f) 0.37 s (13 time intervals)
(g) 3.64 m s^{-1}
(h) 0
(i) 3.64 m s^{-1}
(j) 30 cm
(k) 0.8 m s^{-1}
(l) 3.73 m s^{-1} at 77.4° to the horizontal

3.
(a) 0.5 m s^{-1}
(b) 0.6 s
(c) 0.1 s
(d) 5.94 m s^{-1}
(e) 5.96 m s^{-1} 85° to the horizontal
(f) 1.68 s
(g) 0.18 m s^{-1}
(h) 1.28 m s^{-2}
(i) 2.15 m s^{-1}
(j) 7.42 m s^{-1}
(k) 7.34 m s^{-1}
(l) 0.48 s
(m) 0.08 s
(n) About 0.62 m s^{-1}
(o) 7.33 m s^{-1} at 85° to the horizontal

4.
(a) 2.6 s
(b) About 15.5 m s^{-1}
(c) About 20.2 m s^{-1}
(d) About 8.6 m

5.
(a) 19.6 m
(b) About 5 frames per second
(c) About 14 m
(d) About 7 m s^{-1}

Set 11 Uniform Circular Motion

1. The acceleration of the moving object is perpendicular to the direction of its motion (directed to the centre of curvature of the motion).

2.
(a) 465.23 m s^{-1} (taking one day as 23 hours, 56 minutes, 4 seconds)
(b) 3.39×10^{-2} m s^{-2}
(c) The gravitational field at the equator would be reduced by this amount, but the field at the poles would be unaffected. A point on the equator has this speed but a point at the poles rotates in a circle with an infinitely small radius, so has negligible speed.

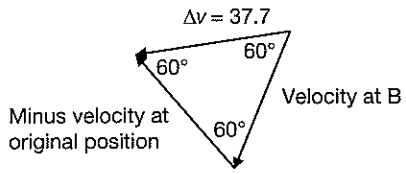
3. (a) 37.7 m s^{-1}

(b) 948 m s^{-2}

(c) 2369 N

(d) 0.042 s

(e) Remember instantaneous velocities are given by the tangents to the point on the circle.



(f) The change in velocity from vector diagram is 37.7 m s^{-1}, time for change is 0.042 s which gives an acceleration of 898 m s^{-2}. The difference can be attributed to rounding off errors.

4. (a) 8.9686 m s^{-2} towards the centre of the Earth.

(b) 8.9682 m s^{-2} towards the centre of the Earth.

(c) The values are the same, the slight difference being due to rounding off the orbital speed (it is actually 7740.1 m s^{-1}), and they should be the same because the centripetal acceleration is the gravitational acceleration/field strength.

(d) 3587.4 N towards the centre of the Earth.

(e) 3587.3 N towards the centre of the Earth.

(f) The values are the same, the slight difference being due to rounding off the orbital speed (it is actually 7740.1 m s^{-1}) and they should be the same because the centripetal force is provided by the gravitational force.

5. (a) 29 877 m s^{-1} (30 000 is a better answer)

(b) 5.95×10^{-3} m s^{-2}

(c) 3.570×10^{22} N (3 dp's so answer can be compared with (e))

(d) 5.95×10^{-3} m s^{-2}

(e) 3.557×10^{22} N

(f) They are the same as they should be, since the centripetal acceleration is the gravitational field strength.

(g) Allowing for rounding off errors, they are the same.

6. 35 N

7. (a) 33.8 m s^{-1}

(b) 32.7 m s^{-2} towards the centre

8. (a) 0.29 m s^{-1}

(b) 1×10^{-3} m s^{-2} towards its centre

9. (a) Their speed is tangential to their circular motion, and in the absence of any force to stop them moving outwards in the direction of their tangential speed, they move. When the reach the drum, the drum provides an inwards centripetal force to them to prevent them moving further outwards.

(b) 56.5 m s^{-1}

(c) 10 670 m s^{-2} towards the centre of the drum

10. (a) 5.8 s

(b) 8.65 m s^{-2} towards the centre.

11. (a) 7.0 m s^{-1}

(b) 875 m s^{-2} towards the centre of the disc

12. From $F = \frac{mv^2}{r}$, ratio of forces = 3 : 2 : 0 (for Tommy, $r = 0$)

13. (a) 2.71 m s^{-1}

(b) 7.75 m s^{-2}

(c) 5.04 N

14. 3933 N

15. 321 N

16. (a) 9.0 m s^{-2} directed towards the centre of curvature of the corner curve

(b) 11 250 N directed towards the centre of curvature of the corner curve

(c) 11 250 N directed towards the centre of curvature of the corner curve

(d) 9.0 N kg^{-1}

17. 12 208 N

18. C

Set 12 Circular Motion on an Inclined Track

1. 30 km h^{-1} = 8.33 m s^{-1}

Total frictional force = 160 × 0.28 = 44.8 N

From $F = \frac{mv^2}{r} = 44.8 = \frac{160 \times 8.33^2}{r}$

r = 247.8 m

2. (a) From $F = \frac{mv^2}{r} = 44.8 = \frac{160 \times 8.33^2}{r}$

$1050 \times 0.8 = 840 = \frac{1050v^2}{70}$

Therefore v = 7.48 m s^{-1}

(b) No. Both the frictional force and the centripetal force depend on the mass and the friction is required to overcome the inertial force of the car opposite the centripetal force. When we solve these problems, we effectively use the equation

$m \times \text{friction} = m \times \frac{v^2}{r}$

The mass cancels out on each side.

3. From the vector diagram:

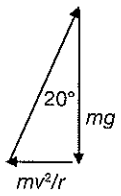
$F_c = \frac{mv^2}{r} = mg \tan 20°$.

Therefore $v = \sqrt{gr \tan 20°}$

$= \sqrt{(9.8 \times 225 \times \tan 20°)}$

= 28.3 m s^{-1}

= 102 km h^{-1}



4. From $v = \sqrt{gr \tan \theta} = \sqrt{(9.8 \times 200 \times \tan \theta)}$

Therefore θ = 11.5°

5. F_c for an object of mass 1 kg moving at 75 km h^{-1} = $1 \times \frac{(75 \div 3.6)^2}{65}$ = 6.68 N

For the same object moving at 90 km h^{-1}, $F_c = 1 \times \frac{(90 \div 3.6)^2}{65}$ = 9.62 N

So friction has to reduce this to a net 6.68 N

Therefore from 9.62 × f = 6.68

f = 0.69 N kg^{-1}

6. From $v = \sqrt{gr \tan 10°} = \sqrt{(9.8 \times 200 \times \tan 10°)}$ = 18.6 m s^{-1}

7. From $v = \sqrt{gr \tan 10°} = \frac{460}{3.6} = \sqrt{(9.8 \times r \times \tan 40°)}$

Therefore r = 1985.5 m

8. 11.5 m s^{-1}

9. 32.5°

10 41.2 m s^{-1} (148 km h^{-1})

11. 17.5°

12. 3.46 km

13. (a) 24.7 m s^{-2}

(b) 68.4°

14. 18.6 m s^{-1} (67 km h^{-1})

15. (a) 11.5 m s^{-1} (41.25 km h^{-1})

(b) 48.7°

16. 60 km $h^{-1} = \frac{60}{3.6}$ = 16.67 m s^{-1}

From v = 16.67 = $\sqrt{gr \tan \theta} = \sqrt{(9.8 \times 200 \times \tan \theta)}$

Therefore θ = 8.06°

17. From $v = \sqrt{gr \tan 31°} = \sqrt{(9.8 \times 316 \times \tan 31°)}$ = 43.1 m s^{-1} = 155.3 km h^{-1}

18. From v = 40 = $\sqrt{gr \tan \theta} = \sqrt{(9.8 \times 250 \times \tan \theta)}$

Therefore θ = 33.15°

19. From $v = \sqrt{gr \tan 10°} = \sqrt{(9.8 \times 75 \times \tan 20°)}$ = 16.36 m s^{-1}

Set 13 Orbital Velocity

1. (a) From $F_G = \frac{Gm_S M_E}{R^2} = F_C = \frac{m_S v_O^2}{R}$

Rearranging we get $v_O^2 = \frac{GM_E}{R}$

So $v_O = \sqrt{\frac{GM_E}{R}}$

(b) 7740 m s^{-1}

2. 3081 m s^{-1}

3. (a) 1.6×10^7 m

(b) 9.628×10^6 m

(c) Answers will be the same because these values are independent of the mass of the satellite.

4. (a) 10 : 5 : 2

(b) 625 : 39 : 1

(c) 625 : 78 : 3

5. (a) 1 : 1 : 1

(b) 1 : 1 : 1

(c) 1 : 2 : 3

6. (a) 1 : 1 : 1

(b) 1 : 4 : 16

(c) 1 : 2 : 4

7. (a) 28 875 m s^{-1}

(b) 33 977 km

(c) 2.27×10^{23} kg

8. (a) 1026 m s^{-1}

(b) 2.73×10^{-3} m s^{-2}

9. (a) 29 822 m s^{-1} (using equation derived from Newton's law of gravity or 29 865 if using 365.25 days to orbit and 86400 sec/day and $\frac{2\pi}{T}$)

(b) 5.93 (or 5.95×10^{-3} m s^{-2})

10. 7914 m s^{-1}

11. 6790 m s^{-1}

12. (a) 2.9×10^3 m s^{-1}

(b) 5.74×10^3 s

13. (a) 7798 m s^{-1}

(b) 4.1×10^7 m

(c) 1.47 hours

14. (a) 7740 m s^{-1}

(b) 1.51 hours

15. 7.71×10^3 m s^{-1}

16. (a) 7257 m s^{-1}

(b) -2.1×10^{11} J

(c) 6.93 N kg^{-1}

(d) 2.77×10^4 N

(e) 2.77×10^4 N

17. 9.6×10^6 m

18. The orbital velocity of the satellite above Earth will be 7364 m s^{-1} while that above Mars will be 3121 m s^{-1}. The Earth satellite is moving 2.36 times faster.

19. (a) 1477 m s^{-1}

(b) 12 m s^{-2}

(c) 447 s

20. B

21. D

22.

(a)	Orbital speed at closest distance (m s^{-1})	30 124	Orbital speed at furthest distance (m s^{-1})	29 625
(b)	Gravitational field at closest distance (N kg^{-1})	6.17×10^{-3}	Gravitational field at furthest distance (N kg^{-1})	5.77×10^{-3}
(c)	Gravitational force at closest distance (N)	3.70×10^{22}	Gravitational force at furthest distance (N)	3.46×10^{22}N
(d)	Gravitational potential energy at closest distance (J)	-5.44×10^{33}	Gravitational potential energy at furthest distance (J)	-5.27×10^{33}

Set 14 Kepler's Laws of Planetary Motion

1. C
2. A
3. B
4. B
5. B
6. D
7. C
8. A
9. C
10. C

Set 15 Kepler's Law of Periods

1. (a) $(2\sqrt{2})T$
   (b) $\frac{v}{\sqrt{2}}$
   (c) The answer would still be the same as orbital speed and period are both independent of the mass of the planet.
   (d) Orbital velocity values would all be multiplied by $\sqrt{2}$, and period values would therefore be decreased by a factor of $\sqrt{2}$.
2. (a) P = 499 200 km
   Q = 2.85 Earth days
   R = 115 000 km
   S = 0.72 Earth days (17.3 hours)
   (b) 1.23×10^{24} kg
3. (a) Orbital period of X : orbital period of Y = 11.15 : 1
   (b) Orbital velocity of planet X : orbital velocity of planet Y = 1 : 2.23 (you may have 5 : 12)
   (c) Gravitational force on planet X : gravitational force on planet Y = 1 : 6.25 (you may have 4 : 25)
4. 6.07×10^{24} kg
5. 2.00×10^{30} kg
6. 35 843 km
7. 1.78×10^{30} kg
8. 307.6 km
9. 2.4×10^{5} km
10. 6.1 hours
11. 164.3 Earth years
12. 6.47×10^{23} kg
13. (a) 1.08×10^{24} kg
    (b) 4.19 N kg^{-1}
    (c) 24 522 N
14. (a) 42 300 km
    (b) 3.1×10^{3} m s^{-1}
    (c) 0.226 m s^{-2}
15. 2.8×10^{11} m
16. (a) Orbital period of X : orbital period of Y = 1 : 8
    (b) Orbital velocity of planet X : orbital velocity of planet Y = 2 : 1
    (c) Gravitational force on planet X : gravitational force on planet Y 24 : 1
17. A
18. A
19. C

Set 16 Escape Velocity

1. (a) Escape velocity is the velocity an object with no power source of its own must be given in order to escape a gravitational field.

(b) Surface escape velocity will be larger than orbital escape velocity because the strength of the gravitational field at the orbital position will be less than the strength of the field on the surface of the planet.

(c) Escape velocity refers to the intitial velocity that must be given to a projectile in order to escape the Earth's gravitational field. The projectile has no form of propulsion. If a rocket has its own motors, then as long as it has fuel, it can rise as slowly as it likes until it is eventually far enough from Earth to be considered out of the gravitational field.

2. (a) From $\Delta E_p = \Delta E_K$

$$-\frac{GmM}{R} = \frac{1}{2}mv_e^2$$

Cancelling m, and rearranging we get $v_e = \sqrt{\frac{2GM}{R}}$

(b) 5.1×10^3 m s^{-1}

(c) 8.1×10^2 N kg^{-1} (or m s^{-2})

(d) The centripetal acceleration $= \frac{v^2}{r} = \frac{\left(\frac{\text{circumference}}{\text{period of rotation}}\right)^2}{R}$

$$= \frac{\left(2\pi \times 1.6 \times 10^4\right)^2}{\left((2 \times 60)^2 \div 1.6 \times 10^4\right)}$$

$= 43.8$ m s^{-2}

This is far less than the surface gravitational field strength, so matter will not be lost due to the rotation.

3. (a) 11 200 m s^{-1}

(b) 11 200 m s^{-1}

(c) Escape velocity is independent of the mass of the projectile, so these answers must be identical.

4. 6.37×10^{23} kg

5. (a) 1684 m s^{-1}

(b) 1.12×10^{22} kg

(c) Because 1000 m s^{-1} is less than the escape velocity, it will rise and slow down because of the pull of gravity backwards until it reaches a maximum height, where momentarily it will stop, then fall back to the surface of the asteroid.

6. (a) 3267 m s^{-1}

(b) 2310 m s^{-1}

(c) The gravitational field strength at the orbital position is less than at the surface, so a lesser velocity is required to overcome the gravitational force from the orbital position.

7. (a) A has the larger escape velocity.

(b) The escape velocity of A will be three times larger than that of B (escape velocity is proportional to the square root of the planet's mass).

8. 21 689 m s^{-1} (21 700 m s^{-1} rounded)

9. 61 345 m s^{-1}

10. 6.18×10^5 m s^{-1}

11. 2372 m s^{-1}

12. 11 030 m s^{-1}

13. (a) Radius about 0.89 cm

(b) (i) 6.17×10^5 m s^{-1}

(ii) Radius about 1482 m

14. (a) 18 262 m s^{-1}

(b) 4.38 Earth years

(c) 7.8 m s^{-1}

15. B

16. C

17. C

18.

(a)	3.38×10^{23} kg	(b)	10 393 m s^{-2}	(c)	12 765 km
(d)	6.42×10^{23} kg	(e)	59 543 m s^{-2}	(f)	120 000 km
(g)	8.68×10^{25} kg	(h)	23 622 m s^{-2}	(i)	2320 km
(j)	2.00×10^{30} kg	(k)	3360 m s^{-1}		

Set 17 Electrostatic Charge

1. (a) Zero or neutral charge.

(b) Negative charge.

(c) Positive charge.

(d) Electrostatic charges are due to excess or deficiency of electrons only because protons are contained within nuclei. While electrons can be easily transferred, it requires atom splitting energies to access protons.

(e) The coulomb, symbol C.

(f) One coulomb is equal to the charge on 6.25×10^{18} electrons (or protons).

(g) -1.6×10^{-19} C

(h) $+1.6 \times 10^{-19}$ C

(i) Electrostatic charge is formed by the transfer of electrons from one object to another, and is a stationary form of electricity. Current electricity is often referred to as dynamic electricity and involves the movement of electrons through a conductor. Note however, that conventional current is deemed to flow in the opposite direction to electron flow – from positive or higher potential to negative or lower potential.

2. (a) $0.6 \times 6.25 \times 10^{18} = 3.75 \times 10^{18}$ electrons have been added to the object.

(b) $0.75 \times 6.25 \times 10^{18} = 4.6875 \times 10^{18}$ electrons have been removed from the object.

(c) $4.0 \times 10^{-3} \times 6.25 \times 10^{18} = 2.6 \times 10^{16}$ electrons have been removed from the object.

(d) $0.6 \times 10^{-3} \times 6.25 \times 10^{18} = 3.75 \times 10^{16}$ electrons have been added to the object.

(e) $0.3 \times 10^{-6} \times 6.25 \times 10^{18} = 1.875 \times 10^{12}$ electrons have been added to the object.

3. (a) $4.0 \times 10^{6} \div 6.25 \times 10^{18} = 6.4 \times 10^{-13}$ C

(b) $2.5 \times 10^{20} \div 6.25 \times 10^{18} = 40$ C

(c) $1.6 \times 10^{10} \div 6.25 \times 10^{18} = 2.56 \times 10^{-9}$ C

(d) $6.4 \times 10^{9} \div 6.25 \times 10^{18} = 1.024 \times 10^{-9}$ C

(e) $7.5 \times 10^{16} \div 6.25 \times 10^{18} = 1.2 \times 10^{-2}$ C

4. 16 C in 0.5 s = 32 C in 1 s = 32 A

5. 1.6 A for 15 s = $1.6 \times 15 = 24$ C $\times 6.25 \times 10^{18} = 1.5 \times 10^{20}$ electrons

6. From $Q = It$, $I = \frac{Q}{t} = 8.0 \times 10^{16} \div (6.25 \times 10^{18} \div 0.25) = 0.05$ A

7. From $Q = It$, $t = \frac{Q}{I} = 6.023 \times 10^{23} \div (6.25 \times 10^{18} \div 32 \times 10^{-3}) = 3.01 \times 10^{6}$ s = 836.53 hours = 34.86 days

8. (a) 0.25 A = 0.25 C of charge per second

(b) $0.25 \times 6.25 \times 10^{18} = 1.5625 \times 10^{18}$ electrons per second

(c) Total charge = $0.25 \times 6 \times 3600 = 5400$ C

(d) Total number of electrons = $5400 \times 6.25 \times 10^{18} = 3.375 \times 10^{22}$ electrons

(e) Total charge for the month = $5400 \times 30 = 162\,000$ C

(f) Total number of electrons in the month = $3.375 \times 10^{22} \times 30 = 1.0125 \times 10^{24}$ electrons

9. (a) From $Q = It$, $I = \frac{Q}{t} = 6.4 \times 10^{19} \div 6.25 \times 10^{18} \div 1.0 = 10.24$ A

(b) From $Q = It$, $I = \frac{Q}{t} = 1.6 \times 10^{17} \div 6.25 \times 10^{18} \div 1.0 = 25.6$ mA

(c) From $Q = It$, $I = \frac{Q}{t} = 3.2 \times 10^{15} \div 6.25 \times 10^{18} \div 60 = 8.53 \times 10^{-6}$ A = 8.53 μA

(d) From $Q = It$, $I = \frac{Q}{t} = 8.0 \times 10^{19} \div 6.25 \times 10^{18} \div 60 = 0.213$ A

(e) From $Q = It$, $I = \frac{Q}{t} = 4.0 \times 10^{12} \div 6.25 \times 10^{18} \div 0.1 = 6.4 \times 10^{-6} = 6.4$ μA

10. (a) From $Q = It$, number of electrons = $It \times 6.25 \times 10^{18} = 0.3 \times 1 \times 6.25 \times 10^{18} = 1.875 \times 10^{18}$ electrons

(b) Number of electrons = $It \times 6.25 \times 10^{18} = 2.0 \times 1 \times 6.25 \times 10^{18} = 1.25 \times 10^{19}$ electrons

(c) Number of electrons = $It \times 6.25 \times 10^{18} = 4.5 \times 10^{-3} \times 1 \times 6.25 \times 10^{18} = 2.8 \times 10^{16}$ electrons

(d) Number of electrons = $It \times 6.25 \times 10^{18} = 3.6 \times 10^{-6} \times 1 \times 6.25 \times 10^{18} = 2.25 \times 10^{13}$ electrons

(e) Number of electrons = $It \times 6.25 \times 10^{18} = 8.0 \times 10^{-6} \times 1 \times 6.25 \times 10^{18} = 5.0 \times 10^{13}$ electrons

Set 18 Coulomb's Law

1. Like static charges repel each other.
Unlike static charges attract each other.

$$F = \frac{kQ_1Q_2}{d^2}$$

Where F = force between Q_1 and Q_2 (N)
Q_1 = charge on first object (C)
Q_2 = charge on second object (C)
d = distance between charges (m)
k = Coulomb's constant
$= 9 \times 10^9$ (N m^2 C^{-2})

2. $F = 9 \times 10^9 \times 3.0 \times 10^{-6} \times 4.0 \times 10^{-6} \div (5 \times 10^{-3})^2 = 0.432$ N repulsion

3. $F = 9 \times 10^9 \times 3.0 \times 10^{-6} \times -5.0 \times 10^{-6} \div (10 \times 10^{-3})^2 = 1350$ N attraction

4. $F = 9 \times 10^9 \times 4.0 \times 10^{-6} \times 9.0 \times 10^{-6} \div (6 \times 10^{-2})^2 = 90$ N repulsion

5. $F = 9 \times 10^9 \times -3.5 \times 10^{-3} \times 8.0 \times 10^{-6} \div (14 \times 10^{-2})^2 = 12\,857$ N attraction

6. $F = 9 \times 10^9 \times 4.0 \times 10^{-9} \times 12.0 \times 10^{-9} \div (4 \times 10^{-3})^2 = 0.027$ N repulsion

7. (a) $F = 4 \times 10^{-8} = 9 \times 10^9 \times Q^2 \div (6 \times 10^{-3})^2$, from which $Q = 1.26 \times 10^{-11}$ C

(b) One will be positive, the other negative because the force is an attraction

8. $F = 4.5 \times 10^{-1} = 9 \times 10^9 \times 3Q \times Q \div (5 \times 10^{-3})^2$, from which $Q = 2.04 \times 10^{-6}$ C either both positive or both negative because of the repulsion.

9. (a), (b), and (c) Force on each charge = vector sum of forces due to the other two charges as shown in the diagram below (not drawn to scale). The magnitude of each force is calculated using the same formula: $F = \frac{kQ_1Q_2}{d^2}$

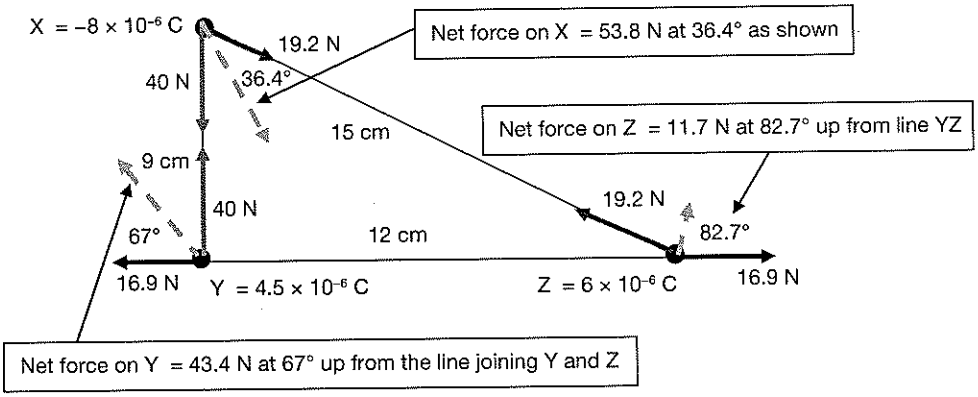
Then using the cosine rule to find the magnitude of each resultant force (dashed lines) and the sine rule to find a relevant angle of action for each of these forces, we get:

Force between X and Y (attraction) = 40 N

Force between X and Z (attraction) = 19.2 N

Force between Y and Z (repulsion) = 16.9 N

Net forces and directions given in the box labels on the diagram



10. (a) 2.0×10^{-5} N attraction

(b) 0.5×10^{-5} N repulsion

(c) 4.0×10^{-5} N attraction

(d) 18.0×10^{-5} N repulsion

(e) 10.0×10^{-5} N repulsion

11. A

12. A

13. A

14. D

15. C

16. D

17. D

Set 19 Electric Field

1. (a) An electric field is a region in which a charged object will experience a force.

(b) Depending on the situation, N C^{-1} or, for parallel plates, V m^{-1}.

(c) $E = \frac{F}{Q}$ and $E = \frac{V}{d}$

(d) Positive charges experience a force in the same direction as the field, so they move in the same direction as the field.

(e) Negative charges experience a force in the opposite direction to the field, so they move in the opposite direction to the field.

(f) Because fields are vector quantities and the arrows we use to represent them are vector arrows.

(g) The relative strength of any field is indicated by how close the field vectors are. The closer the field vectors, the stronger the field.

(h) Same direction as the force on the positive charge.

(i) In the opposite direction to the force on the charge.

(j) Electric field is directed away from higher potential towards lower potential.

(k) Electric field is directed towards lower potential.

2. (a) Towards the left.

(b) Towards the left.

(c) It will not be affected by the field because it carries no charge.

(d) Towards the right.

(e) Towards the left.

3. We should say that the field accelerates the charge because there is a net force acting on the charge due to the field, and therefore its motion will be accelerated motion.

4. D

5. C

6. D

7. B

8. (a) Left-hand side.

(b) Left-hand side.

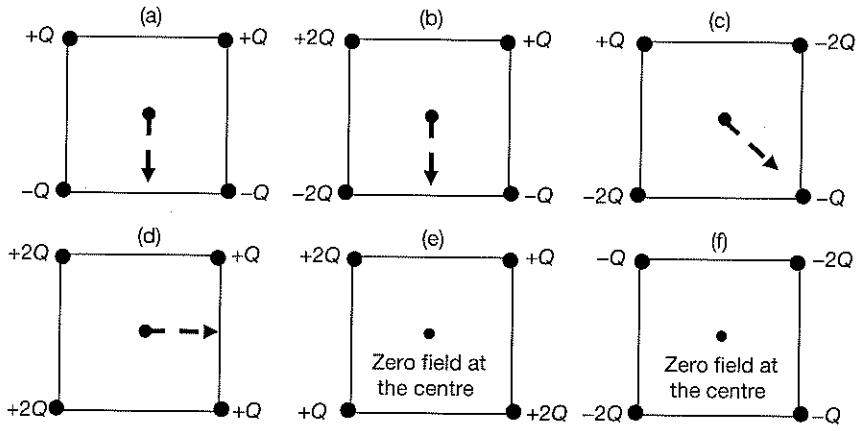
(c) Positive.

(d) Negative.

9. A

10. B

11.



12. W could be +, in which case X must be –

W could be –, in which case X must be +

Y could be +, in which case Z must be +

Y could be –, in which case Z must be –

Set 20 Electric Field Strength

1. (a) At W, field = 24 000 N C^{-1} away from the charge.
At X, field = 13 500 N C^{-1} away from the charge.
At Y, field = 2667 N C^{-1} away from the charge.
At Z, field = 96 000 N C^{-1} away from the charge.

(b) At W, field = 12 000 N C^{-1} towards the charge.
At X, field = 6750 N C^{-1} towards the charge.
At Y, field = 1333 N C^{-1} towards the charge.
At Z, field = 48 000 N C^{-1} towards the charge.

2. B

3. D

4. A

5. 1.875×10^3 N C^{-1} towards the charge.

6. -2.2×10^{-10} C

7. (a) 1.897 m

(b) Charge is positive as field is directed away from it.

8. 20 000 N C^{-1} away from the charge.

9. -1.1×10^{-17} C

10. (a) 3.21 m

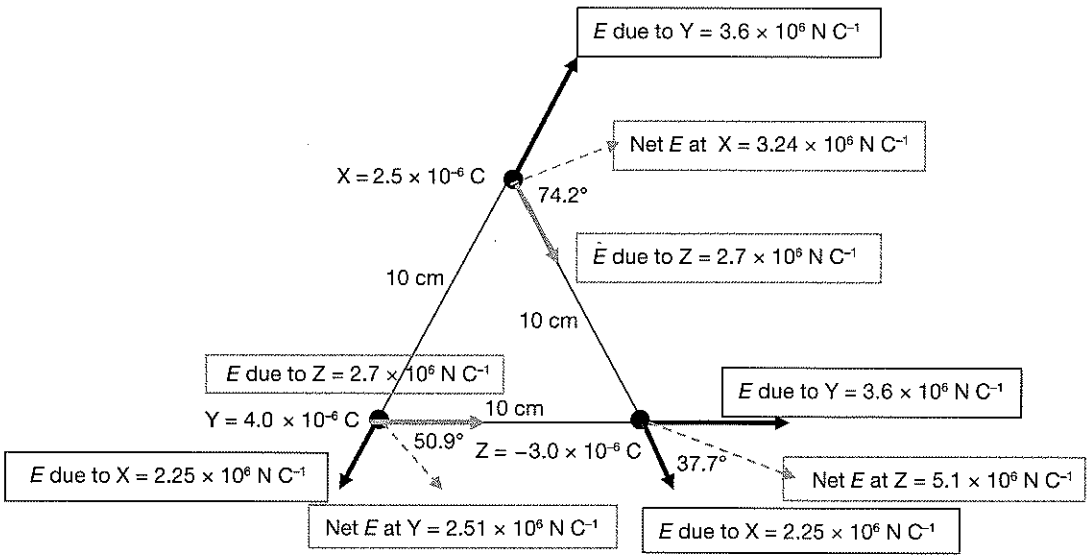
(b) Charge is negative as field is directed towards it.

11. The directions of the fields at the corners are shown in the diagram along with the resultant vector (all not to scale). Field strengths are calculated using $E = \frac{kQ}{d^2}$, net fields using the cosine rule, and directions using the sine rule.

(a) Net E at X = 3.24×10^6 N C^{-1} at 74.2° as shown on the diagram.

(b) Net E at Y = 2.51×10^6 N C^{-1} at 50.9° as shown on the diagram.

(c) Net E at Z = 5.11×10^6 N C^{-1} at 37.7° as shown on the diagram.

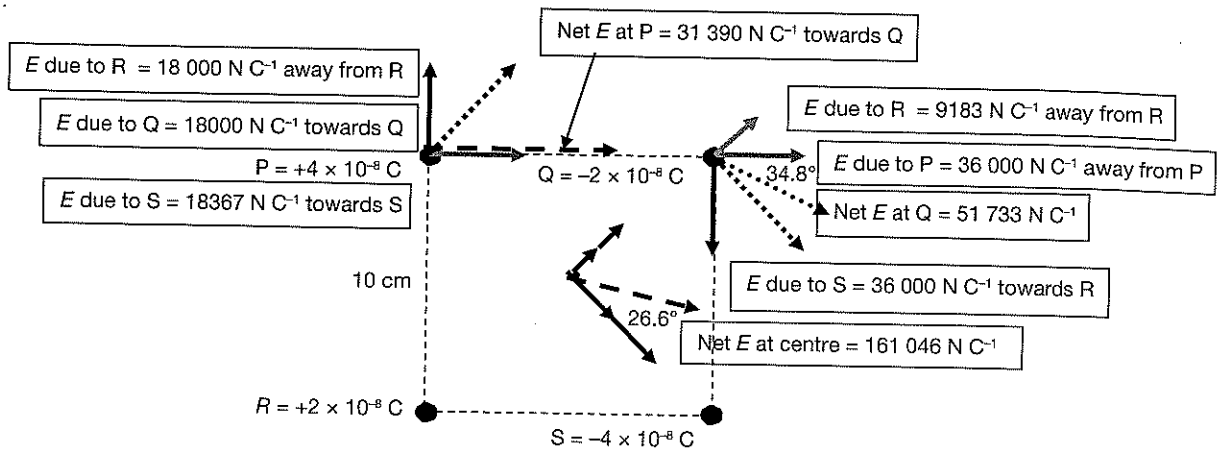


12. The directions of the fields at the corners are shown in the diagram along with the resultant vector (all not to scale). Field strengths are calculated using $E = \frac{kQ}{d^2}$, net fields using the cosine rule, and directions using the sine rule. Diagonal = 14.14 cm, distance to centre = 7.07 cm.

(a) The resultant of E due to R and E due to Q (Pythagoras) = 25 456 N C^{-1} at 45° (dotted vector).
This resultant is at 90° to E due to S, so their resultant = 31 390 N C^{-1} directed towards Q (dashed vector).

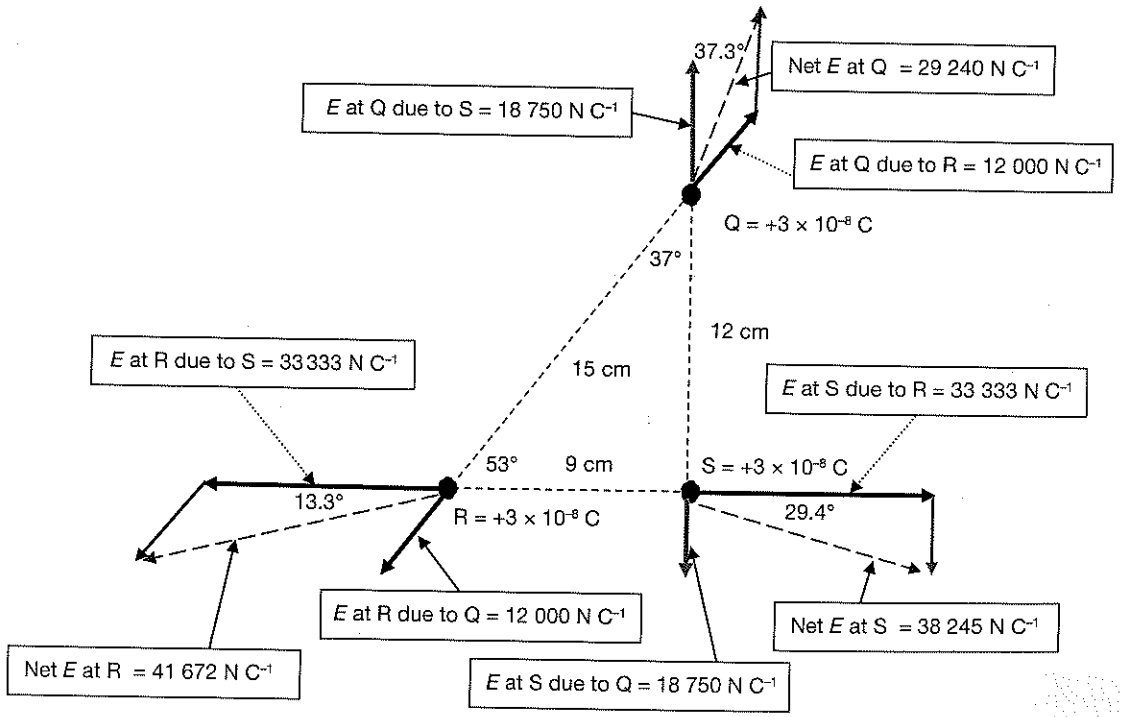
(b) The resultant of E due to P and E due to S (Pythagoras) = 50 912 N C^{-1} at 45° (dotted vector).
This resultant is at 90° to E due to R, so their resultant = 51 733 N C^{-1} directed as shown (dashed vector).

(c) At the centre, field due to P = 72 022 N C^{-1} towards S.
At the centre, field due to Q = 36 011 N C^{-1} towards Q.
At the centre, field due to R = 36 011 N C^{-1} away from Q.
At the centre, field due to S = 72 022 N C^{-1} towards S.
Net field at centre = $\sqrt{(72022)^2 + (144044)^2}$ = 161 046 N C^{-1} at 26.6° as shown.



13. The directions of the fields at the corners are shown in the diagram along with the resultant vector (all not to scale). Field strengths are calculated using $E = \frac{kQ}{d^2}$, net fields using the cosine rule, and directions using the sine rule.

(a) See labels on the diagram Interior angles in vector triangle for resultant on charge Q are 143°, 14.3° and 22.7°.
(b) See labels on the diagram.
(c) See labels on the diagram.



Set 21 Force on a Charge in an Electric Field

1. D
2. B
3. C
4. B
5. A
6. A
7. D
8. C
9. C
10. B
11. A
12. C
13. D
14. C
15. C
16. C
17. C
18. A
19. B
20. C
21. B
22. D
23. C
24. (a) 5×10^{-9} C
(b) Negative – force is acting in the opposite direction to the field.
25. 1.875×10^{4} N C^{-1} north
26. 3832 m s^{-2}
27. 2.5×10^{2} N C^{-1} east
28. 1.4×10^{3} N C^{-1} north
29. 6.0×10^{-9} N north
30. 1.92×10^{-8} N east
31. –1.0 C
32. $+3.33 \times 10^{-10}$ C
33. 3.75×10^{4} N C^{-1} north
34. 2.4×10^{3} m s^{-2} in direction of field
35. 2.4×10^{-17} N
36. (a) Forces will be equal in magnitude but opposite in direction. Equation shows force is independent of mass and all other variables are equal, so force equal. Signs of charges are opposite, so forces in opposite directions.
(b) Forces will be the same magnitude, but opposite in direction.

Set 22 Electric Field Strength Between Parallel Plates

1. (a) Uniform in intensity and directed from higher to lower potential.
(b) Plate X.
(c) Plate X.
2. (a) Only D.
(b) A – field lines are not drawn evenly spaced to indicate a uniform field.
B – field lines are in the wrong direction.
C – field lines are in the wrong direction.
3. (a) In each case, field is 25 000 V m^{-1}.
(b) The distance apart is the same for each as is the potential difference between them ($E = \frac{V}{d}$).

4. (a) 40 000 V m^{-1}
   (b) 32 000 V m^{-1}
   (c) 5000 V m^{-1}
   (d) 50 000 V m^{-1}
5. A
6. C
7. B
8. C
9. C
10. (a) From $mg = \frac{QV}{d}$

    $Q = \frac{mgd}{V} = 2.6 \times 10^{-8}$ C

    (b) The top plate must be negative to attract the charge upwards and so counteract the gravitational force, therefore the bottom plate is positive.
11. (a) E
    (b) E
    (c) $0.5E$
    (d) $0.5E$
    (e) E
    (f) 1 : 1 ignoring forces due to their own charges.
12. A Plates W, X will deflect the beam upwards and plates Y, Z will deflect it out of the page – therefore region A.
13. A
14. D The force on each charge is given by $F = Eq$, and since the charges are identical, and the field between parallel plates is constant, the forces on the three charges will be the same.
15. D
16. B
17. (a) $\frac{E}{3}$
    (b) $2E$
    (c) $\frac{2E}{3}$
    (d) Has no effect on the field, therefore still E.
    (e) Has no effect on the field, therefore still E.
    (f) E
18. (a) 1.2×10^{-16} N
    (b) Electron will accelerate towards the positively charged plate while continuing through the plates with constant velocity parallel to them – i.e. it will follow a parabolic path towards the positive plate.

Set 23 Work Done By a Field

1. B
2. B
3. (a) Electric field is directed from Y to X, the same direction as the electric potential (Y is connected to the positive terminal of the power supply), and in the opposite direction to the movement of the electrons through the field.
   (b) From $I = \frac{Q}{t} = \frac{\left(2.5 \times 10^{15}\right) \div \left(1.6 \times 1.0^{-19}\right)}{1} = 4 \times 10^{-4}$ A = 0.4 mA
   (c) From $W = QV$, work done on each electron = $1.6 \times 10^{-19} \times 500 = 8 \times 10^{-16}$ J
4. (a) At point Q because it is closest to the charge, or because the field lines are closest together here.
   (b) Moving from P to R involves moving to a higher potential, so external work would need to be done on the charge. Moving from P to T involves moving slightly further from the central charge, or to a slightly lower potential, so the field will do work on the charge. The work done from P to R will be greater than the work done from P to T.
   (c) Moving to point Q because it is at the highest potential.
   (d) The statement is correct in that P and S are in the same relative field strength positions being equidistant from the central charge.
   (e) Yes. The same logic applies.
5. A

6. (a) The relevant equation is $W = Fd$

(b) $W = QV$

(c) $F = EQ$

(d) $E = \frac{V}{d}$

(e) From this equation, N m^{-1}, from the equation in (c) above, N C^{-1}.

7. From $W = QV = 3.75 \times 10^{8}$ J, i.e answer D.

8. From $W = QV = 4.8 \times 10^{-15}$ J, i.e. answer D.

9. (a) The bottom plate is positive because field lines are directed away from positive towards negative.

(b) Towards the bottom plate.

(c) Force will not change. $F = EQ$, and both E and Q do not change in values (field between parallel plates is constant).

(d) Work will need to be done on the charge by an external force because it is being moved closer to the negative plate (it is a negative charge and would not do this naturally).

(e) Work will be done on the charge by the field because it is being moved closer to the negative plate, and being positive, this is the natural direction it will move as a result of the field.

10. (a) X will accelerate to the right, Y will accelerate to the left.

(b) Yes, by an external force.

(c) Yes, by the field.

(d) Yes, by the field.

(e) Yes, by an external force.

(f) In each case the work done on Y will be double the work done on X for equal distances moved.

11. (a) B

(b) D

12. (a) $E = \frac{V}{d} = \frac{20}{(50 \times 10^{-3})} = 4000$ V m^{-1}

(b) The electron will accelerate towards the positive plate.

(c) $F = EQ = 400 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-16}$ N

(d) $a = \frac{F}{m} = \frac{(6.4 \times 10^{-17})}{(9.1 \times 10^{-31})} = 7.03 \times 10^{14}$ m s^{-2}

(e) $W = QV = 1.6 \times 10^{-19} \times 20 = 3.2 \times 10^{-18}$ J (= $Fs = 6.4 \times 10^{-17} \times 50 \times 10^{-3}$)

(f) $W = \Delta E_K = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2 = 3.2 \times 10^{-18}$

From which $v = 2.65 \times 10^{6}$ m s^{-1}

13. (a) An external force because in moving closer to a negative charge, the electron is moving to a position which for it, is a higher potential energy position.

(b) $W = QV = 4.8 \times 10^{-15} = 1.6 \times 10^{-19} \times V$

From which $V = 30\,000$ V

(c) 30 000 V

(d) 0 V

Set 24 Magnetic Field Around Straight Conductors

1. (a) Out of the page

(b) Out of the page

(c) Out of the page

(d) Into the page

(e) Into the page

(f) Into the page

(g) Out of the page

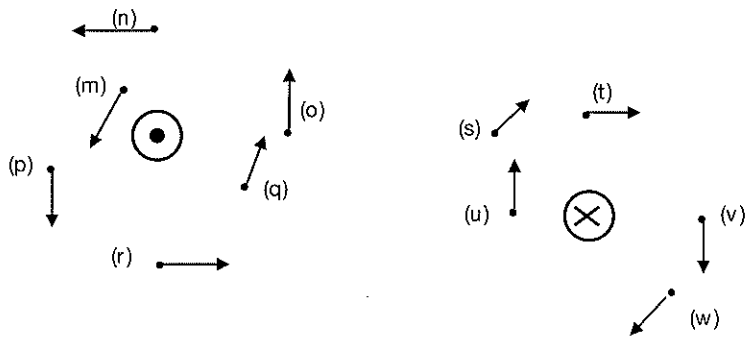
(h) Out of the page

(i) Into the page

(j) Into the page

(k) Out of the page

(l) Out of the page



(m) to (r) Field is anticlockwise. In each case, the direction of the field is in the direction of the tangent to the circles – not shown on diagram around the current (right hand grip rule – RHGR).

(s) to (w) Magnetic field is clockwise and direction is tangential at each point as shown on the diagram.

2. In each case, from $B = \frac{kI}{d}$, and using the RHGR:

(a) 6.25×10^{-6} T

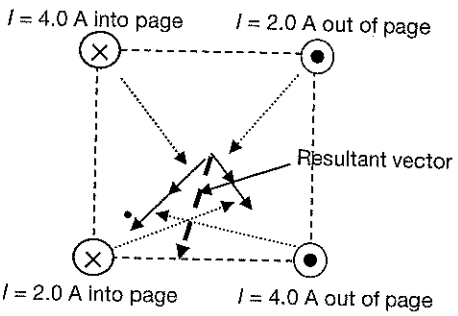
(b) 3.0×10^{-9} T

(c) 5.33×10^{-6} T

(d) 6.67×10^{-7} T

(e) 8.0×10^{-11} T

3. Using the RHGR, the directions of the four fields at the centre are shown in the diagram along with the resultant vector (all not to scale).

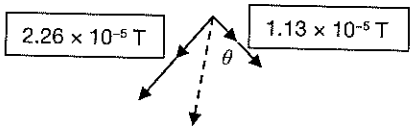


Distance from each charge to the centre = $\sqrt{\frac{(10^2 + 10^2)}{2}} = 7.07$ cm

Field due to each 4.0 A currents = 1.13×10^{-5} T

Field due to each 2.0 A currents = 5.66×10^{-6} T

Therefore field vectors at the centre are:



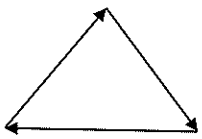
Resultant = net field at the centre = $\sqrt{(2.26\times10^{-5})^2 + (1.13\times10^{-5})^2} = 2.53 \times 10^{-5}$ T

$\theta = \tan^{-1}\frac{(2.26\times10^{-5})}{(1.13\times10^{-5})} = 63°$

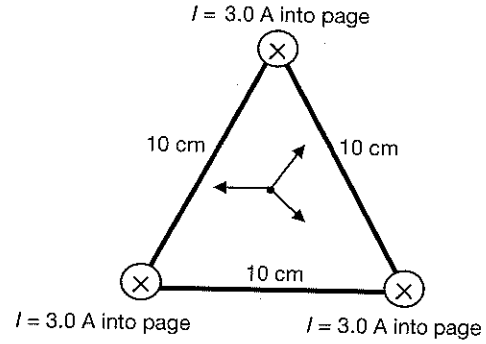
So, field at the centre is 2.53×10^{-5} T at 63° left from the line joining the top left hand corner and the bottom right hand corner.

4. Using the RHGR, the directions of the four fields at the centre are shown in the diagram along with the resultant vector diagram (all not to scale)

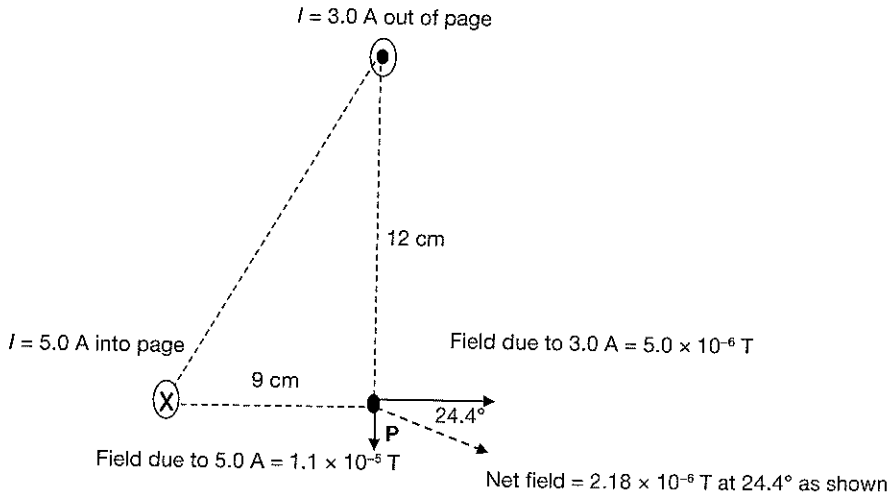
Given the figure is an equilateral triangle, and that the currents in the conductors are identical, the three field vectors are at 120° to each other and when added will form a closed vector triangle:



The field in the centre of the triangle is therefore zero.



5. Using the RHGR, the directions of the two fields at point P are shown in the diagram along with the resultant vector (all not to scale).



6. C

7. (a) I_P = into the page
   I_Q = out of the page
   I_R = out of the page
   (b) $I_P > I_R > I_Q$ as indicated by the relative closeness of the field lines around each.
   (c) Only those around R have been drawn correctly – becoming further apart further from the conductor. The other two sets have been drawn equidistant and do not reflect the weakening field further from the conductors.

Set 25 Magnetic Fields and Solenoids

1. B
2. B
3. C
4. (a) A solenoid is a coil of wire having *n* turns.
   (b) The magnetic domains within the soft iron core are lined up by the induced magnetic field around the solenoid and this intensifies the magnetic field.
   (c) No. Copper is not magnetic.
5. Both B and D.

Set 26 Magnetic Flux and Flux Density

1. (a) Magnetic flux is a measure of the magnetic field permeating an area.
   (b) Flux density is the magnetic field per unit area.
   (c) Magnetic field intensity.
   (d) Flux = webers = Wb
   (e) Flux density = webers per square metre = Wb m^{-1} = teslas = T
2. (a) $B = \frac{\phi}{A} = \frac{4}{0.12^2} = 277.8$ T
   (b) Flux would not change = 4.0 Wb
   Flux density would halve = 138.9 T
   (c) 69.4 T
   (d) Flux stays the same = 4.0 Wb
   Flux density = $\frac{4}{36 \times 10^{-4}} = 1.1 \times 10^3$ T (= 4 × 277.8 as area is quarter what it was)
   (e) Flux = 4.0 Wb
   Flux density = $\frac{4}{(324 \times 10^{-4})} = 123.5$ T
3. A
4. B
5. C
6. B
7. B
8. C
9. D
10. D
11. A
12. B
13. C

Set 27 Moving Charges in a Magnetic Field 1

1. (a) Up
   (b) Out of the page
   (c) Up
   (d) Into the page
   (e) Out of the page

(f) Left
(g) Out of the page
(h) Down
(i) Right
(j) Into the page
(k) Down
(l) Into the page
(m) Into the page
(n) Left
(o) Left
(p) Out of the page
(q) Up
(r) Into the page
(s) No force
(t) Down
(u) Out of the page
(v) Up
(w) Right
(x) No force
(y) Out of the page
(z) Into the page
(aa) Out of the page
(bb) Right

Set 28 Moving Charges in a Magnetic Field 2

1. (a) 2.9×10^{7} m s^{-1}
(b) 4.5×10^{14} N
(c) 1.7 cm

2. (a) 3×10^{-12} N
(b) Left (west initially, then in a circular path)

3. (a) 0.5 T
(b) East (right)

4. (a) 1.6×10^{-18} C
(b) Need to know the directions of *F*, *B* and the velocity of the charge.

5. (a) 1.1 T
(b) Towards the east

6. (a) 3×10^{-12} N
(b) Left (west initially, then in a circular path)

7. (a) 0.5 T
(b) East (right)

8. (a) Force on electron from the east = 8×10^{-16} N out of the page (circular path while in field).
(b) Force on electron from the west = 8×10^{-16} N into the page (circular while in field).
(c) Force on electron from the south = 0 (parallel to the field) – travels in straight line.

9. (a) X is positively charged as determined by applying the right hand rule.
Y is negatively charged – it is deflected in the opposite direction to X.
(b) X is deflected more, so if the masses and speed are equal, it has a higher charge.
Or, if the charges and masses are equal, it has a higher velocity.
Or, if the speeds and charges are equal, it has less mass.
Or, there is some combination of two of the variables or all three that result more deflection for X.

10. D

11. B Apply the right hand palm rule (remember cathode rays are negatively charged so your thumb needs to point in the opposite direction to their direction of motion).

12. C
13. B
14. C
15. D
16. C

Set 29 Moving Charges in a Magnetic Field 3

1. C
2. C
3. B
4. (a) 4.99×10^{-17} N up out of the page
   (b) 2.3×10^{-4} m
5. (a) 1.6×10^{-12} N perpendicular to motion so that particle moves in a circle
   (b) 0.018 m
   (c) 2.5×10^{6} m s^{-1}
   (d) 0.086 m from bottom corner
6. (a) X deflected up the page, Y deflected down the page
   (b) Radius of path of X: 11.1 m, radius of path of Y: 22.2 m
7. (a) $+3.46 \times 10^{-18}$ C
   (b) 2.16×10^{-16} N north
   (c) 111 m s^{-1} into the page
   (d) $+6.25 \times 10^{-18}$ C
   (e) 0.0167 T south
   (f) 6×10^{-16} N south
   (g) 0.7 T south
   (h) 0.3 T east
   (i) $+1.6 \times 10^{-19}$ C
   (j) 9.72×10^{-17} N east
   (k) 600 m s^{-1} south
   (l) 13.3 m s^{-1} west
8. (a) $+6.9 \times 10^{-19}$ C
   (b) 1.87×10^{-18} N west
   (c) 1.57 m s^{-1} out of the page
   (d) $+1.8 \times 10^{-17}$ C
   (e) 0.019 T east
   (f) 2.05×10^{-16} N north
   (g) 2.87°
   (h) 49.8°
   (i) 60 m s^{-1} out of the page
   (j) 3.0 T east
   (k) 20°
   (l) -4.5×10^{-18} C

Set 30 The Motor Effect 1

1. (a) Up the page
   (b) Out of the page
   (c) Down the page
   (d) Into the page
   (e) Into the page
   (f) Left

(g) Into the page
(h) Down the page
(i) Right
(j) Into the page
(k) Left
(l) Into the page
(m) Out of the page
(n) Right
(o) Towards bottom right hand corner of the field
(p) No force
(q) Towards bottom left corner of field
(r) Out of the page
(s) No force
(t) Up the page
(u) Out of the page
(v) Towards top right corner of field
(w) Towards to left corner of field
(x) Out of the page
(y) Into the page
(z) No force
(aa) Out of the page
(bb) Towards bottom left corner of field

Set 31 The Motor Effect 2

1. (a) A current carrying conductor placed in and at an angle to a magnetic field will experience a force.
(b) The length of the conductor, the flux density of the field, the sine of the angle between the direction of the current and the field (and the number of turns in the conductor if it is a coil)
(c) $F = nBIL \sin \theta$
(d) The direction of the magnetic flux and the direction of the current.
(e) The interaction of two (or more) magnetic fields.

2. (a) Electric currents do interact with each other as per $F = \frac{kI_1I_2l}{d}$.
(b) Electric currents do interact with magnetic fields.
(c) Stationary charges do not interact with magnetic fields.
(d) Moving charges do interact with magnetic fields.
(e) Magnetic fields do interact with each other.
(f) A magnetic field is induced around all current carrying conductors.
(g) A magnetic field is induced around all moving charges.

3. B
4. C
5. B
6. A
7. (a) The black bar will roll (accelerate) to the left.
(b) The current through the bar is down towards the bottom of the page, so by the RHPR, the force on it is towards the left.
(c) Because both the current and the field direction have been changed, the bar will roll to the left still.
8. C
9. B
10. B
11. C
12. D
13. 1.28×10^{-5} N out of the page

Set 32 Forces on Straight Conductors in Magnetic Fields

1. (a) $2F$
(b) $0.5F$
(c) $0.5F$
(d) $1.73F$
(e) New force will be $0.87F$

2. (a) 17.45°
(b) 90° – Force is maximum when sin θ is maximum (=1) and this is when $\theta = 90°$.
(c) 0° – Force is minimum when sin θ is minimum (=0) and this is when $\theta = 0°$.
(d) 0.3 N – With half the resistance, double the current will flow, so force will double.

3. 0.13 N into the page

4. $3F$

5. Using $F = BIL \sin \theta$ for each, and the RHPR for direction, we get:
(a) 2.6×10^{-4} N out of the page
(b) 4.8×10^{-2} N out of the page
(c) 8.0×10^{-2} N out of the page
(d) 4.3×10^{-2} N into the page
(e) 1.9×10^{-3} N into the page
(f) 5.2×10^{-8} N into the page

Set 33 Torque on a Coil 1

1. (a) A torque is a turning force caused by two equal and opposite forces and is equal to the product of the magnitude of those forces and the perpendicular distance between them.
(b) $\tau = Fr$
τ = torque measured in newton metres (N m)
F = refers to the equal and oppositely acting forces in newtons
d = perpendicular distance between the two forces in metres.
and
$\tau = nBIA \cos \theta$
τ = torque measured in newton metres (N m)
n = number of turns in the coil
B = magnetic field strength (flux) in teslas
I = current in coil (amperes)
$A \cos \theta$ = area of coil parallel to the magnetic field

2. (a) Torque will be halved.
(b) Direction of the torque will be reversed.
(c) Torque will be doubled.
(d) Torque increased four times.
(e) Torque increased (current in coil increases assuming voltage remains constant).

3. (a) From $F = nBIL$ = 4.5 N at right angles to both the field and the current.
(b) $\tau = Fd = 4.5 \times 0.04 = 0.18$ N m
(c) Zero
(d) From $\tau = Fd \cos \theta = 0.156$ N m
(e) Force does not change regardless of position in the rotation, so F = 4.5 N

4. (a) 0.11 N into the page (vertically down)
(b) 5.6×10^{-3} N m anticlockwise
(c) 2.8×10^{-3} N m clockwise
(d) $\Delta\phi = nBA = 1500 \times 0.6 \times 0.05 \times 0.05 = 2.25$ Wb

5. B

6. B

7. A

8. D

9. (a) 7.81 N down

(b) 0.195 N m

(c) $\Delta\phi$ needed = $nBA = 250 \times 0.5 \times 0.025 \times 0.025 = 0.078$ Wb

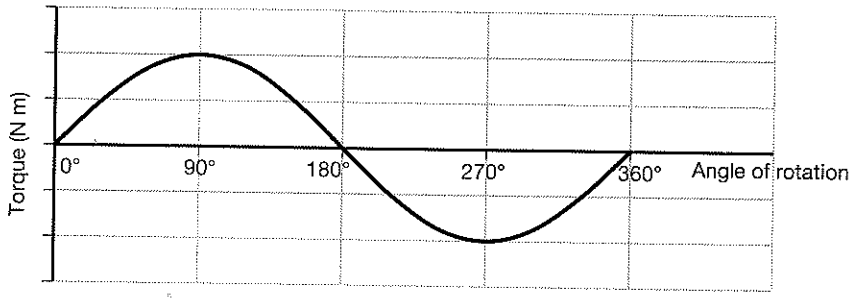
(d) Its inertia will carry it slightly through the vertical position and then the torque will be in the opposite direction, so the coil will oscillate with small, decreasing amplitude at the vertical position until it comes to rest.

10. (a) 1.04×10^6

(b) The torque on the coil depends on the angle the area makes with the coil. As it rotates, the torque varies according to the cosine of the angle of the coil from parallel to the field – or – since torque is the product of two identical forces acting in opposite directions and the perpendicular distance between their lines of action, as the coil rotates, this distance decreases to zero (perpendicular to the field) and then increases as it continues to rotate, varying again, as the cosine of the angle of rotation.

(c) The torque varies according to the cosine of the angle of rotation as per (coil starting parallel to the magnetic field):

The graph should start with torque at a maximum.



Set 34 Torque on a Coil 2

1. (a) 0.72 N m

(b) 3.6×10^{-2} N m

(c) 8×10^{-4} m^2

(d) 1.6×10^{-3} m^2

(e) 1.5 A

(f) 2.4 A

(g) 0.15 T

(h) 0.02 T

(i) 20

(j) 30

2. (a) 0.39 N m

(b) 0.014 N m

(c) 1.8×10^{-3} m^2

(d) 2.77×10^{-3} m^2

(e) 0.7 A

(f) 0.52 A

(g) 0.17 T

(h) 0.68 T

(i) 70

(j) 954

(k) 29.8°

(l) 60°

Set 35 Simple DC Motors

1. A simple motor works because the current flowing through the coil induces a magnetic field around the coil. This field interacts with the magnetic field produced by the stator magnet to produce forces on each side of the coil perpendicular to the field (the motor effect). These forces produce a torque which rotates the coil.

2. (a)

Part	Name of part	Function(s) of part
A	Connection to negative terminal of external circuit	Completes circuit and allows current to flow through coil.
B	Insulating ring	Ensures axle does not short circuit the split ring commutator halves.
C	Split ring commutator	Ensures current is always in the same direction relative to the magnetic field so that torque on the coil is in a constant direction.
D	Axle	Provides pivot line for rotation of the system.
E	Coil	Carries current to interact with magnetic field and produces torque to cause rotation.
F	Carbon brushes	Provide electrical contact between split ring commutators and external circuit.
G	Connection to positive terminal of external circuit	Completes circuit and allows current to flow through coil.

(b) Clockwise as you look at it.

(c) There are typically two different answers to this question. The most common is:

To change the direction of the current in the coil so that the rotor continues to turn in the same direction.

A better answer, which is technically more correct is:

To ensure that the current in the coil is always in the same direction relative to the magnetic field so that the coil rotates in a constant direction. The key phrase here is 'relative to the magnetic field', and the current needs to be in the same direction in the coil relative to the magnetic field (i.e. clockwise or anticlockwise). To do this, the split rings change the direction of the current through the coil 'relative to the external circuit' but this effectively makes it in a constant direction relative to the magnetic field.

Note: Make sure you are clear on the difference here – it is the subtle difference in the language used that is important.

(d) Use a radial magnetic field so the torque on the coil is constant.

Use two or more separate coils. This will also make the torque on the system more uniform, and will also increase the net torque on the system.

(e) Electromagnets produce a stronger field than bar magnets and the soft iron core intensifies the electromagnetic induction, so both these features will result in a stronger magnetic interaction and therefore a stronger force on the sides of the coil, and therefore a stronger torque. Increasing the number of turns in the coil will also increase the net force on the rotor ($F = nBIL$), and therefore the net torque on the rotor.

(f) Increase the number of turns in the field electromagnets, increase the current flowing through the electromagnets.

3. C

4. A

Set 36 Simple AC Motors

1. Their structure is identical with the exception of the split ring commutator in the DC motor and the slip ring electrical connectors in the AC motor.

2. (a) U = slip rings which provide the electrical connection between the motor and the external circuit.

V = carbon brushes which provide electrical contact from the external circuit to the slip rings.

W = bar magnet which provides the stator magnetic field.

X = circuit wire connecting brushes to the external power supply.

Y = circuit wire connecting brushes to the external power supply.

Z = axle on which rotor is assembled and which allows rotor to rotate.

(b) To provide electrical connection between the external power supply and the motor coil.

(c) It is an AC motor so it oscillates clockwise/anticlockwise.

(d) It is an AC motor so the polarity of the connections change each cycle of the AC.

3. Student X: Statement is incorrect because the AC supply automatically does this.

Student Y: Technically correct, but in doing this, because the supply is AC, the slip rings act purely as an electrical connector so that the changing phase of the AC input is not altered.

Student Z: Statement is correct.

4. D

5. C (Torque must remain positive or it will not rotate through 360 degrees.)

Set 37 Faraday and Induction

1. (a) First law: When there is relative movement between a conductor and a magnetic field (either physical movement or change in magnitude of the magnetic field) a potential difference is generated. If the conductor is part of an electric circuit, a current is induced in the circuit.

Second law: The magnitude of the induced potential difference is directly proportional to the rate at which the conductor 'cuts through' the magnetic field.

(b) In his first experiment, he simply moved a magnet into and out of a coil and noticed that the ammeter in the circuit registered a current whose direction depended on the direction of movement of the magnet. When the magnet was stationary, regardless of whether it was inside the solenoid or not, no current was detected.

Faraday also experimented with two coils of wire wrapped around opposite sides of a soft iron ring. He noticed that when a current was flowing steadily in the first loop nothing happened, but when he switched the current on and off continuously, AC was induced in the second circuit.

2. An alternating current would be observed in the ammeter. This would be produced by the alternating emf induced in the coil due to the alternating directions of movement of the magnet relative to the coil.

3. (a) 1 – To study the relationship between the strength of a magnetic field moving with the same relative motion to a coil and the emf induced in the coil.

2 – To study the relationship between the rate of the relative motion of constant strength magnets to a coil and the emf induced in the coil.

(b) 1 – Pair of magnets P and R could be used to do this study.

2 – Pairs of magnets P and Q as well as R and S could be used to do this study. Preferably, both pairs would be used separately to increase the reliability of the results.

(c) It may or may not be correct in that it depends on whether the strong magnet (R) produces a larger emf from the lower height than the weaker magnet (Q) from the higher height. If this is the case, then the prediction will be correct.

(d) P and Q alone would show that the speed at which the magnet moves into the coil affects the emf produced (the faster, the greater the emf).

(e) Any three of the height and number of turns in the coil, the diameter of the coils and the two lower heights the magnets were dropped from.

(f) It wouldn't. Pair P and R have not changed and can still be used to study the strength at a constant speed. Purpose 2, speed and emf is strengthened by having three different speeds instead of two as previously.

(g) P < R < S < Q

4. The AC current in coil X induces a changing magnetic field around coil X and this induces a current in coil Y. Remember that the concept of 'relative motion' includes this situation – the coils remain stationary but the magnetic field near them constantly changes – this constitutes 'relative motion'.

5. (a) An emf is induced in both coils because there is relative movement between a conductor (the wire in the coils) and a magnetic field.

(b) Because there is not a closed circuit in solenoid X, no current will flow. A charge separation only will result. A current will flow in coil Y because the thick copper wire joins its ends to form a closed circuit.

(c) X will make a very slight movement (perhaps negligible) away from the magnet as the charge separation is set up and a N pole forms on the right hand side of the coil. It will then simply swing back and forth with a low amplitude because of its inertia.

Y will move away from the magnet whenever the magnet continues to move towards it. If/when the magnet stops, Y will simply swing back and forth.

6. (a) $-I$

(b) $2I$

(c) $-\frac{I}{3}$

(d) $2I$

7. (a) The galvanometer needle will flicker momentarily to one side and return to zero indicating a brief induced current as the magnetic field increases in the coil containing the switch.

(b) The galvanometer needle will flicker momentarily to the opposite side then return to zero indicating a brief induced current as the magnetic field decreases in the coil containing the switch.

(c) The galvanometer needle will swing back and forth indicating an induced AC current as the magnetic field increases and decreases in the coil containing the switch.

(d) Faraday concluded that it was the changing current in the first coil that caused the induced current in the second coil.

(e) ε = induced emf in the coil (V)

$\Delta\phi$ = change in magnetic flux (Wb)

Δt = time for change in flux to occur (s)

n = number of turns in the solenoid

The minus sign takes the law of conservation of energy into account.

Set 38 Lenz's Law and Electromagnetic Induction

1.
(a) Left
(b) Up (towards top of page)
(c) Right
(d) Left
(e) Up
(f) Down
(g) Left
(h) Right
(i) Down
(j) To top right
(k) Down
(l) Up
(m) Left
(n) Up the page
(o) Down to bottom left
(p) No induction
(q) Down to bottom right
(r) Up
(s) No induction
(t) Left
(u) Left
(v) To top left
(w) To bottom left
(x) To top left
(y) Up to top right
(z) No induction
(aa) Up to top right
(bb) Up to top left

2.
(a) Left to right
(b) Into the page
(c) Down
(d) Down
(e) Right to left
(f) Out of the page
(g) Up the page
(h) Up the page
(i) Down the page
(j) Into the page
(k) Left to right
(l) Right to left

3.
(a) To the left
(b) Into the page
(c) Up
(d) Up
(e) Right
(f) Into the page
(g) Up
(h) Down
(i) Up
(j) Out of the page
(k) Left
(l) Right

Set 39 Lenz's Law and Coils

1. (a) Clockwise
(b) No induction
(c) Clockwise
(d) Anticlockwise
(e) No induction
(f) No induction
(g) Anticlockwise
(h) Clockwise
(i) No induction
(j) No induction
(k) No induction
(l) Anticlockwise
(m) Clockwise
(n) No induction
(o) Anticlockwise

2. (a) BGA
(b) AGB
(c) AGB
(d) BGA
(e) AGB
(f) BGA
(g) No induction
(h) BGA
(i) AGB

3. B

4. (a) B
(b) Remember that the geographic North Pole of Earth is a magnetic south pole, so the field will be directed vertically down into the pole. Using the RHPR we determine the direction of the induced emf in the plane will be from right to left. Since this is the direction of conventional current flow, electrons will move the other way. Electrons move from the right to build up on the right side of the wing (and tail).

5. A

6. B

7. D

Set 40 Eddy Currents

A **In general**

1. (a) An eddy current is a current which has been induced in a solid sheet of metal produced by the relative motion of the sheet through a magnetic field.
(b) Eddy currents are produced whenever there is relative movement between a sheet conductor and a magnetic field.

2. (a) Nothing – it is not subjected to a changing magnetic field because the current in the circuit is constant.
(b) The current in the circuit is anticlockwise and increasing, so the disc is subjected to a changing magnetic field out of the page. There will be an induced emf in the disc which will induce a second magnetic field to oppose the changing field due to the circular loop. The field due to the loop is out of the page, so the induced eddy current in the disc will be constant value anticlockwise.
(c) The current in the circuit is anticlockwise and decreasing, so the disc is subjected to a changing magnetic field which is directed out of the page but decreasing in value. There will be an induced emf in the disc which will induce a second magnetic field to oppose the change in the field due to the circular loop. The field due to the loop is out of the page, so the induced eddy current in the disc will be constant value anticlockwise, to induce a field out of the page.
(d) The disc is subjected to a constantly changing emf in both magnitude and direction, so an AC eddy current will be induced in the disc (180° out of phase to the AC from the power supply).

B Electromagnetic brakes

3. Electromagnetic brakes have no friction to wear brake linings away, are cheaper to maintain, strength of brake is proportional to rate of movement, so braking is smooth and reliable.

4. Braking force depends on back emf produced and this depends on the rate at which magnetic field is cut by the wheels, and this depends on the speed of the object, so there will always be a braking force as long as there is movement. However, the slower the speed, the less intense the braking force.

5. The movement of the metal wheels or fins of the moving object passes through a strong magnetic field and this induces an eddy current in the wheel or fin. The direction of this current, by Lenz's law, is such that the magnetic field it induces will react with the applied magnetic field to produce a force on the wheel or fin to oppose the forward motion.

6. The intensity of the braking depends on the rate at which the wheel cuts magnetic field and this reduces automatically as the train slows down. Mechanical brakes rely on pressure applied by a driver, who may jam on the brakes applying too much pressure and causing a dangerous skid.

7. If operational (switched on), they are independent of human reaction time, human concentration, tiredness, foot pressure on brake pedal, surface conditions, weather, so they will be more reliable and safer.

8. (a) As the plate moves into the magnetic field there will be an induced emf which will result in an eddy current in the plate because the amount of field through the plate increases the further it passes into the field – the flux through the plate is changing. This eddy current induces a magnetic field which interacts with the static field to produce a Lenz's law force to oppose the motion of the plate – i.e. the Lenz's law force is upwards, against gravity, so the rate of fall of the plate decreases.

   (b) The thing to remember here is that the production of an induced emf (which results in the second magnetic field to interact with the applied magnetic field to produce a force) is determined by the rate at which magnetic flux is cut. If the sheet of metal is totally within the magnetic field then the magnetic field through the sheet is constant. So there is no change in flux so there will be no induced emf and therefore no induced eddy current and therefore no second magnetic field and therefore no Lenz's law force upwards. The plate will accelerate through the field with gravitational acceleration.

   (c) As the plate moves out of the magnetic field there will be an induced emf which will result in an eddy current in the plate because the amount of field through the plate decreases the further it passes out of the field – the flux through the plate is changing. This eddy current induces a magnetic field which interacts with the static field to produce a Lenz's law force to oppose the motion of the plate – i.e. the Lenz's law force is upwards, against gravity, so the rate of fall of the plate decreases.

9. (a) Faraday's laws of electromagnetic induction and Lenz's law which is really an application of the law of conservation of energy.

   (b) So that the law of conservation of energy is not violated.

10. X would fall fastest as the slots will prevent the formation of large eddy currents. Slots in Z are parallel to the eddy currents, so they will form in the metal 'fingers' and interact with the magnetic field to produce a force to oppose the fall of the sheet. Z will therefore fall more slowly than X. Because Y is a solid plate, eddy currents will be at maximum value so the upwards force will be the largest. Plate Y will fall the slowest.

11. (a, b) It can be argued that because rings X and Z are not complete, there is no circuit for eddy current to form, so all three rings will fall at the same rate.

    It could also be argued that as they fall, the rings are moving parallel to the magnetic field and therefore not cutting magnetic field, and therefore there will be no induction in any rings as it falls.

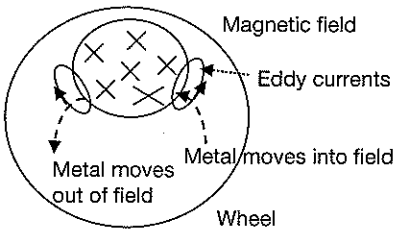
    It can also be argued however, that at the start of their fall the rings do cut magnetic field (the field curves outwards from the north pole and it therefore perpendicular to the direction of motion initially) and therefore an eddy current will certainly form in ring Y but not the other two rings as they do not have complete circuits – so ring Y will certainly fall more slowly initially than rings X and Z and therefore reach the bottom slightly behind those two rings.

    However, initially there will be a charge separation in rings X and Z as their fall cuts magnetic field initially, so while a current will not form, there is charge movement making the sides of the gaps oppositely charged initially. This charge movement will interact with the magnetic field to produce a Lenz's law type force to slow rings P and R very slightly – so it can also be argued that on this consideration, Z will reach the ground first, X second and Y last.

12. (a) The relative motion of X through the magnetic field will induce eddy currents in the disc which will induce a second magnetic field which will interact with the applied field to produce a force which opposes the motion (Lenz's law). So X slows down.

    (b) Y will also slow down, but will take significantly longer than X to stop.

    (c) There will be two eddy currents , one at the leading edge of each disc (exiting the magnetic field) and one at the trailing edge (entering the magnetic field) – see diagram. The directions are obtained using the RHGR for the induction of an emf in a solenoid (coil) and Lenz's law. The diagram looks at the wheels straight on from the right (magnetic field into the discs).



    (d) Eddy currents in Y will be in the same directions as in X but not continuous because of the spaces between the 'teeth' in the disc when no currents will be induced.

13. (a) Z first, then X then Y.

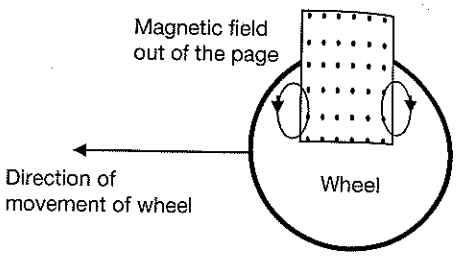
    (b) The rings all fall at the same rate, therefore they all cut magnetic field at the same rate. Therefore the eddy currents induced in each will be the same and the opposing electromagnetic force (Lenz's law force) on each will be the same. The thicker rings will have more mass, so this force will have less effect on them.

14. (a) Clockwise

    (b) No induction

    (c) Anticlockwise

15. Above the magnet clockwise eddy currents will be induced in the cylinder to induce a magnetic south pole to oppose the movement of the magnet downwards. Below the magnet anticlockwise eddy currents will be induced in the cylinder to induce a magnetic south pole to oppose the movement of the magnet downwards. The net result because the magnet will keep falling, albeit slightly slower, will be spiralling eddy currents in the directions stated above and below the magnet.



16. (a) There will be two sets of eddy currents induced in the wheel. At X on the diagram, the induced current will be anticlockwise to replace the decreasing magnetic field as the wheel moves out of the field and at Y it will be clockwise to oppose the increasing field out of the page as the wheel moves into the field. The directions of these eddy currents are such that the magnetic field they induce in the wheel opposes the changing magnetic field (from the applied field) as the wheel moves out of or into the applied field.

(b) The wheel will slow down and eventually stop.

C Induction cooktops

17. AC currents in the coils under the cooktop induce constantly changing magnetic fields which induce eddy currents in the base of the saucepan.

18. Iron will be the best. Aluminium will not work very efficiently and copper will not work at all. The base of the saucepan has to be metallic in order to induce eddy currents, but also has to have a relatively high resistance so that the electrical energy in the eddy currents is converted to sufficient heat energy to cook food. The iron based saucepan will also be more efficient because iron is a magnetic substance and will result in the production of stronger magnetic fields internally and therefore larger eddy currents.

19. No hot coils, base of saucepan is much cooler than conventional hotplates, take saucepan away and cooktop is cool. In induction cooktops temperatures reach only about 100°C and heating occurs only within the saucepan base, so the only energy lost is radiated from the saucepan itself whereas in conventional cooktops, the hot rings, at temperatures up to about 300°C to 400°C radiate much more wasted energy all the time as well as the hot saucepan.

20. No wasted energy into surroundings, temperature of saucepan base much easier to control, less energy needed to heat saucepan, base coils do not need to get hot, so no heat loss there either.

21. No. Changing emf is needed to induce a changing magnetic field around the coils and this changing magnetic field is needed to induce the eddy currents in the saucepan base. A constant magnetic field induced by DC current would not induce eddy currents.

22. B

23. C

24. B

25. A

26. D

Set 41 Back Emf in Motors

1. (a) A back emf is an induced potential difference which opposes any supply potential difference to a motor.

(b) In all motors (AC or DC) and coils of wire which have an AC supply emf connected to them.

(c) Back emf is an example of Faraday's law of induction. It is produced when the coil of a motor rotates within the magnetic fields of the motor.

(d) Any current carrying coil moving inside a magnetic field will have an emf induced in it (Faraday's law). The direction of this emf will be such that it opposes the change causing it (i.e. the motion caused by the supply emf – Lenz's law). The induced emf will therefore oppose and reduce the supply emf. The law of conservation of energy dictates that back emf must be in the opposite direction to supply emf.

(e) The back emf will reduce the net effective supply emf, and therefore the current flow through the coil. This means the rate at which the motor coil accelerates decreases until at maximum speed, when the back emf equals the supply emf, the motor runs at constant speed.

(f) Back emf cannot be reduced. Because it is a consequence of the rotation of the coil of a motor, then as long as the coil rotates, a back emf will be produced. However, you may say that the slower the coil rotates, the smaller the back emf will be.

(g) The soft iron core intensifies the magnetic field in the rotor space in a motor and therefore will increase the rate of rotation of the motor coil. There will therefore be a stronger induction of back emf.

(h) If the speed of the motor decreases then the rate at which its coil is cutting flux also decreases so the induced back emf will decrease.

2. (a) The back emf reduces both the supply emf and an included protective resistor to zero ohms as it reaches maximum speed. If the motor jams and the protective resistor is not effectively in the circuit, then the suddenly increased net supply emf can result in excess current in the coil and can cause it to heat up and perhaps burn out.

(b) Motors often contain a variable internal resistor which reduces supply current when the motor coil is not turning (to prevent the coil overheating and burning out) and reduces to zero resistance when the motor is operating normally and is producing a back emf. The back emf effectively assumes the role of this resistor. If the resistor 'kicks in' as soon as the motor jams, then it will be protected.

3. (a) The back emf.

(b) If the speed of the motor decreases then the rate at which its coil is cutting flux also decreases so the induced back emf will decrease.

4. Because there is no changing magnetic field in the DC circuit, there will be no induction in the coil, so the brightness of the globe is not affected. In the AC circuit, the AC current induces a changing magnetic field which the soft iron core intensifies and this changing field induces a (back) emf in the coil. This opposes the supply emf, so the brightness of the globe decreases.

5. Any current carrying coil moving inside a magnetic field will have an emf induced in it (Faraday's law). The direction of this emf will be such that it opposes the change causing it (i.e. the motion caused by the supply emf – Lenz's law). The induced emf will therefore oppose and reduce the supply emf. The voltage across the coil will therefore be less than 240.

6. (a) The relative motion of the coil (produced by the interaction of the induced magnetic field in the coil due to the supply current interacting with the stator magnetic field) in the stator magnetic field induces an emf in the coil (Faraday's law) to oppose the change causing it (Lenz's law). This reduces the net supply emf in the circuit.

   (b) From $V = RI$, the net supply emf = 2.0 × 3.5 = 7.0 V. So the back emf = 5 V.

7. As motor slows down back emf decreases, which allows for a greater net supply emf and a greater current through the coil. The motor coil in the drill will get warmer. The drill may overheat.

8. (a) A changing current in a conductor (coil X) induces a changing magnetic field around it, and this in turn induces an emf in coil Y.

   (b) Coil X has a decrease in current as the iron core intensifies the back emf. Coil Y has an increase in current as the iron core in it intensifies the magnetic field produced in it by induction.

   (c) Both cores would get warm due to the induction of eddy currents in them and the interaction of these eddy currents with their resistance to convert electrical energy into heat energy ($P = I^2R$).

9. C

Set 42 Transformers

1. (a) To step voltages up or down.

   (b) A step-up has more turns in secondary to boost voltage for reducing losses in long distance transmission. A step-down has more turns in primary coil so reduces voltage to the needed level in home appliances.

   (c) No changing magnetic field so no current can be induced nor voltage changed.

   (d) Step-up and step-down.

   (e) See (b).

2. (a) Step-up.

   (b) 9375

3. (a) Step-up.

   (b) 400 V

   (c) 5 A

4. (a) 300

   (b) 80 mA

5. (a) 24 : 500

   (b) 500 : 24

   (c) The 5000 V coil.

   (d) Microwave ovens use step-up transformers to generate the microwaves (3000 to 5000 V).

6. (a) Is actually correct physics, but does not refer to household circuits as required by the question.

   (b) Appliances are built to work most efficiently at the level of power consumption they are designed for. It would be potentially dangerous to cause them to run on higher power ratings as they would overheat and possibly fail, or cause a fire.

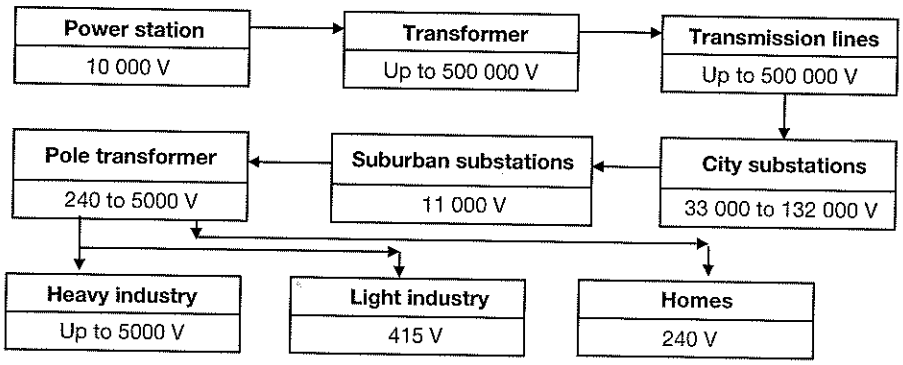
   (c) Is correct statement.

   (d) When voltage is stepped down, current is proportionally increased, so power input and output from a transformer is equal provided it is an ideal transformer – using them does not save power. Internal resistors may be used in them to reduce the current delivered to the appliance, so yes, the appliances will use less energy, but energy will be wasted inside the transformer (converted to heat energy) – power overall must be conserved (law of conservation of energy).

7. C
8. A
9. B
10. B
11. B
12. D
13. C
14. A
15. B

16. D
17. C
18. A
19. C
20. A

Set 43 Transformers and Electricity Transmission

1. Transformers are used at generating stations to step up transmission voltage to mimimise energy losses by heating effects during transmission. Step-down transformers lower the voltage at the consumer end to safer and more usable sizes.
2. A
3. D
4. 

5.

Transmission voltage	400 kV	300 kV	200 kV	100 kV
Current in lines ($I = P/V$)	100 A	133.3 A	200 A	400 A
Power loss ($P = I^2R$)	1 250 000	2 221 111	5 000 000	20 000 000
Loss (%)	3.13	5.55	12.5	50

6. C
7. (a) 0.83 A
   (b) 9960 V
   (c) 240 040 V
   (d) 8267 W
8. (a) Six – there are six output connection terminals plus the common (bottom) terminal.
   (b) (i) 350 V
   (ii) 300 V
   (iii) 200 V
   (iv) 100 V
   (c) 250 V
9. A
10. Answers will vary, for example:

Atmospheric pollution has increased enormously as demand for electrical power increased over the years. This will have contributed to asthma problems for many people and reduced the quality of their lifestyle, diverting money into medical expenses and reduced spending on other areas such as leisure activities or consumables.

Pollution from the burning of fossil fuels to work steam turbines produces heat pollution as well as air pollution. Heat pollution from coolant water into natural streams has affected water life adversely. This reduces the ability of all organisms in the waters to survive and affects the quality of the water. This in turn reduces the enjoyment many people get from water activities (fishing, swimming) and may have long-term effects on larger food chains.

Ugly power lines cross the countryside, power poles and lines in streets also offer impact sites for accidents so increasing the death toll from transport accidents. Ugly transformers in substations and high voltage transmission systems may produce possible harmful high frequency electromagnetic radiations associated with high voltage transmission.

The development of hydroelectricity schemes impacted severely on specific areas of the environment, requiring huge dams, flooding of forests, valleys and even towns. Wind turbines are controversial too – the blades kill birds and bats. Photovoltaic arrays cover arable land.

11. Answers will vary, for example:

The major impact transformers have had on society centres on their use to transmit high voltages/low current electricity to consumers and step it down at the consumer end for factory or domestic use. Lifestyles of people have changed particularly for people in isolated areas that have grid power connected because of this ability to transmit over long distances economically.

People have power tools and electrical appliances to enable work to be done more quickly and efficiently in homes and workshops. Handymen can do their own repairs and make things saving money on hiring others to do some jobs. Money saved can be directed towards holidays and other lifestyle activities, e.g. leisure, education, clothes, food, computers, electronic games, TVs. Families can spend less time on household chores, having electrical devices to do many of these for them, e.g. blenders, vacuum cleaners, microwave ovens, washing machines – all of which save time and enable them to spend their time on other activities or with their children, thus improving the quality of family life.

Electronic devices enable people to access data from all over the world, giving school assignments a totally new dimension and expanding the educational resources available to children enormously.

Both private and business transactions have expanded enormously via the internet. The variety and availability of all sorts of products has expanded from a local shopping centre to a worldwide market for both sellers and buyers. These opportunities increase business profitability and buyer satisfaction and improve lifestyles for all.

Set 44 The DC Generator

1. (a) To act as a source of DC emf (current).

(b) Mechanical energy is changed into electrical energy.

(c) All generators work on the principle of electromagnetic induction where if there is relative motion between a conductor and a magnetic field, an emf will be induced in the conductor.

(d) The rate at which the rotor turns, the strength of the applied magnetic field, the number of turns in the rotor coil.

(e) The volt.

2. (a) P = coil

Q = lead connecting to external circuit (this acts as the negative terminal of the generator)

R = insulating disc

S = split ring commutator

T = rotor axle

U = electrical connection to the external circuit (positive terminal)

V = carbon brush

W = applied magnetic field

(b) The split ring commutator changes the AC emf induced within the generator coil to a DC emf for delivery into the external circuit.

(c) Emf would be more constant in value.

3. (a) 2 V because each turn contributes independently to the emf generated.

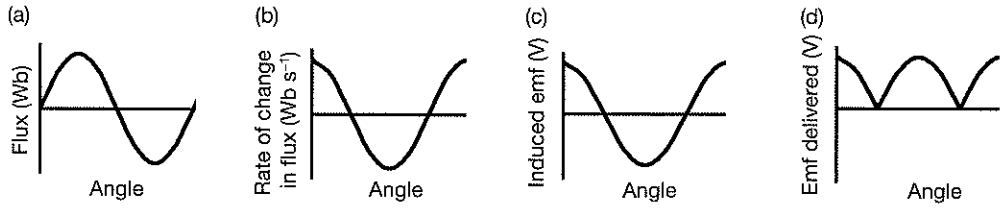
(b) 2 V because flux will be cut at twice the rate and emf = rate of cutting flux.

(c) 0.5 V because emf is directly proportional to the strength of the magnetic field used.

(d) Greater than 2 V because the length of the conductors in the coil cutting field would be longer.

(e) Emf would be larger as both coils would contribute to the emf produced (it will not be doubled) and more constant in value rather than fluctuating from zero to maximum.

4.



5. A

6. A

7. B

8. D

9. (a) Generator X is rotating twice as fast as generator Y but producing only half the emf. Generator X must therefore have only 25% of the number of turns in its coil as generator Y, or perhaps a coil with 25% the area, or perhaps an applied magnetic field only one quarter as strong, or some combination of these three factors which results in the emf being only half that of Y.

(b) Generator Y is certainly rotating at only half the speed of generator X. Its greater emf is due to structural differences as in (a).

10. D

11. C

Set 45 The AC Generator

1. (a) We know it is AC because of the presence of the slip rings, and we know it is a generator by the presence of Q, an input handle for mechanical energy.

(b) P Magnet
Q Handle to input mechanical energy
R Coil
S Magnet
T Slip rings
U Carbon brushes

(c) Mechanical energy into electrical energy.

2. D

3. (a) The pointer would oscillate back and forth indicating the production of an alternating current in the coil.

(b) As above, indicating the production of an alternating emf.

4. A

5. Graph X indicates about five complete rotations to every three rotations of Y, so X is turning faster than Y. Amplitude of graph for X is about the same as for Y, so, accepting that X is turning faster than Y, it must have fewer turns in its coil.

6. Generator X output shows 5.25 rotations in 0.01 s, so its frequency will be 525 Hz.

Generator Y shows 3.0 rotations in 0.01 s, so its frequency will be 300 Hz.

7. For P, the coil is parallel to the magnetic field (cutting field at maximum rate).

For Q, the coil is perpendicular to the magnetic field (not cutting field at this instant).

8. 5.25 : 3.0 = 21 : 12 = 7 : 4

9. The coil in X has fewer turns, or X has a weaker applied magnetic field, or the length of the side of the coil cutting flux is smaller, or some combination of these three factors resulting in an emf the same amplitude as Y.

10. D

11. The gradient of the graph represents the rate of change in flux and it is this that determines the amplitude of generated emf. The gradient is zero at the crests and troughs of the graph.

12. (a) It models a simple AC generator.

(b) The drill turns the coil within the magnetic field and this relative motion induces an AC emf in the coil.

(c) This works on the basis of Faraday's law of electromagnetic induction, which states that whenever there is relative motion between a conductor and a magnetic field, an emf will be induced in the conductor.

13. (a) A

(b) This is when output voltage is the greatest which indicates flux being cut at a maximum rate.

(c) D

14. A

Set 46 Comparing DC and AC Generators and Motors

1. (a) Motors are designed to produce mechanical energy while generators produce electrical energy.

(b) Motors convert electrical energy into mechanical energy while generators convert mechanical energy into electrical energy.

(c) Structurally, a motor and a generator are identical except for the source of their input power. A motor has electrical input whereas a generator has mechanical input. The output and input electrical connections on the generator and motor are identical – in a generator they output electrical energy and in a motor they input electrical energy.

(d) Motors have a supply emf, so the emf generated inside the motor as a result of the relative movement between its coil and the magnetic field, will, by the law of conservation of energy (as expressed in Lenz's law), oppose the supply emf – therefore it is called a 'back' emf. In a generator, the induced emf is the desired product, and in addition, there is no supply emf so it opposes nothing.

(e) The motor is an example of the motor effect, but both also use Faraday's principle of electromagnetic induction (the motor to produce the back emf, the generator to produce the product emf).

(f) The only structural difference is the use of slip rings in the AC generator to deliver AC to the external circuit and a split ring commutator on the DC to change the produced AC into DC for the external circuit.

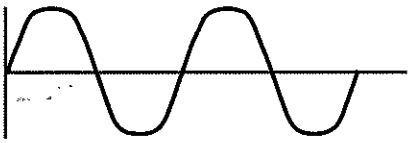
2. (a) There is no evidence in the diagrams to enable this identification – we would need to know the source of the energy supply – mechanical (or electrical) in order to identify them as generators (or motors). Perhaps you could answer that there is no handle evident so assume they are motors.

(b) Both are DC devices. They both have split ring commutators.

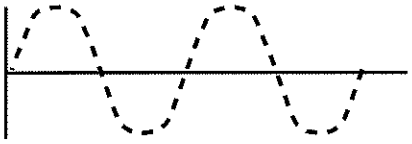
(c) Generator X has two coils at right angles to each other and two sets of split ring commutators – one set for each coil. Apart from that they are identical in structure.

(d) Having two independent coils, Y needs to have electrical contact points for each coil.

(e) (i) Emf produced internally from coil in X.

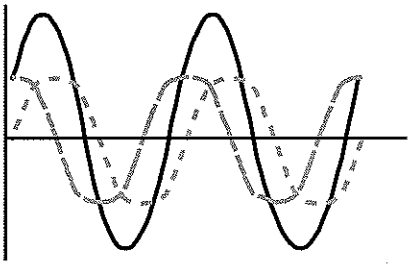


(ii) Emf produced internally from coil Q in Y.

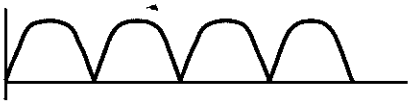


(iii) Emf produced internally from coil R in Y.

(iv) The net emf generated in device Y – this will be the sum of the emf's from each coil Q and R, so we need to superimpose the graphs, remembering that the coils are 90° out of phase with each other.

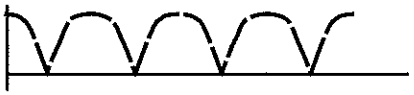


(f) (i) Emf delivered by coil in X.



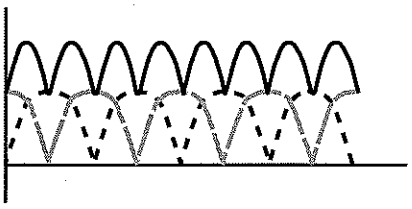
(ii) Emf delivered by coil Q in Y on its own.

(iii) Emf delivered by coil R in Y on its own.



(iv) The net emf generated in device Y.

(Note that the amplitude of the resultant wave is not twice that of each component and that the frequency is doubled).



(g) A commutator is a component in a machine that changes something. This is what the split ring commutator does – it changes produced or delivered AC into DC. However, the slip rings act as pure electrical contacts only, they change nothing.

(h) It is an electrical insulator between the metal axle of the motor/generator and the metal split ring commutator. Without it, the metal axle would short circuit the circui and no current would pass through the coil.

Set 47 The AC Induction Motor

1. An induction motor is an AC electric motor in which the electric current in the rotor needed to produce torque is induced by electromagnetic induction due to motion of the magnetic field of the stator windings relative to the rotor.

2. The most common AC induction motor is the squirrel cage motor. In this motor, the rotor consists of a cylinder made of copper or aluminium bars connected by two end rings and encased in a laminated iron armature. The end rings connect the bars together in series and allow current to flow from one side of the cage to the other. The bars of the squirrel cage are at an angle of about 30 degrees to the axle to keep the torque more uniform as the cage rotates.

3.
- A rotating magnetic field is set up in the electromagnets in the stator with the same frequency as the supply AC. The stator field magnets change polarity as the AC supply changes phase.
- This rotating field over the bars has the same effect as the bars rotating in the opposite direction between fixed magnets, and induces currents in the bars.
- These currents induce a second magnetic field around the squirrel cage bars which interacts with the rotating stator field.
- The direction of this interaction (by Lenz's law) is such that it opposes the relative motion of the bars backwards – i.e. the magnetic field induced around the bars tries to stop the stator field rotating.
- It cannot do this, so, by Newton's third law, the magnetic force pulls the rotor forwards – it 'chases' the rotating stator field.

4. AC induction motors differ from simple AC motors in that there is no coil on the rotor and no direct feed current into the rotor. Because there is no current feed into the rotor, there are also no slip rings required.

5. Induction motors are cheaper to make and run than other motors, require little maintenance and are long lasting. They produce no electrical interference when running. Induction motors are slow to start but the relatively larger velocity of the stator field compared to the slowly moving rotor increases the induced emf and provides a large starting torque.

6. Induction motors have a maximum speed of 1500 rpm, draw a large starting current, are relatively inefficient and large for the power they develop. This reduces as the motor picks up speed.

7. Over 90% of the world's large power output motors are AC induction motors. They are also used extensively in power tools and other electrical appliances (like washing machines, vacuum cleaners, hair dryers, food processors, fans, heaters) where the energy losses are not as economically important.

8. B

9. Bit of a trick question – Faraday for the electromagnetic induction, Lenz for the application of the law of conservation of energy in the system, and Sir Isaac Newton (laws of motion governing the resultant movement of the rotor).

10. D

11. C

12. Remembering that it is *relative motion of a conductor in a magnetic field* that we use to determine directions here, then we have to consider the motion of the field clockwise as being equivalent to the field being stationary, and the squirrel cage rotating anticlockwise. Using the RHPR (field from left to right, relative motion of the bar on the left down the page, and therefore the Lenz's law force acting on it up the page) we get induced emf in this bar out of the page. Applying the RHGR, this gives us an induced magnetic field which is anticlockwise, which is as shown.

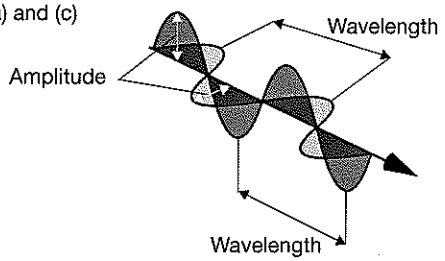
Using the RHPR (field from left to right, relative motion of the bar on the right up the page, and therefore the Lenz's law force acting on it down the page), we get induced emf in this bar into the page. Applying the RHGR, this gives us an induced magnetic field which is clockwise, which is as shown.

Set 48 Electromagnetic Waves

1. (a) Electromagnetic waves are transverse waves made up of mutually induced perpendicular changing electric and magnetic fields.
(b) Electromagnetic waves are produced by moving charges – for example, oscillating electrons in a conductor connected to AC emf.
(c) They are the only waves that can travel through a vacuum in which they all travel at 3×10^8 m s^{-1} which is much faster than any other wave form in any medium.
(d) The electromagnetic spectrum.
(e) Either electromagnetic energy or radiant energy.
(f) Gamma radiation.
(g) Wavelength is inversely proportional to frequency.
(h) Wavelength is inversely proportional to energy carried.
(i) Frequency is directly proportional to energy as per $E = hf$.
(j) They are considered to have both particle and wave nature depending on which property is considered.
(k) Photon.

2. (a) P = X-rays
Q = microwaves
(b) Right hand end (TV waves).
(c) TV waves.
(d) Microwaves.
(e) P
(f) All those to the left – ultraviolet, X-ray and gamma radiation.
(g) Gamma.
(h) Infra-red.
(i) They all travel at the same speed = 3×10^8 m s^{-1}
(j) 3×10^8 m s^{-1}

3. (a) and (c)



(b) The two wavelengths will be the same.

(d) The two amplitudes will be the same.

Set 49 Superconductors

1. (a) A superconductor is a material which conducts electricity with zero electrical resistance.

(b) Metals, non-metals, alloys, organic compounds, ceramics.

(c) Resistance is caused by energy loss due to collisions between current electrons and lattice particles in the conductor.

(d) The resistance decreases because the collisions between current electrons and lattice particles are less energetic.

(e) There will be a temperature at which the resistance of the material drops to zero.

(f) The resistance dropped with temperature as expected, but at a particular temperature for superconducting substances it suddenly dropped to zero resistance rather than continuing a gradual decrease.

(g) Zero.

(h) The critical temperature of a substance is the temperature at which its electrical resistance drops to zero.

(i) It implies this, however, the proposed situation is that collisions still occur, but the energy involved in these collisions is negligible.

(j) Type I superconductors are those that have critical temperature in the order of liquid helium temperature.

Type II superconductors are those that have critical temperature in the order of liquid nitrogen temperature.

Type I superconductors lose their superconductivity when exposed to strong magnetic fields.

Type II superconductors retain their superconductivity when exposed to strong magnetic fields (to a limit).

2. (a) Superconducting technology is only feasible costwise if a room temperature superconductor can be discovered.

(b) Electrical energy would be cheaper as none would be lost in transmission and use due to resistance heating effects. Electrical devices would run cooler and this would reduce fire hazards. If electrical energy is produced and transmitted more efficiently, there will be a decrease in the fossil fuels needed to produce the same amounts. This will benefit the environment and lessen health hazards due to pollution effects.

3. Large scale use is impractical due to the low temperatures needed to sustain superconductivity, and infrastructure costs to replace current technologies would be enormous.

4. D

5. D

6. B

7. A

8. C

9. (a) In conductors above the critical temperature, valence electrons are relatively free to move when a potential difference is applied.

(b) When moving through the metal lattice the electrons collide with atoms and lose energy as heat.

(c) It is these collisions which give rise to the resistance of the material.

(d) It is thought that valence electrons become detached from atoms and are free to move when a voltage is applied.

(e) Lattice atoms become positively charged.

(f) At low temperature a lattice has little movement.

(g) As conduction electrons pass through the lattice there is a Coulomb force of attraction between the lattice nuclei and the electrons.

(h) This Coulomb force distorts the lattice.

(i) This distortion increases the force on the electrons.

(j) Because electrons move so slowly, they now approach each other within the Coulomb repulsive force range.

(k) There is now a Coulomb force of attraction between them.

(l) Electrons moving in opposite directions exchange packets of momentum (phonons).

(m) The phonons hold them together to form Cooper pairs.

(n) The energy of the Cooper pairs is below a critical quantum level (called the Fermi level).

(o) When they interact (collide) with other electrons or lattice particles, there is no transfer of energy.

(p) This holds as long as the temperature remains below the critical temperature for that substance.

(q) The Cooper pairs are the carriers of superconductivity.

10. (a) Yes. Its temperature/resistivity graph shows a sharp drop to zero resistivity.

(b) About 97 K.

(c) Any temperature below the critical temperature, that is 97 K and below.

(d) The discovery shows that the target of a room temperature superconducting material is getting closer. Currently superconductors with critical temperature in the order of liquid nitrogen are used in electronics and MRI machines, and some maglev trains. Higher critical temperature superconductors will make these applications more feasible and cheaper and perhaps open the doors for other immediate applications.

Set 50 Magnetic Levitation

1. (a) Magnetic levitation refers to the supporting of the weight force of an object using magnetic repulsion or attraction.

(b)
- They have a minimum of moving parts which minimises maintenance.
- There is no friction to wear the tracks or wheels.
- They are much faster than conventional systems, needing only to overcome air resistance to travel.
- They do not touch the tracks, so vibrations do not throw the tracks out of alignment.
- Because they do not touch the tracks, acceleration and braking are faster and smoother and they operate much better in icy and wet conditions.
- They are also energy efficient, requiring only half the energy costs per passenger of conventional trains, however, they still require a lot of electricity to produce the strong magnetic fields required.
- They produce less air and noise pollution.

(c) Magnetic attraction. This is shown by the orientation of the magnets below the 'T' of the track. If the mechanism was repulsion, the magnets would be mounts above the track.

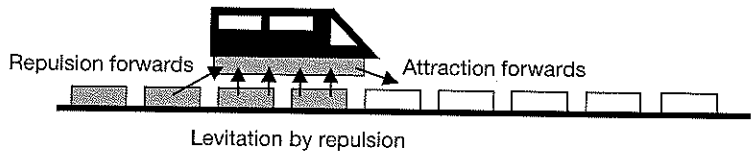
2. (a) Using two magnets, one in the base, one on the car (actually it is visible in the photo) with like poles facing each other. This arrangement however, would result in permanent levitation, so the magnet in the base must be an electromagnet which works only when the current flows.

(b) If the four corners also contain repulsive magnetic poles, then any deviation of the car from the central position will result in a stronger force opposing its displacement from the central position. It will therefore move back until the four corner forces are balanced.

(c) Field decreases in intensity usually as the inverse of distance or distance squared. So, assuming the poles remain at constant intensity, then there will be a particular distance of levitation such that the weight force is balanced by the magnetic forces of repulsion.

3. (a) Like magnetic poles mounted on the track and under the train repel and provide the levitation (see diagram).

(b) Attraction from the magnets in front of the train assist propulsion forwards (see diagram).

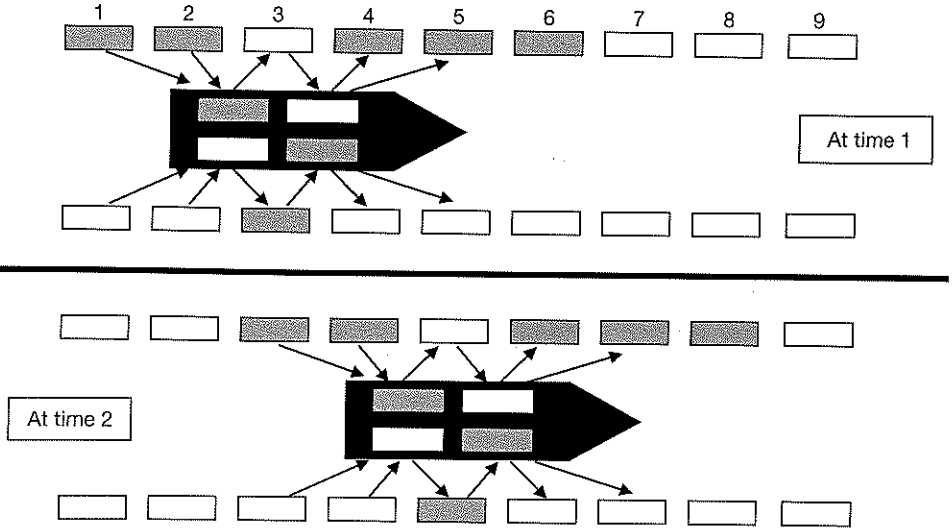


Levitation by repulsion

(c) Repulsion from magnets behind the train assist propulsion forwards (see diagram).

(d) Changing too soon could result in an attraction backwards which would decrease the efficiency of the forwards motion.

4. (a) The arrows on the diagram indicate the forces of attraction between magnets on the track and train at the front of the train puling the train forwards, and forces of repulsion at the back of the train pushing it forwards.



(b) In the time 1 diagram, magnets 3 and 4 need to reverse polarity when the opposite poles in the train are alongside them to ensure forwards motion. In diagram for time 2, magnets 5 and 6 need to change.

5. D

Set 51 Using Superconducting Technology for Maglev

1. (a) The Meissner effect refers to the proposal that superconducting materials below their critical temperatures change from being paramagnetic to being diamagnetic and repel all magnetic fields from within them causing them to levitate on their own magnetic field.

(b) A paramagnetic material allows magnetic field to pass through it.

(c) A diamagnetic material does not allow magnetic field to pass through it.

(d) A superconducting magnet is an electromagnet made from coils of superconducting wire that must be cooled to its critical temperature or below during operation.

(e) The magnetism developed by superconducting magnets is much stronger than ordinary normal conducting electromagnets.

(f) Overall, superconducting magnets are less expensive to operate (than ordinary electromagnets) because no energy is lost with no electric resistance of the superconducting wire during use.

2. SCMaglev trains are much lighter than other high-speed maglev trains and can travel in much longer train sets, increasing energy efficiency levels dramatically. SCMaglev has been estimated to consume 30% less energy than a traditional maglev when travelling at similar speeds, up to 311 mph or 500 km h^{-1}. Further, SCMaglev has been estimated to consume 50% less energy than a Boeing 777-200ER widebody aircraft.

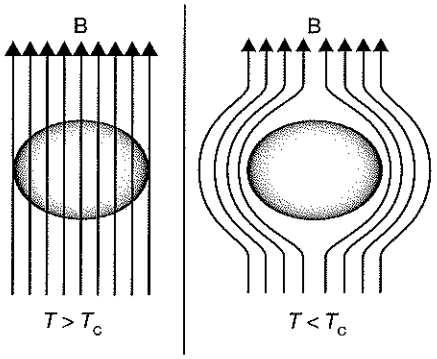
3. A

4. D

5. D is the best answer as we have no indication of the total energy required for both levitation and propulsion.

6. This train (Japan's 'bullet' train) is so shaped to reduce air resistance especially at high speeds. This will allow increased speeds for less expenditure of energy.

7.



8. (a) Because a superconductor below the critical temperature is diamagnetic, as the magnet approaches the superconductor, its magnetic field cannot enter it, so the magnet will be levitated on its own magnetic field.

(b) Because there is relative motion between the magnet and the superconductor, then according to Faraday's law, an emf could be produced in the superconductor, *however*, because the superconductor is diamagnetic, then no magnetic field will penetrate it, so *it is not subject to a changing magnetic field* which is the essential requirement for electromagnetic induction of an emf. Therefore, it would seem that eddy currents will not form within the superconductor, and will not be responsible for the levitation.

9. (a) As the superconductor reaches its critical temperature, it will become diamagnetic and repel any magnetic field from within it. The magnet is therefore levitated as its own magnetic field is excluded from the superconductor.

(b) Initially there is no relative motion between the magnet and the conductor, so no emf will be induced to produce eddy currents. However, as the superconductor changes from being paramagnetic to diamagnetic and the magnetic field is expelled from it, it is subjected to a changing magnetic field. According to Faraday's law, this will result in the induction of an emf. *However*, the direction of this emf, and subsequent eddy currents will be such that the second magnetic field they produce will oppose the motion of the magnet upwards. So in this case, if eddy currents form, they oppose and therefore do not produce the elevation.

10. (a) In general this is true, but given that magnetic fields cannot permeate superconductors at or below their critical temperatures, the statement may not apply to superconductors. (*May* is used here, because we are actually dealing with quantum particles and effects here, so normal rules of physics may not apply.)

(b) Again, the statement is correct for normal conductors, but the argument in (a) applies for superconductors.

(c) According to Faraday's law of induction, this would be a correct statement, although superconductors below their critical temperatures may be an exception.

(d) This is a true statement for all maglev.

(e) This is an incorrect statement for all maglev if we accept that magnetic fields cannot permeate superconductors at or below their critical temperatures.

(f) Accepting that we cannot get induced magnetic fields inside superconductors below their critical temperatures, this statement would be correct.

(g) This is a true statement.

11. C

Set 52 Magnetic Resonance Imaging (MRI)

1. (a) Magnetic resonance imaging is a technique used in radiology to gain detailed computer generated images of internal structures of the body.

(b) MRI provides good contrast between the different soft tissues of the body, which makes it especially useful in imaging the brain, muscles, the heart, and cancers compared with other medical imaging techniques

(c) MRI provides high resolution images of soft body tissue with none of the risks associated with using X-rays or ionised radioisotopes which can both damage that tissue.

(d) Nuclear magnetic resonance imaging (NMRI), or magnetic resonance tomography (MRT).

(e) MRI does not use ionising radiation, so they pose no risk of damage to DNA that can subsequently cause cancer.

(f) The disadvantages of MRI are its high operational cost compared to other scan techniques as well as the cost of the equipment itself (millions of dollars).

2. Any three of the following:

- The strong magnetic fields used result in a loud hammering noise when the machine is in use (up to 130 decibels) which could damage a person's hearing.
- Both CT and MR images may be enhanced by using contrast agents. Contrast agents are chemicals injected into the bloodstream to give higher contrasting images. Commonly used MRI contrast agents may be retained in people with significant permanent or transient kidney dysfunction.
- The strong magnetic fields and radio pulses produced during MRI can affect metal implants, including cochlear implants and cardiac pacemakers. In the case of cardiac pacemakers, the results can sometimes be lethal, so patients with such implants are generally not eligible for MRI.
- MRI machines use superconducting technologies to produce the strong magnetic fields required. The cryogenic liquids used could be extremely dangerous if their containers are broken or accidents occur.
- Patients undergoing an MRI scan have to be placed into a narrow, confined tube. Anyone with claustrophobia may find this uncomfortable.

3. Because they use superconducting coils to produce the very strong magnetic fields needed, it takes a long time to cool the superconducting material to below its critical temperature. If they were turned off and were required for an emergency scan, it would take up to 2 days to get the machine to operating temperatures. So, they are never turned off. This situation will change when higher temperature superconductors are discovered and used in them. The computing components of the machine can, of course, be turned off when not in use.

4. (a) MRI machines make use of the fact that some nuclei behave like small magnetic tops.

(b) Being charged, protons, for example, can be aligned by a strong magnetic field.

(c) When a person is inside the strong magnetic field of the MRI, the magnetic field causes many protons to become aligned with its direction.

(d) A radio frequency electromagnetic pulse is then turned on, producing a varying electromagnetic field.

(e) This electromagnetic pulse has the right frequency to be absorbed by the protons.

(f) The absorbed pulse changes the direction of the spin of the protons relative to the magnetic field.

(g) This frequency is known as the Larmor frequency.

(h) In effect, the protons absorb energy and move to a higher energy level.

(i) After the electromagnetic field is turned off, the spin of the protons returns to their original directions.

(j) The protons then realign with the applied magnetic field.

(k) They are said to 'relax'.

(l) During this relaxation they re-emit the energy they previously absorbed as a radio frequency electromagnetic radiation.

(m) The rate at which they relax depends on the type of tissue the protons are in.

(n) Relaxation times can be measured and plotted by computers connected to the MRI machine.

(o) Because the protons in different tissues in the body relax at different rates, the type of tissue can be determined.

(p) In general the relaxation time is shorter for denser tissues.

(q) The differences in relaxation times for various tissues are known, so they can be easily identified from the scan.

Data Sheet

Acceleration of free fall, g	$9.81\ m\ s^{-2}$
Gravitational constant, G	$6.67 \times 10^{-11}\ N\ m^2\ kg^{-2}$
Avogadro constant, N_A	$6.02 \times 10^{23}\ mol^{-1}$
Gas constant, R	$8.31\ J\ K^{-1}\ mol^{-1}$
Boltzmann constant, k	$1.38 \times 10^{-23}\ J\ K^{-1}$
Stefan-Boltzmann constant, σ	$5.67 \times 10^{-8}\ W\ m^{-2}\ K^{-4}$
Coulomb constant, k	$8.99 \times 10^{9}\ N\ m^2\ C^{-2}$
Permittivity of free space, ε_0	$8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$
Permeability of free space, μ_0	$4\pi \times 10^{-7}\ T\ m\ A^{-1}$
Speed of light in a vacuum, c	$3.00 \times 10^{8}\ m\ s^{-1}$
Planck constant, h	$6.63 \times 10^{-34}\ J\ s$
Elementary charge, e	$1.60 \times 10^{-19}\ C$
Electron rest mass, m_e	$9.110 \times 10^{-31}\ kg = 0.000549\ u = 0.511\ MeV\ c^{-2}$
Proton rest mass, m_p	$1.673 \times 10^{-27}\ kg = 1.007276\ u = 938\ MeV\ c^{-2}$
Neutron rest mass, m_n	$1.675 \times 10^{-27}\ kg = 1.008665\ u = 940\ MeV\ c^{-2}$
Atomic mass unit, u	$1.661 \times 10^{-27}\ kg = 931.5\ MeV\ c^{-2}$
Magnetic constant ($\mu_0 / 2\pi$)	$2 \times 10^{-7}\ T\ m\ A^{-1}$
Wien's displacement constant (b)	$2.898 \times 10^{-3}\ m\ K$

1 light year (ly) = 9.46×10^{15} m

1 parsec (pc) = 3.26 ly

1 astronomical unit (AU) = 1.50×10^{11} m

1 radian (rad) = $\frac{180°}{\pi}$

1 kilowatt hour (kWh) = 3.60×10^{6} J

1 atm = $1.01 \times 10^{5}\ N\ m^{-2}$ = 101 kPa = 760 mmHg

Gravity and Motion

- $w = mg$
  w = weight force, m = mass,
  g = acceleration due to gravity (gravitational field strength)

- $F = \frac{GMm}{r^2}$ and $g = \frac{F}{m} = \frac{GM}{r^2}$
  F = gravitational force, G = universal constant of gravitation (6.67×10^{-11} N m^2 kg^{-2}),
  M = mass of first body, m = mass of second body,
  r = separation between the centres of mass of the two bodies,
  g = acceleration due to gravity

- $v_y = gt + u_y$, $y = \frac{1}{2} gt^2 + u_y t$, $v_y^2 = 2gy + u_y^2$, $v_x = u_x$ and $x = u_x t$
  y = vertical displacement, x = horizontal displacement,
  u_y = initial vertical velocity, v_y = vertical velocity at time t,
  u_x = initial horizontal velocity, v_x = horizontal velocity at time t,
  g = speed of light acceleration due to gravity,
  t = time into flight

- $v = \frac{2\pi r}{T}$
  v = tangential velocity, T = period

- $a_c = \frac{V^2}{r}$
  a_c = centripetal acceleration, v = tangential velocity,
  r = radius of the circle

- $F_{net} = \frac{mv^2}{r}$
  F_{net} = net force, m = mass of body undergoing uniform circular motion, v = tangential velocity,
  r = radius of the circle

- $\frac{T^2}{r^3} = \frac{4\pi^2}{GM}$
  T = period of satellite, M = mass of the central body,
  r = orbital radius, G = universal constant of gravitation (6.67×10^{-11} N m^2 kg^{-2})

Electromagnetism

- $F = \frac{1}{4\pi\varepsilon_o} \frac{Qq}{r^2}$
  F = force, $\frac{1}{4\pi\varepsilon_o}$ = Coulomb constant (9×10^9 N m^2 C^{-2}),
  q = charge on the first object,
  Q = charge on the second object,
  r = separation between the charges

- $E = \frac{F}{q} = \frac{1}{4\pi\varepsilon_o} \frac{g}{r^2}$
  E = electric field strength, F = force,
  q = charge, r = distance from the charge,
  $\frac{1}{4\pi\varepsilon_o}$ = Coulomb constant (9×10^9 N m^2 C^{-2})

- $V = \frac{\Delta U}{q}$
  V = electrical potential difference,
  ΔU = change in potential energy, q = charge

- $B = \frac{\mu_0 I}{2\pi r}$
  B = magnetic flux density, I = current in wire,
  r = distance from the centre of the wire,
  $\frac{\mu_0}{2\pi}$ = magnetic constant (2×10^{-7} T A^{-1} m)

- For a straight, current carrying wire perpendicular to a magnetic field $F = BIL$
  B = magnetic flux density, F = force on the wire,
  L = length of wire in the magnetic field,
  I = current in the wire

- For a charge moving perpendicular to a magnetic field $F = qvB$
  F = force on a charge moving in an applied magnetic field,
  q = charge, v = velocity of the charge, B = magnetic flux density

- $\Phi = BA_\perp$
  Φ = magnetic flux, $A_\perp$ = area of current loop perpendicular to the applied magnetic field,
  B = magnetic flux density

- Emf $= -\frac{n\Delta(BA_\perp)}{\Delta t} = -n\frac{\Delta\Phi}{\Delta t}$
  Emf = induced potential difference, $\Delta\Phi$ = change in magnetic flux, n = number of windings in the loop,
  $A_\perp$ = area of current loop perpendicular to the applied magnetic field, Δt = time interval over which the magnetic flux change occurs, B = magnetic flux density

- $\frac{V_p}{V_s} = \frac{n_p}{n_s}$
  V_p = potential difference across the primary coil,
  V_s = potential difference across the secondary coil,
  n_p = number of turns on primary coil,
  n_s = number of turns on secondary coil

- $I_p V_p = I_s V_s$
  I_p = current in primary coil, V_p = potential difference across primary coil, I_s = current in secondary coil,
  V_s = potential difference across secondary coil